Zeros of polyanalytic polynomials

A consolidated research note,
with exact local fibers and a linear sharp region

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Abstract. We study distinct zeros of polynomials in z and $\bar{z}$ with prescribed analytic and conjugate degrees. An unrestricted splitting recurrence and translated coefficient reduction prove $E_{kr,k(2r+1)} = k^2(2r+1)^2$ for all $r, k \geq 1$. Inversion and circle addition give $M_{m,n} = m^2 + n^2$ throughout $m < n \leq 3m$, except for the line $n = 3m - 1$, whose first three cases are known separately. The main families have integer extremizers with maximal real-axis counts. The bound $\min\{m^2 + n^2, (4m+1)n - 5m\}$ and radial constructions identify the sharp fixed-m asymptotic coefficient. We also classify all fibers of the local splitting model, determine attainable-count spectra, and prove the harmonic distinct minimum $n-m+1$ with a sharp total-fiber-deficit bound. For $z^n - z^k + \varepsilon\bar{z}^m$, $1 \leq k < m < n$, a monotone critical-value argument gives the complete parameter-plane count, the exact maximum $n+2m+k+1$, and the uniform integer maximizer $\varepsilon = 3$. The endpoint $k = m$ has a separate exact maximum and spectrum governed by $\gcd(n-m, 2m)$, including every unit-circle fiber. Divisor formulas determine the sharp three-term maxima in conjugate degrees one and two, including the distinct answer obtained by requiring every zero to be nondegenerate. Sharp factorization deficits classify reducible equality cases and pure-head factor-count strata. Optimizing completion over the proved cone further improves pure-head lower bounds. These family maxima, optimal constants, and asymptotic statements are distinguished from the global bidegree endpoints that remain unresolved. A restriction-map construction also proves $M_{2,n} = 9n - 10$ for every even $n \geq 10$, reduces the large odd-degree gap to two, and completely determines the degree tradeoff for at least eighty-four zeros: $(m, n) = (1, 18), (2, 12), (3, 10)$. A critical shared seed further reduces the degree-seven endpoint to $E_{2,7} \in \{45, 47\}$.

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1 The problem, the literature, and the scope of the conclusions

Allowing a polynomial to depend on both z and $\bar{z}$ changes its zero set fundamentally. It may have no zeros, finitely many isolated zeros, or an entire real algebraic arc of zeros. Even when the zero set is finite, an analytic degree is not by itself the total degree, and different coefficient spaces have different extremal counts. The purpose of this note is to put the successive results of the research discussion into one proof structure, with those distinctions made explicit.

1.1 Notation and the original question

Throughout, the independent analytic variable is written first:

$$f \in \mathbb{C}[z,w], \qquad F_f(z) = f(z,\bar{z}), \qquad f^*(z,w) = \overline{f(\bar{w},\bar{z})}, \qquad N(f) = \#F_f^{-1}(0).$$

The ordered bidegree is $(\deg_w f, \deg_z f) = (m,n)$: the conjugate degree is listed first, although z is written first in $f(z,w)$. For a multivariate polynomial, an unqualified deg denotes total degree. The notation $N(F)$ is used interchangeably with $N(f)$ when $F = F_f$. Every zero count in this note counts distinct points, unless a local algebraic multiplicity is explicitly mentioned. We write $f^{\#}(w) = \overline{f(\bar{w})}$ only for a univariate polynomial.

The original rectangular problem is the following.

Problem 1.1. *Fix $m \geq 1$. For polynomials of bidegree (m,n), $n > m$, with $N(f) < \infty$, does there exist a bound $N(f) \leq C_1(m)n + C_2(m)$ independent of the coefficients and of n?*

As of the source check for this note, Eremenko's maintained problem list marks the non-holomorphic-equation estimate as solved using AI by reduction to the Thom genus inequality [Ere26b]. That status notice is not a replacement for the proof below, and does not certify the stronger finite-degree assertions in this research record.

This is the polynomial conjecture on page 1 of Eremenko's note dated September 7, 2026 [Ere26, p. 1], with the finiteness assumption stated explicitly. It is not the elliptic-system question on the following pages of that note. Two corrections are necessary when quoting the source literally. First, without finiteness, $z^{n-1}(zw-1)$ is a counterexample: its real-slice zero set contains the unit circle. Second, the rough n^2 estimate printed there is not valid for a general rectangle of bidegrees. For $0 < a < 1$,

$$f(z,w) = (z^2 - a^2)w - z$$

has bidegree $(1,2)$ and exactly the five zeros

$$0, \quad \pm\sqrt{1+a^2}, \quad \pm i\sqrt{1-a^2}.$$

For a nonzero zero, division by z gives $|z|^2 - 1 = a^2\bar{z}/z$; its real and imaginary parts give precisely the listed possibilities. The correct rectangular intersection bound is m^2+n^2. The repeated symbol C_1 in the constant term of Eremenko's display is immaterial to the qualitative growth question; we use the two-constant notation of Problem 1.1.

To avoid transferring a sharp result to a larger class without proof, we use the following separate quantities:

Notation	Coefficient class over which the maximum is taken
$M_{m,n}$	Exact bidegree (m,n) and finite zero set. No bound on total degree other than $m+n$.
$E_{m,n}$	$cz^n + \sum_{i+j\le n-1,\ j\le m} c_{ij} z^i \bar{z}^j$, where $c \neq 0$. Its highest homogeneous part is purely analytic.
$M^{\mathrm{harm}}_{m,n}$	$p(z) + \overline{q(z)}$, where $\deg p = n > \deg q = m$.
$S_{m,n}$	The restricted harmonic family $\varepsilon z^n + q(z) + \overline{q(z)}$, where $\varepsilon \neq 0$ and $\deg q = m$.
$A_{m,n}$	$\varepsilon z^n + R(x,y)$, where $R \in \mathbb{R}[x,y]$ has total degree $m < n$.
$K_{m,n,r}$	The support-restricted space in Section 6, with $1 \le m < r \le n$.
s_n	The least number of nonzero monomials among total-degree-n polynomials with n^2 distinct zeros.

The notation $\mathcal{E}_{m,n}$ denotes the coefficient space defining $E_{m,n}$. Its conjugate degree is at most m; all exact constructions below can be arranged to have degree exactly m. The total-degree class $p(z) + Q(z,\bar{z})$, $\deg p = n$, $\deg_{\mathrm{tot}} Q \le m$, is different from both the rectangular class and $\mathcal{E}_{m,n}$.

1.2 Established results and exact source locations

The harmonic problem was raised by Sheil-Small in 1992; the contemporary discussion is in Khavinson–Świątek [KS03, pp. 409–410]. Wilmshurst proved the quadratic upper bound and constructed extremizers when the smaller degree is one less than the larger [Wil98, pp. 2077–2080]. Bshouty, Hengartner, and Suez gave an independent extremal construction [BHS95, pp. 207–218]. Peretz and Schmid independently treated the quadratic zero bound by algebraic methods [PS97, pp. 203–208]; the original page 204 is also explicitly cited in the proof discussion of [SZ24, p. 6]. A direct copy of the Peretz–Schmid article was not available in this audit; this attribution and its locator are reported through those authors' published references, not represented as a page-by-page inspection.

For conjugate degree one, Khavinson–Świątek proved $3n-2$ as an upper bound [KS03, Theorem 1, p. 410], and Geyer proved sharpness [Gey08, Corollary 2, p. 549]. For a rational analytic term of degree n, Khavinson–Neumann established $5n-5$ [KN06, Theorem 1, author version, p. 1]; Rhie's construction gives sharp examples [Rhi03, pp. 3–4]. Singular zeros and degree-dependent refinements were subsequently treated by Luce–Sète–Liesen [LSL15, Theorem 1.1, author version, p. 2] and Liesen–Zur [LZ18, Theorem 1.1 and Theorem 3.5, author version, pp. 2, 7]. Lee–Makarov's prescribed-pole construction [LM14, Theorem C, p. 3] supplies the sharp degree-dependent endpoint recorded in Section 6.6. Those endpoint results are not new here.

Wilmshurst's proposed refinement $3n-2+m(m-1)$ is false in general. Lee, Lerario, and Lundberg constructed counterexamples and explicitly proposed fixed-m linear growth [LLL15, author version, pp. 1–4]. Their stronger conjecture bounds the number of orientation-reversing zeros by $m(n-1)$ [LLL15, author version, p. 3]. Neither that conjecture nor the exact unrestricted harmonic maximum is proved below. Khavinson, Lee, and Saez give further constructions and warn, in effect, against assigning at most one zero to each orientation-reversing component [KLS18, Theorems 2–3, p. 2]. Lee and Saez analyze the regime $m = n-2$ in detail [LS17, unnumbered Theorem, author version, p. 1]; their count is $n^2 - 2n + 2 + 4k_{\max}(n)$ with the trigonometric integer $k_{\max}$ defined there, not just n^2-2n+2. Their theorem explicitly counts multiplicities, a convention that

should not silently be replaced by distinct zeros.

Certified numerical constructions were developed by Hauenstein, Lerario, Lundberg, and Mehta [HLLM15, Section 1.3, author version, p. 3]. Lundberg's probabilistic construction gives examples with at least $\lceil n\sqrt{m} \rceil$ zeros [Lun23, Theorem 1.1, author version, p. 2]. The concrete deterministic question relevant here is in his Section 4.2, page 9 of arXiv:2201.00788v2: can a strategic choice of the coefficients in the monomial-head family produce order mn instead? The exact restricted-family result in Section 8 answers that question, while Section 9 changes the analytic head and improves the constructed linear coefficient.

Average-case asymptotics should be kept separate from worst-case valence. Lerario–Lundberg establish $n^{3/2}$ growth for the truncated Gaussian model with $m = \alpha n$, $0 < \alpha < 1$, and asymptotic n for fixed m [LL16, Theorem 1, author version, p. 3]. For independent identically distributed Kac coefficients, Lu–Wang prove bounds of order n when $m = \alpha n$ [LW25, Theorem 2.1 and inequalities (2.1)–(2.4), author version, p. 4]. The latter article appeared in 2025, although its accepted author manuscript was posted to arXiv in September 2026. Neither ensemble estimate is a coefficient-uniform upper bound on the maximal number of zeros.

For general polyanalytic polynomials, Khavinson, Lundberg, and Perry formulate the bidegree problem and give the coprime intersection estimate $m^2 + n^2$ [KLP24, Problem 1.9 and Proposition 1.10, author version, pp. 3–4]. In the distinct logharmonic subclass with one linear factor, their $3n - 1$ estimate is attained by Lazebnik–Lundberg [LL26, Theorem 1.3, author version, p. 2]. Sète–Zur give a systematic account of finiteness, total-degree bounds, possible zero counts, and constructions [SZ24, Theorems 2.7, 2.12, 2.16 and 4.1, pp. 5–8, 12–14]. Their Remark 4.4 on page 16 conjectures that at least n nonzero coefficients are needed for n^2 zeros; Section 11 gives a three-term quartic counterexample and proves that degree four is the first failure. Their Section 5 concerns a total-degree-restricted mixed class, not the full rectangle.

The algebraic trace method in Section 7 builds on Furchì's Hermitian trace form [Fur25, Theorem 3.13, Corollary 3.14, and Lemma 4.3, author version, pp. 9–10, 16–17]. We prove the additional trace identity and isotropic-space estimate needed here. The surface arguments use the Thom genus theorem of Kronheimer–Mrowka [KM94, Theorem 1, author version, p. 1]; the version allowing only negative ordinary double points is stated explicitly by Orevkov [Ore24, Theorem 3.1, author version, p. 8]. Finally, separated-coefficient arguments have antecedents in the real-rootedness and tropical Descartes theory of Forsgård–Novikov–Shapiro [FNS17, Proposition 1 and Section 2, author proof, pp. 2, 4]. That proposition attributes the sign-independent real-rootedness criterion to Passare–Rojas–Shapiro [PRS11, Definition 1 and Theorem 1, author version, p. 2]. We give the elementary sign argument rather than presuppose independence of the restrictions to different lines.

All references marked "author version" use the specific version in the bibliography; other page references use the published pagination. Publication metadata and mathematical locators were checked separately. Standard textbook citations identify the chapters or propositions actually used, but this audit does not claim visual inspection of every page of every cited book. The literature review concerns the zero-counting questions in this research record; it is not a proof of priority against every unpublished manuscript.

1.3 The principal conclusions

The first conclusion settles Problem 1.1 and identifies its optimal leading constant.

Theorem 1.2. *For $n > m \geq 1$ and every polynomial of bidegree (m, n) with finite zero set,*

$$N(f) \leq \min\{m^2 + n^2, (4m+1)n - 5m\}. \tag{1.1}$$

For every fixed $m \geq 1$, $\lim_{n\to\infty} M_{m,n}/n = 4m + 1$.

The strengthened constructions in Section 5.1 give the following additional main conclusions. With $U_{m,n} = \min\{m^2 + n^2, (4m+1)n - 5m\}$, every $n > m \geq 1$ satisfies

$$U_{m,n} - 4m(m-1) \leq M_{m,n} \leq U_{m,n}.$$

For $n \geq 2m + 1$ the more precise constructed lower count is $(4m+1)(n-m) + (3m-n)_+(3m+1-n)_+$. The exact maximum in the two-frequency family with radial degrees a, b is $k\min\{2(a+b)+1, k+2a-\mathbf{1}_{2|k}\}$ whenever $k + 2a > 2b$. Optimizing the unequal radial degrees also proves $E_{m,4m+2} \geq 9m^2 + 7m + 2$ and the general bound (5.11). The rectangular and pure-head estimates are lower bounds for their full spaces; only the stated two-frequency maximum is asserted to be exact throughout its parameter range.

The finite-degree results now include a linear region of exact rectangular maxima. The stronger pure-head conclusions have a separate range. The new recurrence and inversion proof are entirely analytic; part of the older pure-head boundary $n = 2m + 2$ retains its certificate dependency.

Theorem 1.3. *For $m \geq 2$ and $m < n \leq 2m + 2$,*

$$E_{m,n} = n^2, \qquad M_{m,n} = m^2 + n^2. \tag{1.2}$$

The same equalities hold when $n = 2m + 3$ and $3 \mid m$. More generally, the pure-head equality holds at

$$(m, n) = (kr, k(2r+1)), \qquad r \geq 1, \quad k \geq 1,$$

and remains valid after increasing m while keeping $m < n$. For the rectangular class,

$$M_{m,n} = m^2 + n^2 \quad (1 \leq m < n \leq 3m,\ n \neq 3m - 1). \tag{1.3}$$

The cases $(1, 2), (2, 5), (3, 8)$ also satisfy the rectangular equality. All these extrema admit Gaussian rational coefficients, exact conjugate degree m, and nondegenerate zeros.

Theorem 19.1 removes the earlier restriction $r \geq k - 1$. The translated construction in Theorem 19.16 also closes the pure-head boundary $r = 1$: $E_{m,3k} = 9k^2$ and $M_{m,3k} = m^2 + 9k^2$ for $k \leq m < 3k$. Theorem 19.2 then proves (1.3) by at most two circle additions. For example, $E_{8,20} = 400$, $M_{8,20} = 464$, $E_{4,12} = 144$, $M_{4,12} = 160$, and $M_{10,30} = 1000$. At $m = 10000$ the proved range beyond $n = 2m$ includes every $1 \leq d \leq 9998$ and $d = 10000$, replacing the earlier sufficient range $1 \leq d \leq 46$. The line $n = 3m - 1$, $m \geq 4$, remains unresolved here. The paired-seed construction in Theorem 21.3 strictly improves its first missing lower bound to $129 \leq M_{4,11} \leq 137$. Equality on the rectangular cone is not asserted for E. Corollary 19.39 also proves that the conjugate degrees of the cubic-boundary extremizers are minimal at total degrees $3, 6, \ldots, 24$. Reducing the conjugate degree by one then loses at least six zeros; the possible degree reductions at $n = 10, 11, 13, 27$ remain explicitly bracketed.

For the pure-head class, Corollary 19.18 gives $E_{m,n} \geq 9m^2 + n - 3m$ when $n \geq 3m$, and the uniform deficit $n^2 - E_{m,n} \leq 4n - 6$ throughout $m < n \leq 3m$. Thus its quadratic asymptotic bound is sharp on the entire cone. Theorem 19.25 improves the first pure-head case beyond the cubic

boundary to $88 \le E_{3,10} \le 100$, with an exact eighty-eight-zero construction and ten real-axis zeros. Total degree ten and, at that degree, conjugate degree three are minimal for this target. The global maximum remains open. The critical-seed construction in Theorem 19.37 improves the smaller endpoint to $E_{2,7} \in \{45, 47\}$, with exactly forty-five regular zeros and seven real-axis zeros. Total degree seven and conjugate degree two are minimal for this target. Proposition 19.38 also makes forty-five the sharp maximum throughout a small full coefficient neighborhood of its critical limiting seed. The global endpoint forty-seven remains open. Proposition 19.27 proves that 41 and 88 are nevertheless the exact maxima in sufficiently small full coefficient neighborhoods of the two fixed limiting seeds. Proposition 19.24 gives an explicit integer coefficient table for the forty-one-zero example, with all 49 complex intersections certified at fixed parameter values.

The restriction-map construction in Theorem 19.34 closes an infinite family of global rectangular endpoints: $M_{2,n} = 9n-10$ for every even $n \ge 10$. For odd $n \ge 13$ it gives $M_{2,n} \in \{9n-12, 9n-10\}$; the smaller cases satisfy $M_{2,8} \in \{60, 62\}$, $65 \le M_{2,9} \le 71$, and $85 \le M_{2,11} \le 89$. Theorem 19.35 narrows the pure-head gaps to two in every even degree $n \ge 8$ and four in every odd degree $n \ge 9$. In particular, $61 \le E_{2,9} \le 65$, $90 \le E_{2,12} \le 92$, and $115 \le E_{2,15} \le 119$. Corollary 19.36 also determines the minimum degrees for whole intervals of target zero counts. For the eighty-four-zero threshold, Corollary 19.32 determines every minimum allowed pure degree at a fixed conjugate degree; the complete undominated choices are $(1, 18), (2, 12), (3, 10)$.

For splitting order two, Lemma 19.13 sharpens the coefficient stability radius to the optimal value one. The unadjusted cubic cofactor then gives the explicit analytic family (19.30), proving $E_{2,6} = 36$ independently of the earlier rational-box certificate. The seed and its split have respectively nine and thirty-six simple zeros; the latter also attains the real-axis bound of six.

An independent certified contact-six construction attains 39 zeros in Corollary 22.9. Its seed has exactly 26 regular zeros and all 36 complex intersections are certified. The fixed choice $\varepsilon = 10^{-24}$ in (22.7) also has its own certificate for all 49 intersections and exactly 39 real zeros. This does not determine the rectangular endpoint $M_{2,7}$.

For splitting order three, Theorem 19.10 determines the exact shear disk for every real constant term. The canonical radius is $1 - (3/4)^{1/3}$; the unique optimized constant gives radius $1/4$, with all first-boundary parameters and fold counts explicitly classified.

Theorem 19.3 gives the exact number of preimages of every complex target under the local folding map, including all degenerate targets. Corollary 19.5 and Lemma 19.4 give its unique largest target disk over all complex centers. Dropping an unnecessary constant in the recurrence upgrades the new families and rectangular cone to real rational, hence integer, coefficients. The same extremizers attain the real-axis upper bounds n in the pure-head class and $m + n$ in the rectangular class (Corollary 19.44). These improvements do not close the remaining global degree gaps.

Theorem 20.1 upgrades each known maximal count to a complete spectrum: with nonzero rectangular corner the simple counts are $n - m, n - m + 2, \dots, M_{m,n}$; in the pure-head class they are $n, n + 2, \dots, E_{m,n}$. Allowing degeneracies gives every positive integer up to the respective maximum. *Exact* positive conjugate degree in the pure-head class instead gives precisely $2, \dots, E_{m,n}$ by Theorem 20.8. The fixed-degree harmonic no-gap theorem above n is already due to Sète–Zur [SZ21]. Theorem 20.5 determines the sharp distinct minimum $n - m + 1$ and proves that the total deficit below n, summed over all targets of one harmonic map, is at most $\deg \gcd(p', q')$. Thus its full distinct spectrum is $\{n - m + 1, \dots, M^{\mathrm{harm}}_{m,n}\}$; at $m = n - 1$ this is exactly $\{2, \dots, n^2\}$. Integer

trinomials realize its lower end and have the exact parameter disk in Theorem 20.9. Separately, Proposition 20.25 improves the pure-head lower bound to $E_{m,4m+1} \geq 10m^2+2m+1$. Theorem 17.2 also optimizes the entire real constant-term parameter of the nine-zero cubic, with unique maximal uniform radius $(5\sqrt{5}-11)/2$.

Theorem 13.7 determines the individual unit-triple power constant in every order not divisible by three, with all equality traces. Along these orders, $C_k/k \to \Gamma \approx 1.11562380$; the constant Γ is characterized exactly in Corollary 13.8. Proposition 13.9 also determines C_{12} and C_{15} and their six equality traces, so every individual order through seventeen is settled.

The trinomial parameter analysis extends beyond the first transition. For $1 \leq k < m < n$, Theorem 20.11 proves

$$\max_{\varepsilon\in\mathbb{C}} N(z^n - z^k + \varepsilon\bar{z}^m) = n + 2m + k + 1,$$

with the universal integer maximizer $\varepsilon = 3$. Proposition 20.13 gives the exact count in every regular parameter region and at its fold and cusp boundaries. The full distinct spectrum in this family has step $\gcd(n-k, m+k)$. The optimal outer radius is at most two. The endpoint $k = m$ is completed by Theorem 20.15. Writing $\ell = \gcd(n-m, 2m)$, its maximum is $n+m+1$ when $2m/\ell$ is even and $n+m+\ell+1$ when it is odd. Its full spectrum and all unit-circle fibers are explicit; complex parameters sometimes exceed the real-parameter maximum.

There is also a complete sparsity-restricted result in conjugate degree one. If $D(N)$ is the largest proper divisor of N, then Theorem 20.20 proves

$$H^{(3)}_{1,n} = n + 2 + 2D(n+1) \qquad (n \geq 4).$$

The low-degree values are $H^{(3)}_{1,2} = 4$ and $H^{(3)}_{1,3} = 7$. Thus a harmonic trinomial attains the full $3n-2$ maximum exactly in degrees $2, 3, 5$. For $z^n - z^k + \varepsilon\bar{z}$, $2 \leq k < n$, the exact family maximum is $n+2+2\gcd(n+1, k+1)$, with an explicit open interval of positive rational maximizing parameters.

The support classification now also closes conjugate degree two. Theorem 20.24 proves, for $n \geq 7$,

$$H^{(3)}_{2,n} = \max\{n+6, n+3+2D(n+2)\}, \qquad H^{(3),\mathrm{reg}}_{2,n} = n+6.$$

The four smaller distinct maxima are $9, 10, 11, 13$ in degrees $3, 4, 5, 6$, and the regular formula holds from degree three. For even $n \geq 8$ the distinct maximum is $2n+5$. The underlying separated-component theorem gives $n+m+1+2\gcd(n+m, k+m)$ for $n > k > m$ under the explicit criterion (20.43), which always holds for $m = 2$. All nonzero maximizing zeros are regular; the multiple origin is retained when $m > 1$. Separately, Proposition 22.5 strengthens the pure-head upper bound by $2(q-2m)$ under contact order $q > 2m$. Its maximal-contact degree-one endpoint has sharp value $3n-2$, including singular members, by Corollary 22.10. The threefold-symmetric contact-five subspace has the smaller sharp distinct maximum 19, including singular parameters (Proposition 22.7); an integer example attains it with all zeros regular.

Factorization now has its own sharp theory. For $n > m \geq 2$, Theorem 25.1 bounds every reducible count by $n^2 + (m-1)^2 + 1$ and characterizes equality by a conjugate-linear factor times a saturated neighboring rectangle. For $n \geq m^2+2$, $M^{\mathrm{red}}_{m,n} = M_{m,n-1} + 1$; the reducible fixed-$m$ asymptotic coefficient is also $4m+1$. In the pure-head class, exact conjugate degree m permits at most $n-m$ factors, and Theorem 25.5 gives sharp factor-count strata. The three-candidate optimization in Theorem 20.26 also improves the pure-head constructions below $n = 4m+1$; for example, $E_{100,399} \geq 99005$ instead of the radial count 89799.

For harmonic polynomials, the strongest general estimates established in the record are most compactly stated together. The endpoints $m = 1$ and $m = n - 1$ are classical, as explained above.

Theorem 1.4. *For $1 \le m < n$, put $g = \gcd(m, n)$. Then*

$$(m+2)n - m - g \le M^{\mathrm{harm}}_{m,n} \le \min\left\{ n^2 - 2\left\lfloor \frac{(n-m)^2}{4} \right\rfloor, (3m+2)n - m^2 - 4m - g \right\}. \tag{1.4}$$

Moreover, $M^{\mathrm{harm}}_{1,n} = 3n - 2$ and $M^{\mathrm{harm}}_{n-1,n} = n^2$.

In particular $mn \le M^{\mathrm{harm}}_{m,n} \le 4mn$ uniformly in both degrees. For fixed $m \ge 2$, the lower coefficient in the constructions is $m + 2$, whereas the upper coefficient is $3m + 2 < 4m + 1$. These are not matching constants. The exact formulas for $S_{m,n}$, $A_{m,n}$, and the cyclic family are separate theorems, not formulas for the unrestricted harmonic maximum.

The proof organization follows dependencies rather than the order of the chat. Sections 2–5 establish intersection bounds, regularization, the surface argument, and the matching asymptotic construction. Sections 6–7 refine the upper bounds. Sections 8–11 treat harmonic, real-valued, and sparse constructions. Sections 13–20 prove the exact finite-degree ranges and their attainable-count spectra. The final sections collect exact certificates, an audit of superseded assertions, and the questions not settled by these arguments.

> **Scope of "sharp."** An exact maximum for a specified subspace, an optimal asymptotic coefficient, and a full formula for every bidegree are different claims. This note proves the first two in the places explicitly stated. It does not prove $M_{m,n} = \min\{m^2+n^2, (4m+1)n-5m\}$ for every pair, and it does not determine $M_{2,7}$.

2 Intersection bounds and nondegenerate zeros

The elementary intersection statements in this section will also certify completeness of constructions. They are established before regularization so that later extremal examples do not depend on the surface argument.

At a real-slice point, the real Jacobian is

$$J_f = |f_z|^2 - |f_w|^2 = \det \begin{pmatrix} f_z & f_w \\ \overline{f_w} & \overline{f_z} \end{pmatrix}. \tag{2.1}$$

A zero is *nondegenerate* if this number is nonzero. We denote the numbers of positive and negative nondegenerate zeros by $N_+(f)$ and $N_-(f)$, and put $N_{\mathrm{nd}} = N_+ + N_-$. Rank zero of the real differential is equivalent to $f_z = f_w = 0$.

The following total-degree statement is particularly useful. It also follows from [SZ24, Corollary 2.14, p. 7] together with local intersection multiplicity.

Lemma 2.1. *If $f = z^N + q(z, w)$ with $\deg_{\mathrm{tot}} q < N$, then (f, f^*) is a zero-dimensional complete intersection of length N^2. In particular, $N(f) \le N^2$. If N^2 distinct real-slice zeros occur, every one is nondegenerate.*

Proof. In a graded monomial order, the leading monomials are z^N and w^N. They are relatively prime, so the two polynomials form a Gröbner basis. The residue classes of $z^i w^j$, $0 \le i, j < N$,

form a basis of the quotient, proving the length assertion and finiteness. Equivalently, the projective curves have no common point at infinity. If N^2 distinct real points occur, each local intersection has length one; the two gradients are independent. Its determinant is (2.1), which proves the last assertion. □

For the rectangle we remove the common-factor hypothesis explicitly.

Proposition 2.2. *If f has bidegree (m,n) and finite real-slice zero set, then $N(f) \le m^2+n^2$.*

Proof. Let h be an irreducible factor, of bidegree (a,b). If h^* is not proportional to h, their curves in $\mathbb{CP}^1 \times \mathbb{CP}^1$ are coprime and have intersection number a^2+b^2, as in [KLP24, Proposition 1.10, author version, p. 4]. Every real-slice zero is an intersection.

If h^* is proportional to h, then $a=b=d\ge 1$. Multiplication by a nonzero constant makes $h(z,\bar z)$ real-valued. At an isolated zero its real gradient vanishes, so $h=h_z=h_w=0$. In characteristic zero, irreducibility and the strict drop in w-degree imply $\gcd(h,h_w)=1$. Their bidegrees are (d,d) and at most $(d-1,d)$, and their intersection number is at most $2d^2-d\le 2d^2$.

Write $f=c\prod h_\nu^{e_\nu}$ with distinct irreducible factors of bidegrees (a_ν,b_ν). Each factor has a finite real-slice zero set. Thus

$$N(f) \le \sum_\nu (a_\nu^2+b_\nu^2) \le \Big(\sum_\nu a_\nu\Big)^2 + \Big(\sum_\nu b_\nu\Big)^2 \le m^2+n^2.$$

Univariate factors satisfy the same bound directly, including degree one. This also covers repeated factors. □

A related estimate requires no global finiteness assumption and is useful when a certificate has already supplied many simple zeros.

Proposition 2.3. *Let $f\in\mathbb{C}[z,w]$ have bidegree (m,n), where $n\ge m\ge 1$, and set $D=\gcd(f,f^*)$. There is an integer $0\le\delta\le m$ such that $\deg_z D=\deg_w D=\delta$, and*

$$N_{\mathrm{nd}}(f) \le (n-\delta)^2+(m-\delta)^2. \tag{2.2}$$

Proof. Unique factorization implies D^* is a constant multiple of D, proving equality of its two degrees. Write $f=Du$, $f^*=Dv$, with u,v coprime. At $D=0$ their differentials are multiples of dD, so (2.1) vanishes. Nondegenerate real-slice zeros therefore lie outside $D=0$ and are intersections of $u=0$ and $v=0$. Their bidegrees are $(m-\delta,n-\delta)$ and $(n-\delta,m-\delta)$; the intersection product gives (2.2). □

In particular, more than $(n-1)^2+(m-1)^2$ nondegenerate zeros force f,f^* to be coprime. Their entire complex intersection is then finite. This is stronger than checking only that a list of real zeros is long.

Saturation also excludes factorization of f itself.

Corollary 2.4. *If $n>m\ge 1$ and $N(f)=m^2+n^2<\infty$, then f is irreducible over $\mathbb{C}$.*

Proof. For $f=uv$ with bidegrees (a,b) and $(m-a,n-b)$, Proposition 2.2 gives a strict deficit $2\{a(m-a)+b(n-b)\}$ unless the factor degrees are $(m,0)$ and $(0,n)$, in either order. In that exceptional case the actual univariate counts give $N(f)\le m+n<m^2+n^2$. Thus no nontrivial factorization is possible. □

For comparison, a polynomial of arbitrary total degree D with finite zero set has $N(f) \le D^2$ [SZ24, Theorem 2.12, p. 6]. This estimate uses D, not its analytic degree. The examples in Section 1 show why the distinction cannot be omitted.

3 Removing degenerate zeros

For a polynomial $Q \in \mathbb{C}[z, w]$, write Q_w and Q_z for its complex partial derivatives. The real differential of $F_Q(z) = Q(z, \bar{z})$ is

$$dF_Q = (Q_z + Q_w)\,dx + i(Q_z - Q_w)\,dy, \qquad z = x + iy,$$

where both partial derivatives are evaluated at $(z, \bar{z})$. Hence

$$J_Q(z) = |Q_z(z, \bar{z})|^2 - |Q_w(z, \bar{z})|^2, \qquad \operatorname{rank} dF_Q(z) = 0 \iff Q_w = Q_z = 0. \tag{3.1}$$

We first isolate the local fact needed for the perturbation argument. It concerns real analytic maps, without any polynomial hypothesis.

Lemma 3.1. *Let $F : (\mathbb{R}^2, 0) \to (\mathbb{R}^2, 0)$ be real analytic, with an isolated zero at 0 and $\operatorname{rank} DF(0) = 1$. Let G be real analytic near 0, and suppose that $G(0) \notin \operatorname{im} DF(0)$. For every sufficiently small $t > 0$, the total number of nondegenerate zeros of $F + tG$ and $F - tG$ in a fixed neighborhood of 0 is at least two. This assertion remains valid when G varies in a sufficiently small finite-dimensional neighborhood.*

Proof. A real analytic change of source coordinates and an invertible real linear change of target coordinates put F in the form $F(x, y) = (x, g(x, y))$, with $Dg(0, 0) = 0$. Isolation gives

$$g(0, y) = cy^\ell + O(y^{\ell+1}), \qquad c \neq 0, \quad \ell \ge 2.$$

Write the components of the transformed G as G_1, G_2, and set $\beta = G_2(0, 0) \neq 0$. The first equation of $F \pm tG = 0$ has a unique solution $x = X_\pm(y, t) = O(t)$ near the origin. After substituting $y = t^{1/\ell}u$ and dividing the second equation by t, the result converges in C^1 on bounded u-intervals to $cu^\ell \pm \beta$. For completeness, the mixed terms of g contain a factor x and have order at least two; after substitution they contribute $O(t^{1/\ell}) + O(t)$, as do their u-derivatives. This proves the claimed C^1 convergence.

If ℓ is odd, each limiting polynomial has one simple real zero. If ℓ is even, one of them has two simple real zeros. The implicit function theorem preserves those zeros and the nonvanishing Jacobian of the two-dimensional system. All estimates and simple-zero conditions are uniform for G sufficiently close to the given one. □

A constant perturbation need not preserve the number of isolated zeros. The following use of derivatives avoids that difficulty and also handles repeated factors. Coefficient neighborhoods refer to the usual finite-dimensional topology.

Lemma 3.2. *Let $Q \in \mathbb{C}[z, w]$ have finitely many zeros on $w = \bar{z}$. Every coefficient neighborhood of Q contains a polynomial R with*

$$\deg_w R \le \deg_w Q, \qquad \deg_z R \le \deg_z Q, \qquad N(R) \ge N(Q),$$

such that every zero of F_R is nondegenerate.

Proof. Let

$$S_Q = \{z : Q(z,\bar{z}) = Q_w(z,\bar{z}) = Q_z(z,\bar{z}) = 0\}, \qquad Q_\xi = Q + \xi_1 Q_w + \xi_2 Q_z.$$

By (3.1), S_Q is precisely the set of rank-zero zeros, and it is finite. At a zero of $Q_\xi(z,\bar{z})$ with $z \notin S_Q$, at least one of Q_w, Q_z is nonzero. Thus the differential with respect to $\xi \in \mathbb{C}^2$ is onto $\mathbb{C}$. The parametric transversality theorem [Hir76, Chapter 3, Section 2] shows that, for a residual set of parameters, all zeros outside S_Q are nondegenerate. We may impose this condition simultaneously on ξ and $-\xi$. An exhaustion by compact sets justifies the assertion on the whole plane. Every such polynomial has finitely many zeros: an infinite polynomial zero set contains an arc [SZ24, Theorem 2.7, p. 5], whereas an arc cannot consist of nondegenerate zeros outside a finite set.

We next choose parameters for which one sign does not decrease the number of zeros. At every rank-one zero of F_Q, the map $(a,b) \mapsto aQ_w + bQ_z$ is onto $\mathbb{C}$. We can therefore choose (a,b) so that the value of $aQ_w + bQ_z$ avoids the one-dimensional real image of dF_Q at every such zero. Only finitely many proper real linear subspaces are excluded. Lemma 3.1, applied in disjoint neighborhoods, shows that the two perturbations $Q \pm t(aQ_w + bQ_z)$ contribute at least two nondegenerate zeros in total near each rank-one zero. Rank-two zeros persist for both signs, and all points of S_Q remain zeros for both signs. The local conclusions hold in an open cone about the chosen direction. Consequently, an arbitrarily small parameter ξ can satisfy both these conclusions and the preceding transversality conditions. It then satisfies

$$N(Q_\xi) + N(Q_{-\xi}) \geq 2N(Q). \tag{3.2}$$

There is one more finite collection of restrictions on the direction. At $z_0 \in S_Q$, let H_k be the first nonzero homogeneous term of Q at the complex point $(z_0, \bar{z}_0)$; here $k \geq 2$. Choose (a,b) so that $a(H_k)_w + b(H_k)_z \not\equiv 0$ for every $z_0 \in S_Q$. Each excluded set is a proper complex linear subspace. For every nonzero sufficiently small parameter in a suitable cone,

$$\operatorname{ord}_{(z_0,\bar{z}_0)} Q_{\pm\xi} = k - 1.$$

Choose the sign in (3.2) with at least $N(Q)$ zeros and repeat the procedure.

After one step, every degenerate zero belongs to S_Q. At the points that remain rank zero the complex vanishing order drops strictly at every subsequent step. When the order reaches one, the point either is nondegenerate or is rank one; in the latter case the next step removes its degeneracy. Thus the procedure terminates in finitely many steps. Derivatives do not increase either degree. Choosing all successive perturbations sufficiently small keeps the final polynomial in the prescribed neighborhood of Q. □

We now pass from the affine plane to a compact surface. Set

$$X = \mathbb{CP}^1_z \times \mathbb{CP}^1_w, \qquad \tau(z,w) = (\bar{w}, \bar{z}), \qquad L = \operatorname{Fix}\tau.$$

Let $\alpha, \beta \in H^2(X;\mathbb{Z})$ be the positive generators pulled back from the first and second factors, respectively. Thus $\alpha^2 = \beta^2 = 0$ and $\int_X \alpha\beta = 1$. A polynomial of bidegree (a,b) defines a divisor with class $b\alpha + a\beta$.

The next elementary genericity statement records the hypotheses needed later; in particular, transversality at the fixed set is not left implicit.

Lemma 3.3. *Suppose $a, b \geq 1$. An arbitrarily small perturbation within the space of sections of $\mathcal{O}_X(b, a)$ can be chosen so that its zero curve C is smooth and connected, avoids (∞, ∞), and is transverse both to L and to τC.*

Proof. Here $\mathcal{O}_X(b, a)$ is the tensor product of the pullbacks of $\mathcal{O}_{\mathbb{CP}^1}(b)$ and $\mathcal{O}_{\mathbb{CP}^1}(a)$ from the two factors. The complete linear system separates first jets and separates any two distinct points. Its evaluation map is a submersion, so parametric transversality gives smoothness of C and transversality to L. On $X \setminus L$, the points x and τx are distinct, and the values of a section at those points can be prescribed independently. Apply the same theorem to the two-point evaluation map to obtain $C \pitchfork \tau C$ off L.

At a fixed point, failure of this last transversality would mean that the complex line $T_x C$ is invariant under the antilinear involution $d\tau_x$. An antilinear involution on a complex line has a nonzero fixed vector. Such a vector would belong to $T_x C \cap T_x L$, contradicting transversality to L. Hence transversality holds there as well. Avoiding (∞, ∞) excludes one coefficient hyperplane, namely the vanishing of the $w^a z^b$ coefficient.

Finally, a disconnected smooth divisor would have disjoint irreducible components. For component bidegrees (a_i, b_i) and (a_j, b_j), disjointness gives $a_i b_j + a_j b_i = 0$. All entries are nonnegative, so all components would have to belong to the same ruling, unless there is only one component. Since the total degrees are both positive, disconnection is impossible. □

The proof has a useful support-preserving form. Its importance is that an arbitrary generic perturbation in a larger coefficient space would not justify the sharper estimates for a smaller space.

Corollary 3.4. *Let $V \subset \mathbb{C}[z, w]$ be a finite-dimensional complex linear space stable under ∂_z and ∂_w. If $f \in V$ has finitely many zeros, then every neighborhood of f in V contains $\widetilde{f}$ with $N(\widetilde{f}) \geq N(f)$ and all zeros nondegenerate.*

Proof. Every perturbation in Lemma 3.2 is a complex linear combination of the current polynomial and its two derivatives, so it stays in V. All local and transversality arguments in that proof are unchanged. Polynomial zero sets are semialgebraic, which supplies the same finiteness argument at intermediate steps. □

In spaces with a prescribed nonzero highest coefficient, the coefficient is unchanged by these derivative perturbations. A final small generic perturbation can therefore be made within the same degree class.

4 The linear bound from the genus of a surface

We prove the second term of (1.1). By Lemmas 3.2 and 3.3, it suffices to work with a smooth connected curve $C \subset X$ satisfying all the conclusions of Lemma 3.3, with bidegree (m, n). Set $N = \#(C \cap L)$. The point (∞, ∞) is absent, so this is the number of affine zeros under consideration.

Consider the smooth map

$$\Phi : X \longrightarrow \mathrm{Sym}^2(\mathbb{CP}^1) \simeq \mathbb{CP}^2, \qquad \Phi(z, w) = \{z, \bar{w}\}. \tag{4.1}$$

In homogeneous coordinates it is

$$\Phi([z_0 : z_1], [w_0 : w_1]) = [z_0 \bar{w}_0 : z_1 \bar{w}_0 + z_0 \bar{w}_1 : z_1 \bar{w}_1].$$

The fibers are precisely the τ-orbits. The restriction to C therefore identifies pairs of points, but its behavior at a fixed point requires a separate check.

Lemma 4.1. *The map $\Phi|_C$ is an immersion. Its only multiple points are transverse positive double points, and their number is*

$$D = \frac{m^2 + n^2 - N}{2}. \tag{4.2}$$

Proof. Off L, the map Φ is a local diffeomorphism. Near a point of L choose compatible affine coordinates and put

$$u = \frac{z + \bar{w}}{2}, \qquad v = \frac{z - \bar{w}}{2}.$$

A smooth change of target coordinates puts (4.1) in the form $(u, v) \mapsto (u, v^2)$. If J denotes the original complex structure on X, then $\ker d\Phi = J(TL)$ at L. Since TC is a complex line,

$$TC \cap TL = \{0\} \quad \Longrightarrow \quad TC \cap J(TL) = \{0\}.$$

Thus $\Phi|_C$ is immersive also at fixed points.

Two distinct points of C have the same image exactly when they are $x, \tau x$, with $x \in (C \cap \tau C) \setminus L$. Transversality of C and τC makes the image branches transverse. No fiber has more than two points. To check signs, identify the branch from τx with the branch of τC at x. The map τ reverses the complex orientation of this curve, whereas Φ reverses the ambient four-dimensional orientation off L. These two sign changes cancel, leaving the positive intersection sign of the complex curves C and τC.

Their intersection number is $m^2 + n^2$. Exactly N intersection points are fixed by τ, and the rest occur in pairs. This proves (4.2). □

We now resolve the double points provided by Lemma 4.1. Resolving a double point replaces two disks by an annulus. In local oriented coordinates the positive intersection is modeled on the two coordinate planes in $\mathbb{C}^2$, and the replacement is the usual smoothing $xy = \delta$, rounded near the boundary. Thus the replacement is embedded and preserves the relative homology class. Performing it in disjoint balls produces a smooth closed connected oriented surface $\Sigma \subset \mathbb{CP}^2$.

By the adjunction formula [Har77, Chapter V, Proposition 1.5, p. 361],

$$g(C) = (m - 1)(n - 1).$$

Each annular replacement decreases the Euler characteristic by two. If $H \in H^2(\mathbb{CP}^2; \mathbb{Z})$ is the positive hyperplane class, restriction to the two factors gives $\Phi^* H = \alpha - \beta$. We have therefore proved the following two identities:

$$g(\Sigma) = (m - 1)(n - 1) + D, \qquad \langle H, [\Sigma] \rangle = \int_C (\alpha - \beta) = m - n. \tag{4.3}$$

The remaining input is the Thom genus theorem of Kronheimer and Mrowka [KM94, Theorem 1, author version, p. 1]: a smooth closed connected oriented surface embedded in $\mathbb{CP}^2$ and representing $d[\mathbb{CP}^1]$, $d > 0$, has genus at least $(d - 1)(d - 2)/2$. No holomorphic or symplectic hypothesis is required. Applying this theorem completes the proof of Theorem 1.2.

Proof. Proposition 2.2 gives the quadratic term. Reverse the orientation of the surface in (4.3); its degree is $d = n - m > 0$. The Thom theorem and (4.2) yield

$$(m-1)(n-1) + \frac{m^2 + n^2 - N}{2} \geq \frac{(n-m-1)(n-m-2)}{2}.$$

Rearranging gives $N \leq (4m+1)n - 5m$. The initial regularization and generic perturbation do not decrease the number of zeros, so this bound applies to the original P. □

The proper mapping degree gives a useful arithmetic restriction on extremal counts. It concerns the maximum, not every degenerate polynomial individually.

Corollary 4.2. *The maximum $M_{m,n}$ is attained by a polynomial with nonzero $w^m z^n$ coefficient and only nondegenerate zeros. Moreover,*

$$M_{m,n} \equiv n - m \pmod 2.$$

Proof. The class is nonempty, for instance by $P = w^m z^n + 1$, and its finite counts are bounded integers. A largest count therefore exists. Apply the two perturbation lemmas to a maximizer. Maximality forces the perturbed count to remain equal to $M_{m,n}$. If the highest coefficient is $c \neq 0$, then $F_P(z) = c\bar{z}^m z^n + O(|z|^{m+n-1})$ at infinity. Thus F_P is proper and has degree $n-m$. The signed regular-value count gives

$$N_+(P) - N_-(P) = n - m, \tag{4.4}$$

where N_+ and N_- count the zeros with positive and negative Jacobian. This is the ordinary degree formula [Hir76, Chapter 5, Section 1]. Combining (4.4) with $N(P) = N_+(P) + N_-(P)$ proves the parity assertion. □

5 The optimal asymptotic coefficient

The upper bound just proved has coefficient $4m + 1$ in front of n. A family depending on wz supplies the matching coefficient. Its extremal count can be determined exactly, not only asymptotically.

We first record an elementary consequence of Descartes' rule of signs. For a sequence of nonzero real numbers, let V denote its number of sign changes. The argument is the coefficient-dominance principle used in the study of sign-independent real-rootedness; see [FNS17, Proposition 1 and its proof, author version, pp. 2, 9–10]. The form below permits gaps in the set of exponents, which will also be important for the harmonic construction.

Lemma 5.1. *Let $0 \leq e_0 < \cdots < e_s$ be integers and let $f(r) = \sum_{i=0}^{s} a_i r^{e_i}$ with $a_i \in \mathbb{R} \setminus \{0\}$. Suppose there exist $0 < \rho_0 < \cdots < \rho_s$ such that*

$$|a_i| \rho_i^{e_i} > \sum_{j \neq i} |a_j| \rho_i^{e_j} \qquad (0 \leq i \leq s).$$

The numbers of positive and negative roots of f are, respectively,

$$V(a_0, \ldots, a_s), \qquad V((-1)^{e_0} a_0, \ldots, (-1)^{e_s} a_s).$$

Every nonzero real root of f is simple.

Proof. At ρ_i the sign of f is the sign of a_i. Every change of sign therefore supplies a different root in the corresponding open interval. Descartes' rule bounds the number of positive roots, counted with multiplicity, by exactly this number of sign changes. Thus all positive roots have been found and are simple. Apply the same argument to $f(-r)$ to treat negative roots. □

A single quantitative choice enforces all the hypotheses. If $0 < \mu \le |b_i| \le B$ and $0 < t < 1$, set

$$f_t(r) = \sum_{i=0}^{s} b_i t^{e_i^2} r^{e_i}, \qquad \rho_i = t^{-2e_i}.$$

At the ith radius one has

$$t^{e_i^2} f_t(\rho_i) = b_i + \sum_{j \neq i} b_j t^{(e_j - e_i)^2}. \tag{5.1}$$

The sum in (5.1) has absolute value at most sBt, so $sBt < \mu$ guarantees strict dominance. In particular, this applies to $b_i \in \{-1, 1\}$ and $0 < t < 1/(s+1)$. We also use the immediate consequence of Descartes' rule that a polynomial with $s + 1$ nonzero monomials has at most s positive roots, counted with multiplicity.

We next state the precise result for this family.

Proposition 5.2. *Let $k > 2m \ge 2$. Among the polynomials*

$$P(z, w) = B(wz) + z^k A(wz), \qquad \deg A = m, \quad \deg B \le m, \quad B(0) \neq 0,$$

with finitely many zeros on $w = \bar{z}$, the largest zero count is $(4m+1)k$. It is attained by an irreducible polynomial with only nondegenerate zeros.

Proof. Put $t = |z|^2$. No positive t can be a common zero of A and B, since that would give a circle of zeros. The possible squared radii are exactly the positive roots of

$$\Psi(t) = t^k A(t)\overline{A(\bar{t})} - B(t)\overline{B(\bar{t})}. \tag{5.2}$$

Every such radius gives k values of the argument of z. The two exponent ranges of (5.2) are $0, \dots, 2m$ and $k, \dots, k + 2m$. They are disjoint, so Ψ has at most $4m + 2$ nonzero monomials. The preceding sparse polynomial bound gives $N(P) \le (4m + 1)k$.

Choose $0 < \delta < 1/(4m + 4)$ and set

$$\begin{aligned} B(t) &= \sum_{j=0}^{m} (-1)^j \delta^{4j^2} t^j, \\ A(t) &= \sum_{j=0}^{m} (-1)^{m+1+j} \delta^{(k+2j)^2} t^j. \end{aligned} \tag{5.3}$$

Apply Lemma 5.1 to the ordered exponents $0, 2, \dots, 2m, k, k + 2, \dots, k + 2m$. The coefficients of $B(r^2) + r^k A(r^2)$ alternate in sign; it therefore has at least $2m + 1$ distinct positive zeros. For $B(r^2) - r^k A(r^2)$, each of the two blocks has m changes of sign and there is no change between the blocks. It has at least $2m$ distinct positive zeros.

Applying the same lemma to each block separately shows that all m zeros of $B(r^2)$ with $r > 0$ lie between 1 and δ^{-4m}, whereas all m positive zeros of $A(r^2)$ lie between δ^{-2k} and $\delta^{-2(k+2m)}$. In each case these account for all roots of the degree-m polynomial in r^2 and are simple. Since $k > 2m$, the two intervals are disjoint. In particular, A and B are coprime and the two expressions with

signs + and − have no common positive zero. Their product, up to sign, is $\Psi(r^2)$. The upper bound already proved forces exactly $4m+1$ simple positive roots of Ψ, and hence exactly $(4m+1)k$ zeros of $P(z,\bar{z})$.

At any such zero, $e^{ik\theta} = s \in \{-1,1\}$. The radial derivative of the real part is $(B(r^2) + sr^kA(r^2))' \neq 0$, while the angular derivative of the imaginary part is $skr^kA(r^2) \neq 0$. The two other entries of this polar-coordinate Jacobian are zero. All zeros are nondegenerate.

Finally, choose any simple root b of B. Since $A(b) \neq 0$, $A(t)z^k + B(t)$ is Eisenstein at the prime $t-b$ in $\mathbb{C}[t][z]$. It is primitive because A, B are coprime, and is therefore irreducible in $\mathbb{C}[t,z]$. It remains irreducible after inverting z. The substitution $t = wz$ identifies this localization with $\mathbb{C}[w,z,z^{-1}]$. The polynomial in (5.3) satisfies $P(0,w) = 1$, so it has no factor z and is irreducible already in $\mathbb{C}[z,w]$. □

With $k = n-m$, Proposition 5.2 proves

$$(4m+1)(n-m) \le M_{m,n} \le (4m+1)n - 5m, \qquad n > 3m. \tag{5.4}$$

Dividing (5.4) by n proves the asymptotic assertion of Theorem 1.2. The difference between the two bounds is the constant $4m(m-1)$. For $m = 1$ it vanishes; together with Section 13, this recovers $M_{1,n} = 5n-5$ for every $n \ge 2$.

5.1 An exact theorem for two angular frequencies

The restriction $k > 2m$ in Proposition 5.2 is useful for that construction, but is not necessary for an exact theorem. The two radial degrees can also be different. This flexibility improves the finite-degree lower bounds for both principal coefficient spaces.

For integers $a, b \ge 0$ and $k \ge 1$ satisfying $k + 2a > 2b$, define

$$R(a,b;k) = \min\{2(a+b)+1,\ k+2a-\mathbf{1}_{2|k}\}. \tag{5.5}$$

The indicator in this formula is one when k is even and zero otherwise.

Theorem 5.3. *Among all polynomials*

$$F(z) = B(|z|^2) + z^kA(|z|^2), \qquad \deg A = a, \quad \deg B = b, \quad B(0) \neq 0,$$

with finite zero set, the exact maximum is $kR(a,b;k)$. The coefficients of A, B may be chosen real and rational, with every zero nondegenerate. In particular, when $a = b = m$, the exact answer for every $k \ge 1$ is

$$k\min\{4m+1,\ k+2m-\mathbf{1}_{2|k}\}. \tag{5.6}$$

Sharpness in this theorem refers to the displayed two-frequency space. It is not a formula for the full rectangular maximum.

Proof. Put $t = |z|^2$ and $A^{\#}(t) = \overline{A(\bar{t})}$, with the analogous notation for B. The origin is not a zero. A positive common zero of A, B would produce a circle of zeros and is excluded by finiteness. Thus each positive root of

$$\Psi(t) = t^kA(t)A^{\#}(t) - B(t)B^{\#}(t)$$

corresponds to exactly k zeros of F. The polynomial Ψ has degree $d = k+2a$, positive leading coefficient, and negative constant term. It has at most d distinct positive roots, and at most $d-1$

when d is even: d distinct positive roots in even degree would force a positive constant term. Its support is contained in $\{0,\dots,2b\}\cup\{k,\dots,k+2a\}$, so Descartes' rule also gives at most $2(a+b)+1$ positive roots. These observations prove the upper bound, including multiple roots and complex coefficients.

We construct real coefficients. We use the following elementary form of coefficient dominance, also proved in Lemma 5.1. If $e_0<\cdots<e_s$ are nonnegative integers, $1\le|c_j|\le 2$, and $0<\delta<1/(2s+2)$, then

$$C(r)=\sum_{j=0}^{s} c_j\delta^{e_j^2}r^{e_j}$$

has exactly as many positive roots as there are coefficient sign changes, and these roots are simple. Indeed, at $r_i=\delta^{-2e_i}$,

$$\delta^{e_i^2}C(r_i)=c_i+\sum_{j\neq i} c_j\delta^{(e_j-e_i)^2},$$

and the remainder has modulus less than one. Intermediate values give the lower count; Descartes' rule, with multiplicities, gives equality. The same statement holds for any fixed positive lower and upper bounds on the coefficient magnitudes, after decreasing δ.

First suppose $k>2b$. For a small positive rational δ, set

$$\begin{aligned} B(t)&=\sum_{j=0}^{b}(-1)^j\delta^{4j^2}t^j,\\ A(t)&=\sum_{j=0}^{a}(-1)^{b+1+j}\delta^{(k+2j)^2}t^j. \end{aligned} \tag{5.7}$$

The positive roots of $B(r^2)+r^kA(r^2)$ and $B(r^2)-r^kA(r^2)$ number $a+b+1$ and $a+b$, respectively. The separate blocks also show that every root of A and B is positive and simple, and that their positive radial intervals are disjoint. Thus the two expressions have no common positive root. Their product is $-\Psi(r^2)$, yielding $2(a+b)+1$ simple positive roots of Ψ. This is the minimum in (5.5), also at the boundary $k=2b+1$ or $k=2b+2$.

Suppose next that $k\le 2b$ is odd. The exponents

$$\{0,2,\dots,2b\}\cup\{k,k+2,\dots,k+2a\}$$

are distinct. In increasing order, assign alternating signs and magnitudes δ^{e^2}. Interpret the even exponents as coefficients of $B(r^2)$ and the odd exponents as those of $r^kA(r^2)$. For the two signs in $B(r^2)\pm r^kA(r^2)$, an adjacent pair from different blocks contributes one sign change in total; an adjacent pair from the same block contributes two. There are $a+b+1$ adjacent pairs, of which $k+a-b-1$ belong to the same block. The total positive root count of the two expressions is consequently $k+2a$. Dominance makes every such root simple.

For the remaining case write $k=2h\le 2b$ and $D=a+h>b$. Put $c_j=(-1)^j\delta^{j^2}$ and define polynomials in t by

$$C_+(t)=\sum_{j=0}^{D}c_jt^j,\qquad [t^j]C_-=\begin{cases} c_j, & j<h,\\ -2c_j, & h\le j\le b,\\ -c_j, & b<j\le D. \end{cases}$$

Then C_+ has D positive simple roots, and C_- has $D-1$: there is exactly one missing sign change, between positions $h-1, h$. Set

$$B = (C_+ + C_-)/2, \qquad A = (C_+ - C_-)/(2t^h).$$

These are polynomials of exact degrees b, a, with $B(0) = 1$. Their amplitude polynomial is $-C_+C_-$, giving the required $2D - 1 = k + 2a - 1$ positive roots.

In the last two cases a possible common root of A, B can be removed without losing any of the strict dominance inequalities. Hold A fixed and make an arbitrarily small rational change to $B(0)$. Only finitely many values are forbidden by the roots of A. Degree and nonzero-constant conditions persist, and all positive roots just counted remain simple. Thus the two factors have no common positive root, so their counts add.

Finally, at a constructed zero $z = re^{i\theta}$ one has $e^{ik\theta} = s \in \{1, -1\}$. The real radial derivative is the nonzero derivative of $B(r^2) + sr^kA(r^2)$; the imaginary angular derivative is $skr^kA(r^2) \neq 0$. The other two entries of this polar Jacobian vanish. This proves planar nondegeneracy and completes the attainment proof. □

5.2 A stronger rectangular lower bound in every degree

Two local counting facts make the exact radial theorem useful beyond its own coefficient space. We state the completeness arguments, since preserving some zeros alone would not prove the asserted exact counts.

Lemma 5.4. *Suppose a polynomial map $P(z, \bar{z})$ has a finite zero set, has a unique highest-total-degree monomial, and all its nonzero zeros are nondegenerate. At the origin suppose*

$$P(z, \bar{z}) = c\bar{z}^s + O(|z|^{s+1}), \qquad c \neq 0, \quad s \geq 1.$$

A perturbation using only monomials of total degree less than s replaces the origin by exactly s^2 nondegenerate zeros and preserves exactly one zero near each nonzero old zero. Rational coefficients can be retained.

Proof. First choose $H(z, \bar{z}) = \bar{z}^s + \sum_{i+j<s} h_{ij} z^i \bar{z}^j$ with s^2 nondegenerate zeros and rational coefficients. Here is an elementary construction from the radial theorem itself. If $s = 2r + 1$ is odd, take $cz^s + R(|z|^2)$ with $\deg R = r$ from the disjoint-block case of Theorem 5.3; it has $s(2r+1) = s^2$ zeros. If $s = 2r$ is even, take $cz^s + \bar{z}R(|z|^2)$ with $\deg R = r - 1$. The same radial construction, now with radial exponent $s - 1$, gives $s - 1$ simple positive radii and $s + 1$ arguments at each radius. The origin is a simple zero with negative Jacobian, so the count is $(s-1)(s+1) + 1 = s^2$. The polar derivative test proves simplicity away from zero. In either case conjugate and divide by the leading coefficient to obtain H. This uses no numerical seed. Put

$$P_t = P + c \sum_{i+j<s} h_{ij} t^{s-i-j} z^i \bar{z}^j, \qquad t > 0.$$

After $z = t\zeta$, division by ct^s gives C^1 convergence to H on compact sets. The implicit function theorem produces all s^2 local zeros. To exclude a further scale, shrink a fixed disk until $|P(z)| \geq |c||z|^s/2$. If $|z| \geq Ct$ inside that disk, the perturbation has modulus at most $|c||z|^s \sum_{i+j<s} |h_{ij}| C^{-(s-i-j)}$. Choose C so this sum is less than $1/2$, enlarging it to contain all zeros of H. Every new zero in the disk lies in $|z| < Ct$. Compactness and local uniqueness in the scaled disk now exclude any

additional zeros. The old simple zeros persist uniquely. On the remaining compact region there are no zeros, and the unchanged unique highest monomial gives a uniform nonvanishing estimate near infinity. These facts prove completeness and nondegeneracy. Take t rational when the input coefficients are rational. □

Theorem 5.5. *For $n \geq 2m+1$, $m \geq 1$, there is a polynomial of exact bidegree (m,n) with rational complex coefficients and exactly*

$$L_{m,n} = (4m+1)(n-m) + (3m-n)_+(3m+1-n)_+ \tag{5.8}$$

nondegenerate zeros. Here $x_+ = \max\{x,0\}$. Equivalently, in the range $2m+1 \leq n \leq 3m+1$ this count is

$$L_{m,n} = m^2 + n^2 - 2m(n-2m-1).$$

Consequently, for every $n > m \geq 1$, with $U_{m,n} = \min\{m^2+n^2, (4m+1)n - 5m\}$,

$$U_{m,n} - 4m(m-1) \leq M_{m,n} \leq U_{m,n}. \tag{5.9}$$

The explicit lower bound (5.8) is to be combined with the stronger exact ranges in Theorem 1.3 and the other certified examples. It does not determine $M_{2,7}$.

Proof. Put $k = n-m$ and $s = \max\{0, 3m+1-n\}$. Then $0 \leq s \leq m < k$. Take A, B of degrees $m, m-s$ from the disjoint-block construction (5.7), using exponent $k-s$. It is indeed disjoint: if $s > 0$, then $k - s = 2(m-s)+1$; if $s = 0$, then $k > 2m$. Consider

$$P(z) = z^k A(|z|^2) + \bar{z}^s B(|z|^2).$$

At a nonzero point, division by $\bar{z}^s$ changes the radial equation to $B(r^2) + r^{k-s}e^{i(k+s)\theta}A(r^2) = 0$. There are $4m - 2s + 1$ simple positive radii and $k+s$ arguments at each radius. The same polar derivative check as in Theorem 5.3 proves nondegeneracy. The highest monomial is a nonzero multiple of $z^n\bar{z}^m$. If $s > 0$, the origin has initial term $B(0)\bar{z}^s$; apply Lemma 5.4. All added monomials have total degree less than $s \leq m$, so the bidegree is unchanged. If $s = 0$ there is no origin to replace. The exact total count is

$$(k+s)(4m-2s+1) + s^2 = (4m+1)(n-m) + (6m+1-2n)s - s^2.$$

For $s > 0$, the last two terms equal $s(s-1)$; for $s = 0$ they vanish. This proves (5.8) and its alternate expression.

The upper bound in (5.9) is Theorem 1.2. For $n \geq 2m+1$, use $L_{m,n} \geq (4m+1)(n-m)$ and $U_{m,n} \leq (4m+1)n - 5m$; the difference is $4m(m-1)$. For $m < n < 2m+1$, the algebraic seeds and the transfer construction of Sections 13, 16 give the quadratic maximum. For $m = 1, n = 2$ the five-zero example in Section 1.1 suffices. No numerical input is needed for this uniform sandwich. □

5.3 Optimizing the pure analytic-head construction

The next lemma permits different conjugate degrees in the lower terms and in the highest monomial. That distinction is essential for using unequal radial degrees.

Lemma 5.6. *Let $P(z,\bar z) = cz^u\bar z^v +$ lower-total-degree terms, where $c \neq 0$ and $d = u+v \geq 1$. Suppose all zeros of P are nondegenerate. For an integer $N > d$ and sufficiently small positive ε, $P(z,\bar z) + \varepsilon z^N$ has exactly*

$$N(P) + N - u + v$$

zeros, all nondegenerate. The $N-u+v$ new zeros approach infinity and have positive Jacobian. No bound on the conjugate degree of the lower-total-degree terms is required.

Proof. Set $t = \varepsilon^{1/(N-d)}$ and $z = \zeta/t$. Multiplication by t^d gives the limit

$$\zeta^N + c\zeta^u\bar\zeta^v = \zeta^u(\zeta^{N-u} + c\bar\zeta^v).$$

Its nonzero zeros have radius $|c|^{1/(N-d)}$ and exactly $N-u+v$ arguments. At such a point, writing $q = N-u > v$, the Jacobian of $\zeta^q + c\bar\zeta^v$ is $(q^2-v^2)|\zeta|^{2q-2} > 0$. For $v = 0$ the same assertion follows directly from the analytic derivative. Multiplication by ζ^u preserves its sign. The implicit function theorem supplies the new simple zeros. For $|z|$ large, the original unique leading monomial gives

$$\tfrac{1}{2}|c| \leq \varepsilon|z|^{N-d} \leq \tfrac{3}{2}|c|$$

at every possible zero. Scaled new zeros therefore lie in a fixed compact annulus. Compactness and local uniqueness account for all of them. On a fixed disk, the old simple zeros persist uniquely and compactness excludes additional zeros. This proves the exact count. □

Theorem 5.7. *For $n \geq 2m+1$, the count $(2m+1)n$ is attained in $\mathcal{E}_{m,n}$ by a polynomial of exact conjugate degree m, with rational coefficients and only nondegenerate zeros. For $n \geq 2m+2$, put*

$$a_{m,n} = \min\left\{m, \left\lfloor \frac{n-2m}{4} \right\rfloor\right\}. \tag{5.10}$$

Then

$$E_{m,n} \geq \max\left\{(2m+1)n,\ n + 2(m+a_{m,n})(n-2a_{m,n}-1)\right\}. \tag{5.11}$$

Each of the two counts in the maximum is realized exactly by such a polynomial. In particular,

$$E_{m,4m+2} \geq 9m^2 + 7m + 2. \tag{5.12}$$

Compared with the lower bound (6.5) at the same degree, (5.12) gains $m(m-1)$ zeros. For every $n > m \geq 1$, writing $V_{m,n} = \min\{n^2, (4m+1)n - 2m(m+2)\}$, one also has

$$V_{m,n} - 6m^2 \leq E_{m,n} \leq V_{m,n}. \tag{5.13}$$

These are improved lower bounds and uniform error estimates for the full space; neither (5.11) nor (5.12) is asserted to be its exact maximum.

Proof. Apply Theorem 5.3 with radial degrees $(a,b) = (0,m)$ and exponent $k = n > 2m$. The unique highest homogeneous term is then analytic of degree n, and the exact count is $(2m+1)n$.

For the second construction, write $a = a_{m,n}$ and $k = n-2a-1$. Since $n-2m \geq 2$ and $a \leq \lfloor (n-2m)/4 \rfloor$, one has $k > 2m$. Use the disjoint-block example with radial degrees (a,m) and exponent k. Its count is $(2a+2m+1)k$, its conjugate degree is exactly m, and its unique highest monomial is $cz^{k+a}\bar z^a$ of total degree $k+2a = n-1 > 2m$. Add εz^n. Lemma 5.6 adds exactly $2a+1$ zeros, yielding

$$(2a+2m+1)(n-2a-1) + 2a + 1 = n + 2(m+a)(n-2a-1).$$

The resulting polynomial belongs to $\mathcal{E}_{m,n}$ and has exact conjugate degree m. A sufficiently small rational ε retains rational coefficients.

The quadratic expression in a has its real vertex at $(n-2m-1)/4$. A nearest integer, taking the smaller one in a tie, is $\lfloor (n-2m)/4 \rfloor$; clipping at m gives (5.10). Thus this choice optimizes that construction over the allowed radial degrees. When $n = 4m+2$, direct substitution gives $9m^2+7m+2$ for both parities of m. The earlier expression is $8m^2+8m+2$, so the improvement is $m(m-1)$.

For (5.13), the upper estimate is Theorem 6.2. If $n \le 2m+1$, the algebraic seed gives $E_{m,n} = n^2$. If $2m+1 < n \le 4m+1$, use the first construction and

$$V_{m,n} - (2m+1)n \le 2mn - 2m(m+2) \le 6m^2.$$

If $n \ge 4m+2$, the same completion construction with $a = m$ is admissible, whether or not it is optimal. Its count is $(4m+1)n - 4m(2m+1)$, whose difference from the linear upper bound is $6m^2$. This proves the uniform estimate in every degree. □

For illustration, the following comparison uses the original lower expression (6.5), not a claim about the best result in all other subclasses or in the wider literature.

m	$n = 4m+2$	Original expression	New constructed count	Gain
2	10	50	52	2
3	14	98	104	6
5	22	242	262	20
10	42	882	972	90
20	82	3362	3742	380

The exact result (5.5), the central replacement, and the completion count have analytic proofs for every parameter. The accompanying verification script checks representative rational constructions by exact sign-dominance and coprimality calculations, and checks the integer optimizations over a stated finite range. These checks are diagnostic; the infinite families follow from the proofs above. The unrestricted harmonic coefficient and the unresolved rectangular pair $(2,7)$ are unchanged by these constructions.

6 Bounds that depend on the Newton support

Fix integers $1 \le m < r \le n$ and let $\mathcal{T}_{m,n,r}$ consist of polynomials with $[z^n]f \neq 0$ and support contained in

$$\{(i,j) \in \mathbb{Z}^2_{\ge 0} : j \le m,\ mi + rj \le mn\}. \tag{6.1}$$

Here $[z^n]f$ denotes the coefficient of $z^n w^0$. The convex hull of the allowed exponents is the trapezoid with vertices $(0,0), (n,0), (n-r,m), (0,m)$; for $r = n$ it is a triangle. The coefficient of w^m need not have the largest permitted degree. All estimates below include this possibility.

The upper bound records this sloping edge, rather than only the enclosing rectangle. Put $g = \gcd(m,r)$.

Theorem 6.1. *For $f \in \mathcal{T}_{m,n,r}$ the zero set of F is finite and*

$$N(F) \le \min\{n^2,\ (4m+1)n - m^2 - 4m - (m-1)r - g\}. \tag{6.2}$$

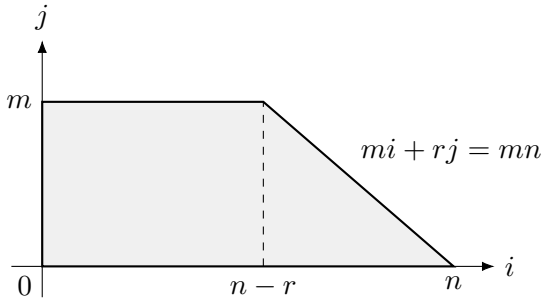


Figure 1: The prescribed support. The slope parameter r records how far the upper right corner is removed. The picture is schematic.

One endpoint of Figure 1 is particularly useful. Let $\mathcal{E}_{m,n}$ be the entire coefficient space

$$F(z) = cz^n + \sum_{\substack{i,j\geq 0,\ i+j\leq n-1\\ j\leq m}} c_{ij} z^i \bar{z}^j, \qquad c \neq 0. \tag{6.3}$$

The only restriction on its top homogeneous part is that it be holomorphic; the lower terms are neither real-valued nor harmonic in general. Denote its largest zero count by $E_{m,n}$. This is a finite integer by Theorem 6.1.

The asymptotic coefficient in this full space is sharp.

Theorem 6.2. *For $n > m \geq 1$,*

$$E_{m,n} \leq \min\{n^2,\ (4m+1)n - 2m(m+2)\}. \tag{6.4}$$

For $n > 4m+1$,

$$E_{m,n} \geq (4m+1)n - 4m(2m+1). \tag{6.5}$$

The lower bound is attained by a polynomial all of whose zeros are nondegenerate. For every fixed m,

$$\lim_{n\to\infty} \frac{E_{m,n}}{n} = 4m+1. \tag{6.6}$$

Indeed, the lattice condition in (6.1) with $r = m+1$ is exactly (6.3). For $1 \leq j \leq m$ it is equivalent to $i + j \leq n-1$; for $j = 0$ it is $i \leq n$. This proves (6.4) from (6.2). The construction proving (6.5) is given in Section 6.5. The optimized lower bound in Theorem 5.7 improves (6.5) in a finite-degree range, and (5.13) gives an all-degree error estimate. The gap between the two older linear expressions is $6m^2$. Thus (6.6) does not assert equality in (6.4) for each pair (m, n).

The other endpoint includes the total-degree class studied by Sète and Zur [SZ24, Section 5]. It is substantially smaller than (6.3).

Corollary 6.3. *If $\deg p = n > m \geq 1$ and $\deg_{\mathrm{tot}} Q \leq m$, then*

$$\#\{z : p(z) + Q(z,\bar{z}) = 0\} \leq \min\{n^2,\ (3m+2)n - m^2 - 4m - \gcd(m,n)\}. \tag{6.7}$$

Proof. Every mixed monomial in Q satisfies $mi + nj \leq n(i+j) \leq nm$. All analytic terms of p also satisfy this inequality. Apply Theorem 6.1 with $r = n$. □

In particular, the second estimate in (6.7) is not restricted to harmonic polynomials. This extension does not identify the maximum in the total-degree class with the harmonic maximum.

Nor does the asymptotic result (6.6) apply to that smaller class: the allowed mixed terms in (6.3) can have degrees tending to n.

For comparison with prior work, the case of conjugate degree one is a rational harmonic equation. Its sharper degree-dependent bounds are due to Liesen and Zur [LZ18, Theorem 1.1, author version v2, p. 2]; sharpness with prescribed pole multiplicities follows from Lee and Makarov [LM14, Theorem C, p. 3]. We record this known endpoint in Section 6.6, rather than count it as a new result.

6.1 Genericity within the support restriction

The inequality $r > m$ makes cz^n the unique highest-total-degree term. Lemma 2.1, after scaling, gives finiteness and $N(F) \le n^2$. The support in (6.1) is downward closed, so Corollary 3.4 applies without enlarging it. Thus we may preserve all zeros through regularization and then make a small generic perturbation in the same support.

A sufficiently small further perturbation preserves any finite set of nondegenerate zeros. We may consequently assume that the affine curve $C = \{f = 0\}$ is smooth and irreducible, that its finite intersections with τC are transverse, and that C is transverse to the fixed set of

$$\tau(z, w) = (\bar{w}, \bar{z}).$$

These are genuine generic conditions in the indicated sparse space. The monomials $1, z, w$ separate first-order jets at affine points and separate two different affine points. They give the required transversality both off and on the fixed set. Irreducibility is a nonempty algebraic open condition: $w^m + z^n + z$ belongs to the space and is irreducible by Eisenstein at z (the case $m = 1$ is immediate). We may also assume the leading coefficient in w has simple zeros, the next coefficient is nonzero there, and the polynomial on the sloping Newton edge has distinct nonzero roots. These are compatible open conditions. All subsequent arguments concern such a perturbation.

6.2 The normalization genus

Write

$$f(z, w) = \sum_{j=0}^{m} a_j(z) w^j, \qquad \deg a_j \le \left\lfloor n - \frac{rj}{m} \right\rfloor.$$

Let $\widetilde{C}$ be the normalization of the closure of C in $X = \mathbb{CP}^1_z \times \mathbb{CP}^1_w$, and let γ be its genus. The projection $z : \widetilde{C} \to \mathbb{CP}^1$ has degree m.

The genus can be bounded directly from the discriminant; no nonsingularity of the curve at infinity is being assumed. Lemma 6.4 provides the precise estimate used in the surface argument.

Lemma 6.4. *For the generic curve just described,*

$$2\gamma \le (2m - 2)n - (m - 1)r - m - g + 2. \tag{6.8}$$

Above (∞, ∞) there are exactly g points of the normalization, with local parametrizations

$$u = t^{m/g}, \qquad v = c_\ell t^{r/g}(1 + O(t)), \qquad c_\ell \neq 0, \quad u = 1/z,\ v = 1/w. \tag{6.9}$$

Proof. The discriminant $\operatorname{Disc}_w f$ is homogeneous of degree $2m-2$ in the coefficients a_j, and homogeneous of weight $m(m-1)$ when a_j has weight j. Therefore

$$\deg_z \operatorname{Disc}_w f \le (2m-2)n - r(m-1).$$

When $m=1$, the discriminant is 1 and this bound is zero. Over a finite value of z with $a_m \neq 0$, its order is the total ramification multiplicity, because the affine curve is smooth. At a zero of a_m, the branch at $w=\infty$ is unramified: in $v=1/w$ the equation is $a_m(z)+a_{m-1}(z)v+O(v^2)=0$, and both $a_m'(z)$ and $a_{m-1}(z)$ are nonzero there. The other branches are generically unramified there as well. Thus the finite ramification contribution is at most $(2m-2)n-r(m-1)$.

Near (∞,∞), multiply $f(1/u,1/v)$ by $u^n v^m$. Put $a=m/g$, $b=r/g$. Its terms of least (a,b)-weight form

$$c\prod_{\ell=1}^{g}(v^a-\lambda_\ell u^b), \qquad c\neq 0,$$

where the λ_ℓ are distinct and nonzero. All other terms have strictly larger weight. Substituting $u=t^a$, $v=t^bV$ and dividing by the least power of t gives a polynomial in V with m simple nonzero roots at $t=0$. The holomorphic implicit function theorem gives the local expansions. Grouping the roots by the action $t\mapsto e^{2\pi i/a}t$ yields exactly g branches and gives (6.9). There are no other points above $z=\infty$. Their total ramification is $g(a-1)=m-g$.

Riemann–Hurwitz now gives

$$2\gamma-2 \le -2m+(2m-2)n-r(m-1)+(m-g),$$

which is (6.8). □

The right side of (6.8) is twice the number of interior lattice points in Figure 1. Only the displayed genus inequality is needed here.

6.3 The quotient surface and its end

The symmetric-product map

$$\Phi : X \longrightarrow \operatorname{Sym}^2(\mathbb{CP}^1)=\mathbb{CP}^2, \qquad \Phi(z,w)=\{z,\bar w\},$$

has fibers equal to the τ-orbits. In affine coordinates it is $[1:z+\bar w:z\bar w]$. Its restriction to the smooth affine curve is an immersion. To check this at a fixed point, use real coordinates $s=(z+\bar w)/2$, $t=(z-\bar w)/2$. The quotient has local form $(s,t)\mapsto(s,t^2)$ and kernel $J(T\operatorname{Fix}(\tau))$. A complex tangent line transverse to $T\operatorname{Fix}(\tau)$ is also transverse to its J-image. Away from the fixed set the quotient is a local diffeomorphism.

The ordinary projective closures of $f=0$ and $f^*=0$ have no common point at infinity and have degrees n,n. In generic position they therefore have n^2 distinct affine intersections. Of these, $N(F)$ are fixed and the rest occur in pairs. The affine image has

$$D_+=\frac{n^2-N(F)}{2} \tag{6.10}$$

ordinary double points. They are positive: the intersection of the two complex curves is positive, transport of the second sheet by the anti-holomorphic involution reverses its two-dimensional orientation, and Φ reverses the ambient four-dimensional orientation. The two changes cancel. There are no triple points, since fibers have size at most two.

If α, β are the positive cohomology generators of the two factors and H is the hyperplane class, then $\Phi^* H = \alpha - \beta$. The coordinate projections on $\widetilde{C}$ have degrees m, n. Thus the image homology class has degree $m - n$. Reversing the source orientation makes it positive, of degree $d = n - m$. This reversal does not change double-point signs.

The finite points of C on $w = \infty$ are smooth, unramified points with distinct images. Their partners under τ are not on C, since C has no point with $z = \infty, w \neq \infty$. They introduce no additional double points. It remains to treat (∞, ∞).

The required filling is elementary at the level of parametrized complex curves. We spell out why its double points have the correct sign.

Lemma 6.5. *The mirror torus link $T(a, -b)$, with $a, b > 0$, bounds $\gcd(a, b)$ immersed oriented disks in a four-ball, with only negative ordinary double points.*

Proof. The normalization of the complex curve $Y^a = X^b$ in a small ball is a union of $e = \gcd(a, b)$ disks, parametrized by $(X, Y) = (t^{a/e}, \zeta_\ell t^{b/e})$. Perturb these parametrizations by sufficiently small holomorphic polynomial maps. Generic such perturbations have nonzero derivative, only transverse double intersections, and no triple intersections. This follows from jet transversality: polynomial variations of sufficiently large degree interpolate the first jets at one point and the values at two or three distinct points. The respective bad conditions have the required real codimensions $4, 4, 8$. All transverse intersections are positive because the disks are complex curves. Near the boundary the unperturbed maps are embedded and transverse to the sphere; their perturbed inverse images of the ball remain disks and the boundary link remains isotopic to $T(a, b)$. Reflect one complex coordinate of the ambient ball. It reverses its orientation, turns every double point negative, and mirrors the boundary link. A collar realizing the boundary isotopy gives the claimed filling. □

We next identify the actual link, including a Newton edge with more than two monomials. Put $a = m/g$, $b = r/g$, so $b > a$. Near $\Phi(\infty, \infty)$ use coordinates

$$X = u + \bar{v}, \qquad Y = u\bar{v}.$$

By (6.9), $|v| = o(|u|)$ and $X = u(1 + o(1))$. For the branch associated with λ_ℓ the boundary on $|X| = \rho$ satisfies, after $Y = \rho^{1+b/a}\eta$,

$$\eta^a = \overline{\lambda_\ell} e^{-i(b-a)\theta} + o(1), \qquad X = \rho e^{i\theta}. \tag{6.11}$$

The convergence is in C^1 in the angular parametrizations. The roots of the limiting equations are distinct and nonzero, so the small error in (6.11) does not alter the braid type.

Move the g distinct nonzero constants $\overline{\lambda_\ell}$ through a path of distinct nonzero configurations to a scaled set of gth roots of unity. The displayed roots never collide, so this gives a link isotopy. The resulting union is

$$\eta^m = Ce^{-i(r-m)\theta}, \qquad C \neq 0,$$

which is $T(m, -(r-m))$. It has $\gcd(m, r-m) = g$ components, in agreement with the g normalization disks. A rounded bidisk in the (X, Y) chart is a four-ball whose boundary contains this braid; by taking ρ small, the curve meets its boundary only on this part. Formula (6.9) also shows that its inverse image consists of g disks. Their induced orientations agree with the complex-parameter orientations of the mirrored torus link. If the whole source is reversed, reverse the orientations of all filling disks as well.

Replace these disks by the fillings in Lemma 6.5, matching collars by the link isotopy. No source handle is introduced, all new double points are negative, and a replacement inside a ball does not change the global homology class. We obtain a closed connected immersed surface of genus γ, degree $d = n - m$, and exactly the positive double points in (6.10).

We use the immersed Thom inequality in the following precise form:

$$\gamma + D_+ \geq \frac{(d-1)(d-2)}{2}. \tag{6.12}$$

The version allowing only negative double points is stated in Orevkov [Ore24, Theorem 3.1, author version v2, p. 8], with attribution to Kronheimer–Mrowka [KM94]. Resolving each positive double point by an oriented annulus increases the genus by one, leaves the homology class unchanged, and reduces the general case to that version. Thus it is (6.12), not an inequality for a singular image, that applies.

Combining (6.8), (6.10), and (6.12) yields

$$N(F) \leq n^2 + (2m-2)n - (m-1)r - m - g + 2 - (n-m-1)(n-m-2).$$

The expression simplifies to the second bound in (6.2). Lemma 3.4 and the preservation of nondegenerate zeros under the subsequent small perturbation transfer it to every initial polynomial. This proves Theorem 6.1.

6.4 An exact count of zeros approaching infinity

Adding a higher analytic monomial changes both the leading homogeneous part and the degree at infinity. In the following situation every added zero can be counted, without an uncontrolled intermediate scale.

Lemma 6.6. *Let $k > m \geq 1$, $r > m$, and let $P \in \mathbb{C}[z, w]$ have degrees at most k, m in z, w, respectively, with $[z^k w^m]P = c \neq 0$. Suppose all zeros of $P(z, \bar{z})$ are nondegenerate. For all sufficiently small real $\varepsilon > 0$, the polynomial*

$$F_\varepsilon(z) = P(z, \bar{z}) + \varepsilon z^{k+r}$$

has exactly $N(P(z, \bar{z})) + r + m$ zeros, all nondegenerate. The $r + m$ new zeros approach infinity and have positive Jacobian.

Proof. The term $cz^k w^m$ is the unique term of total degree $k + m$. All old zeros are confined to a compact set, and each persists uniquely under the small perturbation. Set $t = \varepsilon^{1/(r-m)}$ and $z = \zeta/t$. On compact ζ-sets,

$$t^{k+m} F_\varepsilon(\zeta/t) = \zeta^k(\zeta^r + c\bar{\zeta}^m) + O(t), \tag{6.13}$$

with C^1 convergence. The nonzero zeros of the limit satisfy

$$|\zeta|^{r-m} = |c|, \qquad (r+m)\arg\zeta = \arg(-c) \pmod{2\pi}.$$

There are exactly $r + m$. At such a zero the Jacobian of $\zeta^r + c\bar{\zeta}^m$ is

$$(r^2 - m^2)|\zeta|^{2r-2} > 0.$$

Multiplication by ζ^k preserves its sign. The implicit function theorem applied to (6.13) gives one simple positive zero near each.

For completeness, choose R so large that

$$|P(z,\bar{z}) - cz^k\bar{z}^m| \le \tfrac{1}{2}|c|\,|z|^{k+m} \quad (|z| \ge R).$$

Any zero of F_ε there must satisfy

$$\tfrac{1}{2}|c| \le \varepsilon|z|^{r-m} \le \tfrac{3}{2}|c|.$$

Its scaled coordinate $\zeta = tz$ therefore lies in a fixed compact annulus. Every limit point in that annulus is one of the above $r+m$ simple zeros; compactness and local uniqueness exclude further zeros there. On $|z| \le R$, compactness away from disjoint old-zero neighborhoods excludes all other zeros for small ε. This proves the exact count. □

The lemma applies to the full rectangle of exponents $i \le k, j \le m$. After adding z^{k+r}, this rectangle and the new exponent all belong to (6.1) with $n = k+r$, because $mi+rj \le mk+rm = mn$.

6.5 The optimal coefficient in the support-restricted space

The radial polynomial in Proposition 5.2, with analytic degree $k > 3m$, has $(4m+1)(k-m)$ nondegenerate zeros and a nonzero coefficient at $z^k w^m$. Its coefficients may be chosen real rational. We combine it with the preceding exact count at infinity.

Denote the largest zero count in $\mathcal{T}_{m,n,r}$ by $K_{m,n,r}$. The combination of the preceding lemmas proves a slightly more general asymptotic statement than Theorem 6.2.

Proposition 6.7. *If $n > r+3m$ and $r > m$, then*

$$K_{m,n,r} \ge (4m+1)n - 4m(r+m). \tag{6.14}$$

For fixed m, r with $r > m$, $\lim_{n\to\infty} K_{m,n,r}/n = 4m+1$.

Proof. Put $k = n - r > 3m$. Lemma 5.2 gives $(4m+1)(k-m)$ zeros. Lemma 6.6 adds exactly $r+m$, giving

$$(4m+1)(n-r-m) + (r+m) = (4m+1)n - 4m(r+m).$$

The resulting polynomial lies in $\mathcal{T}_{m,n,r}$ by the observation at the end of Section 6.4. This proves (6.14); dividing it and (6.2) by n proves the limit. □

For $r = m+1$ this gives (6.5). Together with (6.4), it proves (6.6) and completes Theorem 6.2. All constructed zeros are nondegenerate, and the final ε can be chosen positive rational.

The distinction between the two support restrictions is substantive. For example, when $m \ge 2$ is fixed, Corollary 6.3 bounds the leading coefficient for $p+Q$ with $\deg_{\mathrm{tot}} Q \le m$ by $3m+2$, whereas Theorem 6.2 gives the strictly larger optimal coefficient $4m+1$ in (6.3). Both classes have holomorphic highest homogeneous term. That condition alone is therefore insufficient to obtain the lower coefficient.

6.6 A known sharp endpoint and the remaining gap

There is a complete degree-dependent answer when $m = 1$. We include it to locate the new bounds relative to established sharp results.

Proposition 6.8. *For $2 \le r \le n$,*

$$K_{1,n,r} = 5n - 2r - 2. \tag{6.15}$$

In particular, $E_{1,n} = 5n - 6$ for every $n \ge 2$.

Proof. Write $f = p(z) - wq(z)$, with $\deg p = n$, $\deg q \le n - r$. If $q \equiv 0$, the elementary analytic bound suffices. Otherwise, remove a greatest common divisor of degree s, leaving numerator and denominator degrees $a = n-s$ and $b \le n-r-s$. The rational harmonic bound in [LZ18, Theorem 1.1, author version v2, p. 2] gives

$$N(f(z, \bar{z})) \le s + 3a + 2b - 2 \le 5n - 2r - 2.$$

Here $a - b \ge r \ge 2$; denominator zeros are not rational zeros because the reduced pair is coprime.

For equality, apply Lee–Makarov's Theorem C [LM14, p. 3] with rational degree n, one pole of order r at infinity, and $n - r$ simple finite poles. There are $n - r + 1$ distinct poles. Their theorem gives

$$\min\{3n + 2(n - r + 1) - 3, 5n - 5\} = 5n - 2r - 1$$

fixed points on the sphere. Infinity is one, leaving $5n - 2r - 2$ finite zeros. The numerator and denominator degrees are $n, n-r$. Regularity of an extremizer also follows from [LZ18, Theorem 3.5, author version v2, p. 7]. Taking $r = 2$ proves the last assertion. □

Proposition 6.8 is already implicit in those results and is not claimed here as new. It also shows that the upper bound in Theorem 6.1 is not always the finite-degree maximum: for $m = 1, r > 2$ it gives $5n - 6$, whereas (6.15) is smaller.

The finite-degree equalities proved later in Theorem 1.3 settle part of this range. The general expression in Theorem 6.1 is an upper bound, not a claim of equality for every triple (m, n, r).

7 A bound depending on the total-degree gap

Let $1 \le m < n$, let $p \in \mathbb{C}[z]$ have degree n, and let $Q \in \mathbb{C}[z, w]$ have total degree at most m. We write

$$F(z) = p(z) + Q(z, \bar{z}), \qquad d = n - m, \qquad N(F) = \#\{z \in \mathbb{C} : F(z) = 0\}. \tag{7.1}$$

In this note the first variable of Q is the analytic variable. The restriction on Q is a *total-degree* restriction, not merely a bound on its degree in w. Thus the class in (7.1) contains every harmonic polynomial $p + \overline{q}$ with $\deg q \le m$, but it does not contain all polyanalytic polynomials of bidegree (m, n).

The classical harmonic estimate is n^2; see Wilmshurst [Wil98, pp. 2077–2079]. The Hermitian trace approach developed by Furchì gives $n^2 - 2$ when $m \le n - 2$, subject in his statement to a possible $n^2 - 1$ exception determined by two analytic coefficients [Fur25, Theorem 1.1, author version v2, p. 2]. The proof below removes that exception and permits mixed lower-degree terms. The estimate below takes account of the entire gap between the two degrees.

This is the main result of the present note.

Theorem 7.1. *Under the assumptions in (7.1), the zero set of F is finite and*

$$N(F) \le n^2 - 2\left\lfloor \frac{(n - m)^2}{4} \right\rfloor. \tag{7.2}$$

For $d = 2$, (7.2) is $n^2 - 2$. For $d = 3, 4, 5, 6$ the deficits from n^2 become $4, 8, 12, 18$, respectively. For large n with m fixed, this is still a quadratic estimate; it is not a replacement for the linear bound of Corollary 6.3. Rather, the two bounds are effective in different regimes. No claim of sharpness for every d is made.

The proof uses the standard Hermitian trace form, whose signature counts real solutions in the appropriate real structure. That correspondence is Furchì's Theorem 3.13 and Corollary 3.14 [Fur25, author version v2, pp. 9–10]; we include the required argument, including multiple complex roots. The additional ingredient is a totally isotropic subspace of dimension $\lfloor d^2/4 \rfloor$.

7.1 The quotient algebra and its Hermitian form

Divide F by the leading coefficient of p, so that p is monic. For a univariate polynomial a, put $a^{\#}(w) = \overline{a(\bar{w})}$. For $a \in \mathbb{C}[z, w]$, put

$$a^*(z, w) = \overline{a(\bar{w}, \bar{z})}.$$

The latter is a conjugate-linear involution. Define

$$f = p(z) + Q(z, w), \qquad f^* = p^{\#}(w) + Q^*(z, w), \qquad \mathcal{A} = \mathbb{C}[z, w]/(f, f^*). \tag{7.3}$$

With any graded monomial order, the leading monomials of f and f^* are z^n and w^n. Since these are relatively prime, the two polynomials form a Gröbner basis. Consequently

$$\{z^i w^j : 0 \le i, j < n\} \tag{7.4}$$

is a basis of $\mathcal{A}$ and $\dim_{\mathbb{C}} \mathcal{A} = n^2$. This proves, in particular, the finiteness assertion in Theorem 7.1. We identify a polynomial with its class when no ambiguity can arise.

For $a \in \mathcal{A}$, let M_a denote multiplication by a, and set

$$T(a) = \operatorname{Tr} M_a, \qquad \mathcal{H}(a, b) = T(ab^*). \tag{7.5}$$

The ideal in (7.3) is invariant under $*$, so these expressions are well defined. Moreover $T(a^*) = \overline{T(a)}$; hence $\mathcal{H}$ is Hermitian. The involution satisfies $\mathcal{H}(a^*, b^*) = \overline{\mathcal{H}(a, b)}$.

The signature is the positive index minus the negative index; the radical is the nullspace. The following form of the signature theorem retains all multiplicities throughout the algebraic argument.

Lemma 7.2. *The signature of $\mathcal{H}$ is $N(F)$. If R is the number of distinct common zeros of f, f^* and s is the number of two-element orbits of $(z, w) \mapsto (\bar{w}, \bar{z})$ on this set, then*

$$\operatorname{rank} \mathcal{H} = R, \qquad \nu_-(\mathcal{H}) = s, \qquad N(F) = R - 2s.$$

Here ν_- denotes the number of negative eigenvalues, counted with multiplicity.

Proof. Decompose the finite algebra into its local algebras. On a local algebra of length μ supported at x, multiplication by a has trace $\mu a(x)$: the difference from scalar multiplication is nilpotent. At a fixed point of the involution the resulting trace form has a single positive eigenvalue, with coefficient μ, and is zero on the maximal ideal. At a nonfixed pair x, x^*, the two local lengths agree; on their constant coordinates the matrix of the form is

$$\begin{pmatrix} 0 & \mu \\ \mu & 0 \end{pmatrix}.$$

It has one positive and one negative eigenvalue. The remaining local coordinates are in the radical. Different orbits are orthogonal. Adding these contributions proves the assertions and identifies the fixed points with the zeros of F. □

A single isotropic subspace can lose dimension in the radical. The involution provides precisely the control needed to handle this loss.

Lemma 7.3. *Let V be a finite-dimensional complex vector space with a Hermitian form H and a conjugate-linear involution J satisfying $H(Ju, Jv) = \overline{H(u, v)}$. If $U \subset V$ is totally isotropic and $U \cap JU = \{0\}$, then*

$$\operatorname{signature} H \le \dim V - 2 \dim U. \tag{7.6}$$

Proof. Put $r = \dim \operatorname{rad} H$ and $k = \dim(U \cap \operatorname{rad} H)$. The radical is J-invariant. The subspaces $U \cap \operatorname{rad} H$ and $J(U \cap \operatorname{rad} H)$ are disjoint, so $r \ge 2k$. In the nondegenerate quotient $V/\operatorname{rad} H$, the image of U is isotropic of dimension $\dim U - k$. Projection onto the negative eigenspace is injective on any isotropic subspace of a nondegenerate Hermitian space. Therefore $\nu_-(H) \ge \dim U - k$. It follows that

$$\operatorname{signature} H = \dim V - r - 2\nu_-(H) \le \dim V - 2 \dim U - r + 2k \le \dim V - 2 \dim U,$$

as required. □

We shall apply (7.6) to (7.5). No simplicity assumption on the roots of the complexified system will be needed.

7.2 An exact trace identity below degree $2d$

Write

$$Q(z, w) = \sum_{r+s \le m} c_{rs} z^r w^s.$$

For a univariate polynomial u, let

$$S(u) = \operatorname{Tr}_{\mathbb{C}[z]/(p)} M_u, \qquad D_r(u) = [z^{n-1}]\big(z^r u'(z) \bmod p(z)\big).$$

The remainder in this definition has degree less than n. Use $S^{\#}, D_r^{\#}$ for the corresponding functionals defined by $p^{\#}$ in the variable w. These are complex-linear functionals. In particular,

$$D_r(u) = 0 \quad \text{if } \deg u + r < n, \tag{7.7}$$

and the same assertion holds for $D_r^{\#}$. Also $S^{\#}(u^{\#}) = \overline{S(u)}$.

The next identity, rather than just a single vanishing trace, is the algebraic input to the proof.

Lemma 7.4. *If $u \in \mathbb{C}[z]$, $v \in \mathbb{C}[w]$, and $\deg u + \deg v < 2d$, then*

$$\begin{aligned} T(u(z)v(w)) = {} & S(u)S^{\#}(v) \\ & - \sum_{r+s \le m} c_{rs} D_r(u) S^{\#}(w^s v) \\ & - \sum_{r+s \le m} \bar{c}_{rs} S(z^s u) D_r^{\#}(v). \end{aligned} \tag{7.8}$$

Proof. Introduce a real parameter t and replace Q, Q^* by tQ, tQ^* in (7.3); denote the multiplication trace by T_t. The standard basis (7.4) is unchanged for every t.

We first show that $T_t(z^a w^b)$ is affine in t when $a+b<2d$. To compute a diagonal entry of the multiplication matrix, reduce $z^{a+i}w^{b+j}$ to standard monomials. A substitution using a lower term of p or $p^{\#}$ decreases total degree by at least one. A substitution using a term of tQ or tQ^* decreases total degree by at least $n-m=d$ and contributes one factor of t. A reduction path ending in $z^i w^j$ can decrease degree by only $a+b$. It therefore uses at most one term containing t. All reductions strictly decrease total degree, so they terminate, and each diagonal entry is affine in t. Linearity gives the same assertion for $T_t(uv)$.

Suppose first that p has distinct roots $\alpha_1,\dots,\alpha_n$. At $t=0$, the roots of the complexified system are the simple grid points $(\alpha_a,\bar\alpha_b)$. Their derivatives satisfy

$$z'_{ab}(0)=-\frac{Q(\alpha_a,\bar\alpha_b)}{p'(\alpha_a)},\qquad w'_{ab}(0)=-\frac{Q^*(\alpha_a,\bar\alpha_b)}{(p^{\#})'(\bar\alpha_b)}.$$

For small t, the trace is the sum of the values of uv at these n^2 simple roots. Differentiating this sum at zero gives (7.8), because Lagrange interpolation yields

$$D_r(u)=\sum_{a=1}^{n}\frac{\alpha_a^r u'(\alpha_a)}{p'(\alpha_a)}. \tag{7.9}$$

Indeed, the right side is the coefficient of z^{n-1} in the remainder of $z^r u'$ modulo p. The constant term in t is $S(u)S^{\#}(v)$, and the two sums in (7.8) are its first derivative. Affineness in t makes this first-order identity exact at $t=1$.

Finally, all expressions in (7.8), with D_r defined by remainders, are polynomial functions of the coefficients of p, Q and their conjugates. The identity therefore extends from polynomials p with simple roots to all monic p. Formula (7.9) is used only in the proof on the dense simple-root locus. □

The threshold $2d$ is important. At total degree $2d$, two substitutions using the perturbation can contribute to a diagonal entry, and the trace need not be affine in Q. We do not use (7.8) at that threshold or beyond it.

7.3 A large isotropic subspace

For the moment assume that the leading moment matrices

$$\big(S(z^{a+b})\big)_{0\le a,b\le k},\qquad 0\le k<d, \tag{7.10}$$

are nonsingular. This is a nonempty Zariski-open condition on the coefficients of p. To see nonemptiness, take p with n distinct real roots: the matrices are positive definite for $k<n$, by evaluation of a nonzero real polynomial of degree at most k at those roots.

Apply Gram–Schmidt to $1, z, \dots, z^{d-1}$ for the complex bilinear pairing $(a,b)\mapsto S(ab)$. The result is a sequence of monic polynomials r_i of degree i, satisfying

$$S(r_i a)=0\quad(\deg a<i),\qquad 0\le i<d. \tag{7.11}$$

This is bilinear orthogonality, not Hermitian orthogonality. Positivity is not assumed for the general complex polynomial p.

We can now construct the subspace needed in Lemma 7.3.

Proposition 7.5. *Under (7.10), the subspace*

$$U = \operatorname{span}_{\mathbb{C}}\{r_i(z)r_j^{\#}(w) : 0 \le j < i,\ i+j \le d-1\} \subset \mathcal{A} \tag{7.12}$$

is totally isotropic for $\mathcal{H}$, satisfies $U \cap U^* = \{0\}$, and has complex dimension $\lfloor d^2/4 \rfloor$.

Proof. The products $r_i(z)r_j^{\#}(w)$ with $0 \le i, j < d$ are linearly independent in (7.4). The involution exchanges their two indices. The index sets defining U and U^* are disjoint, which proves $U \cap U^* = \{0\}$. Counting the pairs with $i > j$ and $i + j \le d - 1$ gives

$$\dim U = \begin{cases} t^2, & d = 2t, \\ t(t+1), & d = 2t+1, \end{cases} = \lfloor d^2/4 \rfloor. \tag{7.13}$$

Take indices $(i, j), (k, \ell)$ occurring in (7.12) and put

$$u = r_i r_\ell, \qquad v = r_j^{\#} r_k^{\#}.$$

Their total degree is at most $2d - 2$, so Lemma 7.4 applies. The constant term $S(r_i r_\ell)S^{\#}(r_j^{\#} r_k^{\#})$ is zero: by (7.11), both factors could be nonzero only if $i = \ell$ and $j = k$, contradicting $i > j$ and $k > \ell$.

Consider a term with coefficient c_{rs} in the first sum of (7.8). If $D_r(u) \neq 0$, then (7.7) implies $i + \ell + r \ge n$. Since $i + j \le n - m - 1$, it follows that

$$\ell - j \ge m + 1 - r, \qquad k - j \ge m + 2 - r > s.$$

Thus $j + s < k$, and (7.11) implies $S^{\#}(w^s r_j^{\#} r_k^{\#}) = 0$. If $D_r(u) = 0$, the same term is zero directly. Hence every term in the first sum vanishes.

For the second sum, a nonzero $D_r^{\#}(v)$ would imply $j + k + r \ge n$. Using $k + \ell \le n - m - 1$ and $i > j$, we obtain

$$j - \ell \ge m + 1 - r, \qquad i - \ell \ge m + 2 - r > s.$$

Now $\ell + s < i$, so $S(z^s r_i r_\ell) = 0$. Every term in the second sum also vanishes. By (7.5), the resulting identity $T(uv) = 0$ is precisely the required pairwise isotropy of U. □

Combining Lemmas 7.2 and 7.3 with Proposition 7.5, including (7.13), proves (7.2) under (7.10). Notice that the common zeros of f, f^* may have arbitrary multiplicities: the radical estimate in Lemma 7.3 has already accounted for them.

7.4 Removing the auxiliary moment condition

We finish the proof of Theorem 7.1. The coefficient space $p(z) + Q(z, w)$, $\deg p \le n$, $\deg_{\mathrm{tot}} Q \le m$, is stable under both derivatives. The highest analytic coefficient remains nonzero. Corollary 3.4 therefore gives an arbitrarily small perturbation in this space, without decreasing the zero count, whose zeros are all nondegenerate. A further small change of the coefficients of p preserves these zeros and satisfies the nonempty Zariski-open moment condition (7.10). The estimate proved using Proposition 7.5 applies to that last polynomial, and hence to the original one. This proves (7.2) with no assumption on multiplicities or moment matrices.

For a gap of two the bound is $n^2 - 2$, and it is attained by the real-valued construction in Theorem 10.1, with $m = n - 2$. Consequently the total-degree statement from the earlier drafts is

retained in its strongest form: for $\deg p = n \geq 3$ and $\deg_{\mathrm{tot}} Q \leq n-2$, the maximum is exactly n^2-2. For larger gaps the formula (7.2) is an upper bound, not an asserted sharp formula. Combining it with (6.7) gives the two upper bounds in Theorem 1.4 for the harmonic subclass.

8 Harmonic polynomials with many zeros

We now turn to (8.1). Unlike the examples in Section 5, every polynomial constructed in this section is harmonic. The upper bound refines the Descartes argument of [Lun23, Section 4.1, author version, pp. 8–9]; the lower bound makes that argument exact on every line at once. The coefficients on different lines are not independent, and we keep track of their common angular dependence throughout.

The full harmonic maximum also equals the maximal valence in that class, since subtracting a constant preserves both degrees.

Lundberg's probabilistic construction gives $M^{\mathrm{harm}}_{m,n} \geq \lceil n\sqrt{m} \rceil$ by using the family

$$h(z) = \varepsilon z^n + q(z) + \overline{q(z)}, \qquad \deg q = m < n, \quad \varepsilon \in \mathbb{C} \setminus \{0\}. \tag{8.1}$$

His concluding question asks for a deterministic choice of q giving mn-order many zeros in this same family [Lun23, Section 4.2, author version, p. 9]. We determine its exact maximal zero count, denoted by $S_{m,n}$. Put $g = \gcd(m, n)$ and define

$$\Delta_{m,n} = \begin{cases} g, & \text{if } m/g \text{ and } n/g \text{ are both odd,} \\ 0, & \text{otherwise.} \end{cases} \tag{8.2}$$

The correction in (8.2) comes from the parity of the exponent gap and a count of equally spaced directions.

Theorem 8.1. *For every $1 \leq m < n$,*

$$S_{m,n} = n(m+1) + \Delta_{m,n}. \tag{8.3}$$

The maximum is attained with $\varepsilon \in \mathbb{Q}_{>0}$ and $q \in \mathbb{Q}(i)[z]$, and all zeros of the extremizer are nondegenerate.

8.1 Real restrictions and the last exponent gap

For real polynomials the contribution of the final gap can be bounded more precisely than by two in all cases.

Lemma 8.2. *Let $N > M \geq 0$, let $a \in \mathbb{R} \setminus \{0\}$, and let $u \in \mathbb{R}[r]$ have degree at most M. Write $b = [r^M]u$ for the coefficient of r^M. The number of nonzero real roots of $ar^N + u(r)$, counted with multiplicity, is at most*

$$\begin{cases} M+1, & N-M \text{ odd}, \\ M + 2\mathbf{1}_{\{ab<0\}}, & N-M \text{ even}, \end{cases}$$

where the indicator $\mathbf{1}_{\{ab<0\}}$ equals one when $ab < 0$ and zero otherwise.

Proof. There is nothing to prove if $u = 0$. Otherwise, put $d = \deg u$ and list its nonzero coefficients in increasing exponent order. An odd gap between successive exponents contributes exactly one sign change in total to the coefficient lists of $u(r)$ and $u(-r)$. An even gap contributes zero or two sign changes, which is at most its size. The total contribution within u is therefore at most d.

Appending the degree-N coefficient contributes one when $N - d$ is odd. When $N - d$ is even, it contributes two precisely if the two leading coefficients have opposite signs, and otherwise zero. If $d = M$, this proves the assertion by Descartes' rule on the two half-lines. If $d < M$ and $N - M$ is odd, use $d+2 \le M+1$. If $N - M$ is even, the case $d = M-1$ has odd last gap and gives $d+1 = M$; for $d \le M-2$, use $d+2 \le M$. These last two cases also explain why a vanishing coefficient b causes no exception. □

A rotation $z = e^{i\theta}\zeta$ with $\varepsilon e^{in\theta} > 0$ reduces (8.1) to $\varepsilon > 0$. The transformed polynomial $q(e^{i\theta}\zeta)$ still has degree m. After this normalization, the imaginary part of h vanishes exactly on the lines

$$\Lambda_j = e^{ij\pi/n}\mathbb{R}, \qquad 0 \le j < n. \tag{8.4}$$

Write $q(z) = \sum_{k=0}^{m} a_k z^k$, with $a_m \neq 0$, and put $\theta_j = j\pi/n$. The restriction to Λ_j is the real polynomial

$$f_j(r) = \varepsilon(-1)^j r^n + 2\sum_{k=0}^{m} \operatorname{Re}(a_k e^{ik\theta_j}) r^k. \tag{8.5}$$

Its leading coefficient never vanishes. In particular, every member of (8.1) has a finite zero set, and the possible integer counts are bounded. Thus the maximum defining $S_{m,n}$ exists.

8.2 The exact upper bound

Assume first that $\operatorname{Re} a_0 \neq 0$. The origin is not a zero, so counts on the distinct lines (8.4) may be added. If $n - m$ is odd, Lemma 8.2 gives at most $m + 1$ roots per line. The resulting bound is $n(m + 1)$, and in this case $\Delta_{m,n} = 0$.

Suppose that $n - m = 2d$ is even, and put

$$G = \gcd(n, d), \qquad \ell = n/G, \qquad a_m = |a_m| e^{i\beta}.$$

The sign controlling the final gap in (8.5) is that of

$$(-1)^j \operatorname{Re}(a_m e^{imj\pi/n}) = |a_m| \cos\left(\beta - \frac{2\pi dj}{n}\right).$$

By Lemma 8.2,

$$N(h) \le nm + 2B_\beta, \qquad B_\beta = \#\left\{0 \le j < n : \cos\left(\beta - \frac{2\pi dj}{n}\right) < 0\right\}. \tag{8.6}$$

As j varies, the angles $2\pi dj/n$ traverse the vertices of a regular ℓ-gon, each with multiplicity G. An open semicircle contains at most $\ell/2$ such vertices when ℓ is even and at most $(\ell + 1)/2$ when ℓ is odd. The first assertion follows by pairing antipodal vertices. For the second, $(\ell + 3)/2$ consecutive vertices would span an angle $\pi + \pi/\ell$, which is longer than a semicircle. Therefore

$$2B_\beta \le \begin{cases} n, & \ell \text{ even}, \\ n + G, & \ell \text{ odd}. \end{cases} \tag{8.7}$$

Boundary vertices have zero cosine and are not counted, so no nonvanishing assumption on the intermediate coefficients is used.

We identify the parity alternative in (8.7) with (8.2). Write $m = gu$ and $n = gv$, where $g = \gcd(m, n)$ and $\gcd(u, v) = 1$. If u, v are both odd, then

$$G = \gcd\big(gv, g(v-u)/2\big) = g, \qquad \ell = v \text{ is odd.}$$

Here $\gcd(v, (v-u)/2) = 1$ because v is odd and coprime to u. Otherwise u, v have opposite parity. Since $n - m = g(v-u)$ is even, g is even, and

$$G = \frac{g}{2}\gcd(2v, v-u) = \frac{g}{2}, \qquad \ell = 2v \text{ is even.}$$

The last greatest common divisor is one because $v - u$ is odd and coprime to v. Equations (8.6) and (8.7) now give $N(h) \le n(m+1) + \Delta_{m,n}$.

It remains to treat $\operatorname{Re} a_0 = 0$. Each f_j in (8.5) is then divisible by r. Apply Lemma 8.2 to f_j/r, using $(N, M) = (n-1, m-1)$. The last-gap parity and the two relevant leading signs are unchanged, whereas each nonzero-root bound decreases by one. The common origin must be restored once, not n times. This actually gives the stronger estimate

$$N(h) \le n(m+1) + \Delta_{m,n} - n + 1.$$

The upper bound in Theorem 8.1 is therefore valid for all members of the family, including those with a zero at the origin, missing coefficients, or degenerate zeros.

8.3 Rational coefficients attaining the bound

The lower bound uses the same lines but forces every restriction to attain its Descartes bound. Fix $m < n$ and set

$$g = \gcd(m, n), \qquad \eta = \frac{1}{4n}, \qquad t = \frac{1}{16n(m+1)}, \qquad \gamma = 1 + i\eta,$$
$$\sigma = \begin{cases} -1, & \Delta_{m,n} = g \text{ and } n/g \equiv 1 \pmod 4, \\ 1, & \text{otherwise.} \end{cases} \tag{8.8}$$

Define

$$q(z) = \frac{\sigma\gamma}{2}\sum_{k=0}^{m} t^{k^2} z^k, \qquad p(z) = t^{n^2} z^n + q(z), \qquad h = p + \overline{q}. \tag{8.9}$$

All coefficients in (8.9) belong to $\mathbb{Q}(i)$, the two degrees are exactly m, n, and $h(0) = \sigma \neq 0$. The restriction of this h to Λ_j is

$$f_j(r) = (-1)^j t^{n^2} r^n + \sigma \sum_{k=0}^{m} \left(\cos\frac{kj\pi}{n} - \eta \sin\frac{kj\pi}{n}\right) t^{k^2} r^k. \tag{8.10}$$

The factors in parentheses are uniformly separated from zero:

$$\frac{1}{4n} \le \left|\cos\frac{kj\pi}{n} - \eta\sin\frac{kj\pi}{n}\right| < 2. \tag{8.11}$$

Indeed, a zero cosine gives absolute value exactly η. For a nonzero cosine, the angle's distance from a zero of cosine, modulo π, is at least $\pi/(2n)$. Hence its absolute value is at least $\sin(\pi/(2n)) \ge 1/n$.

Subtracting a quantity of absolute value at most η gives at least $3/(4n)$. The upper bound in (8.11) follows from $1+\eta<2$.

The coefficient factor of r^n in (8.10) has absolute value one. Use (5.1) with the support $\{0,1,\dots,m,n\}$, $\mu=1/(4n)$, and $B=2$. The sum of the other normalized monomials at every prescribed radius is bounded by

$$2(m+1)t=\frac{1}{8n}<\frac{1}{4n}.$$

Lemma 5.1 therefore applies simultaneously to every f_j. It gives exact positive and negative root counts, and all these roots are simple.

For each of the consecutive exponent pairs $(0,1),(1,2),\dots,(m-1,m)$, exactly one of the sign lists for $f_j(r)$ and $f_j(-r)$ changes sign. These pairs contribute exactly m roots on each line. If $n-m$ is odd, the final gap contributes one more, proving (8.3) in this case.

For $n-m=2d$, retain the integers $G=\gcd(n,d)$ and $\ell=n/G$ from the upper-bound proof. Write $\gamma=|\gamma|e^{i\alpha}$, so $0<\alpha=\arctan\eta<1/(4n)$. The final gap contributes two roots on Λ_j precisely when

$$\sigma\cos\left(\alpha-\frac{2\pi dj}{n}\right)<0, \tag{8.12}$$

and contributes none otherwise. When ℓ is even, antipodal pairing gives exactly $n/2$ negative signs. None is zero, by (8.11), so the final gaps contribute n roots in total.

When ℓ is odd, the preceding greatest-common-divisor calculation gives $G=g$ and $\ell=n/g$. At phase zero, a regular ℓ-gon has $(\ell-1)/2$ vertices in the open negative semicircle if $\ell\equiv 1 \pmod 4$, and $(\ell+1)/2$ if $\ell\equiv 3 \pmod 4$. This follows directly by counting integers strictly between $\ell/4$ and $3\ell/4$. Every vertex is at least $\pi/(2\ell)$ from the boundary of that semicircle. Since $\alpha<1/(4n)<\pi/(2\ell)$, the phase shift does not change these counts. The sign σ in (8.8) selects $(\ell+1)/2$ negative cosines, each repeated g times. Thus (8.12) contributes $g(\ell+1)=n+g$ roots. Adding the nm roots supplied by the consecutive low exponents proves (8.3) in every case.

Finally, all zeros are nonzero and different lines meet only at the origin. At a zero $z=e^{i\theta_j}r_0$, choose the oriented local coordinates $z=e^{i\theta_j}(r+is)$. Along $s=0$,

$$\partial_r \operatorname{Re} h=f_j'(r_0)\neq 0,\qquad \partial_r \operatorname{Im} h=0,\qquad \partial_s \operatorname{Im} h=t^{n^2}n(-1)^j r_0^{n-1}\neq 0.$$

The real Jacobian is nonzero. This completes the proof of Theorem 8.1, including the exact count, rationality of the coefficients, and nondegeneracy of every zero.

8.4 Interpretation and an exact small example

The construction proves degree-uniform growth from below: $M^{\mathrm{harm}}_{m,n}\geq S_{m,n}\geq n(m+1)$. For fixed m, $S_{m,n}/n\to m+1$; for $m\to\infty$ the ratio $S_{m,n}/(mn)$ tends to one uniformly over $n>m$, because $0\leq\Delta_{m,n}\leq m$. These statements answer Lundberg's deterministic question [Lun23, Section 4.2, author version, p. 9]. They do not identify the maximum for arbitrary analytic heads, nor the maximum over arbitrary target values of one fixed member of this restricted family. The exact subclass count does not settle the best coefficient for fixed $m\geq 2$ in the unrestricted harmonic class.

The general construction uses very different coefficient scales. For one small pair, short rational coefficients suffice.

Example 8.3. The harmonic polynomial

$$h(z) = 10^{-8}z^6 - \operatorname{Re}\big((1+i)(z^2+10z+1)\big)$$

has exactly twenty nondegenerate zeros. Its analytic and conjugate degrees are 6 and 2, respectively.

Proof. Here $q(z) = -(1+i)(z^2+10z+1)/2$ and $p(z) = 10^{-8}z^6 + q(z)$. On Λ_j, with $\theta_j = j\pi/6$ and $0 \le j \le 5$, the equation is

$$f_j(r) = 10^{-8}(-1)^j r^6 - \big(\cos 2\theta_j - \sin 2\theta_j\big)r^2 - 10\big(\cos\theta_j - \sin\theta_j\big)r - 1 = 0.$$

The absolute values of the linear coefficients lie between $7/2$ and 14, and those of the quadratic coefficients between $7/20$ and $7/5$. These inequalities follow from $17/10 < \sqrt{3} < 9/5$. At the radii $1/100, 1, 50, 1000$, the constant, linear, quadratic, and degree-six monomials, respectively, dominate all the others. For the quadratic monomial the estimate is

$$\frac{7}{20}\,50^2 = 875 > 1 + 14\cdot 50 + 10^{-8}50^6 = \frac{3429}{4}.$$

For the other three monomials the same bounds give, respectively,

$$1 > \frac{14}{100} + \frac{7}{5\cdot 10^4} + 10^{-20}, \qquad \frac{7}{2} > 1 + \frac{7}{5} + 10^{-8},$$
$$10^{10} > 1 + 14\cdot 10^3 + \frac{7}{5}\cdot 10^6.$$

Lemma 5.1 makes the following counts exact; signs are listed in exponent order $0, 1, 2, 6$.

j	Coefficient signs	Positive roots	Negative roots	Total
0	$-\ -\ -\ +$	1	3	4
1	$-\ -\ +\ -$	2	2	4
2	$-\ +\ +\ +$	1	1	2
3	$-\ +\ +\ -$	2	2	4
4	$-\ +\ -\ +$	3	1	4
5	$-\ +\ -\ -$	2	0	2

The roots are simple, and $h(0) = -1$, so there is no overlap between lines. The tangential and normal derivative calculation above proves planar nondegeneracy. □

Example 8.3 attains $S_{2,6} = 20$; it is not asserted to maximize the unrestricted harmonic class. An independent exact Sturm check over $\mathbb{Q}(\sqrt{3})$ is included in the supplement as `verify_harmonic_20.py`. The analytic proof above does not depend on that computation.

9 A harmonic construction with a Chebyshev analytic part

The monomial-head family has now been counted exactly. Changing its analytic head yields a larger lower bound for the unrestricted harmonic class. Here T_j denotes the Chebyshev polynomial specified by $T_j(\cos\theta) = \cos(j\theta)$.

Theorem 9.1. *Fix $1 \le m < n$ and $a > 2$. For all sufficiently small $\varepsilon > 0$,*

$$h_\varepsilon(z) = \varepsilon T_n(z) - 2\operatorname{Im} T_m(z-a) \tag{9.1}$$

has exactly $(m+2)n - m - \gcd(m,n)$ distinct nondegenerate zeros.

The harmonic decomposition in this theorem is

$$p_\varepsilon(z) = \varepsilon T_n(z) + iT_m(z-a), \qquad q(z) = iT_m(z-a).$$

Thus both degrees are exact. Rational $a > 2$ and sufficiently small positive rational ε give coefficients in $\mathbb{Q}(i)$. In particular,

$$M^{\mathrm{harm}}_{m,n} \geq (m+2)n - m - \gcd(m,n). \tag{9.2}$$

For fixed m, the leading coefficient of this lower bound is $m+2$. It is a lower bound for the unrestricted harmonic class, not a general formula for its maximum.

We prove Theorem 9.1 by accounting for every zero on the nodal set of $\operatorname{Im} T_n$. The identity $T_n(\cos\zeta) = \cos(n\zeta)$ follows from the defining recurrence and holds for complex ζ.

9.1 The nonreal nodal curves

Every point of the upper half-plane has a unique representation

$$z = \cos(\theta - it) = \cos\theta\cosh t + i\sin\theta\sinh t, \qquad 0 < \theta < \pi, \quad t > 0. \tag{9.3}$$

For example, inverting $z = (\zeta + \zeta^{-1})/2$ selects the unique root outside the unit circle, whose imaginary part is positive. Its polar coordinates give (9.3). The coordinate map has nonzero derivative in this open half-strip. Since $\operatorname{Im} T_n(z) = \sin(n\theta)\sinh(nt)$, the upper-half-plane zero set consists of the disjoint regular curves

$$z_j(t) = \cos\theta_j\cosh t + i\sin\theta_j\sinh t, \qquad \theta_j = j\pi/n, \quad 1 \leq j \leq n-1. \tag{9.4}$$

The lower curves are their reflections; all remaining points of the nodal set lie on the real axis.

The curve of angle θ has real coordinate at height $y > 0$

$$X_\theta(y) = \cot\theta\sqrt{y^2 + \sin^2\theta}.$$

The upper nodal curves of $\operatorname{Im} T_m(z-a)$ have the same form translated by a, with angles $\phi_\ell = \ell\pi/m$, $1 \leq \ell \leq m-1$.

Lemma 9.2. *For $a > 2$, a curve of angle θ and a translated curve of angle ϕ have one transverse intersection if $\theta < \phi$, and none if $\theta \geq \phi$.*

Proof. An intersection is a solution of $X_\theta(y) - X_\phi(y) = a$. For each y, $X_\theta(y)$ is strictly decreasing in θ, so $\theta \geq \phi$ is impossible. If $\theta < \phi$, the difference at zero is $\cos\theta - \cos\phi \in (0,2)$, and the difference tends to infinity. It is strictly increasing for $y > 0$, because

$$\frac{\partial}{\partial\theta}\frac{\cot\theta}{\sqrt{y^2+\sin^2\theta}} = -\frac{1}{\sin^2\theta\sqrt{y^2+\sin^2\theta}} - \frac{\cos^2\theta}{(y^2+\sin^2\theta)^{3/2}} < 0$$

and $X'_\theta(y) = y\cot\theta/\sqrt{y^2+\sin^2\theta}$. This proves existence, uniqueness, and transversality. □

Let k_j count the zeros of $\operatorname{Im} T_m(z_j(t) - a)$ for $t > 0$ on the curve (9.4). They are simple by Lemma 9.2, and

$$k_j = \#\{1 \leq \ell \leq m-1 : j/n < \ell/m\}, \qquad \sum_{j=1}^{n-1} k_j = \frac{(n-1)(m-1)-(g-1)}{2}, \quad g = \gcd(m,n). \tag{9.5}$$

For the sum, there are $g-1$ equality pairs, and $(j,\ell) \mapsto (n-j, m-\ell)$ exchanges the two strict inequalities. No triple intersection is possible, since the curves within either upper family are disjoint.

9.2 Small levels at the two ends

Put $r(z) = T_m(z - a)$ and

$$A_j(t) = \frac{2 \operatorname{Im} r(z_j(t))}{\cosh(nt)}, \qquad t > 0.$$

The zero equation for (9.1) on the upper curve is $A_j(t) = \varepsilon(-1)^j$, and on its lower reflection it is $A_j(t) = -\varepsilon(-1)^j$. Their combined count is the number of solutions of $A_j(t) = \pm\varepsilon$.

At zero,

$$A_j(t) = 2r'(\cos\theta_j) \sin\theta_j \, t + O(t^2),$$

with nonzero coefficient. The roots of T_m' lie in $(-1, 1)$ when $m \geq 2$, whereas $\cos\theta_j - a < -1$; for $m = 1$ the derivative is one. At infinity write $r = \sum_{k=0}^{m} b_k T_k$ with real coefficients. Then

$$\operatorname{Im} r(z_j(t)) = \sum_{k=1}^{m} b_k \sin(k\theta_j) \sinh(kt).$$

This is not identically zero, by its derivative at zero. Let κ_j be the largest index with nonzero coefficient in this sum. It satisfies $1 \leq \kappa_j \leq m < n$. For a nonzero real C_j,

$$\begin{aligned} A_j(t) &= C_j e^{-(n-\kappa_j)t}(1 + o(1)), \\ A_j'(t) &= -(n - \kappa_j) C_j e^{-(n-\kappa_j)t}(1 + o(1)). \end{aligned} \tag{9.6}$$

Thus A_j is eventually monotone and tends to zero. Keeping κ_j in this argument is important: it need not equal m when $\sin(m\theta_j) = 0$.

It follows that, for sufficiently small $\varepsilon > 0$, the two levels together have exactly $2k_j + 2$ simple solutions. Indeed, choose disjoint intervals around the k_j simple interior zeros in (9.5), a one-sided interval at zero, and a terminal half-line on which (9.6) is monotone. Each interior zero contributes one solution to each level. The initial and terminal intervals each contribute one solution to one of the two levels. The remaining set is compact and $|A_j|$ has a positive minimum there, so it contributes none. This argument proves both the exact count and the nonvanishing of A_j' at every solution.

9.3 Completion and nondegeneracy

On the real axis, (9.1) has exactly the n simple zeros of T_n. Their Jacobians are

$$|\varepsilon T_n'(x) + ir'(x)|^2 - |ir'(x)|^2 = \varepsilon^2 T_n'(x)^2 > 0.$$

At a nonreal zero, the derivative of the real part along its curve is a nonzero multiple of $A_j'(t)$. The imaginary part is zero along the curve and has transverse derivative $\varepsilon n(-1)^j \sinh(nt) \neq 0$. The coordinate map in (9.3) is regular, so the planar Jacobian is nonzero.

One common small positive threshold for ε works on the finitely many curves. Using (9.5), the exhaustive count is

$$\begin{aligned} N(h_\varepsilon) &= n + \sum_{j=1}^{n-1} (2k_j + 2) \\ &= n + (n-1)(m-1) - g + 1 + 2(n-1) \\ &= (m+2)n - m - g. \end{aligned}$$

This proves Theorem 9.1 and (9.2).

9.4 Comparison with the monomial-head construction

The preceding section gives

$$S_{m,n} = n(m+1) + \Delta_{m,n}, \qquad \Delta_{m,n} = \begin{cases} g, & m/g \text{ and } n/g \text{ both odd}, \\ 0, & \text{otherwise}, \end{cases} \qquad g = \gcd(m,n).$$

The count in (9.2) exceeds it by $n - m - g - \Delta_{m,n}$, which is nonnegative. If $\Delta = 0$, use $g \mid (n-m)$; if $\Delta = g$, then $(n-m)/g$ is a positive even integer. For fixed m, the difference is $n - O_m(1)$.

Wilmshurst's proposed estimate $3n - 2 + m(m-1)$ was disproved in general by Lee–Lerario–Lundberg [LLL15, author version v1, pp. 2–3]. The next consequence is a uniform family of counterexamples, not a claim to the first disproof of that conjecture.

Corollary 9.3. *If $2 \le m \le n-2$ and $(m,n) \ne (2,4)$, there is a harmonic polynomial of degrees (m,n) with more than $3n - 2 + m(m-1)$ distinct zeros.*

Proof. Subtract the proposed count from Theorem 9.1. The difference is $(m-1)(n-m-1)+1-g$. If $n - m \ge 3$, it is at least $(m-1)(n-m-2) > 0$, since $g \le m$. If $n - m = 2$, it is $m - \gcd(m,2)$, positive except when $m = 2$. □

Corollary 9.3 does not determine the corresponding harmonic maxima. For example, it gives $M_{2,5}^{\mathrm{harm}} \ge 17$ but not equality. The endpoint counts for $m = 1$ and $m = n-1$ in Theorem 9.1 recover already known sharp values. The rectangular equality range in Theorem 1.3 concerns a different coefficient space. In particular, its mixed-term constructions do not give harmonic extremizers; the alternatives $M_{2,7} \in \{49, 51, 53\}$ also remain unresolved.

The construction, Corollary 6.3, and Theorem 7.1 prove (1.4). The exact endpoints in Theorem 1.4 follow from [KS03, Gey08, Wil98]. For $m \ge 2$, the coefficient $3m + 2$ is at most $4m$; for $m = 1$, use the stronger classical $3n - 2$. This proves the uniform bound $mn \le M_{m,n}^{\mathrm{harm}} \le 4mn$ claimed in the introduction. No assertion that the Chebyshev lower bound is the exact maximum is needed.

10 Real-valued lower-degree perturbations

The zero-counting problem for $p(z) + \overline{q(z)}$ distinguishes the degrees of its analytic and anti-analytic parts. One particularly useful family is $\varepsilon z^n + q(z) + \overline{q(z)}$. In this family, the imaginary part vanishes on n lines, and one may count zeros by restricting the real part to those lines. Lundberg used this reduction to obtain the bound $n(m+2)$ and asked for deterministic constructions with order mn zeros; see [Lun23, Sections 4.1–4.2, author version, pp. 8–9].

Here we retain the monomial analytic term but allow the real-valued perturbation to be an arbitrary real polynomial, not necessarily a harmonic one. Throughout, $1 \le m < n$ and

$$F(z) = \varepsilon z^n + R(x,y), \qquad z = x + iy, \quad \varepsilon \in \mathbb{C} \setminus \{0\}, \quad R \in \mathbb{R}[x,y], \quad \deg R = m. \tag{10.1}$$

Write $N(F) = \#F^{-1}(0)$, counting distinct points. Let $A_{m,n}$ be the largest possible value of $N(F)$ in (10.1). A zero is *nondegenerate* if the real Jacobian of $F : \mathbb{R}^2 \to \mathbb{R}^2$ is nonzero there. Finiteness, including for singular zeros, will follow directly from the line restrictions.

The exact answer depends only on the parity of the degree gap.

Theorem 10.1. *For every $1 \le m < n$,*

$$A_{m,n} = n(m+1) + \begin{cases} 0, & n-m \text{ is odd}, \\ m, & n-m \text{ is even}. \end{cases} \tag{10.2}$$

The maximum is attained with $\varepsilon \in \mathbb{Q}_{>0}$, $R \in \mathbb{Q}[x,y]$, and all zeros nondegenerate.

There is a finer answer when the gap is even. Let H_m be the leading homogeneous part of R. Denote by $s(H_m)$ the number of distinct real projective zeros of H_m having odd multiplicity. Equivalently, this is the number of real linear factors occurring with odd exponent, with proportional factors identified. Factoring a real binary form shows that

$$0 \le s(H_m) \le m, \qquad s(H_m) \equiv m \pmod 2.$$

No condition is imposed on its other real or complex factors.

The next statement gives both the coefficient-dependent upper bound and its exact extremal meaning.

Theorem 10.2. *Suppose $n-m$ is even. Every polynomial in (10.1) satisfies*

$$N(F) \le n(m+1) + s(H_m). \tag{10.3}$$

For each integer s with $0 \le s \le m$ and $s \equiv m \pmod 2$, the maximum among such polynomials with $s(H_m) = s$ equals $n(m+1)+s$. It is attained with rational coefficients and only nondegenerate zeros.

Theorem 10.2 explains the extra term in (10.2): the leading real homogeneous form can change sign on at most m projective directions. In the harmonic subclass those directions are equally spaced, which imposes an additional arithmetic restriction. The exact difference from the harmonic subclass is recorded after the proof.

The two real-root counting ingredients have already been proved: Lemma 8.2 is the joint Descartes estimate, and Lemma 5.1 makes it exact under separated coefficients. We now use the fact that all leading coefficients of the line restrictions come from a single real homogeneous form.

10.1 The upper bound and the leading form

A rotation $z = e^{i\omega}\zeta$ with $\varepsilon e^{in\omega} = |\varepsilon|$ preserves the number of zeros, the degree of the real perturbation, and $s(H_m)$. We may therefore assume $\varepsilon > 0$. Put

$$\theta_j = \frac{j\pi}{n}, \qquad v_j = (\cos\theta_j, \sin\theta_j), \qquad L_j = e^{i\theta_j}\mathbb{R}, \qquad 0 \le j < n. \tag{10.4}$$

Every zero lies on one of these lines because $\Im F(z) = \varepsilon \Im z^n$. On L_j the equation is the real equation

$$f_j(r) = \varepsilon(-1)^j r^n + R(rv_j) = 0. \tag{10.5}$$

Each f_j has degree n and nonzero leading coefficient. In particular, there are finitely many zeros without any separate hypothesis.

Suppose first that $R(0,0) \neq 0$. The lines in (10.4) meet only at the origin, which is not a zero, so their root counts add. If $n-m$ is odd, Lemma 8.2 gives $N(F) \le n(m+1)$ immediately.

Now let $n-m$ be even. Define

$$B(H_m) = \#\{0 \le j < n : (-1)^j H_m(v_j) < 0\}.$$

Equations (10.5) and the estimate in Lemma 8.2 give

$$N(F) \le nm + 2B(H_m). \tag{10.6}$$

The missing ingredient is a constraint relating the leading coefficients on different lines.

Lemma 10.3. *If $n-m$ is even and H_m is a nonzero real homogeneous polynomial of degree m, then*

$$2B(H_m) \le n + s(H_m). \tag{10.7}$$

Proof. Regard $0, \dots, n-1$ as the vertices of a cycle, and mark the vertices counted by $B(H_m)$. Let E be the number of edges with both endpoints marked. At most $2(n - B(H_m))$ edges meet an unmarked vertex, so $E \ge 2B(H_m) - n$.

Each edge with two marked endpoints forces a sign change of $H_m(\cos\theta, \sin\theta)$ in the corresponding open interval of length π/n. For the last interval, use $H_m(v_n) = (-1)^m H_m(v_0)$ and $m \equiv n \pmod 2$; marking is then consistent across the endpoint of $[0, \pi]$. A sign change requires a real projective zero of odd multiplicity in that interval. Distinct edges give disjoint intervals, so $E \le s(H_m)$. This proves (10.7). Zeros at the sample directions do not create an exception: such vertices are unmarked, whereas an edge counted by E has nonzero values at both endpoints. □

Combining (10.6) and (10.7) proves (10.3) in Theorem 10.2, and hence the even case of Theorem 10.1, whenever the origin is not a zero.

If $R(0,0) = 0$, divide each restriction in (10.5) by r. The leading exponent becomes $n-1$, the lower degree is at most $m-1$, and the gap retains its parity. Count the nonzero roots with Lemma 8.2, then add the common origin only once. For odd gap the resulting bound is $nm+1$. For even gap it is

$$n(m-1) + 2B(H_m) + 1 \le nm + s(H_m) + 1.$$

Both are strictly below the corresponding asserted maximum. This completes the upper-bound proofs for all coefficients and all multiplicities.

10.2 Simultaneous extremal constructions

We now attain the bounds. In the even case we prescribe an admissible integer s, with $0 \le s \le m$ and $s \equiv m \pmod 2$. Set

$$\eta = \frac{1}{4n}, \qquad L(x,y) = x + \eta y, \qquad \tau = \frac{1}{2(m+1)(8n)^m}.$$

For $0 \le k < m$, put $H_k = L^k$. If $n-m$ is odd, also put $H_m = L^m$. If $n-m$ is even, define

$$\begin{aligned} \phi_\nu &= \frac{(n - s + \nu + \frac{1}{2})\pi}{n}, \qquad 0 \le \nu < s, \\ H_m(x,y) &= -(x^2+y^2)^{(m-s)/2} \prod_{\nu=0}^{s-1} (\sin\phi_\nu\, x - \cos\phi_\nu\, y). \end{aligned} \tag{10.8}$$

An empty product equals one. This leading form has exactly s real projective zeros, all simple; the other displayed factor has no real projective zeros. Finally define

$$R_\tau(x,y)=\sum_{k=0}^{m}\tau^{k^2}H_k(x,y),\qquad F_\tau(z)=\tau^{n^2}z^n+R_\tau(\Re z,\Im z). \tag{10.9}$$

The perturbation has degree exactly m, and $F_\tau(0)=1$.

10.2.1 One choice of scale works on every line

For the sample directions in (10.4),

$$\frac{1}{4n}\le|L(v_j)|<2.$$

Indeed, if $\cos\theta_j=0$, the left side is an equality. Otherwise $|\cos\theta_j|\ge\sin(\pi/(2n))\ge 1/n$, so subtracting at most η still leaves at least $3/(4n)$. It follows that the values of $H_k(v_j)$ for $k<m$, and also for $k=m$ in the odd case, are nonzero and have absolute value between $(4n)^{-m}$ and 2^m.

For (10.8), every sine factor at v_j is nonzero and has absolute value at least $\sin(\pi/(2n))\ge 1/n$. Since $|v_j|=1$, it follows that $|H_m(v_j)|\ge n^{-s}\ge(4n)^{-m}$ and $|H_m(v_j)|\le 1$. Thus every line restriction of (10.9), on the support

$$0,1,\dots,m,n,$$

has its unscaled coefficients bounded in absolute value between $\delta=(4n)^{-m}$ and $B_0=2^m$. The leading coefficient is $(-1)^j$. Our scale satisfies

$$(m+1)B_0\tau=\frac{\delta}{2}<\delta.$$

Lemma 5.1, with (5.1), therefore applies simultaneously to the restrictions on all n lines and to both halves of each line.

10.2.2 Exact count and nondegeneracy

The exponent pairs $(0,1),(1,2),\dots,(m-1,m)$ each contribute one to the sum of positive and negative sign variations, regardless of the coefficient signs. Their total contribution on each line is exactly m. If $n-m$ is odd, the last gap contributes one more. Thus (10.9) has exactly $n(m+1)$ zeros in the odd case.

In the even case, the last gap contributes two exactly when $(-1)^jH_m(v_j)<0$. Let $d'=n-s$, which is even. In (10.8), $H_m(v_j)$ is negative for $0\le j\le d'$. For $j>d'$, it changes sign at each step, because exactly one of the linear factors changes sign. Consequently the marked sample indices are the even indices in $0,\dots,d'-1$, together with all indices in $d',\dots,n-1$. The same description covers $s=0$. Hence

$$B(H_m)=\frac{n-s}{2}+s=\frac{n+s}{2}.$$

Lemma 5.1 now gives exactly $nm+2B(H_m)=n(m+1)+s$ zeros. This proves attainment in Theorem 10.2. Taking $s=m$ proves the even case of (10.2).

All the real roots on each line are simple and nonzero. At such a root, the derivative of $\Re F_\tau$ in the direction of the line is nonzero. The derivative of $\Im F_\tau=\tau^{n^2}\Im z^n$ in that direction is zero, but its normal derivative is nonzero. These two facts imply a nonzero real Jacobian. This verifies nondegeneracy in the plane, rather than merely simplicity of a univariate restriction.

10.2.3 Rational coefficients

In the odd case the construction already has rational coefficients. In the even case the displayed coefficients of H_m are real algebraic numbers. Approximate the coefficients of each of its linear factors by rational numbers, keeping the positive quadratic factor unchanged. For sufficiently accurate approximations, the s real projective roots remain distinct, avoid the sample directions, and have the same signs at those directions. All strict dominance inequalities are also preserved. The count and nondegeneracy just proved therefore remain valid, and the leading form still satisfies $s(H_m) = s$. The resulting polynomial has coefficients in $\mathbb{Q}$. This proves all attainment assertions in Theorems 10.1 and 10.2.

10.3 Comparison with the restricted harmonic family

Equations (10.2) and (8.3) give

$$A_{m,n} - S_{m,n} = \begin{cases} 0, & n-m \text{ odd}, \\ m - \Delta_{m,n}, & n-m \text{ even}. \end{cases}$$

Thus arbitrary real-valued lower terms do not change the limiting coefficient $m+1$ for fixed m, although they can change the bounded correction. This compares only the two monomial-head spaces, not the unrestricted harmonic and rectangular maxima.

10.4 A rational quartic with fourteen zeros

The first strict comparison above has a particularly simple witness. Set $t = 1/1000$ and consider

$$F(z) = t^{16} z^4 - t^4(2x^2 + 5xy + 2y^2) + t\left(x + \frac{y}{16}\right) + 1, \qquad z = x + iy. \tag{10.10}$$

All possible zeros lie on $y = 0$, $y = x$, $x = 0$, or $y = -x$. Use respectively the real parametrizations $r, (1+i)r, ir, (-1+i)r$. The restrictions are $at^{16}r^4 + bt^4r^2 + ctr + 1$, with the following exact coefficients:

Line	a	b	c	Number of real roots
$y = 0$	1	-2	1	4
$y = x$	-4	-9	$17/16$	2
$x = 0$	1	-2	$1/16$	4
$y = -x$	-4	1	$-15/16$	4

For the support $0, 1, 2, 4$, all unscaled coefficients have absolute value at least $1/16$ and at most 9. Since $3 \cdot 9t < 1/16$, Lemma 5.1 applies. The two Descartes counts give the last column, and all these roots are simple. The four lines meet only at the origin, where $F(0) = 1$, so there are exactly fourteen distinct zeros. Their real Jacobians are nonzero by the argument already given.

This example is not harmonic: its Laplacian is $-8t^4 \neq 0$. It proves $A_{2,4} = 14$, whereas Theorem 8.1 gives $S_{2,4} = 12$. It does not give an upper bound of twelve for an arbitrary harmonic polynomial with analytic degree four and anti-analytic degree two.

The supplied standard-library Python program independently verifies (10.10) by exact rational Sturm computations. It also checks square-freeness of every line restriction and all the strict dominance inequalities. Its finite combinatorial tests of (10.8) and (10.7) are diagnostic only; the proofs above cover all degrees.

11 Sparse extremizers and a published conjecture

The coefficient-count conjecture in [SZ24, Remark 4.4, p. 16] is false. We give the least-degree counterexample, an exact obstruction for binomials, and sharp zero counts in a full cyclic coefficient space. The size of the support is a different extremal question from the number of zeros at prescribed bidegree. Write $\nu(F)$ for the number of nonzero monomials of F.

11.1 Two monomials and the first possible counterexample

The needed lower bound for the coefficient count does not require a curve-theoretic argument.

Proposition 11.1. *Let $n \geq 2$. A degree-n polyanalytic polynomial with at most two nonzero monomials and finitely many zeros has at most $2n$ distinct zeros. This bound is attained.*

Proof. A single nonconstant monomial has only the origin as a zero. For two distinct monomials write

$$F(z) = \alpha z^{a_1}\bar{z}^{b_1} + \beta z^{a_2}\bar{z}^{b_2}, \qquad \alpha\beta \neq 0, \quad d_j = a_j + b_j, \quad k_j = a_j - b_j.$$

For $z = re^{i\theta} \neq 0$, the equation is

$$r^{d_1-d_2}e^{i(k_1-k_2)\theta} = -\beta/\alpha. \tag{11.1}$$

If $d_1 = d_2$, a nonzero solution generates a whole ray; hence finiteness excludes all nonzero solutions. If $d_1 \neq d_2$ and $k_1 = k_2$, a nonzero solution generates a whole circle. In the remaining case, (11.1) gives one positive radius and exactly $|k_1 - k_2|$ angles. Since d_1, d_2 are distinct integers with maximum n,

$$|k_1 - k_2| \leq d_1 + d_2 \leq 2n - 1.$$

Adding the origin when it is a zero proves the upper bound. The polynomial $z^n - \bar{z}^{n-1}$ has the origin and $2n-1$ distinct unit roots as its zeros, and attains the bound. □

Let s_n be the least support size of a degree-n extremal polynomial. The exact low-degree values follow once Theorem 11.3 is proved.

Corollary 11.2. *One has $s_1 = 1$, $s_2 = 2$, $s_3 = 3$, and $s_4 = 3$.*

Proof. The examples z and $z^2 - \bar{z}$ give the first two values, and a single monomial cannot have four zeros. For degree three use

$$z^3 + z\bar{z} - \frac{1}{8}.$$

Its imaginary part vanishes on the three lines $e^{2\pi ij/3}\mathbb{R}$, $j = 0, 1, 2$. On each line the restriction is

$$r^3 + r^2 - \frac{1}{8} = \left(r + \frac{1}{2}\right)\left(r^2 + \frac{1}{2}r - \frac{1}{4}\right),$$

which has three different nonzero real roots. Thus there are nine distinct zeros. Proposition 11.1 excludes two monomials in degrees three and four; Theorem 11.3 supplies the quartic upper bound for s_4. □

In particular, the proposed inequality $s_n \geq n$ first fails at $n = 4$. This is a statement about the degree of the first counterexample, not a determination of s_n for all n.

Theorem 11.3. *The polynomial*

$$F(z) = z^4 + z\bar{z}^2 - \frac{1}{8}\bar{z} \tag{11.2}$$

has sixteen nondegenerate zeros. Among all degree-four polyanalytic polynomials with sixteen distinct zeros, the least number of nonzero coefficients is three.

There is also a uniform even-degree construction. In this statement the number of coefficients is an attained upper bound for the least support size; minimality is asserted only in the quartic case above.

Theorem 11.4. *For every even integer $n \geq 2$, there is a polynomial $P \in \mathbb{Q}[z, w]$ whose unique leading homogeneous term is a nonzero multiple of z^n, with $\deg_w P = n/2$, exactly $n/2 + 1$ nonzero coefficients, and n^2 nondegenerate zeros on $w = \bar{z}$.*

Section 11.1 gives the elementary obstruction for two monomials. Section 11.2 proves Theorem 11.3 and, in Proposition 11.5, classifies the entire complex one-parameter family containing it. Section 11.3 proves a sharp zero-count formula in a cyclically equivariant coefficient space and yields Theorem 11.4.

11.2 A trinomial with sixteen zeros

We first prove Theorem 11.3 directly. The parameter calculation afterward gives its full range of stability within the trinomial family.

Proof. More generally, for real t put

$$F_t(z) = z^4 + (|z|^2 - t)\bar{z}.$$

The origin is a zero, with $J_{F_t}(0) = -t^2$. For a nonzero zero write $z = re^{i\theta}$ and divide by $\bar{z}$. This gives

$$r^3 e^{5i\theta} + r^2 - t = 0. \tag{11.3}$$

Suppose $0 < t < 4/27$. The positive radii with $e^{5i\theta} = 1$ are the positive roots of $r^3 + r^2 - t$, which has exactly one such root. The radii with $e^{5i\theta} = -1$ are the positive roots of $r^2 - r^3 - t$. The latter function is negative at zero, has its unique positive maximum $4/27 - t$ at $2/3$, and tends to $-\infty$; hence it has exactly two positive roots. These three radii are different and simple, and each gives five angles in (11.3). Together with zero, there are exactly sixteen zeros.

For $t = 1/8$ they are

$$\{0\} \cup \left\{ \frac{\sqrt{5}-1}{4} e^{2\pi i j/5} : 0 \leq j < 5 \right\} \cup \left\{ re^{(2j+1)\pi i/5} : r \in \left\{ \frac{1}{2}, \frac{1+\sqrt{5}}{4} \right\},\ 0 \leq j < 5 \right\}. \tag{11.4}$$

At a nonzero zero put $\epsilon = e^{5i\theta} \in \{1, -1\}$. Direct differentiation and the relation $t = r^2 + \epsilon r^3$ give

$$J_{F_t}(z) = 5r^5(3r + 2\epsilon).$$

The positive-branch root has positive Jacobian. The two negative-branch radii lie strictly on opposite sides of $2/3$, so their Jacobians are nonzero as well. The origin is nondegenerate because $t \neq 0$. The sixteen points in (11.4) are therefore all nondegenerate. Formula (11.2) has three nonzero coefficients; Proposition 11.1 proves minimality. □

The same example admits an exact description for a complex parameter. For $\tau = a + ib$ define

$$F_\tau(z) = z^4 + z\bar{z}^2 - \tau\bar{z}, \qquad \mathcal{D}(a,b) = 4a^3 + (36a - 4)b^2 - 27(a^2 + b^2)^2. \tag{11.5}$$

The two exceptional nonzero parameters below correspond to a triple radial root.

Proposition 11.5. *For the family* (11.5)*,*

$$N(F_\tau) = \begin{cases} 16, & \mathcal{D}(a,b) > 0, \\ 11, & \mathcal{D}(a,b) = 0, \quad \tau \notin \{0,\ 1/6 + i/(6\sqrt{3}),\ 1/6 - i/(6\sqrt{3})\}, \\ 6, & \textit{otherwise.} \end{cases}$$

If $\mathcal{D}(a,b) \neq 0$, every zero is nondegenerate.

Proof. For $z = re^{i\theta} \neq 0$ the equation is $r^3 e^{5i\theta} = \tau - r^2$. With $s = r^2$, its positive radial solutions are precisely the positive roots of

$$\psi_\tau(s) = s^3 - s^2 + 2as - (a^2 + b^2). \tag{11.6}$$

Each positive root gives exactly five different angles, since $\tau - s \neq 0$ there. The discriminant of (11.6) is $\mathcal{D}(a,b)$.

If $\mathcal{D} > 0$, then $a > 0$: for $a \leq 0$ all terms in the formula for $\mathcal{D}$ are nonpositive, with equality only at $a = b = 0$. The cubic has three different real roots, with sum 1, pairwise-product sum $2a > 0$, and product $a^2 + b^2 > 0$. They must all be positive. Indeed, if two were $-u, -v$ with $u, v > 0$, the third would be $1 + u + v$ and the pairwise-product sum would be $uv - (u + v)(1 + u + v) < 0$. Thus $N(F_\tau) = 1 + 5 \cdot 3$. If $\mathcal{D} < 0$, there is one real root. It is positive because $\psi_\tau(0) < 0$ and the leading coefficient is positive. This gives $1 + 5$ zeros.

If $\mathcal{D} = 0$ and $\tau \neq 0$, the same sign argument gives only positive real roots. There are two distinct ones unless the cubic is $(s - 1/3)^3$, which occurs exactly at the two listed nonzero exceptional parameters. These cases give eleven and six zeros, respectively. For $\tau = 0$, the sole positive root is $s = 1$, so again there are six zeros including zero.

At a nonzero zero a direct polar-coordinate computation gives

$$J_{F_\tau}(z) = 5s\psi'_\tau(s). \tag{11.7}$$

Thus every nonzero zero is nondegenerate when $\mathcal{D} \neq 0$. The origin has Jacobian $-|\tau|^2 \neq 0$ in these cases. This proves the last assertion using (11.7). □

11.3 A sharp count under cyclic symmetry

Fix $n \geq 2$, put $k = \lfloor n/2 \rfloor$, and let $\omega = e^{2\pi i/(n+1)}$. Consider all total-degree-n polynomials whose leading homogeneous part is cz^n, $c \neq 0$, and satisfy $F(\omega z) = \omega^{-1} F(z)$. Since a monomial $z^i \bar{z}^j$ transforms by ω^{i-j}, this full coefficient space is exactly

$$\mathcal{C}_n = \{F(z) = cz^n + \bar{z}R(|z|^2) :\ c \in \mathbb{C} \setminus \{0\},\ R \in \mathbb{C}[s],\ \deg R \leq k - 1\}.$$

The zero polynomial R is allowed. The parity of n changes the exact maximum in this space.

Theorem 11.6. *For every $n \geq 2$,*

$$\max_{F \in \mathcal{C}_n} N(F) = \begin{cases} n^2, & n \text{ even}, \\ n^2 - n - 1, & n \text{ odd}. \end{cases} \tag{11.8}$$

The maximum is attained with rational coefficients, exactly $k+1$ nonzero monomials, and all zeros nondegenerate.

We separate the amplitude count from the elementary construction that attains it. This also handles missing coefficients and multiple radial roots in the upper bound.

Lemma 11.7. *For $F = cz^n + \bar{z}R(|z|^2) \in \mathcal{C}_n$, let $Z_+(\Phi)$ be the number of distinct positive roots of*

$$\Phi(s) = |c|^2 s^{n-1} - R(s)\overline{R(\bar{s})} \in \mathbb{R}[s]. \tag{11.9}$$

Then $N(F) = 1 + (n+1)Z_+(\Phi)$. At a nonzero zero with $s = |z|^2$,

$$J_F(z) = (n+1)s\Phi'(s). \tag{11.10}$$

Proof. The origin is always a zero. Dividing the equation at a nonzero point $z = re^{i\theta}$ by $\bar{z}$ gives

$$cr^{n-1}e^{i(n+1)\theta} + R(r^2) = 0.$$

Its modulus equation is exactly (11.9). At a positive root, $R(s) \neq 0$, and the phase equation has exactly $n+1$ solutions. The polynomial Φ has degree $n-1$ and positive leading coefficient, since $2 \deg R \leq 2k - 2 < n - 1$. Thus the zero count is finite.

For the Jacobian write $F = \bar{z}G$ off the origin, where $G = cr^{n-1}e^{i(n+1)\theta} + R(r^2)$. At a zero, multiplication by $\bar{z}$ contributes determinant s, while conversion from polar to Cartesian coordinates contributes $1/r$. Substituting $cr^{n-1}e^{i(n+1)\theta} = -R(s)$ into the two derivatives of G gives

$$J_F = (n+1)\big((n-1)|R(s)|^2 - 2s\Re(R'(s)\overline{R(s)})\big).$$

Using $|R(s)|^2 = |c|^2 s^{n-1}$ yields (11.10). □

For the lower bound we use a coefficient-dominance observation. It is just a quantitative way of making Descartes' rule of signs exact.

Lemma 11.8. *Let $0 \leq e_0 < \cdots < e_q$ be integers and $\sigma_j \in \{-1, 1\}$. Let $t > 0$ satisfy $qt < 1$. Then*

$$p_t(r) = \sum_{j=0}^{q} \sigma_j t^{e_j^2} r^{e_j}$$

has exactly as many positive roots as there are sign changes in $(\sigma_0, \ldots, \sigma_q)$. Every positive root is simple.

Proof. At $\rho_j = t^{-2e_j}$,

$$t^{e_j^2} p_t(\rho_j) = \sigma_j + \sum_{\ell \neq j} \sigma_\ell t^{(e_\ell - e_j)^2}.$$

The last sum has modulus at most $qt < 1$. Thus the values at the increasing radii ρ_j have the prescribed signs. Each sign change produces a different positive root. Descartes' rule, counting roots with multiplicity, gives the matching upper bound and forces all of these roots to be simple. The case $q = 0$ is immediate. □

We can now prove the sharp count and the even-degree sparsity claim.

Proof. If $R = 0$, then $N(F) = 1$. Otherwise let h be the order of vanishing of R at zero, so $0 \le h \le k-1$. Factoring s^{2h} from Φ leaves a real polynomial with positive leading coefficient, negative constant term, and degree $D = n-1-2h$. If n is even, D is odd, and there are at most $D \le n-1$ different positive roots. If n is odd, D is even. There cannot be D different positive roots, since their product would make the constant term positive. Thus there are at most $D-1 \le n-2$. Lemma 11.7 proves the two upper bounds in (11.8), including all degeneracies and missing terms.

For attainment, choose the explicit rational parameter

$$t = \frac{1}{2(k+1)}, \qquad c = t^{(n-1)^2}, \qquad R_t(s) = \sum_{j=0}^{k-1} (-1)^j t^{4j^2} s^j,$$

and put

$$G_{n,t}(z) = t^{(n-1)^2} z^n + \bar{z} \sum_{j=0}^{k-1} (-1)^j t^{4j^2} |z|^{2j}. \tag{11.11}$$

The positive radial solutions are the positive roots of the two real polynomials

$$R_t(r^2) + cr^{n-1}, \qquad R_t(r^2) - cr^{n-1}.$$

Their common support is $0, 2, \dots, 2k-2, n-1$. In both coefficient lists the first k signs alternate, contributing $k-1$ changes. Their last signs are opposite, so their last intervals contribute one change in total. Lemma 11.8, with $q = k$, shows that they have exactly k and $k-1$ positive roots, in some order, all simple. They have no common positive root because their difference is $2cr^{n-1}$. Consequently, Φ has exactly $2k-1$ positive simple roots. Lemma 11.7 now gives

$$N(G_{n,t}) = 1 + (n+1)(2k-1),$$

which is (11.8). Formula (11.10) proves nondegeneracy at every nonzero zero, and the origin has Jacobian -1. Finally, (11.11) has exactly $k+1$ rational nonzero coefficients and conjugate degree k. □

For even n the construction (11.11) proves Theorem 11.4. It is useful to compare the parity classes:

n	2	4	6	8	10
$N(G_{n,t})$	4	16	36	64	100
$\nu(G_{n,t})$	2	3	4	5	6

The general zero bound shows that each listed even-degree count is extremal, not merely maximal among the displayed examples.

For completeness, the analogous odd-degree radial construction gives a uniform support estimate across both parities.

Corollary 11.9. *For every $n \ge 2$, there is a rational-coefficient polynomial of total degree n, with holomorphic leading term, $\lceil n/2 \rceil + 1$ nonzero coefficients, and n^2 nondegenerate zeros.*

Proof. For even n use Theorem 11.4. For $n = 2k+1$ choose a rational $0 < t < 1/(k+1)$ and set

$$H_{n,t}(z) = t^{n^2} z^n + \sum_{j=0}^{k} (-1)^j t^{4j^2} |z|^{2j}.$$

Its imaginary part confines the zeros to n lines. On one such line, with real coordinate r, its restriction is $\pm t^{n^2} r^n + \sum_{j=0}^{k} (-1)^j t^{4j^2} r^{2j}$. Applying Lemma 11.8 to its positive and negative half-lines gives $2k+1 = n$ simple nonzero real roots: the k even exponent gaps contribute $2k$ sign changes in total, and the final odd gap contributes one. The lines meet only at zero, which is not a root. Thus there are n^2 distinct zeros. The real part has nonzero tangential derivative and the imaginary part has nonzero normal derivative at every zero, proving nondegeneracy. There are $k+2$ nonzero coefficients. □

11.4 The earlier four-term quintic

For comparison with the first sparsity counterexample in the research record, we retain its explicit factorization. The even-degree construction above has superseded the earlier even-degree induction by saving one term. Set

$$F_5(z) = 99z^5 + 1031z^2\bar{z}^2 - 4132z\bar{z} + 3200. \tag{11.12}$$

The pure-head intersection bound is Lemma 2.1; no new upper-bound argument is needed.

For clarity, the next proposition locates every zero, rather than merely counting them.

Proposition 11.10. *Let F_5 be given by (11.12), and set*

$$\rho_\pm = \frac{128 \pm 4\sqrt{34}}{99}.$$

Its zero set consists of the 25 distinct points

$$\left\{\rho_\pm e^{2\pi i j/5} : 0 \le j < 5\right\} \cup \left\{re^{(2j+1)\pi i/5} : r \in \{1, 2, 10\},\ 0 \le j < 5\right\}. \tag{11.13}$$

Every zero is nondegenerate.

Proof. The real polynomial

$$\begin{aligned} f(s) &= 99s^5 + 1031s^4 - 4132s^2 + 3200 \\ &= (s+1)(s+2)(s+10)(99s^2 - 256s + 160) \end{aligned} \tag{11.14}$$

has five distinct nonzero real roots: $-1, -2, -10, \rho_-, \rho_+$. The quadratic factor has discriminant $64 \cdot 34 > 0$, and both its roots are positive.

Since $\operatorname{Im} F_5(z) = 99 \operatorname{Im} z^5$, every zero lies on one of the five lines $L_j = e^{ij\pi/5}\mathbb{R}$, $0 \le j < 5$. On L_j, with $z = se^{ij\pi/5}$ and $s \in \mathbb{R}$, the restriction of F_5 is $f(s)$ when j is even and $f(-s)$ when j is odd. Thus (11.14) supplies exactly five simple roots on every line. The lines meet only at 0, where $F_5(0) \neq 0$; hence there are exactly 25 zeros. Rewriting the signed real roots in polar coordinates gives (11.13).

At any of these zeros, the derivative of $\operatorname{Re} F_5$ along its line is nonzero. The derivative of $\operatorname{Im} F_5$ along that line is zero, whereas its derivative in the normal direction is nonzero, since $z \neq 0$. The two real gradients are therefore independent. □

The four monomials in (11.12) are distinct, and all four coefficients are nonzero. Proposition 11.10 consequently contradicts the coefficient-count assertion in [SZ24, Remark 4.4, p. 16] in degree five, with no additional hypothesis or limiting argument. The polynomial is not harmonic: its terms $z\bar{z}$ and $z^2\bar{z}^2$ are genuinely mixed.

The odd-degree radial family and the smaller even-degree support are already covered by Corollary 11.9. Thus no second odd-degree proof is needed here. For every $n \geq 2$,

$$s_n \leq \lceil n/2 \rceil + 1.$$

Minimality in all higher degrees remains unproved; the exact low-degree values are those of Corollary 11.2.

12 A local four-zero splitting

We turn to the finite-degree equalities (1.2). The total-degree saturation lemma and the common-factor criterion proved earlier will exhaust each construction, rather than leaving an unproved global finiteness assumption. The following quadratic model is elementary.

Lemma 12.1. *If $\alpha, \beta \in \mathbb{C}$ and $|\alpha| < |\beta|$, then $u^2 + \alpha u - \beta\bar{u}$ has exactly four zeros, all nondegenerate.*

Proof. The origin has negative Jacobian $|\alpha|^2 - |\beta|^2$. A nonzero zero satisfies $|u + \alpha| = |\beta|$. If $|2u + \alpha| \leq |\beta|$, the triangle inequality gives

$$|u + \alpha| \leq \tfrac{1}{2}\big(|2u + \alpha| + |\alpha|\big) < |\beta|,$$

a contradiction. Every nonzero zero has positive Jacobian. The map is proper, all its zeros are isolated, and its degree is two because its highest homogeneous part is u^2. The signed zero count is two; hence there are three positive zeros and the one negative zero at the origin. □

A common perturbation can realize this model at every zero simultaneously.

Lemma 12.2. *Suppose $f = z^N +$ lower-total-degree terms has N^2 distinct zeros. Let $A, B \in \mathbb{C}[z, w]$ satisfy $\deg(Af - Bf^*) < 2N$ and*

$$|A(z, \bar{z})| < |B(z, \bar{z})| \quad \text{whenever } F_f(z) = 0.$$

For all sufficiently small positive ε, the polynomial

$$g_\varepsilon = f^2 + \varepsilon(Af - Bf^*) \tag{12.1}$$

has exactly $4N^2$ real-slice zeros, all nondegenerate and lying in arbitrarily small neighborhoods of the zeros of F_f.

Proof. At a zero z_0, Lemma 2.1 makes $\xi = F_f(z)$ a local real-analytic coordinate. On the real slice, $f^* = \overline{f}$. Substituting $\xi = \varepsilon u$ in (12.1) and dividing by ε^2 gives a C^1 limit on bounded u-sets equal to

$$u^2 + A(z_0, \bar{z}_0)u - B(z_0, \bar{z}_0)\bar{u}.$$

Lemma 12.1 and the implicit function theorem give four nondegenerate zeros near each of the N^2 seed zeros. The highest homogeneous part of g_ε is z^{2N}, so Lemma 2.1 excludes every additional zero. This also proves the assertion about neighborhoods, after taking ε sufficiently small. □

13 Explicit seeds and two successive squarings

Fix $a = 1 + 2i$, $b = 1 - 2i$, and $c = \sqrt{5}$. Define

$$\begin{aligned} &S_0 = 3, \qquad S_1 = z, \qquad S_2 = z^2 + 2bw, \\ &S_k = zS_{k-1} + bwS_{k-2} - ab^2 S_{k-3} \quad (k \geq 3), \\ &H_k = S_k + (-a)^k. \end{aligned} \tag{13.1}$$

The S_k are the power sums of the roots of

$$\lambda^3 - z\lambda^2 - bw\lambda + ab^2 = 0. \tag{13.2}$$

We record both the roots and the support of these polynomials.

Proposition 13.1. *For every $k \geq 2$, the polynomial H_k has highest homogeneous part z^k, support contained in $i + 2j \leq k$, and exactly k^2 nonzero nondegenerate real-slice zeros. They are*

$$z = b(u^{-1} + v^{-1}) - auv, \qquad u^k = 1, \quad v^k = -1. \tag{13.3}$$

Its conjugate degree is $\lfloor k/2 \rfloor$. For $k = 2r + 1 \geq 3$,

$$[w^r]H_{2r+1} = (2r + 1)b^r z. \tag{13.4}$$

Proof. Assign weight one to z and weight two to w in (13.1). Induction gives the support condition and shows that the only term of total degree k is z^k. The highest-weight coefficients are $[w^r]S_{2r} = 2b^r$ and $[zw^r]S_{2r+1} = (2r + 1)b^r$. All weights occurring in S_k are congruent to k modulo three. Thus no constant multiple of w^r occurs in S_{2r+1}, which proves (13.4).

At a point (13.3), set $w = \bar{z} = a(u + v) - b/(uv)$. The three roots of (13.2) are

$$b/u, \qquad b/v, \qquad -auv.$$

Their kth power sum is $-(-a)^k$, so $H_k = 0$. To check distinctness, the numbers u, v are roots of $aT^3 - wT^2 - zT + b = 0$, whose third root is $-b/(auv)$. The kth power of the third root is neither 1 nor -1: otherwise b/a would be a root of unity. But $b/a = (-3 - 4i)/5$ and its sum with its inverse is $-6/5$, not a rational algebraic integer. Thus u and v can be uniquely recovered, proving that all k^2 points are distinct. Lemma 2.1 gives completeness and nondegeneracy. If their sum z were zero, the three equal-modulus roots of (13.2) would form an equilateral triple. Then v/u would be a cube root of unity, contrary to $(v/u)^k = -1$. Hence no zero is the origin. □

13.1 Sharp constants in the unit-triple estimate

We determine the individual best constants in every order not divisible by three, as well as orders $3, 6, 9, 12, 15$, with every equality case. We also determine the best linear coefficient valid simultaneously in all orders $k \geq 3$. Individual constants at multiples of three starting at eighteen remain unresolved here.

Lemma 13.2 (Trace geometry near the origin)**.** *For every $s \in \mathbb{C}$ with $|s| < 1$, the polynomial $X^3 - sX^2 + \bar{s}X - 1$ has three distinct unit roots. If $0 < R = |s| < 1$, they can be numbered as*

$$\eta_j = \omega_j e^{i\delta_j}, \qquad \omega_j^3 = 1, \quad \sum_j \delta_j = 0, \quad |\delta_j| \leq \tfrac{2}{3} \arcsin R < \tfrac{\pi}{3}.$$

In particular, if $h = \max_j |\delta_j|$, then $h > 0$, $\sum_j |\delta_j| = 2h$, and $R \geq \sin(3h/2)$.

Proof. The roots are invariant under reciprocal conjugation. A repeated nonunit root would require its reciprocal conjugate with the same multiplicity, impossible in degree three. A repeated unit root would give $s = 2\eta + \eta^{-2}$ and $|s| \geq 1$. Thus the cubic has no repeated root in the open unit trace disk. Starting at $s = 0$ and continuing along the segment to s, reciprocal conjugation fixes each continued root, since it has the same initial value. All three roots are unit. Their equation gives $|\eta_j^3 - 1| \leq 2R$. The three resulting arcs are disjoint, so continuation assigns one root to each cube root of unity and proves the phase bound. The product of the roots is one; since $|\sum \delta_j| < \pi$, their sum is zero. The three absolute phases are therefore $h, a, h - a$ for some $0 \leq a \leq h$. The remaining conclusions follow from the phase bound and $s \neq 0$. □

Proposition 13.3 (Best constants for orders two through five)**.** *Let* $|\eta_j| = 1$, $\eta_1\eta_2\eta_3 = 1$, *and* $s = \eta_1 + \eta_2 + \eta_3 \neq 0$. *The least constants* C_k *for which*

$$\left|\eta_1^k + \eta_2^k + \eta_3^k - 3\mathbf{1}_{3|k}\right| \leq C_k|s|$$

holds for every such triple are

$$C_2 = 3, \qquad C_3 = 4, \qquad C_4 = \frac{52 + 20\sqrt{10}}{27}, \qquad C_5 = 5 + 5\rho - 5\rho^2 - 5\rho^3 - \rho^4, \tag{13.5}$$

where ρ *is the unique positive root of* $4\rho^3 + 15\rho^2 + 10\rho - 5 = 0$*; in fact* $3/10 < \rho < 1/3$. *Writing* $\omega = e^{2\pi i/3}$, *equality holds precisely at the traces*

k	*equality traces*
$2, 3$	$s^3 = -1$
4	$s = \dfrac{4 - \sqrt{10}}{3}\,\omega^j, \quad j = 0, 1, 2$
5	$s = -\rho\omega^j, \quad j = 0, 1, 2.$

Each trace determines the unordered triple uniquely as the roots of $X^3 - sX^2 + \bar{s}X - 1$. *For* $k = 2, 3$, *the triples are the permutations of* $(q, q, -q)$ *with* $q^3 = -1$. *For* $k = 4, 5$, *all three roots are distinct.*

Proof. Put $s = Re^{i\theta}$, $0 < R \leq 3$, and $p_k = \sum_j \eta_j^k$. Newton's identities give

$$\begin{aligned} p_2 &= s^2 - 2\bar{s}, \qquad p_3 - 3 = s(s^2 - 3\bar{s}), \\ p_4 &= s^4 - 4s^2\bar{s} + 2\bar{s}^2 + 4s, \\ p_5 &= s^5 - 5s^3\bar{s} + 5s\bar{s}^2 + 5s^2 - 5\bar{s}. \end{aligned}$$

For $k = 2$, use $|p_2|/R \leq R + 2$ when $R \leq 1$, and $|p_2|/R \leq 3/R$ when $R \geq 1$. The common upper bound is three, and equality forces $R = 1$ and $e^{3i\theta} = -1$. For $k = 3$, use $s^2 - 3\bar{s} = \frac{1}{2}\sum_{j<\ell}(\eta_j - \eta_\ell)^2$ and $\sum_{j<\ell}|\eta_j - \eta_\ell|^2 = 9 - R^2$. These identities and the triangle inequality give

$$|p_3 - 3|/R \leq R^2 + 3R, \qquad |p_3 - 3|/R \leq (9 - R^2)/2.$$

Use the first for $R \leq 1$ and the second for $R \geq 1$. Their common upper bound is four, with the same equality traces.

For $k = 4$, the range $R \geq 1$ gives $|p_4|/R \leq 3$. On $0 < R < 1$,

$$\frac{p_4}{s} = 4 - 4R^2 + R^3e^{3i\theta} + 2Re^{-3i\theta}, \qquad |p_4|/R \leq f_4(R) := 4 + 2R - 4R^2 + R^3.$$

All three terms in this last triangle inequality have positive coefficients, so equality requires $e^{3i\theta} = 1$. The unique maximum of f_4 on $[0, 1]$ occurs at $R = (4 - \sqrt{10})/3$ and equals the stated $C_4 > 4$.

For $k = 5$, the range $R \geq 3/5$ gives $|p_5|/R \leq 5$. For $0 < R < 3/5$, multiply p_5/s by $e^{2i\theta}$ to obtain $a + bv + cv^2$, where

$$v = e^{3i\theta}, \quad a = -5 + 5R^2 < 0, \quad b = 5R(1 - R^2) > 0, \quad c = R^4 > 0.$$

With $t = \Re v$, its squared modulus is

$$(a - c)^2 + b^2 + 2b(a + c)t + 4act^2.$$

This is a strictly concave quadratic with vertex

$$t_* = -\frac{5 - 5R^2 - R^4}{4R^3} < -1.$$

Indeed, $5 - 5R^2 - 4R^3 - R^4$ is strictly decreasing for $R > 0$ and is $1379/625 > 0$ at $R = 3/5$. Thus the angular maximum is attained only at $v = -1$, and equals

$$f_5(R) = 5 + 5R - 5R^2 - 5R^3 - R^4.$$

Its derivative has exactly one positive zero, namely ρ, because $4R^3 + 15R^2 + 10R - 5$ is strictly increasing on $R > 0$. The maximum exceeds $f_5(3/10) > 5$, excluding the outer range.

Finally all displayed equality traces are admissible. For $R = 1$ this follows from $(q, q, -q)$; for $R < 1$, Lemma 13.2 proves that $X^3 - sX^2 + \bar{s}X - 1$ has three distinct unit roots. This proves attainment as well as necessity of every equality condition. □

Proposition 13.4 (The sixth-order constant and all equality cases)**.** *With the notation of Proposition 13.3, let y_* be the unique root in $(0, 1)$ of*

$$25y^3 - 50y^2 - 5y + 12 = 0, \tag{13.6}$$

and put $x_* = (5y_*^2 - 11y_* + 4)/2$. Then

$$C_6 = \frac{6}{5\sqrt{5}}\sqrt{545y_*^2 - 1420y_* + 704}. \tag{13.7}$$

Equality holds exactly at the six traces satisfying

$$|s|^2 = y_*, \qquad \operatorname{Re}(s^3) = y_*x_*. \tag{13.8}$$

Each trace determines the unordered triple as the three distinct unit roots of $X^3 - sX^2 + \bar{s}X - 1$. In particular, the equality traces do not lie on the rays $\Im(s^3) = 0$ that supplied the lower-order maxima.

Proof. If $R = |s| \geq 1$, the elementary bound $|p_6 - 3| \leq 6$ gives a quotient at most six. For $0 < R < 1$, put $Y = R^2$ and $X = R\cos(3\arg s)$. Newton's identity is

$$p_6 - 3 = s^6 - 6s^4\bar{s} + 9s^2\bar{s}^2 + 6s^3 - 2\bar{s}^3 - 12s\bar{s}.$$

An expansion therefore gives $|p_6 - 3|^2/R^2 = Q(X, Y)$, where

$$\begin{aligned} Q(X, Y) = {} & -Y(4X - Y - 4)\big(4X^2Y - 8XY^2 + 4XY + 12X \\ & + Y^3 + 14Y^2 - 47Y + 36\big). \end{aligned} \tag{13.9}$$

The compact parameter region is $X^2 \leq Y \leq 1$; at its only point with $Y = 0$ the quotient extends continuously as zero.

We first exclude its boundary. On $Y = 1$,

$$Q(X,1) = 4(5-4X)(X+1)^2 \le 27, \quad 27 - Q(X,1) = (2X-1)^2(4X+7) \ge 0.$$

On $X = \pm\sqrt{Y}$, with $t = \sqrt{Y} \in [0,1]$, the square root of Q is respectively

$$g_+(t) = t(3-t)(1+t)(2-t)^2, \qquad g_-(t) = t(1-t)(t+2)^2(t+3).$$

Both are strictly below six. Here is a rational positivity certificate requiring only expansions of quintics: the minimum degree-five Bernstein coefficient of $6-g$ on each listed interval is

g	interval	minimum Bernstein coefficient
g_+	$[0,1]$	$9/10$
g_-	$[0,1/2]$	$17/32$
g_-	$[1/2,3/4]$	$61/1280$
g_-	$[3/4,1]$	$699/1024$.

Explicitly, for an interval $[a,b]$, write $6 - g(a+(b-a)t) = \sum_{\ell=0}^{5} a_\ell t^\ell$. Its Bernstein coefficients are $\sum_{\ell=0}^{j} a_\ell \binom{j}{\ell}/\binom{5}{\ell}$, $0 \le j \le 5$. The displayed positive minima thus certify the boundary assertion without numerical optimization.

For an interior stationary point, exact elimination gives

$$\operatorname{Res}_X(Q_X, Q_Y) = 110592Y^4(Y-4)^5(Y-1)^2(25Y^3 - 50Y^2 - 5Y + 12). \tag{13.10}$$

The last cubic decreases strictly on $[0,1]$, and its endpoint values have opposite signs. At its unique root, reducing the two stationary equations gives

$$2X = 5Y^2 - 11Y + 4, \qquad Q = \frac{36}{125}(545Y^2 - 1420Y + 704).$$

For clarity, the ideal generated by Q_X, Q_Y and the cubic has the same reduced Gröbner basis as the ideal generated by $2X - 5Y^2 + 11Y - 4$ and that cubic, in lexicographic order $X > Y$. Thus no other interior stationary point has been omitted.

Rational evaluations isolate $503/1000 < y_* < 63/125$. On this interval $x_*^2 < y_* < 1$, and the displayed critical value is greater than 36: its difference from 36 is decreasing there and is $197856/390625 > 0$ at $Y = 63/125$. Direct substitution modulo the cubic verifies stationarity. The maximum is consequently interior and unique in (X,Y), which proves (13.7) and excludes the range $R \ge 1$. Since $|x_*| < \sqrt{y_*}$, equation (13.8) has exactly six trace solutions. All are admissible and have distinct unit roots by Lemma 13.2. This proves both attainment and the complete equality statement. □

Proposition 13.5 (Seventh- and eighth-order constants and equality)**.** *With the notation of Proposition 13.3, put*

$$f_7(R) = 7 + 7R - 21R^2 + 14R^4 - 7R^5 + R^6,$$
$$f_8(R) = 8 + 12R - 24R^2 - 30R^3 + 8R^4 + 20R^5 + 8R^6 + R^7.$$

For $k = 7, 8$, let ρ_k be the unique zero of f_k' in $(0, 3/k)$. Then the individual sharp constants are

$$C_7 = f_7(\rho_7), \qquad C_8 = f_8(\rho_8). \tag{13.11}$$

All equality traces are, respectively,

$$s = \rho_7\omega^j \quad (k=7), \qquad s = -\rho_8\omega^j \quad (k=8), \qquad j = 0,1,2. \tag{13.12}$$

Each trace determines the unordered triple as the three distinct unit roots of $X^3 - sX^2 + \bar{s}X - 1$.

Proof. Write $s = Re^{i\theta}$ and $v = e^{3i\theta}$. The estimate $|p_k| \le 3$ excludes $R \ge 3/k$ once a value greater than k has been exhibited. Newton's recurrence gives

$$\begin{aligned} A_7(R,v) &:= \frac{p_7}{s} = 7(1-R^2)(1-2R^2) + 7R^3(1-R^2)v + R^6v^2 + 7R(1-R^2)v^{-1}, \\ A_8(R,v) &:= e^{2i\theta}\frac{p_8}{s} = -8(1-R^2)(1-2R^2) + 4R(1-R^2)(3-5R^2)v \\ &\qquad + 8R^4(1-R^2)v^2 + R^7v^3 + 2R^3v^{-1}. \end{aligned}$$

All four coefficients in A_7 are positive for $0 < R \le 3/7$. The triangle inequality therefore gives $|A_7| \le f_7(R)$, and equality holds exactly when $v = 1$.

For the eighth order, let $H_8(R,x) = |A_8(R,v)|^2$ with $x = \Re v$. This is a polynomial: in the Laurent product $A_8(R,v)A_8(R,v^{-1})$, replace $v^j + v^{-j}$ by $2T_j(x)$, where T_j is the ordinary Chebyshev polynomial. The polynomial

$$D(R,x) = -R^{-1}\partial_x H_8(R,x)$$

has bidegree $(12,3)$ and is strictly positive on $[0,3/8]\times[-1,1]$. Here is an exact coefficient certificate. Expand $D(3u/8, 2t-1) = \sum_{h=0}^{12}\sum_{\ell=0}^{3} d_{h\ell}u^h t^\ell$. The tensor Bernstein coefficients

$$b_{ij} = \sum_{h=0}^{i}\sum_{\ell=0}^{j} d_{h\ell}\frac{\binom{i}{h}}{\binom{12}{h}}\frac{\binom{j}{\ell}}{\binom{3}{\ell}}, \qquad 0 \le i \le 12, \quad 0 \le j \le 3,$$

have the exact positive minimum

$$\min b_{ij} = \frac{303335857569}{4294967296}. \tag{13.13}$$

Since the Bernstein basis is nonnegative and sums to one, this proves the claimed positivity by rational polynomial arithmetic. For $R > 0$, H_8 is consequently strictly decreasing in x. Its unique angular maximum is $v = -1$, where $A_8(R,-1) = -f_8(R)$. The latter polynomial is positive on $[0,3/8]$, since it is at least $8 - 24(3/8)^2 - 30(3/8)^3 > 0$.

It remains to optimize the two radial polynomials. Both are strictly concave on their respective intervals. More explicitly, the minimum Bernstein coefficients of $-f_7''$ on $[0,3/7]$ in degree four, and of $-f_8''$ on $[0,3/8]$ in degree five, are respectively $50784/2401$ and 48. The same coefficient conversion as above, in one variable, verifies these values. Direct evaluation gives $f_k'(0) > 0 > f_k'(3/k)$. Thus there is exactly one critical point ρ_k, and it is the unique radial maximizer. Since $f_k(0) = k$ and $f_k'(0) > 0$, its value exceeds k, as needed to exclude the outer range. Finally, all traces in (13.12) have modulus below one and hence are admissible by Lemma 13.2. The angular strictness and radial uniqueness prove both attainment and the complete equality statement. □

Proposition 13.6 (The ninth-order constant and all equality cases)**.** *With the notation of Proposition 13.3, define*

$$\begin{aligned} F_9(Y) &= 512Y^9 - 7616Y^8 + 43848Y^7 - 123203Y^6 + 172055Y^5 \\ &\quad - 97935Y^4 - 9396Y^3 + 23040Y^2 - 189Y - 972, \\ L_9(Y) &= 15207936Y^8 - 197849152Y^7 + 915234552Y^6 \\ &\quad - 1698080469Y^5 + 594170401Y^4 + 1588963305Y^3 \\ &\quad - 1428234222Y^2 + 164181132Y + 22342419. \end{aligned}$$

Let y_* be the unique zero of F_9 in $(0, 49/100)$, and put $x_* = -L_9(y_*)/22654044$. For $Y > 0$ set

$$\begin{aligned} a &= YX - 3Y + 3, \qquad b = a^2 + Y^2(Y - X^2), \\ Q_9(X, Y) &= \frac{b(b^2 + 9b - 24a^3 + 18ab)}{Y}. \end{aligned} \tag{13.14}$$

Then the sharp constant and all equality traces are

$$C_9 = \sqrt{Q_9(x_*, y_*)}, \qquad |s|^2 = y_*, \quad \operatorname{Re}(s^3) = y_* x_*. \tag{13.15}$$

There are exactly six such traces, each determining the unordered triple as the three distinct unit roots of $X^3 - sX^2 + \bar{s}X - 1$.

Proof. Newton's identities applied to the triple $(\eta_1^3, \eta_2^3, \eta_3^3)$ give $p_9 - 3 = p_3^3 - 3p_3\bar{p}_3$, while $p_3 = s^3 - 3|s|^2 + 3$. Put $R = |s|$, $Y = R^2$, and $X = R\cos(3\arg s)$. Then $a = \operatorname{Re} p_3$ and $b = |p_3|^2$, so

$$|p_9 - 3|^2/R^2 = b\{b^2 + 9b - 6\operatorname{Re}(p_3^3)\}/Y = Q_9(X, Y).$$

Cancellation in (13.14) yields a polynomial divisible by Y. At $Y = 0$ it therefore has value zero. For $R \geq 7/10$, the elementary estimate $|p_9 - 3| \leq 6$ gives a quotient at most $60/7$. It suffices to maximize Q_9 on the compact region $X^2 \leq Y \leq 49/100$ and exhibit a value greater than $(60/7)^2$.

On $Y = 49/100$ the same estimate applies; every trace in this region is admissible by Lemma 13.2. On $X = \pm\sqrt{Y}$, with $0 \leq t = \sqrt{Y} \leq 7/10$, the square roots of Q_9 are respectively

$$g_+(t) = t(3 - t)(t^3 - 3t^2 + 3)^2, \qquad g_-(t) = t(3 + t)(3 - 3t^2 - t^3)^2.$$

The following positive minima of the degree-eight Bernstein coefficients of $60/7 - g_\pm$ prove that both are below $60/7$:

branch	interval	minimum coefficient
$+$	$[0, 7/10]$	$5381613/5600000$
$-$	$[0, 7/20]$	$126955805813/179200000000$
$-$	$[7/20, 7/10]$	$9172266721/89600000000.$

The coefficient conversion is the one in the proof of Proposition 13.4, with degree eight in place of five. Thus the whole boundary is bounded by $(60/7)^2$.

For an interior stationary point, exact elimination gives

$$\operatorname{Res}_X((Q_9)_X, (Q_9)_Y) = 429981696Y^8(Y - 3)^5(Y^3 - 9Y^2 + 18Y - 9)^5 F_9(Y). \tag{13.16}$$

The cubic factor is increasing and strictly negative on $[0, 49/100]$. The Sturm sequence of F_9 has respectively seven and six sign variations at 0 and $49/100$; hence it has exactly one zero in this interval. Moreover, the ideals

$$((Q_9)_X, (Q_9)_Y, F_9) \quad \text{and} \quad (22654044X + L_9(Y), F_9)$$

have the same reduced Gröbner basis in lexicographic order $X > Y$. This both excludes additional stationary points and verifies the one stated in the proposition.

Rational isolation and polynomial interval evaluation give

$$\frac{244822607}{10^9} < y_* < \frac{244822608}{10^9}, \qquad -\frac{475279}{10^7} < x_* < -\frac{475277}{10^7}, \qquad Q_9(x_*, y_*) > 76 > \frac{3600}{49}.$$

These intervals place the point strictly inside $X^2 < Y < 49/100$. It is consequently the unique global maximizer in (X, Y). All calculations above are over the rationals: the accompanying verifier checks the displayed resultant and ideal equality, the Sturm variations, every Bernstein coefficient, and the last interval inequalities. Finally $|x_*| < \sqrt{y_*} < 1$ gives exactly six admissible traces, each with distinct unit roots. This proves sharpness and every equality case. □

Theorem 13.7 (Every order not divisible by three)**.** *For $k \geq 4$ with $3 \nmid k$, define*

$$\sigma_k = \begin{cases} 1, & k \equiv 1 \pmod 3, \\ -1, & k \equiv 2 \pmod 3, \end{cases} \qquad F_k(R) = \frac{T_k(\sigma_k R, \sigma_k R)}{R},$$

where T_k is the integer polynomial in (19.5). The quotient is a polynomial, with $F_k(0) = k$. Its derivative has exactly one zero ρ_k in $(0, 3/k)$. The individual sharp constant is

$$C_k = F_k(\rho_k) > k. \tag{13.17}$$

Equality holds precisely at the three traces

$$s = \sigma_k \rho_k \omega^j, \qquad j = 0, 1, 2. \tag{13.18}$$

Each trace determines the unordered triple as the three distinct unit roots of $X^3 - sX^2 + \bar{s}X - 1$. Thus no finite-order cutoff is required for orders not divisible by three. Orders $3, 6, 9$ were treated above; this theorem does not determine the individual constants at multiples of three. Proposition 13.9 below additionally settles orders twelve and fifteen; multiples starting at eighteen remain open.

Proof. Write $s = Re^{i\theta}$, $v = e^{3i\theta}$ and $p_k = \sum \eta_j^k$. Put $e = 0$ if $\sigma_k = 1$, and $e = 2$ otherwise. Define the Laurent polynomial

$$A_k(R, v) = e^{i(e-1)\theta} \frac{p_k}{R}, \qquad H_k(R, x) = |A_k(R, v)|^2, \quad x = \operatorname{Re} v.$$

The latter is a real polynomial: replace $v^j + v^{-j}$ by $2T_j(x)$, using ordinary one-variable Chebyshev polynomials in this replacement. We shall prove, on $0 < R \leq 3/k$,

$$\sigma_k \partial_x H_k(R, x) > 0 \quad (-1 \leq x \leq 1), \qquad F_k''(R) < 0, \qquad F_k'(3/k) < 0 < F_k'(0). \tag{13.19}$$

For $k = 4, 5, 7, 8$ the earlier proofs already give the same angular maximizers and unique radial maxima, so those cases are settled.

First consider $10 \leq k \leq 35$, $3 \nmid k$. Newton's recurrence computes A_k, H_k, F_k over the integers. For each of these eighteen orders, every tensor Bernstein coefficient of

$$\frac{\sigma_k}{R} \partial_x H_k(R, x) \quad \text{on } [0, 3/k] \times [-1, 1]$$

is strictly positive, in bidegree $(2k-4, \lfloor k/2 \rfloor - 1)$. Likewise every degree-$(k-3)$ Bernstein coefficient of $-F_k''$ on $[0, 3/k]$ is strictly positive. The endpoint derivatives have the signs in (13.19). These are finite exact rational certificates, reconstructed by `verify_nonmultiple_power_constants.py`; their coefficients' minima, degrees and hashes are recorded in `nonmultiple_power_constants_verification.json`. No numerical sign comparison is used.

Here is a uniform certificate for every $k \geq 36$ in the two remaining residue classes. It includes a bound for the entire uncomputed tail, rather than extrapolating the finite cases. Set $r = kR$,

$u = r/3$, and $h = 36/k$, so $0 \le u, h \le 1$. For $\sigma \in \{1, -1\}$ let $a = 1$ or 2, respectively, and let $e = 0$ or 2 as above. Form the finite Laurent polynomial

$$\mathcal{A}_{\sigma,20}(u,h,v) = \sum_{\substack{1\le d\le 20, i+j=d \\ i+2j\equiv a \ (3)}} \frac{(-1)^j u^{d-1}}{i!j!} \prod_{\ell=1}^{d-1} \left(1 + \frac{3\ell - i - 2j}{36} h\right) v^{(i-j-1+e)/3}. \tag{13.20}$$

An empty product is one. The exponent of v is always an integer. Define

$$D_{\sigma,20}(h,u,x) = \frac{\sigma}{3u} \partial_x |\mathcal{A}_{\sigma,20}(u,h,v)|^2,$$
$$G_{\sigma,20}(h,u) = \sigma \mathcal{A}_{\sigma,20}(u,h,\sigma), \qquad x = \operatorname{Re} v.$$

The first expression has a removable factor u and is a polynomial. On replacing x by $2t - 1$, the following exact tensor Bernstein bounds hold on the indicated unit cubes, for both signs σ:

Polynomial	Bernstein degrees	Every coefficient exceeds
$D_{\sigma,20}$	$(38, 37, 12)$ in (h, u, t)	$1/9$
$-\partial_u^2 G_{\sigma,20}/9$	$(19, 17)$ in (h, u)	$3/40$
$-\partial_u G_{\sigma,20}(h, 1)/3$	19 in h	$1/6$

This table consists of 39292 rational coefficients. To reproduce it, expand each polynomial on the unit cube as $\sum_\alpha c_\alpha x^\alpha$; for multidegree n, the coefficient with index ν is

$$\sum_{\alpha \le \nu} c_\alpha \prod_j \frac{\binom{\nu_j}{\alpha_j}}{\binom{n_j}{\alpha_j}}.$$

The verifier constructs (13.20) directly and checks all entries and the displayed strict inequalities by rational arithmetic. Since the Bernstein basis is nonnegative and sums to one, these are uniform polynomial bounds.

We now justify their application to the full power sum. The coefficient of $s^i \bar{s}^j$ in p_k, with $i + 2j + 3c = k$ and $d = i + j > 0$, is

$$\frac{k(-1)^j (d + c - 1)!}{i!j!c!}.$$

This follows by expanding $-k[t^k] \log(1 - st + \bar{s}t^2 - t^3)$, or by Newton's recurrence. After division by kR and substitution $R = 3u/k$, it gives exactly the summand in (13.20). If c is a negative integer, that summand vanishes because its product contains a zero factor. Thus the formula remains valid for every indicated (i, j).

For a term of total degree d, the sum of absolute values of its normalized coefficients at $r = 3$ is at most

$$b_d = \frac{2^d}{d!} \left(1 + \frac{2d}{36}\right)^{d-1}.$$

Indeed $|3\ell - i - 2j| \le 2d$ and $\sum_{i+j=d} 1/(i!j!) = 2^d/d!$. At $0 \le r \le 3$ the same bound has the additional factor $(r/3)^{d-1}$. Put $d_0 = 21$ and, for $j = 0, 1, 2$, set

$$q_j = \left(\frac{6}{22} + \frac{12}{36}\right) \left(\frac{22}{21}\right)^j < 1, \qquad B_j = \frac{21^j b_{21}}{1 - q_j}, \qquad M_j = \sum_{d=1}^{20} d^j b_d + B_j. \tag{13.21}$$

The elementary inequality $(1+x)^n < e^{nx}$ and $e < 3$ give, for $d \geq 21$,

$$\frac{b_{d+1}}{b_d} \leq \frac{6}{d+1} + \frac{12}{36}.$$

Consequently $\sum_{d\geq 21} d^j b_d \leq B_j$, while the corresponding full sums are at most M_j.

For the angular derivative, a product of degree-d and degree-e' terms has angular frequency at most $(d+e')/3$. Using $|T'_\ell(x)| \leq \ell^2$ for $|x| \leq 1$, the error in the normalized derivative $\sigma\partial_x H_k/(k^3 R)$ is at most

$$\frac{2}{27}(M_2B_0 + 2M_1B_1 + M_0B_2) < \frac{1}{4000}. \tag{13.22}$$

The factor two covers the two choices of which term lies in the tail; counting their overlap twice only increases the bound. The power of r after division by r is nonnegative, so its maximum is bounded at $r = 3$. The errors in the normalized radial second derivative and the derivative at $r = 3$ are respectively bounded by

$$B_2/9 < 1/30000, \qquad B_1/3 < 1/200000.$$

All three inequalities are rational evaluations of (13.21), also checked by the verifier. They are strictly smaller than the corresponding Bernstein margins. This proves the first three inequalities in (13.19) for every $k \geq 36$; the fourth follows from the exact degree-two coefficient, $F'_k(0) = k(k-\sigma_k)/6 > 0$.

Strict angular monotonicity forces $v = \sigma_k$. At that phase $|p_k|/R = |F_k(R)|$. Strict concavity and the endpoint derivative signs give a unique radial maximum $F_k(\rho_k) > F_k(0) = k$. On the other hand, $|F_k(3/k)| \leq k$, because $3/k < 1$ is an admissible trace radius and $|p_k| \leq 3$. Concavity therefore gives $F_k(R) \geq -k$ throughout the interval, so maximizing the absolute value introduces no additional maximizers. The range $R \geq 3/k$ also has $|p_k|/R \leq 3/R \leq k$. Finally the three traces (13.18) have modulus less than one and are admissible by Lemma 13.2. This proves the sharp constant, attainment and every equality case. □

Corollary 13.8 (The sharp asymptotic constant in these orders)**.** *Let t_* be the unique root in $(0, \pi/3)$ of*

$$t(\sin t + \sqrt{3}\cos t) = 1 - \cos t + \sqrt{3}\sin t.$$

Then, along integers not divisible by three,

$$k\rho_k \longrightarrow \sqrt{3}\,t_*, \qquad \frac{C_k}{k} \longrightarrow \Gamma := \frac{1 - \cos t_* + \sqrt{3}\sin t_*}{\sqrt{3}\,t_*}. \tag{13.23}$$

Numerically, $\sqrt{3}\,t_* \approx 1.35878949$ and $\Gamma \approx 1.11562380$. These are asymptotic values for the specified residue classes, not a replacement for the all-order optimal coefficient $4/3$.

Proof. For a real trace s, the three unit roots are $1, e^{i\vartheta}, e^{-i\vartheta}$, where $\cos\vartheta = (s-1)/2$. It follows that, uniformly in C^2 for $0 \leq r \leq 3$,

$$\frac{F_k(r/k)}{k} \longrightarrow g(r) := \frac{1 - \cos(r/\sqrt{3}) + \sqrt{3}\sin(r/\sqrt{3})}{r}.$$

At zero the removable value is one. The formula follows by expanding $\arccos((\sigma_k r/k - 1)/2)$ at $r/k = 0$; the two residue classes give the same limit. Equivalently,

$$g(r) = \int_0^1 \left\{ \cos\frac{tr}{\sqrt{3}} + \frac{1}{\sqrt{3}}\sin\frac{tr}{\sqrt{3}} \right\} dt.$$

The expression in braces is positive for $0 \le tr/\sqrt{3} \le \sqrt{3} < 2\pi/3$, so $g''(r) < 0$. Moreover $g'(0) = 1/6 > 0$, while $g'(\pi/\sqrt{3}) < 0$ is equivalent to $\pi < 2\sqrt{3}$. Thus g has a unique maximizer $r_* \in (0, \pi/\sqrt{3})$. Its derivative equation is the one displayed for $t_* = r_*/\sqrt{3}$. Uniform convergence and uniqueness give both limits in (13.23). □

Proposition 13.9 (The twelfth and fifteenth orders)**.** *The remaining individual constants through order seventeen are also sharp. More precisely,*

$$11.5606419718 < C_{12} < 11.5606419719, \tag{13.24}$$

$$14.3757909243 < C_{15} < 14.3757909244. \tag{13.25}$$

The following algebraic description specifies the constants exactly and gives every equality case. For $k \in \{12, 15\}$ put

$$a = 3 - 3Y + YX, \qquad b^2 = Y^2(Y - X^2),$$

and define the integer polynomial

$$Q_k(X, Y) = \frac{[T_{k/3}(a + ib, a - ib) - 3][T_{k/3}(a - ib, a + ib) - 3]}{Y}. \tag{13.26}$$

The numerator is even in b; substitute the displayed value of b^2 and cancel the factor Y. Let

$$(L_{12}, b_{12}) = (23/2, 12/23), \qquad (L_{15}, b_{15}) = (57/4, 8/19).$$

For each of these orders there is exactly one solution (X_k, Y_k) of

$$\partial_X Q_k = \partial_Y Q_k = 0, \qquad X^2 < Y < b_k^2. \tag{13.27}$$

Then $C_k = \sqrt{Q_k(X_k, Y_k)}$. Equality holds at exactly six traces, characterized by

$$|s|^2 = Y_k, \qquad \mathrm{Re}(s^3) = X_k Y_k. \tag{13.28}$$

Each determines the unordered triple as the distinct unit roots of $X^3 - sX^2 + \bar{s}X - 1$.

Proof. Writing $s = Re^{i\theta}$, put $X = R\cos 3\theta$ and $Y = R^2$. Newton's identity gives

$$p_3 = s^3 - 3|s|^2 + 3 = a + ib, \qquad \frac{|p_k - 3|^2}{|s|^2} = Q_k(X, Y).$$

All points with $X^2 \le Y < 1$ are admissible, by Lemma 13.2. For $R \ge b_k$, the elementary estimate $|p_k - 3| \le 6$ bounds the ratio by L_k. It therefore suffices to examine $X^2 \le Y \le b_k^2$.

The supplied verifier reconstructs the following exact rational certificates directly from Newton's recurrence. Factor the resultant $\mathrm{Res}_X(\partial_X Q_k, \partial_Y Q_k)$ over $\mathbb{Q}$. Exactly one irreducible factor has any root in $(0, b_k^2)$, and it has exactly one. Its degree and Sturm variations at the two endpoints are

k	$\deg F_k$	$V(0)$	$V(b_k^2)$
12	18	14	13
15	29	24	23.

The other factors have degrees at most two when $k = 12$, and at most four when $k = 15$; their endpoint variations are equal. The factor Y contributes no root in the open interval. Exact lexicographic elimination gives

$$\langle \partial_X Q_k, \partial_Y Q_k, F_k \rangle = \langle X - h_k(Y), F_k(Y) \rangle \quad \text{in } \mathbb{Q}[X, Y],$$

where $h_k \in \mathbb{Q}[Y]$. Thus the unique root of F_k determines at most one stationary point. The verifier recomputes this ideal equality and both gradient remainders. Rational isolation establishes $X_k < 0$, $X_k^2 < Y_k < b_k^2$ and $Q_k(X_k, Y_k) > L_k^2$, as well as (13.24)–(13.25). For orientation, $(X_{12}, Y_{12}) \approx (-0.024646, 0.141632)$ and $(X_{15}, Y_{15}) \approx (-0.015190, 0.091782)$; these decimals are not used in the certificate.

On $X = \pm R$, $Y = R^2$, the trace is real up to a cube root of unity. Set

$$g_{k,\pm}(R) = \frac{3 - T_k(\pm R, \pm R)}{R}.$$

For $0 \le R \le b_k < 1$, these are nonnegative: the unit roots of a real trace are $1, e^{i\vartheta}, e^{-i\vartheta}$, so $p_k = 1 + 2\cos(k\vartheta) \le 3$. Every Bernstein coefficient of $L_k - g_{k,\pm}$ is strictly positive on this interval, after one bisection for the minus branch at $k = 12$ and with no bisection in the other three cases. Hence the parabolic boundary has ratio less than L_k. The boundary $Y = b_k^2$ has ratio at most L_k, while the origin has limiting ratio zero. The isolated interior point is consequently the unique global maximizer in (X, Y). Finally, $|X_k| < \sqrt{Y_k} < 1$ gives exactly the six admissible traces in (13.28). This proves sharpness and all equality cases. □

The individual results fit a sharp estimate uniform in the order.

Lemma 13.10 (Optimal uniform linear coefficient)**.** *If $|\eta_j| = 1$, $\eta_1\eta_2\eta_3 = 1$, and $s = \eta_1+\eta_2+\eta_3 \neq 0$, then, for every $k \ge 3$,*

$$\left|\eta_1^k + \eta_2^k + \eta_3^k - 3\mathbf{1}_{3|k}\right| \le \frac{4k}{3}|s| < (2k-1)|s|. \tag{13.29}$$

The coefficient $4/3$ is the smallest real number A for which the bound $Ak|s|$ holds simultaneously for all $k \ge 3$ and all admissible triples. Equality in the first inequality occurs exactly when $k = 3$ and the triple is a permutation of $(q, q, -q)$ with $q^3 = -1$; for every $k \ge 4$ it is strict.

Proof. Proposition 13.3 proves the claim for $k = 3, 4, 5$: $C_3 = 4$, $C_4 < 16/3$, and $C_5 \le \max_{R\ge0}(5 + 5R - 5R^2) = 25/4 < 20/3$. It also supplies the equality case and proves that the coefficient $4/3$ cannot be decreased.

Suppose $k \ge 6$, put $R = |s|$ and $B = 3 + 3\mathbf{1}_{3|k}$, and denote the absolute value on the left of (13.29) by D_k. The trivial bound is $D_k \le B$. Hence $R > 3B/(4k)$ gives the desired strict inequality. For $R \le 3B/(4k) \le 3/4$, use Lemma 13.2 and let h be its maximal absolute phase. Since $\arcsin R \le \pi R/2$,

$$0 < \frac{kh}{2} \le \frac{k}{3}\arcsin R \le \frac{\pi k R}{6} \le \frac{\pi B}{8} \le \frac{3\pi}{4} < \pi.$$

The three absolute phases are $h, a, h - a$. Concavity of sine on $[0, \pi]$ therefore yields

$$\begin{aligned} D_k &= \left|\sum_j \omega_j^k (e^{ik\delta_j} - 1)\right| \\ &\le 2\sin(kh/2) + 2\sin(ka/2) + 2\sin(k(h-a)/2) \\ &\le 2\sin(kh/2) + 4\sin(kh/4). \end{aligned} \tag{13.30}$$

Set $x = 3h/2$ and $b = k/3 \ge 2$. The function $\sin t/t$ is strictly decreasing on $(0, \pi)$, since the derivative of $t\cos t - \sin t$ is $-t\sin t < 0$. As $0 < x \le bx/2 \le bx < \pi$,

$$2\sin(bx) + 4\sin(bx/2) < 2b\sin x + 2b\sin x = \frac{4k}{3}\sin(3h/2) \le \frac{4k}{3}R.$$

This proves strictness also at $R = 3B/(4k)$ and completes the proof. □

To apply the estimate at a seed zero, choose $\rho_0^3 = -ab^2$. The three roots of (13.2), divided by ρ_0, have modulus one, product one, and nonzero sum z/ρ_0. At the origin their unnormalized kth power sum is $3\rho_0^k \mathbf{1}_{3|k}$. Thus

$$|S_k(z,\bar{z}) - S_k(0,0)| \le \frac{4k}{3} c^{k-1}|z| < (2k-1)c^{k-1}|z| \quad \text{at every zero of } H_N, \quad k \ge 3. \tag{13.31}$$

For every $k \ge 3$ with $3 \nmid k$, and also for $k = 3, 6, 9, 12, 15$, the factor $4k/3$ can be replaced by the individual sharp constant C_k from the preceding propositions and Theorem 13.7 and Proposition 13.9; equality requires z/ρ_0 to be one of the corresponding equality traces. These are uniform estimates over all unit triples; optimality on an individual finite seed set is a separate restriction.

The first square covers every even conjugate degree on the new boundary. A second square supplies another complete residue class.

Proposition 13.11. *For every even $m \ge 2$, and for every $m \ge 5$ with $m \equiv 1 \pmod 4$, there is a polynomial in $\mathcal{E}_{m,2m+2}$ with $(2m+2)^2$ nondegenerate zeros and coefficients in $\mathbb{Q}(i)$.*

Proof. Let $L = 2r+1$, $r \ge 1$, and set $H = H_L$,

$$B(z) = Lb^r z, \qquad A(z,w) = \begin{cases} w^2, & r = 1, \\ S^*_{r+1}(z,w) - S^*_{r+1}(0,0), & r \ge 2. \end{cases}$$

The polynomial A is monic of degree $r+1$ in w, with total degree $r+1$. At every zero of H, inequality $|A| < |B|$ follows from (13.31) if $r \ge 2$. If $r = 1$, the three equal-modulus roots of (13.2) are not all equal, so $0 < |z| < 3c$ and $|w|^2 < 3c|z| = |B(z)|$.

By (13.4), the coefficients of w^{2r+1} in AH and BH^* are both B. Hence $D = AH - BH^*$ has conjugate degree at most $2r$ and total degree at most $3r+2 < 2L$. Lemma 12.2 shows that

$$f_\delta = H^2 + \delta D$$

has $(2L)^2$ nondegenerate zeros for every sufficiently small positive δ. Its leading w^{2r} coefficient tends to $B^2 \ne 0$, so its conjugate degree is exactly $2r$. This proves the assertion for even m.

For the second square, put $B_\delta = [w^{L-1}]f_\delta$ and $U = A^2$. The polynomial U is monic in w of degree $L+1$ and has total degree $L+1 < 2L$. All zeros of f_δ tend to zeros of H, and $B_\delta \to B^2$ coefficientwise. The strict inequality $|A| < |B|$ on the finite seed set consequently gives

$$|U| < |B_\delta| \quad \text{at every zero of } f_\delta$$

when δ is sufficiently small. In

$$f_\delta^2 + \eta(U f_\delta - B_\delta f_\delta^*)$$

the terms of w-degree $2L$ cancel. All remaining terms have conjugate degree at most $2L-1$, and the correction has total degree below $4L$. For fixed sufficiently small δ, Lemma 12.2 gives $(4L)^2$ nondegenerate zeros for all sufficiently small positive η. This is the pair $m = 2L-1 = 4r+1$, $n = 4L = 2m+2$. All parameters may be chosen rational. If the conjugate degree in this second construction is smaller than m, adding an arbitrarily small nonzero rational multiple of w^m makes it exact and preserves all $(2m+2)^2$ zeros; Lemma 2.1 again excludes additional zeros. □

14 A squaring criterion that persists under iteration

The remaining odd conjugate degrees require more than a second isolated application of Lemma 12.2. We now keep track of the next coefficient after the leading cancellation. This coefficient allows an auxiliary inequality to pass to every subsequent square.

Suppose $N = 2m + 2$ and

$$f = z^N + \text{terms of total degree below } N, \qquad \deg_w f = m.$$

Let A have total degree below N and be monic in w of degree $N - m = m + 2$. Define

$$\begin{aligned} B &= [w^m]f, \qquad C = [w^{m-1}]f, \qquad D = [w^{N-m-1}]A, \\ e &= [w^{N-1}]f^*, \qquad d = C + B(D - e). \end{aligned} \tag{14.1}$$

The quantities B, C, D, d are polynomials in z. The coefficient e is a constant because the only term of total degree N in f^* is w^N. The role of d is the exact identity

$$[w^N](Af - Bf^*) = 0, \qquad [w^{N-1}](Af - Bf^*) = d. \tag{14.2}$$

The following proposition includes the condition needed for a second application, not just the first one.

Proposition 14.1. *Suppose f has N^2 nondegenerate zeros. Let $S = w + s(z)$, with $\deg S < N$, and suppose that at every zero of F_f,*

$$|A| < |B|, \qquad |S|(|A| + |B|) < |d|. \tag{14.3}$$

For all sufficiently small positive ε, put

$$g = f^2 + \varepsilon(Af - Bf^*), \qquad A_g = Sf^*.$$

Then g has highest homogeneous part z^{2N}, conjugate degree exactly $N-1$, and $(2N)^2$ nondegenerate zeros. With B_g, C_g, D_g, e_g, d_g defined from g, A_g as in (14.1), every zero of F_g satisfies

$$|A_g| < |B_g|, \qquad |w|(|A_g| + |B_g|) < |d_g|. \tag{14.4}$$

Proof. Since $2m = N - 2$, identities (14.2) give

$$B_g = [w^{N-1}]g = \varepsilon d, \qquad C_g = [w^{N-2}]g = B^2 + \varepsilon[w^{N-2}](Af - Bf^*). \tag{14.5}$$

The second inequality of (14.3) forces d to be nonzero at every seed zero, hence not identically zero. Thus g has conjugate degree $N - 1$. Both correction terms have total degree below $2N$, so Lemma 12.2 gives exactly $(2N)^2$ nondegenerate zeros. All of them lie in arbitrarily small fixed neighborhoods of the seed zeros.

The polynomial $A_g = Sf^*$ is monic in w of degree $N+1$ and has total degree below $2N$. These are precisely the required degree conditions for the new pair $(m', N') = (N - 1, 2N)$. At a zero of g, put $x = F_f(z)$. The equation $x^2 + \varepsilon(Ax - B\bar{x}) = 0$ implies

$$|x| \le \varepsilon(|A| + |B|),$$

also when $x = 0$. On a sufficiently small neighborhood of the seed set, the strict second inequality of (14.3) still holds. All zeros of g lie there for small ε. Consequently,

$$|A_g| = |S||x| \le \varepsilon|S|(|A| + |B|) < \varepsilon|d| = |B_g|.$$

This proves the first inequality in (14.4).

For the second one, the next coefficient of A_g is $D_g = e + s(z)$, whereas $e_g = [w^{2N-1}]g^*$ stays bounded as $\varepsilon \to 0$. Equations (14.5) therefore yield the coefficientwise limit

$$d_g = C_g + B_g(D_g - e_g) = B^2 + O(\varepsilon). \tag{14.6}$$

Choose small compact neighborhoods of the finite seed set on which $|B|$ is bounded below by a positive constant. On these neighborhoods all coefficient polynomials occurring above are uniformly bounded. At zeros of g, the bound on x gives $|A_g| + |B_g| = O(\varepsilon)$, while (14.6) bounds $|d_g|$ below by a positive constant. The coordinate $w = \bar{z}$ is bounded there. Taking ε smaller proves the second inequality of (14.4). □

Thus after one successful step the next choice of S is simply w. The construction does not require a parameter that works for infinitely many degrees at once. For each finite number of steps, the parameters are chosen successively, each sufficiently small after the preceding ones have been fixed. If the initial data are rational, all these parameters may be chosen positive and rational.

We next give the finite starting data for the residue class not covered by Proposition 13.11. The degree-eight polynomial f_0 is specified exactly in Table 5. It was used as an isolated extremizer in the preceding degree-$(3, 8)$ construction. The additional polynomials A_0, S_0 in Tables 6 and 7 are what allow that isolated example to propagate.

All three tables, together with the rational boxes in the supplementary files, establish the following finite assertion.

Proposition 14.2. *The polynomial f_0 has highest homogeneous part z^8, conjugate degree three, and 64 nondegenerate zeros. The polynomial A_0 has total degree five and is monic of degree five in w; $S_0 = w + s_0(z)$ with $\deg s_0 = 3$. With*

$$B_0 = [w^3]f_0, \quad C_0 = [w^2]f_0, \quad D_0 = [w^4]A_0, \quad e_0 = [w^7]f_0^*, \quad d_0 = C_0 + B_0(D_0 - e_0),$$

every zero of F_{f_0} satisfies

$$|A_0| < \frac{3}{50}|B_0|, \qquad |S_0|(|A_0| + |B_0|) < \frac{7}{10}|d_0|. \tag{14.7}$$

Proof. The degree assertions are read from the coefficient tables. There are 64 disjoint rational boxes of radius 10^{-30} in each real coordinate. The program `verify_seed.py` verifies in each box a contraction and inclusion bound strictly below 10^{-12}, using only rational arithmetic. It proves the existence of a nondegenerate zero in each box. Lemma 2.1 shows that these are all the zeros.

The program `verify_auxiliary.py` proves the two inequalities in (14.7) on the whole of every box, not just at its center. It also checks that B_0 and d_0 do not vanish in any box. Appendix B gives the rational bounds and every comparison used by the two programs, as well as the exact input tables. The supplied output records that all comparisons pass. □

Because the margins in (14.7) are strictly smaller than one, they imply (14.3) with room to spare. The finite certificate can now be used in every degree generated by the iteration, without any further numerical root finding.

Corollary 14.3. *For every integer $j \geq 0$, there is a polynomial with highest homogeneous part $z^{2^{j+3}}$, exact conjugate degree $2^{j+2}-1$, and 2^{2j+6} nondegenerate zeros. Its coefficients may be taken in $\mathbb{Q}(i)$.*

Proof. Start with f_0, A_0, S_0 from Proposition 14.2. Proposition 14.1 changes (m, N) to $(N-1, 2N)$ and supplies the next auxiliary polynomial. At all later steps take $S = w$. The initial values are $(m, N) = (3, 8)$, and induction gives the asserted degrees. The number of zeros is the square of the total degree at every step. Rationality follows from successive rational choices of the small positive parameters. □

15 Odd-degree composition and the complete boundary

Repeated squaring changes $m+1$ by a factor of two. The next construction changes it by any odd factor, while preserving the relation $N = 2m+2$. It is essential here to use the support of the outer polynomial, rather than only its total degree.

Proposition 15.1. *Suppose $f = z^N +$ lower-total-degree terms has conjugate degree exactly m, where $N > 2m$, and has N^2 nondegenerate zeros. For every odd integer $q = 2s+1 \geq 3$, there is a polynomial with highest homogeneous part z^{qN}, conjugate degree exactly $m + sN$, and $(qN)^2$ nondegenerate zeros. Rational coefficients may be retained when $f \in \mathbb{Q}(i)[z, w]$.*

Proof. Use the seed H_q from Proposition 13.1 and put

$$F_t(z, w) = t^q H_q\big(f(z, w)/t, f^*(z, w)/t\big), \qquad t > 0. \tag{15.1}$$

This is a polynomial: for each monomial $U^i V^j$ of H_q, the factor t^{q-i-j} has a nonnegative exponent. The unique term of total degree qN is z^{qN}, since the unique term of total degree q in H_q is U^q.

The support condition $i + 2j \leq q$ bounds the conjugate degree of a composed term by

$$mi + Nj \leq mq + (N-2m)j \leq m + sN.$$

Since $N > 2m$, equality can occur only at $(i, j) = (1, s)$. Its coefficient in H_q is $qb^s \neq 0$ by (13.4). The corresponding highest w-coefficient in (15.1) is a nonzero scalar times $[w^m]f$. Thus the conjugate degree is exactly $m + sN$; no other term can cancel it.

On the real slice, (15.1) vanishes precisely when $F_f(z)/t$ is one of the q^2 nondegenerate zeros of $H_q(U, \bar{U})$. Near each zero of F_f, the map F_f is a real-analytic local diffeomorphism. For small t, the q^2 distinct target values therefore have q^2 distinct preimages in that neighborhood. Each preimage is nondegenerate, by the chain rule. This gives q^2N^2 zeros, and Lemma 2.1 proves completeness. A sufficiently small positive rational t preserves rational coefficients. □

We can now remove every parity exception on the boundary.

Proposition 15.2. *For every integer $m \geq 2$, there is a polynomial in $\mathcal{E}_{m,2m+2}$ with conjugate degree exactly m and $(2m+2)^2$ nondegenerate zeros. All coefficients may be taken in $\mathbb{Q}(i)$.*

Proof. Even m and $m \equiv 1 \pmod 4$, $m \geq 5$, are covered by Proposition 13.11. If $m \equiv 3 \pmod 4$, write uniquely

$$m + 1 = 2^{j+2}q, \qquad j \geq 0, \quad q \geq 1 \text{ odd}.$$

Corollary 14.3 supplies $m_0 = 2^{j+2} - 1$ and $N_0 = 2^{j+3} = 2(m_0 + 1)$. For $q = 1$ it already gives the desired pair. For $q = 2s + 1 \geq 3$, Proposition 15.1 gives

$$m_0 + sN_0 = q(m_0 + 1) - 1 = m, \qquad qN_0 = 2m + 2.$$

The three cases exhaust all integers $m \geq 2$. □

The congruence decomposition is only an organization of the construction. The conclusion of Proposition 15.2 is a single extremal statement for all $m \geq 2$. In particular, the construction now covers $(m, n) = (5, 12), (7, 16), (9, 20), (11, 24), (15, 32)$, without separate searches for polynomials in those degree pairs.

16 From total degree to rectangular bidegree

The preceding sections construct the n^2 bounded zeros. A local perturbation after inversion supplies the additional m^2 zeros required by the rectangular intersection number.

Proposition 16.1. *Suppose $n > m \geq 2$ and there is a polynomial in $\mathcal{E}_{m,n}$ with n^2 nondegenerate zeros. Then there is a polynomial of exact bidegree (m, n) with n^2+m^2 nondegenerate zeros. Rational coefficients may be retained.*

Proof. Normalize the highest homogeneous part of the given polynomial f to z^n. If necessary, first add a sufficiently small nonzero rational multiple of w^m to make its conjugate degree exactly m. This retains all n^2 zeros, and Lemma 2.1 shows that no other zero occurs. Write

$$H_m^*(u, v) = v^m + Q_m(u, v), \qquad Q_m(u, v) = \sum_{i+j \leq m-1} q_{ij} u^i v^j.$$

Proposition 13.1 gives m^2 nonzero nondegenerate zeros of this real-slice polynomial; its constant term q_{00} is nonzero. For $t > 0$ put

$$g_t(z, w) = f(z, w) + \sum_{i+j \leq m-1} q_{ij} t^{m-i-j} z^{n-i} w^{m-j}. \tag{16.1}$$

All exponents are nonnegative. The term $q_{00} t^m z^n w^m$ is nonzero, so the bidegree is exactly (m, n). As $t \to 0$, the coefficients tend to those of f, and its n^2 nondegenerate zeros persist.

Set $G_t(u, v) = u^n v^m g_t(1/u, 1/v)$. Every nonleading monomial of f has total degree at most $n - 1$, so its inverted contribution has total degree at least $m + 1$. Consequently,

$$t^{-m} G_t(t\zeta, t\eta) \longrightarrow \eta^m + Q_m(\zeta, \eta) = H_m^*(\zeta, \eta)$$

in C^1 on compact real-slice sets. Each of the m^2 nonzero nondegenerate zeros of this limit gives a simple zero of G_t under the scaling. Inverting back yields m^2 zeros of g_t escaping to infinity, distinct from the n^2 bounded zeros.

These $n^2 + m^2$ nondegenerate zeros exceed $(n - 1)^2 + (m - 1)^2$. Lemma 2.3 forces g_t, g_t^* to be coprime and bounds their entire real-slice zero set by $n^2 + m^2$. The count is therefore exact, and all zeros are nondegenerate. Formula (16.1) permits a sufficiently small rational t. □

The low-degree range of the two maxima can now be identified together.

Proof of the low-degree assertion in Theorem 1.3. If $m < n \le 2m+1$, then $\lfloor n/2 \rfloor \le m$. Proposition 13.1 puts H_n in $\mathcal{E}_{m,n}$ with n^2 nondegenerate zeros. A sufficiently small rational w^m perturbation makes the conjugate degree exact when necessary. If $n = 2m+2$, use Proposition 15.2. In either case, Lemma 2.1 gives $E_{m,n} = n^2$. Proposition 16.1 then gives $M_{m,n} \ge n^2 + m^2$, and Lemma 2.3 supplies the reverse inequality. □

The next degree pair beyond the range for $m = 2$ is $(2, 7)$. No step above constructs a 53-zero polynomial in that pair, nor does it rule one out. The proof should therefore be used to enlarge the established exact range, not to assert a sharp formula in all bidegrees. Likewise, the constructed polynomials contain mixed monomials; they do not supply matching extremizers for the harmonic problem.

17 A cubic construction on the next diagonal

A cubic local model gives a new infinite family, beyond the range (1.2). For this section the maximum in the pure analytic-head space is again denoted by $E_{m,n}$.

Theorem 17.1. *For every integer $r \ge 1$,*

$$E_{3r,6r+3} = (6r+3)^2, \qquad M_{3r,6r+3} = (6r+3)^2 + 9r^2. \tag{17.1}$$

The extrema have exact conjugate degree $3r$, rational real coefficients, and only nondegenerate zeros.

The proof of the pure-head equality is given next. The rectangular equality then follows from Proposition 16.1; its auxiliary seed can be chosen over $\mathbb{Q}(i)$. To retain real coefficients, one may instead construct the auxiliary seed inductively from $z-1$ using a small real term sz^j added to the conjugate of the preceding seed. The nonzero model $\zeta^j + \bar{\zeta}^{j-1} = 0$ adds $2j-1$ positive simple zeros at infinity, precisely as in [SZ24, proof of Theorem 4.1, pp. 12–14]. The old constant is nonzero, so zero is not introduced. This gives j^2 nonzero simple zeros at each stage and justifies the real-coefficient assertion.

17.1 A uniform nine-zero cubic

Theorem 17.2 (Optimal uniform disk for the cubic model)**.** *For $0 < \kappa < 4/27$ consider*

$$H_{\kappa,e}(v) = v^3 + |v|^2 + e\bar{v} - \kappa.$$

Let $r \in (1/2, 2/3)$ be the unique solution of $\kappa = r^2(2r-1)$. The largest radius $R(\kappa)$ such that every $|e| < R(\kappa)$ gives nine distinct, nondegenerate zeros is

$$R(\kappa) = \min\{\kappa,\ 2r - 3r^2\}. \tag{17.2}$$

Over all real κ, the largest such positive radius is

$$R_* = \kappa_* = \frac{5\sqrt{5} - 11}{2} = 0.0901699437\ldots, \tag{17.3}$$

attained uniquely at $\kappa = \kappa_*$. In particular, $R(9/100) = 9/100$, so the rational choice used below retains its exact claimed radius, although it is not the unique optimal choice in the larger real-parameter family.

Proof. At $e = 0$, the zero equation on the three lines $v = se^{2\pi ij/3}$ is $s^3 + s^2 - \kappa = 0$. It has three distinct nonzero real roots exactly when $0 < \kappa < 4/27$. Thus this interval is necessary for any positive nine-zero disk centered at $e = 0$.

At a singular zero write $v = \sqrt{x}e^{i\theta}$, $x > 0$. Eliminating e from $H_{\kappa,e} = 0$ and $|3v^2+\bar{v}|^2-|v+e|^2 = 0$ gives

$$\cos(3\theta) = \frac{\kappa^2 - x^2 - 8x^3}{2x^{3/2}(3x + \kappa)}, \tag{17.4}$$

$$|e|^2 = \Psi_\kappa(x) := \kappa^2 + \frac{(x - \kappa)^2(2 - \kappa - 5x)}{\kappa + 3x}. \tag{17.5}$$

The angular feasibility condition is

$$(4x^3 - x^2 - 2\kappa x - \kappa^2)(16x^3 - x^2 + 2\kappa x - \kappa^2) \le 0. \tag{17.6}$$

The first factor has a unique positive root, namely r^2, by Descartes' rule and $\kappa = r^2(2r - 1)$. The second factor minus four times the first is $3x^2 + 10\kappa x + 3\kappa^2 > 0$. Therefore every feasible x satisfies $0 < x \le r^2$.

The derivative factors as

$$\Psi'_\kappa(x) = -\frac{2(x - \kappa)(15x^2 + (9\kappa - 3)x - 5\kappa)}{(\kappa + 3x)^2}.$$

The quadratic factor has exactly one positive root, larger than κ, since its value at κ is $8\kappa(3\kappa-1) < 0$. Hence Ψ_κ decreases to $x = \kappa$, then increases to one local maximum, then decreases. Here $\kappa < r^2$, and consequently its minimum on $[0, r^2]$ is

$$\min\{\Psi_\kappa(\kappa), \Psi_\kappa(r^2)\} = \min\{\kappa^2, (2r - 3r^2)^2\}.$$

Both candidates are achieved by actual singular zeros. For the first take $|v|^2 = \kappa$, $\cos(3\arg v) = -\sqrt{\kappa}$, and $e = -v^3/\bar{v}$. For the second take $v = -r$, $e = 2r-3r^2$. The defining equation and Jacobian vanish in each case. This proves that (17.2) is the exact distance from zero to the singular-parameter set.

Below that radius no zero is singular. Along a segment from $e = 0$ the cubic leading term confines all zeros to a common compact disk, so the nine simple zeros continue and exhaust the fiber. At a nearest singular parameter nine distinct zeros are impossible by Lemma 2.1. Thus the radius cannot be enlarged even if only distinctness is required.

Finally, $\kappa(r)$ is strictly increasing, while $2r - 3r^2$ is strictly decreasing for $1/2 < r < 2/3$. They coincide when $r^2 + r - 1 = 0$, that is, $r = (\sqrt{5} - 1)/2$. Substitution gives (17.3) and uniqueness of its maximum. □

For the explicit global formulas and their rational certificates, retain the convenient rational choice $\kappa = 9/100$.

Lemma 17.3. *For every $e \in \mathbb{C}$ with $|e| < \kappa$, the polynomial*

$$H_e(v) = v^3 + |v|^2 + e\bar{v} - \kappa$$

has exactly nine zeros, all nondegenerate. The radius κ is optimal for this fixed value of the constant term.

Proof. Since $9/100 < \kappa_*$, Theorem 17.2 gives $R(9/100) = 9/100$. □

The form needed in the construction is obtained by a translation and a complex dilation. We state the coefficients explicitly.

Corollary 17.4. *Let $a, b, \delta \in \mathbb{C}$, where $b \neq 0$ and $|\delta| < \kappa|b|^2$. Put*

$$d = \frac{a^2 + b\bar{a}}{3}, \qquad e = \frac{ba}{3} + \delta, \qquad g = \frac{a^3}{27} + \frac{b|a|^2}{9} + \frac{\delta\bar{a}}{3} - \kappa|b|^2 b. \tag{17.7}$$

Then $u^3 + au^2 + bu\bar{u} + du + e\bar{u} + g$ has exactly nine nondegenerate zeros.

Proof. Set $v = u + a/3$. The polynomial becomes $v^3 + b|v|^2 + \delta\bar{v} - \kappa|b|^2 b$. Choose λ with $\lambda^3 = |b|^2 b$, so $|\lambda| = |b|$, and put $v = \lambda y$. After division by $|b|^2 b$, the resulting polynomial is

$$y^3 + |y|^2 + \frac{\delta\bar{\lambda}}{|b|^2 b}\bar{y} - \kappa.$$

The conjugate-linear coefficient has modulus $|\delta|/|b|^2$. Lemma 17.3 applies. Both changes of variables and the nonzero target multiplication preserve nondegeneracy. □

17.2 The general construction

Fix $r \geq 2$, and write

$$N = 2r + 1, \qquad \epsilon = \frac{1}{100(r+1)^N}, \qquad Q(z, w) = \prod_{j=1}^{r} (zw - \epsilon^2 j^2), \qquad f = z^N + Q. \tag{17.8}$$

All coefficients are rational and real, and $Q^* = Q$. Let

$$a = [(zw)^{r-1}]Q = -\epsilon^2 \sum_{j=1}^{r} j^2. \tag{17.9}$$

We first record precise information about the seed.

Lemma 17.5. *The polynomial f in (17.8) has exactly N^2 nondegenerate zeros. Every zero satisfies $|z| \geq 3\epsilon/4$.*

Proof. Consider

$$\phi(s) = s^N + \prod_{j=1}^{r} (s^2 - \epsilon^2 j^2).$$

Under $s = \epsilon x$, its rescaled form is $\epsilon x^N + \prod_{j=1}^{r}(x^2 - j^2)$. At the endpoints of each of the $2r$ disjoint intervals centered at $\pm j$ with radius $1/4$, the product has magnitude at least $7/16$: one factor has magnitude at least $(1/4)(7/4)$, and every other factor has magnitude at least $(3/4)(7/4) > 1$. Meanwhile $|\epsilon x^N| \leq 1/100$. The endpoint signs are therefore unchanged, and there is a sign-changing root in each interval.

These $2r$ roots are simple. A sign-changing root of multiplicity at least three would already consume at least $2r + 2$ degrees. The last root of the real polynomial of degree $2r + 1$ is real and distinct from them. More explicitly, the coefficient of s^{N-1} is one, so the sum of all roots is -1. The sum of the absolute values of the $2r$ small roots is at most

$$2r\epsilon(r+1) \leq \frac{1}{50}.$$

The remaining root belongs to $[-51/50, -49/50]$ and is simple. In particular all roots have magnitude at least $3\epsilon/4$.

Let $\omega = e^{2\pi i/N}$. Because N is odd, the N lines $\omega^j\mathbb{R}$ are distinct. On each of them $F_f(s\omega^j) = \phi(s)$. Moreover, $\operatorname{Im} F_f(z) = \operatorname{Im} z^N$, so every zero lies on one of these lines. Their only common point is zero, which is not a zero of f. Thus there are exactly N^2 zeros. At each one, the derivative of the real part along its line is nonzero, whereas the imaginary part has zero tangential derivative and nonzero normal derivative. This proves nondegeneracy. □

The next coefficients are chosen to remove, rather than merely bound, the forbidden conjugate powers. Define

$$\begin{aligned} A &= w^{r+1}, \qquad B = -z^r, \qquad \Delta = -\frac{a}{3} z^{r-1} w^r, \\ D &= \frac{w^{2r+2} - z^N}{3}, \qquad E = -\frac{z^r w^{r+1} + a z^{r-1} w^r}{3}, \\ L &= \frac{z^r w + a z^{r-1}}{27} - \frac{z^r}{9}. \end{aligned} \tag{17.10}$$

Put

$$\begin{aligned} T_1 &= f(Af + Bf^*), \\ T_2 &= Df + Ef^*, \\ T_3 &= Q\left(-\frac{w^{r+2}}{27} + \frac{w^{r+1}}{9}\right) + \left(\kappa - \frac{a}{9}\right) z^{2r} w^r + Lf^*. \end{aligned} \tag{17.11}$$

Finally, for $t > 0$, define

$$P_t = f^3 + tT_1 + t^2 T_2 + t^3 T_3. \tag{17.12}$$

These are explicit polynomial expressions; no coefficient interpolation is required.

We verify the degree conditions before counting zeros.

Lemma 17.6. *For $r \geq 2$, every T_j in (17.11) has total degree less than $3N$ and conjugate degree at most $3r$. In addition, $\deg_w T_3 \leq 2r$.*

Proof. The two top w-coefficients of Q are z^r and az^{r-1}. In $Af + Bf^*$ the coefficient of w^{2r+1} cancels, so its w-degree is at most $2r$. Multiplication by f gives $\deg_w T_1 \leq 3r$. The total degree of T_1 is at most $5r + 3 < 3N$.

For T_2, the coefficients of w^{3r+2} cancel between Df and Ef^*. The coefficients of w^{3r+1} are $az^{r-1}/3$ and $-az^{r-1}/3$, and also cancel. The term $z^N w^{2r+2}/3$ has conjugate degree at most $3r$ precisely because $r \geq 2$. All other terms have the same bound. The total degree is at most $4r + 3 < 3N$.

In T_3, the coefficient of w^{2r+2} cancels between $-Qw^{r+2}/27$ and $z^r w f^*/27$. The coefficient of w^{2r+1} also cancels, using the remaining two terms of L. Thus $\deg_w T_3 \leq 2r$, and its total degree is at most $3r + 2 < 3N$. □

To identify the local limit, set

$$G = \frac{A^3}{27} + \frac{BAA^*}{9} + \frac{\Delta A^*}{3} - \kappa B^2 B^* = \frac{w^{3r+3}}{27} - \frac{z^N w^{r+1}}{9} + \left(\kappa - \frac{a}{9}\right) z^{2r} w^r. \tag{17.13}$$

The following exact identity will also explain why the reduction used for the constant term has the required effect:

$$T_3 - G = \frac{w^{r+1}}{9} f + \left(L - \frac{w^{r+2}}{27}\right) f^*. \tag{17.14}$$

It follows by substituting $f = z^N + Q$ and $f^* = w^N + Q$.

We can now complete the local count without requiring an exact unperturbed nine-zero model at every seed zero.

Proposition 17.7. *For every fixed $r \geq 2$, the polynomial P_t belongs to $\mathcal{E}_{3r,3N}$, has exact conjugate degree $3r$, and has exactly $9N^2$ nondegenerate zeros for all sufficiently small $t > 0$.*

Proof. The highest homogeneous part of f^3 is z^{3N}. By Lemma 17.6, this is also the highest homogeneous part of P_t. The coefficient of $z^{3r}w^{3r}$ is one at $t = 0$ and hence remains nonzero for sufficiently small t.

Fix a seed zero z_0. By Lemma 17.5, $\xi = F_f(z)$ is a local real-analytic coordinate; write its inverse as $z = \psi(\xi)$. Set $\xi = tu$ and divide (17.12) by t^3. Using (17.10), (17.13), and (17.14), the resulting maps converge in C^1 on compact u-sets to

$$u^3 + a_0u^2 + b_0u\bar{u} + d_0u + e_0\bar{u} + g_0,$$

where $a_0 = A(z_0, \bar{z}_0)$, $b_0 = B(z_0, \bar{z}_0)$ and the last three coefficients are given by (17.7), with $\delta_0 = \Delta(z_0, \bar{z}_0)$. In particular $b_0 \neq 0$.

Let $S_r = \sum_{j=1}^{r} j^2$. Equations (17.9) and (17.10), together with the modulus bound in Lemma 17.5, give

$$\frac{|\delta_0|}{|b_0|^2} = \frac{\epsilon^2 S_r}{3|z_0|} \leq \frac{4\epsilon S_r}{9} \leq \frac{1}{225} < \kappa. \tag{17.15}$$

Here $S_r \leq (r+1)^3$ and $\epsilon S_r \leq (r+1)^{2-2r}/100 \leq 1/100$. The uniform estimate (17.15) permits the use of Corollary 17.4 at every seed zero.

Each limit has nine nondegenerate zeros, and the real implicit function theorem gives nine nearby zeros of P_t. The seed neighborhoods can be chosen disjoint, and there are finitely many of them, so one common positive threshold for t works for all N^2 seed zeros. We have constructed $9N^2 = (3N)^2$ distinct nondegenerate zeros. Lemma 2.1 excludes all further zeros and proves the count. □

The construction has real rational coefficients whenever t is rational. No explicit numerical lower bound for the allowed smallness threshold is needed to obtain rational extremizers.

17.3 The first parameter

The preceding proof uses $2r+2 \leq 3r$ and cannot simply be specialized to $r = 1$. The following separate choice handles that case while using the same uniform cubic lemma.

Set

$$f = z^3 + zw - \frac{1}{8}, \qquad A = w^2, \qquad B = -z, \qquad \Delta = \frac{w - z^2}{12} - \frac{zw^2}{3},$$

and put

$$D = \frac{w^4 - z^3}{3}, \qquad E = \frac{w - z^2}{12} - \frac{2zw^2}{3}.$$

Define $T_1 = f(Af + Bf^*)$ and

$$\begin{aligned} T_2 = &-\frac{2}{3}z^5w^3 - z^6w - \frac{1}{3}z^4w + \frac{1}{24}z^3w \\ &-\frac{1}{3}z^8 - \frac{1}{3}z^6 + \frac{1}{6}z^5 + \frac{1}{24}z^3 - \frac{1}{192}z^2. \end{aligned} \tag{17.16}$$

$$T_3 = \frac{z^2w^2}{27} - \frac{w^2}{36} - \frac{23z^2w}{300} - \frac{zw}{108} + \frac{7z}{288} + \frac{1}{1728}. \tag{17.17}$$

Let $P_t = f^3 + tT_1 + t^2T_2 + t^3T_3$.

The identities required for the local limit can be checked without solving for any roots. Equation (17.16) satisfies

$$T_2 - Df - Ef^* = \frac{w - z^2}{3} f^2 + \frac{w - 2z^2}{3} ff^*. \tag{17.18}$$

Also, the difference between (17.17) and

$$G = \frac{w^6}{27} - \frac{2z^3w^2}{9} - \frac{z^4}{36} + \frac{53z^2w}{450}$$

belongs to (f, f^*). Indeed, use $z^3 = w^3 = 1/8 - zw$ to reduce G; its remainder is exactly T_3. This G is $A^3/27 + BAA^*/9 + \Delta A^*/3 - \kappa B^2B^*$.

Here the seed zeros are $s\omega^j$, where $\omega = e^{2\pi i/3}$ and

$$s \in \left\{ -\frac{1}{2}, \frac{-1+\sqrt{5}}{4}, \frac{-1-\sqrt{5}}{4} \right\}.$$

They are nondegenerate by the same line-restriction argument as in Lemma 17.5. At such a zero,

$$\frac{|\Delta|}{|B|^2} = \left| \frac{2}{3}s^2 + \frac{1}{3}s - \frac{1}{12} \right| = \frac{1}{12} < \kappa.$$

For $s = -1/2$ the expression inside the absolute value is $-1/12$; for the other two values it is $1/12$.

In the coordinate $\xi = F_f(z) = tu$, identity (17.18) contributes only an $O(t)$ term after the rescaling used in Proposition 17.7. The same is true for the difference $T_3 - G$. The limit is therefore again the cubic in Corollary 17.4, with nine nondegenerate zeros. Each of the nine seed zeros supplies nine zeros of P_t.

Direct inspection of T_1, (17.16), and (17.17) shows conjugate degree at most three and total degree at most eight. Thus $P_t \in \mathcal{E}_{3,9}$ and Lemma 2.1 bounds its number of zeros by 81. The constructed 81 zeros are all nondegenerate, and the conjugate degree is exactly three for sufficiently small t, since $[z^3w^3]f^3 = 1$. This proves the first assertion of (17.1) also for $r = 1$.

Applying Proposition 16.1 with $m = 3r$ and $n = 6r + 3$ completes the proof of (17.1), and hence the second family in Theorem 1.3. All choices of sufficiently small parameters are made in a finite sequence; density of the rationals permits each chosen positive parameter to be rational. No uniform numerical value for all r is claimed.

18 Quartic splitting and a further sharp diagonal

A quartic local model enlarges the range of exact maxima. The construction below cancels every forbidden conjugate power. The errors introduced by the cancellations tend to zero after the appropriate local normalization. In particular, no numerical extremizer is used in this section.

Theorem 18.1. *For every integer $r \geq 3$, put $m = 4r$ and $n = 8r + 4$. Then*

$$E_{4r,8r+4} = (8r+4)^2, \qquad M_{4r,8r+4} = 16r^2 + (8r+4)^2. \tag{18.1}$$

Both maxima are attained over $\mathbb{Q}(i)$, with exact conjugate degree $4r$ and only nondegenerate zeros. Thus the quadratic bounds are sharp on the additional diagonal $n = 2m + 4$, for $m \equiv 0 \pmod 4$, $m \geq 12$.

This statement concerns the entire pure-head and rectangular coefficient spaces, respectively. It is stronger than an exact count within a radial subfamily. The restriction $r \geq 3$ belongs to this particular proof. Theorem 19.1 below also proves the cases $r = 1, 2$ in both classes. Its pure-head case $r = 1$ uses the translated construction of Theorem 19.16.

18.1 A quartic with sixteen simple zeros

For $b \neq 0$ define

$$\mathcal{H}_b(v) = v^4 + bv^2\bar{v} + \frac{b^2}{8}\bar{v}^2 - \frac{b^2\bar{b}}{16}v + \frac{\bar{b}^4}{256}. \tag{18.2}$$

Lemma 18.2. *If $b/\bar{b}$ is not a root of unity, then $\mathcal{H}_b$ has exactly sixteen zeros, all nondegenerate.*

Proof. Put $\beta = b/4$ and $\alpha = \bar{\beta}$. For $s^4 = 1$ and $t^4 = -1$ set

$$v = \beta(s^{-1} + t^{-1}) - \alpha st, \qquad \bar{v} = \alpha(s + t) - \beta/(st).$$

The three roots of $X^3 - vX^2 - \beta\bar{v}X + \alpha\beta^2$ are $\beta/s, \beta/t, -\alpha st$. Their fourth power sum is $-\alpha^4$. Newton's identities give

$$v^4 + 4\beta v^2\bar{v} + 2\beta^2\bar{v}^2 - 4\alpha\beta^2 v + \alpha^4 = 0,$$

which is (18.2). These sixteen points are distinct. Indeed s, t are roots of $\alpha T^3 - \bar{v}T^2 - vT + \beta = 0$, whose third root is $-\beta/(\alpha st)$. The fourth power of the third root is $-(\beta/\alpha)^4$, which is neither 1 nor -1 under the hypothesis. Thus s and t can be uniquely recovered from v. The highest homogeneous part of $\mathcal{H}_b$ is v^4. The two complexified equations therefore have intersection length 16, so the sixteen distinct real-slice intersections exhaust that length and are all nondegenerate. This last conclusion is also Lemma 2.1. □

Lemma 18.3. *Let $\mathcal{B}$ be a finite set of complex numbers of modulus one, each satisfying the hypothesis of Lemma 18.2. There is $\eta > 0$ such that, for $b/|b| \in \mathcal{B}$, $|a| \leq 2|b|$, and*

$$|d_2|/|b|^2 + |d_3|/|b|^3 < \eta,$$

the polynomial

$$\mathcal{H}_b(u + a/4) - d_2u\bar{u} - d_3\bar{u} \tag{18.3}$$

has exactly sixteen nondegenerate zeros.

Proof. Use $u = |b|v$ and divide by $|b|^4$. The unperturbed maps become $\mathcal{H}_{b/|b|}(v + a/(4|b|))$. For each of the finitely many phases these are translations of one fixed polynomial with sixteen simple zeros. The translations range over a compact disk. Choose disjoint small neighborhoods of the untranslated zeros and then translate these neighborhoods with the zeros. The real implicit function theorem and compactness give a uniform positive tolerance for C^1 perturbations there. The normalized perturbations are uniformly C^1 small as the displayed sum tends to zero, since all these neighborhoods lie in a fixed disk. Consequently sixteen distinct simple zeros persist. The pure quartic leading part excludes every additional zero, proving the assertion. □

18.2 The seed and the explicit corrections

Fix $r \geq 3$, put $N = 2r + 1$, and take a positive rational ϵ, to be made sufficiently small. Set

$$Q(z, w) = \prod_{j=1}^{r} (zw - \epsilon^2 j^2), \quad f = z^N + Q, \quad f^* = w^N + Q. \tag{18.4}$$

Write

$$Q = (zw)^r + c_1(zw)^{r-1} + c_2(zw)^{r-2} + \cdots, \quad c_1 = -\epsilon^2 S_r, \quad c_2 = \epsilon^4 T_r, \tag{18.5}$$

where $S_r = \sum_{j=1}^{r} j^2$ and $T_r = \sum_{1 \leq j < k \leq r} j^2 k^2$.

For $0 < \epsilon \leq 1/(100(r+1)^N)$ the seed has exactly N^2 simple zeros; each satisfies

$$3\epsilon/4 \leq |z| \leq 2, \qquad \arg z \in \frac{\pi}{N}\mathbb{Z}. \tag{18.6}$$

Here is a direct verification. Since Q is real on the real slice, every zero lies on a line $z = s\omega^j$, $s \in \mathbb{R}$, $\omega = e^{2\pi i/N}$. Along each line the equation is $s^N + \prod_{j=1}^{r}(s^2 - \epsilon^2 j^2) = 0$. With $s = \epsilon x$, the intervals of radius $1/4$ about $\pm 1, \ldots, \pm r$ each contain a sign-changing root: the product has endpoint modulus at least $7/16$, while the remaining term has modulus at most $1/100$. These $2r$ roots are simple, since any odd multiplicity at least three would already exceed the degree $2r + 1$. The remaining root is real; the coefficient of s^{N-1} is one, so the sum of all roots is -1. The sum of the absolute values of the $2r$ small roots is at most $1/50$, placing the last root in $[-51/50, -49/50]$. It too is simple. The N lines are distinct and have no common zero. The radial derivative is nonzero at each root, as is the normal derivative of $\operatorname{Im} z^N$, proving planar simplicity and (18.6).

Fix $\lambda = 1 + 2i$ and define

$$A = \lambda w^{r+1}, \qquad B = -\lambda z^r, \qquad G = Af + Bf^*. \tag{18.7}$$

The star operation on A, B has its usual polynomial meaning; thus $A^* = \bar{\lambda} z^{r+1}$ and $B^* = -\bar{\lambda} w^r$. The two correction coefficients needed below are

$$D_2 = \frac{\lambda^2 c_1}{4} z^{r-1} w^r, \qquad D_3 = \frac{\lambda^3}{16}\left(c_1 z^{r-1} w^{2r+1} + c_2 z^{r-2} w^{2r}\right). \tag{18.8}$$

Define the following four polynomials:

$$T_1 = f^2 G, \tag{18.9}$$

$$T_2 = \frac{G^2}{8} + \frac{AfG}{4} + \frac{BA^* f^2}{4} - D_2 f f^*, \tag{18.10}$$

$$T_3 = \frac{A^2 G}{16} + \frac{BA^*}{16}(Af + G) - \frac{B^2 B^*}{16} f - D_3 f^*, \tag{18.11}$$

$$T_4 = \frac{A^4}{256} + \frac{BA^2 A^*}{64} + \frac{B^2 (A^*)^2}{128} - \frac{B^2 B^* A}{64} + \frac{(B^*)^4}{256} - \frac{\lambda^4 w^{2r+3}}{256} f^*. \tag{18.12}$$

Our polynomial is explicitly

$$P_{\epsilon,t} = f^4 + tT_1 + t^2 T_2 + t^3 T_3 + t^4 T_4, \qquad t > 0. \tag{18.13}$$

Lemma 18.4. *For $r \geq 3$, all four T_j have conjugate degree at most $4r$ and total degree below $4N$. Thus $P_{\epsilon,t} \in \mathcal{E}_{4r,4N}$; its conjugate degree is exactly $4r$ for all sufficiently small t.*

Proof. The terms of w-degree $2r+1$ cancel in G. Its two highest possible w-coefficients are

$$[w^{2r}]G = \lambda c_1 z^{r-1}, \qquad [w^{2r-1}]G = \lambda c_2 z^{r-2}.$$

The term $\lambda z^N w^{r+1}$ does not affect these identities because $r \geq 3$. Hence $\deg_w G \leq 2r$ and $\deg_w T_1 \leq 4r$.

In T_2 the only possible forbidden power is w^{4r+1}, from $AfG/4$. Its coefficient is $\lambda^2 c_1 z^{2r-1}/4$, exactly canceled by $D_2 f f^*$. The other terms have w-degree at most $4r$; here $BA^* = -|\lambda|^2 z^N$ is analytic.

In T_3 only $A^2G/16$ can exceed degree $4r$ in w. Its coefficients of w^{4r+2} and w^{4r+1} are respectively $\lambda^3 c_1 z^{r-1}/16$ and $\lambda^3 c_2 z^{r-2}/16$. They are canceled by $D_3 f^*$. The part $D_3 Q$ has w-degree at most $3r+1 \leq 4r$.

In T_4 combine the first and last terms:

$$\frac{A^4 - \lambda^4 w^{2r+3} f^*}{256} = -\frac{\lambda^4 w^{2r+3} Q}{256}.$$

Its w-degree is $3r+3 \leq 4r$. Every other term has w-degree at most $\max\{2r+2, 2r+1, 4r\} = 4r$.

Direct total-degree bounds for T_1, T_2, T_3, T_4 are, respectively, $7r+4$, $6r+4$, $5r+4$, $4r+4$. They are all strictly less than $4N = 8r+4$. The highest homogeneous part of (18.13) is therefore z^{4N}. Its $z^{4r}w^{4r}$ coefficient tends to one as $t \to 0$, proving the exact conjugate degree assertion. □

18.3 The sixteen-fold local count

Fix a seed zero z_0 and append a subscript zero to coefficient polynomials evaluated at $(z_0, \bar{z}_0)$. Since f is a local real coordinate there, set $f = tu$. On the real slice $f^* = t\bar{u}$. Division of (18.13) by t^4 gives, in local coordinates, C^1 convergence on compact sets to

$$\mathcal{H}_{B_0}(u + A_0/4) - (D_2)_0 u\bar{u} - (D_3)_0 \bar{u}. \tag{18.14}$$

For clarity, expanding the translated quartic gives quadratic coefficients

$$\frac{3A_0^2}{8} + \frac{B_0\bar{A}_0}{4}, \qquad \frac{A_0 B_0}{2}, \qquad \frac{B_0^2}{8},$$

for $u^2, u\bar{u}, \bar{u}^2$. Its linear coefficients are

$$\frac{A_0^3}{16} + \frac{B_0 A_0 \bar{A}_0}{8} - \frac{B_0^2 \bar{B}_0}{16}, \qquad \frac{B_0 A_0^2}{16} + \frac{B_0^2 \bar{A}_0}{16}.$$

Equations (18.10) and (18.11) give exactly these coefficients, apart from the two displayed errors. The value of T_4 at a seed zero is the translated quartic's constant term, since its last summand vanishes there. This proves (18.14) without discarding any term at the local scale.

We verify the hypotheses of Lemma 18.3 uniformly over the seed zeros. First $B_0 \neq 0$ and $|A_0|/|B_0| = |z_0| \leq 2$. The phases $B_0/|B_0|$ belong to the fixed finite set

$$\mathcal{B}_r = \left\{ -\frac{\lambda}{|\lambda|} e^{i\pi r j/N} : 0 \leq j < 2N \right\}.$$

The ratio of any element to its conjugate is $\lambda/\bar{\lambda}$ times a root of unity. But $\lambda/\bar{\lambda} = (-3+4i)/5$ is not a root of unity: its sum with its inverse is $-6/5$, which is not a rational algebraic integer. Every phase therefore satisfies Lemma 18.2.

The two normalized errors admit explicit bounds:

$$\frac{|(D_2)_0|}{|B_0|^2} \le \frac{|c_1|}{4|z_0|} \le \frac{\epsilon S_r}{3}, \tag{18.15}$$

$$\frac{|(D_3)_0|}{|B_0|^3} \le \frac{1}{16}\left(|c_1| + \frac{|c_2|}{|z_0|^2}\right) \le \epsilon^2\left(\frac{S_r}{16} + \frac{T_r}{9}\right). \tag{18.16}$$

Both bounds tend to zero for fixed r. Choose a sufficiently small positive rational ϵ, respecting the earlier seed bound, so that their sum is below the tolerance for $\mathcal{B}_r$. Every local limit (18.14) then has sixteen simple zeros.

Now keep this ϵ fixed and choose $t > 0$ sufficiently small. The real implicit function theorem supplies sixteen distinct simple zeros of $P_{\epsilon,t}$ near each of the N^2 seed zeros. Their neighborhoods can be chosen disjoint. The pure leading part z^{4N} bounds the entire zero set by $(4N)^2 = 16N^2$, so these zeros exhaust it. This proves the pure-head equality in Theorem 18.1. The inversion transfer in Proposition 16.1 supplies m^2 further simple zeros for the rectangle and proves its equality. All coefficients in the displayed formulas are Gaussian rational when ϵ, t are rational, and the transfer also permits rational parameters. The choices are sequential: first r, then ϵ, then t, then the transfer parameter. No fixed numerical parameter is claimed to work for every r.

18.4 A further rectangular equality and scope

Corollary 18.5. *For $r \ge 3$ and $s \ge 0$,*

$$M_{4r+s,8r+4+s} = (4r+s)^2 + (8r+4+s)^2. \tag{18.17}$$

In particular $M_{4r+1,8r+5}$ attains its quadratic bound. This adds rectangular equality cases on $n = 2m+3$ for $m \equiv 1 \pmod 4$, $m \ge 13$, beyond the earlier condition $3 \mid m$ whenever that condition fails.

Proof. Apply the circle-addition inequality of Corollary 22.2 to (18.1). Its increment $2s(12r+4+s)$ is exactly the increment of m^2+n^2. The rectangular upper bound proves equality. □

For example, the new pure-head/rectangular pairs are

$$(E_{12,28}, M_{12,28}) = (784, 928), \qquad (E_{16,36}, M_{16,36}) = (1296, 1552),$$

and circle addition gives $M_{13,29} = 1010$ and $M_{17,37} = 1658$. The latter step is a rectangular statement; it does not assert $E_{13,29} = 29^2$ or $E_{17,37} = 37^2$.

The exact degree cancellations and the local expansion are also checked by the accompanying symbolic verifier. The proof of the infinite family is the argument above, including the explicit normalized error bounds, rather than a finite parameter sweep. This particular quartic proof does not determine $M_{2,7}$ or the unrestricted harmonic maximum. The next section extends the pure-head family to $r = 2$ and the rectangular family to all $r \ge 1$.

19 Unrestricted splitting and a linear sharp region

A complete local fiber count and a single polynomial adjustment replace the phase restriction and order-dependent correction series of the preceding versions. It makes the normalized local error

small at every seed zero while keeping the conjugate degree of the quadratic recurrence coefficient at most twice the seed degree. There is consequently no upper restriction on the splitting order. The first recurrence has a slightly mixed highest homogeneous part at the cubic boundary. A translated coefficient reduction below also yields a pure highest part at that boundary in every order.

Theorem 19.1 (Splitting of arbitrary order)**.** *For all integers $r \geq 1$ and $k \geq 1$,*

$$E_{kr,k(2r+1)} = k^2(2r+1)^2. \tag{19.1}$$

For every $r \geq 1$, $k \geq 1$, and $kr \leq m < k(2r+1)$,

$$M_{m,k(2r+1)} = m^2 + k^2(2r+1)^2. \tag{19.2}$$

All these maxima admit real rational coefficients, exact conjugate degree m (or kr in (19.1)), and only nondegenerate zeros. Clearing denominators gives integer-coefficient extremizers. In particular, $E_{m,3k} = 9k^2$ and $M_{m,3k} = m^2 + 9k^2$ for $k \leq m < 3k$.

Theorem 19.2 (A linear sharp region)**.** *If m, n are integers satisfying*

$$1 \leq m < n \leq 3m, \qquad n \neq 3m-1, \tag{19.3}$$

then

$$M_{m,n} = m^2 + n^2. \tag{19.4}$$

The examples under (19.3) can be chosen over $\mathbb{Q}$, hence over $\mathbb{Z}$ after scaling, with a nonzero coefficient of $z^n w^m$, and all their zeros are nondegenerate. They are irreducible. The previously established cases $(1,2), (2,5), (3,8)$ also satisfy (19.4). Thus the only line not settled here inside $m < n \leq 3m$ is $n = 3m-1$, $m \geq 4$.

The continuous part is $m < n \leq 3m-2$, together with the entire boundary $n = 3m$. This replaces the earlier sufficient width $\sqrt{m} - 5m^{1/4} - 4$ beyond $n = 2m$ by width $m-2$ and the endpoint $d = m$. For example,

$$E_{8,20} = 400, \quad M_{8,20} = 464, \quad E_{4,12} = 144, \quad M_{4,12} = 160.$$

The pure-head endpoint $E_{4,12} = 144$ follows from Theorem 19.16. The exception $n = 3m-1$ is a limitation of the available constructions, not a nonattainment theorem. The value of $M_{2,7}$ remains open here.

19.1 The exact fibers of the local model

The local recurrence is a rescaling of the classical A_2 folding polynomials, or generalized Chebyshev polynomials. This underlying family is not new; see Hoffman–Withers [HW88]. The trace coordinates, cubic discriminant, and deltoid geometry used below are also described by Bowman [Bow20, Sections 3–4, pp. 3–6]. We give the fiber count, including repeated-root targets, explicitly because it identifies the exact phase obstruction and permits a simpler global construction over $\mathbb{Q}$.

Define polynomials with integer coefficients by

$$\begin{aligned} &T_0 = 3, \qquad T_1 = U, \qquad T_2 = U^2 - 2V, \\ &T_j = UT_{j-1} - VT_{j-2} + T_{j-3} \quad (j \geq 3). \end{aligned} \tag{19.5}$$

Thus $T_k(U,V)$ is the kth power sum of the three roots of $X^3 - UX^2 + VX - 1$. Its highest homogeneous part is U^k. Put $\omega = e^{2\pi i/3}$ and

$$D(c) = |c|^4 + 18|c|^2 - 8\,\mathrm{Re}(c^3) - 27, \qquad \delta_{k,j} = \#\{\xi : \xi^3 = 1,\ \xi^k = \omega^j\}. \tag{19.6}$$

The curve $D = 0$ is the deltoid $c = 2q + q^{-2}$, $|q| = 1$, with cusps $3\omega^j$, $j = 0, 1, 2$.

Theorem 19.3 (Complete local fiber count)**.** *For every $k \geq 2$ and every $c \in \mathbb{C}$, the number of distinct solutions of $T_k(u, \bar{u}) = c$ is exactly as follows:*

Target	*All distinct solutions*	*Nondegenerate solutions*
$D(c) < 0$	k^2	k^2
$D(c) > 0$	k	k
$D(c) = 0$, $c \notin \{3, 3\omega, 3\omega^2\}$	$k(k+1)/2$	k
$c = 3\omega^j$	$(k^2 + 3k + 2\delta_{k,j})/6$	$\delta_{k,j}$

Here $\delta_{k,j} = 1$ if $3 \nmid k$; if $3 \mid k$, it is 3 for $j = 0$ and 0 for $j = 1, 2$. In particular, $D(c) < 0$ is necessary and sufficient for k^2 distinct solutions.

Proof. Write $C_c(X) = X^3 - cX^2 + \bar{c}X - 1$. Its root multiset is invariant under $X \mapsto 1/\bar{X}$ and its product is one. With distinct roots, either all three are on the unit circle, or they have the form

$$re^{i\phi}, \quad r^{-1}e^{i\phi}, \quad e^{-2i\phi}, \qquad r > 1.$$

In the first case the discriminant is $-64\prod_{i<j}\sin^2((\theta_i - \theta_j)/2) < 0$. In the second it is $(r - r^{-1})^2(2\cos(3\phi) - r - r^{-1})^2 > 0$. A repeated off-circle root is impossible: invariance would force its distinct partner to be repeated as well. Consequently the repeated-root case consists of q, q, q^{-2} with $|q| = 1$; the triple-root case occurs precisely when $q^3 = 1$. This proves the asserted discriminant classification.

For a real-slice input u, let $\lambda_1, \lambda_2, \lambda_3$ be the roots of C_u. The polynomial having roots λ_i^k is $C_{T_k(u,\bar{u})}$: their sum is $T_k(u, \bar{u})$, their product is one, and their second elementary symmetric function is the conjugate of their sum. Conversely, any unordered triple of lifts of the roots of C_c, invariant under reciprocal conjugation and with product one, gives a solution $u = \sum \lambda_i$. The trace u uniquely recovers its unordered triple through C_u. Thus the following counts neither omit nor duplicate solutions.

If $D(c) < 0$, label its three distinct unit roots μ_i. Choose $\lambda_1^k = \mu_1$ and $\lambda_2^k = \mu_2$ independently; then $\lambda_3 = (\lambda_1\lambda_2)^{-1}$ is forced and has kth power μ_3. All lifts are unit roots. This gives k^2 different unordered triples because the three target labels are distinct. If $D(c) > 0$, designate the target root outside the unit circle. Its lift has k choices, its inside partner is forced by reciprocal conjugation, and the unit lift is forced by the product. There are exactly k triples.

At a smooth boundary target, choose an unordered pair, with repetition, from the k roots of q. The lift of q^{-2} is forced. It cannot coincide with a lift of q, because $q \neq q^{-2}$. There are $k(k+1)/2$ choices. At a cusp, the ordered triples of kth roots of ω^j with product one form a set of size k^2. Under permutation of the three entries, each transposition fixes k triples and each three-cycle fixes $\delta_{k,j}$. Burnside's formula gives $(k^2 + 3k + 2\delta_{k,j})/6$ orbits.

For regularity, let J_k be the real Jacobian of $u \mapsto T_k(u, \bar{u})$. Differentiation of the elementary symmetric functions, with $\lambda_3 = (\lambda_1\lambda_2)^{-1}$, gives

$$J_k(u) = k^2 \prod_{i<j} \frac{\lambda_i^k - \lambda_j^k}{\lambda_i - \lambda_j}, \qquad D(T_k(u, \bar{u})) = \frac{D(u)J_k(u)^2}{k^4}. \tag{19.7}$$

For example, the Jacobian of the trace coordinates in $(\log\lambda_1, \log\lambda_2)$ is a Vandermonde determinant; taking powers multiplies the two logarithmic coordinates by k and replaces that determinant by the one for λ_i^k. Their quotient is the first identity. The second follows by squaring it. Each divided difference extends polynomially across a collision, with value $k\lambda_i^{k-1}$ there. Distinct target roots therefore give a nonzero Jacobian. At a smooth boundary target exactly the k equal-pair lifts are regular; every unequal-pair lift has a zero factor in (19.7). At a cusp only the all-equal lifts are regular, and there are exactly $\delta_{k,j}$ of them. □

Lemma 19.4 (The unique largest disk in the deltoid)**.** *Let $\Omega = \{c \in \mathbb{C} : D(c) < 0\}$. The largest radius of an open Euclidean disk contained in Ω is one, and its unique center is zero. Its boundary contacts with $\partial\Omega$ are exactly the three points $c^3 = -1$.*

Proof. For $r = |c|$ one has

$$D(c) \le (r-1)(r+3)^3, \qquad D(c) \ge (r-3)^3(r+1).$$

Thus the open unit disk is contained in Ω, whereas $\Omega \subset \{|c| < 3\}$. A disk of largest radius exists: maximize the continuous distance to $\partial\Omega$ over $\overline{\Omega}$. Its center a is interior and its radius ρ is positive.

Parametrize the boundary by $\gamma(\theta) = 2e^{i\theta} + e^{-2i\theta}$. A positive-radius disk contained in $\overline{\Omega}$ cannot touch a cusp. Indeed, at the cusp three,

$$\gamma(\theta) = 3 - 3\theta^2 + i\theta^3 + O(\theta^4),$$

so the interior has zero opening angle there; the tangent cone of a disk at a boundary point is a half-plane. The other cusps are rotations of this one. At smooth boundary points,

$$\gamma'(\theta) = -4e^{-i\theta/2}\sin(3\theta/2).$$

No two distinct smooth points therefore have parallel tangent lines: their tangent angles modulo π are $-\theta/2$.

At a largest disk, the vectors from its center to the contact points must have zero in their convex hull. Otherwise a small motion away from all contacts increases the distance to the boundary; compactness controls the remaining boundary points. One contact cannot suffice, and two would have to be antipodal, giving parallel tangents. Hence there are at least three smooth contacts.

Put $q = e^{i\theta}$. Multiplying $|2q + q^{-2} - a|^2 - \rho^2$ by q^3 gives the degree-six polynomial

$$H(q) = 2q^6 - aq^5 - 2\bar{a}q^4 + (5 + |a|^2 - \rho^2)q^3 - 2aq^2 - \bar{a}q + 2. \tag{19.8}$$

Every contact is a double root of H. Consequently there are exactly three contacts q_1, q_2, q_3, and

$$H(q) = 2(q^3 - Sq^2 + Tq - U)^2, \quad S = \sum q_j, \quad U = \prod q_j, \quad T = U\bar{S}.$$

The constant and degree-five coefficients give $U^2 = 1$ and $a = 4S$. The degree-four coefficient then gives

$$S^2 = -(4 + 2U)\bar{S}.$$

If $S \neq 0$, this forces $|a| = 24$ or 8, contradicting $|a| < 3$. Thus $a = S = 0$. The degree-three coefficient gives $\rho^2 = 5 + 4U$, which is one or nine. Radius three is impossible, since the disk would contain $-2 \notin \overline{\Omega}$. Thus $\rho = 1$, and every maximizing disk has center zero. Finally $D(c) = -8(1 + \mathrm{Re}(c^3))$ on $|c| = 1$, proving the complete contact statement. □

Corollary 19.5 (Sharp constant-target radius)**.** *For every $k \geq 2$, all targets $|c| < 1$ have k^2 simple preimages under $T_k(u, \bar{u})$. The radius one is optimal among all open target disks, and the unique maximizing center is zero. On $|c| = 1$ the only exceptions to k^2 simple preimages are $c^3 = -1$, where the exact distinct count is $k(k+1)/2$.*

Proof. For $r = |c|$,

$$D(c) \leq r^4 + 8r^3 + 18r^2 - 27 = (r-1)(r+3)^3.$$

The right side is negative for $r < 1$. Equality holds when $c = -r$, so every disk of radius greater than one contains an exterior target with only $k < k^2$ preimages. On the unit circle, $D(c) = -8(1 + \mathrm{Re}(c^3))$. Lemma 19.4 proves optimality and uniqueness when arbitrary centers are allowed. □

Corollary 19.6 (Globally optimal quartic stability disk)**.** *For the normalized quartic family $F_\tau(z) = z^4 + z\bar{z}^2 - \tau\bar{z}$, the unique largest open parameter disk on which every polynomial has sixteen distinct nondegenerate zeros is*

$$|\tau - 1/9| < 1/27. \tag{19.9}$$

On its boundary, exactly the three parameters

$$\frac{4}{27}, \qquad \frac{5}{54} + \frac{i\sqrt{3}}{54}, \qquad \frac{5}{54} - \frac{i\sqrt{3}}{54}$$

have fewer than sixteen zeros; each has eleven distinct zeros, of which six are nondegenerate. More generally, for a real center $0 < a < 4/27$, its exact maximal radius is

$$R(a) = \min\{a^{3/2},\ 4/27 - a\}. \tag{19.10}$$

Thus the original center $a = 1/8$ has radius $5/216$, while $a = 1/9$ gives the globally optimal radius $1/27$.

Proof. With $\mathcal{D}$ as in (11.5), direct expansion gives

$$D(3 - 27\tau) = -27^3 \mathcal{D}(\mathrm{Re}\,\tau, \mathrm{Im}\,\tau).$$

Proposition 11.5 identifies the sixteen-zero region with $\mathcal{D} > 0$. It is therefore the affine image of Ω under $c \mapsto (3 - c)/27$. Lemma 19.4 proves the radius, uniqueness over all complex centers, and the three boundary contacts. These are smooth discriminant parameters, so the radial cubic has one double and one simple positive root. Equation (11.7) gives five singular and five nondegenerate nonzero zeros, together with the nondegenerate origin.

For the last formula, the entire discriminant boundary is parametrized by

$$\tau_\pm(t) = t - \frac{3}{2}t^2 \ \pm i\, t^{3/2}\sqrt{1 - 9t/4}, \qquad 0 \leq t \leq 4/9.$$

Indeed, a repeated positive root t of (11.6) gives precisely these two parameters; $t = 0$ supplies the remaining boundary point. For real a,

$$|\tau_\pm(t) - a|^2 = a^2 - 2at + (1 + 3a)t^2 - 2t^3, \qquad \frac{d}{dt}|\tau_\pm(t) - a|^2 = -2(3t - 1)(t - a).$$

When $0 < a < 4/27$, the minimum over this interval occurs at $t = a$ or $t = 4/9$. The squared distances are respectively a^3 and $(4/27 - a)^2$, proving (19.10). They cross only at $a = 1/9$; this also follows from $h^3 + h^2 - 4/27 = (h - 1/3)(h + 2/3)^2$ with $h = \sqrt{a}$. □

Corollary 19.7 (The two local orientations)**.** *For $D(c) < 0$, the fiber of T_k has $(k^2+k)/2$ positive and $(k^2-k)/2$ negative zeros. For $D(c) > 0$, all its k zeros are positive. Every nondegenerate zero over a boundary target is positive as well.*

Proof. The highest term u^k makes the map proper with topological degree k. For a regular target, the signed count is therefore $N_+ - N_- = k$. Combine this with Theorem 19.3. A nondegenerate zero over a boundary target continues to nearby exterior targets with the same sign, which proves the last assertion. □

19.2 Removing the phase restriction

Fix $k \geq 2$, $b \neq 0$, $\beta = b/k$, and $\alpha = \bar{b}/k$. The parameter-dependent recurrence used in the global construction is

$$\begin{aligned} &S_0 = 3, \qquad S_1 = U, \qquad S_2 = U^2 + 2\beta V, \\ &S_j = US_{j-1} + \beta V S_{j-2} - \alpha\beta^2 S_{j-3} \quad (j \geq 3). \end{aligned} \tag{19.11}$$

Choose any $d \neq 0$ with $d^3 = -\alpha\beta^2$. Then $|d| = |\beta|$ and $d^2 = -\beta\bar{d}$, so induction gives

$$S_k(dU, \bar{d}V) = d^k T_k(U, V). \tag{19.12}$$

Thus $S_k(u, \bar{u})$ has k^2 simple zeros for every $b \neq 0$. Its full k^2-valent target region is exactly $D(c/d^k) < 0$; its unique largest open target disk is centered at zero and has radius $(|b|/k)^k$. This is an exact target-translation statement, not a quantitative bound for all coefficient perturbations.

The shifted model of the preceding version was

$$\mathcal{H}_{k,b}(U, V) = S_k(U, V) + (-\alpha)^k. \tag{19.13}$$

It has the following precise limitation.

Lemma 19.8 (Necessary and sufficient phase condition)**.** *For $k \geq 2$ and $b \neq 0$,*

$$N(\mathcal{H}_{k,b}) = \begin{cases} k^2, & (b/\bar{b})^{2k} \neq 1, \\ k(k+1)/2, & (b/\bar{b})^{2k} = 1. \end{cases}$$

All zeros in the first case are nondegenerate. In the second case, exactly k are nondegenerate. In particular, the older assumption that $b/\bar{b}$ is not a root of unity was sufficient but unnecessary.

Proof. By (19.12), the target is $\zeta = -(-\alpha/d)^k$, with $|\zeta| = 1$ and $\zeta^3 = -(\alpha/\beta)^{2k}$. Apply Corollary 19.5 and the smooth-boundary row of Theorem 19.3. □

The final constant in (19.13) is not needed for splitting. Omitting it removes the exceptional phases altogether. Every monomial of S_k has the form

$$c\,\beta^{j+2q}\alpha^q U^i V^j, \qquad i + 2j + 3q = k, \tag{19.14}$$

so its only term of total degree k in U, V is U^k.

Lemma 19.9 (Uniform stability without a phase condition)**.** *For each $k \geq 2$ there is a $\delta_k > 0$, independent of both the phase and the magnitude of $b \neq 0$, such that*

$$S_k(u, \bar{u} + \eta u) \tag{19.15}$$

has exactly k^2 nondegenerate zeros whenever $|\eta| < \delta_k$. If b and η are real, precisely k of these zeros are real.

Proof. Set $u = dv$ in (19.12). The normalized expression is $T_k(v, \bar{v} + \eta(d/\bar{d})v)$, and the magnitude of its second-input perturbation is $|\eta|$. The implicit function theorem at the k^2 simple zeros of $T_k(v, \bar{v})$ gives a fixed positive tolerance. Its pure highest term v^k persists, so the bound in Lemma 2.1 excludes additional zeros.

For real b, choose d real. The identity

$$T_k(1 + 2\cos\theta, 1 + 2\cos\theta) = 1 + 2\cos(k\theta) \tag{19.16}$$

follows from the three roots $1, e^{i\theta}, e^{-i\theta}$. It supplies k distinct real roots in $(-1, 3)$; the real restriction has degree k, so there are no others. Under a small real η, their simple continuations remain real by conjugation symmetry. Decrease δ_k if needed to preserve all of them. □

19.3 The sharp cubic shear disk and its optimal constant

The tolerance in Lemma 19.9 can also be determined exactly in order three. Allowing the constant term to vary improves the best uniform shear radius to a rational value. This perturbation changes the analytic quadratic coefficient; the conjugate-linear perturbation of Theorem 17.2 is a different family.

Theorem 19.10 (Optimal cubic shear radius)**.** *For a real constant c define*

$$F_{c,\eta}(v) = v^3 - 3|v|^2 - 3\eta v^2 + c, \qquad \eta \in \mathbb{C}.$$

A positive centered parameter disk on which all fibers have nine distinct zeros exists exactly when $0 < c < 4$. In that interval its largest radius, also when nondegeneracy is required, is

$$R(c) = \min\left\{\sqrt{\frac{c}{27}},\, 1 - \left(\frac{c}{4}\right)^{1/3}\right\}. \tag{19.17}$$

The unique optimal constant and its radius are

$$c_* = \frac{27}{16}, \qquad R(c_*) = \frac{1}{4}. \tag{19.18}$$

For the canonical folding polynomial, $c = 3$, the exact radius is $1 - (3/4)^{1/3} = 0.0914397035\ldots$.

Here are all equality cases on the first parameter circle. Put $\omega = e^{2\pi i/3}$ and

$$\mathcal{I}_c = \left\{\frac{v}{3} : |v|^2 = \frac{c}{3}, \quad \cos(3\arg v) = \frac{\sqrt{c}}{6\sqrt{3}}\right\},$$
$$\mathcal{O}_c = \left\{\left(\left(\frac{c}{4}\right)^{1/3} - 1\right)\omega^j : 0 \leq j < 3\right\}.$$

The exceptional parameters on $|\eta| = R(c)$ are exactly $\mathcal{I}_c$ for $c < c_*$, exactly $\mathcal{O}_c$ for $c > c_*$, and their disjoint union for $c = c_*$. They number six, three, and nine, respectively. Each has exactly eight distinct zeros, of which one is an ordinary fold and seven are nondegenerate. Every other parameter on that circle has nine nondegenerate zeros.

Proof. At $\eta = 0$ the equation is $T_3(v, \bar{v}) = 3-c$. Its target discriminant is $-c^3(4-c)$, so Theorem 19.3 proves the necessity and sufficiency of $0 < c < 4$ for nine simple zeros at the center.

Fix such a c. Every zero is nonzero. Eliminating η gives

$$\eta = \frac{v}{3} - \frac{\bar{v}}{v} + \frac{c}{3v^2}, \qquad (F_{c,\eta})_v = v^2 + 3\bar{v} - \frac{2c}{v} \quad \text{at a zero.}$$

With $x = |v|^2$, singularity is equivalent to $|v^3 + 3x - 2c| = 3x$. The value $x = 2c/3$ cannot satisfy this condition when $0 < c < 4$. Thus the singular parameters satisfy

$$\cos(3 \arg v) = -\frac{x^3 - 12cx + 4c^2}{2x^{3/2}(3x - 2c)}, \tag{19.19}$$

$$|\eta|^2 = Q_c(x) := \frac{-2c^3 + 13c^2x - cx^3 - 24cx^2 + 2x^4 + 9x^3}{3x^2(3x - 2c)}. \tag{19.20}$$

Angular feasibility factors as

$$(x^3 - 4c^2)(x^3 - 36x^2 + 24cx - 4c^2) \le 0. \tag{19.21}$$

Write $B_c(x)$ for the second factor and $a = (4c^2)^{1/3}$. The signs

$$B_c(0) < 0, \quad B_c(c/3) = c^3/27 > 0, \quad B_c(2c/3) = 4c^2(2c/27 - 1) < 0$$

show that B_c has exactly three distinct positive roots $b_1 < b_2 < b_3$. Also $a > 2c/3$ and $B_c(a) = 12a(2c - 3a) < 0$. The feasible set is therefore

$$[b_1, b_2] \cup [a, b_3], \qquad b_1 < c/3 < b_2 < 2c/3 < a < b_3.$$

The derivative of (19.20) is

$$Q_c'(x) = \frac{2(3x - c)A_c(x)}{3x^3(3x - 2c)^2}, \qquad A_c(x) = x^4 - cx^3 + 9cx^2 - 10c^2x + 4c^3.$$

This quartic is strictly positive. To see this, set $t = x/c$. For $t \ge 1$, $A_c(x)/c^3 = ct^3(t - 1) + 9t^2 - 10t + 4 > 0$. For $0 < t < 1$ the exact identity

$$\frac{A_c(x)}{c^3} = (2t^2 - t)^2 + 8(t - 5/8)^2 + 7/8 + (4 - c)t^3(1 - t) > 0$$

proves the same assertion. Hence the minima on the two feasible intervals are respectively at $x = c/3$ and $x = a$, with values

$$Q_c(c/3) = c/27, \qquad Q_c(a) = \left(1 - (c/4)^{1/3}\right)^2.$$

Both are attained. At the first, (19.19) gives the six values in $\mathcal{I}_c$, and the zero equation gives $\eta = v/3$. At the second, it gives $\cos(3 \arg v) = 1$, $v = (2c)^{1/3}\omega^j$, and the three values in $\mathcal{O}_c$. Strictness of the minimization proves that this list contains all nearest singular parameters.

Below their distance from zero, no zero is singular. The pure highest term v^3 confines all zeros uniformly on compact parameter sets. Continuation from $\eta = 0$ therefore preserves all nine simple zeros. At a nearest singular parameter, Lemma 2.1 rules out nine distinct zeros. This proves (19.17), including sharpness for distinct counts. Its first entry strictly increases with c and its second strictly decreases. Their unique intersection is (19.18).

For the precise boundary counts we check the local singularities. At an inner one, rotate v_0 to $r > 0$ and multiply the target by the inverse cubic phase. Write $b = \sin(3 \arg v_0)$, so $b^2 = 1 - r^2/36$ and $r^2 = c/3 < 4/3$. In real coordinates $v = e^{i \arg v_0}(r + X + iY)$ the linear part is $i(6rbX + r^2 Y)$, and the real quadratic part is $r(3X^2 - 5Y^2)/2$. On its kernel $(X, Y) = (r, -6b)t$, the quadratic coefficient is $4r^3 - 90r \neq 0$. Thus the singularity is an ordinary fold. At an outer one, rotation reduces to $v_0 = t = (2c)^{1/3} > 0$ and $\eta = t/2 - 1$. The derivatives in the real and imaginary directions are 0 and $6t$, while the second real derivative in the kernel direction is $3t \neq 0$. This is also a fold.

The inner parameters have positive cosine of three times their argument, whereas the outer ones have cosine -1; the two sets are disjoint. Each listed parameter consequently has exactly one singular zero. Approaching it from the open disk merges precisely two of the nine zeros, and the other seven remain simple. All other boundary parameters are regular. This proves all the stated counts and equality cases. □

Corollary 19.11 (A maximally robust cubic model)**.** *For $b \neq 0$ the polynomial*

$$u^3 + bu\bar{u} + b\eta u^2 - \frac{|b|^2 b}{16}$$

has exactly nine simple zeros whenever $|\eta| < 1/4$. Among all real choices of the multiplier of $-|b|^2 b$, this radius is uniquely maximal. If b and η are real, exactly three zeros are real. For the unmodified model $S_3(u, \bar{u} + \eta u)$, the exact uniform tolerance in Lemma 19.9 is $1 - (3/4)^{1/3}$.

Proof. Choose $d^3 = -|b|^2 b/27$, with $|d| = |b|/3$, and set $u = dv$. Division by d^3 gives $F_{27/16,\eta'}$ with $\eta' = \eta d/\bar{d}$, so $|\eta'| = |\eta|$. The canonical constant of S_3 instead gives $c = 3$. Apply Theorem 19.10. When the coefficients are real the three real roots at $\eta = 0$ continue throughout the stated real interval. Nondegeneracy prevents their loss, and a real cubic has no further real roots. □

The exact symbolic identities, the positive-quartic certificate, and both fold expansions are checked in `verify_cubic_shear_radius.py`, with report `cubic_shear_radius_verification.json`. The feasible-interval and continuation arguments above prove the all-parameter claim.

19.4 One correction at every seed zero

Fix $r \geq 1$, put $N = 2r + 1$, and take

$$Q(t) = \prod_{j=1}^{r} (t - \epsilon^2 j^2), \qquad f = z^N + Q(zw), \qquad f^* = w^N + Q(zw), \tag{19.22}$$

with $\epsilon > 0$ small and rational. This seed has N^2 simple zeros. Indeed, on each of the N distinct lines $z = se^{2\pi ij/N}$, $s \in \mathbb{R}$, its imaginary part vanishes and its real equation is

$$s^N + \prod_{j=1}^{r} (s^2 - \epsilon^2 j^2) = 0.$$

After $s = \epsilon x$, the $2r$ simple roots $\pm 1, \ldots, \pm r$ of the limiting equation persist. The remaining simple root tends to -1 in the original s coordinate, since at $\epsilon = 0$ the polynomial is $s^{2r}(s + 1)$. Degree counting accounts for all roots. None is zero; different lines give disjoint zero sets. The derivative along each line is nonzero, as is the transverse derivative of $\operatorname{Im} z^N$. This proves the asserted nondegeneracy. In particular,

$$\arg z \in \frac{\pi}{N}\mathbb{Z}, \qquad |z| \to 0 \text{ on the inner zeros}, \qquad |z| \to 1 \text{ on the } N \text{ outer zeros}. \tag{19.23}$$

Put $\lambda = 1$ and define

$$B = -\lambda z^r, \qquad \widetilde{A} = \lambda w^{r+1} - Bzw, \qquad G = \widetilde{A} f + B f^*. \tag{19.24}$$

At a seed zero, write $\rho = |z|$. Since $w^N = -Q(\rho^2)$,

$$\eta := \frac{\widetilde{A}}{B} = -\frac{w^{r+1}}{z^r} - \rho^2 = \frac{Q(\rho^2)}{\rho^{2r}} - \rho^2 = \pm\rho - \rho^2 \longrightarrow 0. \tag{19.25}$$

For the inner zeros both terms tend to zero. For the outer zeros $Q(\rho^2) > 0$, so the sign is positive and $\rho \to 1$. The convergence is uniform over all seed zeros for fixed r.

Since the unshifted model has no exceptional phase, the real choice $\lambda = 1$ is admissible at every zero. Lemma 19.9 provides a tolerance depending only on k.

The other purpose of the same adjustment is the degree estimate

$$\deg_w G \le 2r, \qquad \deg_z G \le 3r + 2, \qquad \deg_{\mathrm{tot}} G \le 3r + 3. \tag{19.26}$$

The terms $\lambda w^{r+1} Q(zw)$ and $-\lambda z^r w^N$ cancel at degree w^{2r+1}. Every remaining term of $\lambda w^{r+1} f + B f^*$ has w-degree at most $2r$. The extra term $-Bzwf$ has w-degree $r+1 \le 2r$. The other two estimates follow directly from (19.24).

19.5 A recurrence with bounded conjugate degree

Fix $k \ge 2$ and a real parameter $\tau > 0$. Define

$$L_0 = 3, \qquad L_1 = f, \qquad L_2 = f^2 + \frac{2\tau}{k} G,$$

$$L_j = f L_{j-1} + \frac{\tau}{k} G L_{j-2} - \frac{\tau^3}{k^3} B^2 B^* L_{j-3} \quad (j \ge 3), \tag{19.27}$$

$$P_{\epsilon,\tau} = L_k. \tag{19.28}$$

No division by a polynomial occurs in this definition. The final constant from the older shifted local model is deliberately omitted.

Lemma 19.12. *For each $k \ge 2$ and $r \ge 1$, first choose a sufficiently small positive rational ϵ. Then, for every sufficiently small $\tau > 0$, (19.28) has exactly $(kN)^2$ nondegenerate zeros and*

$$\deg_z P_{\epsilon,\tau} = kN, \qquad \deg_w P_{\epsilon,\tau} = kr, \qquad \deg_{\mathrm{tot}} P_{\epsilon,\tau} = kN.$$

For $r \ge 2$ its highest homogeneous part is exactly z^{kN}. For $r = 1$ that part converges coefficientwise to z^{3k} as $\tau \to 0$. Rational τ gives coefficients in $\mathbb{Q}$. In addition, exactly kN of its zeros are real.

Proof. Since $3r + 2 < 2N$ and $3r + 3 \le 2N$, the estimates (19.26) and $\deg_w(B^2 B^*) = r$, $\deg_z(B^2 B^*) = 2r$ prove by induction that

$$\deg_w L_j \le jr, \qquad \deg_z L_j \le jN, \qquad \deg_{\mathrm{tot}} L_j \le jN.$$

The coefficient of z^{kN} is one. At $\tau = 0$ the polynomial is f^k, whose coefficient of $z^{kr} w^{kr}$ is one; this coefficient remains nonzero for small τ. When $r \ge 2$, $3r + 3 < 2N$ makes every τ-dependent term have total degree strictly less than kN. For $r = 1$ the inequalities are non-strict, but every coefficient depends polynomially on τ and the highest part at $\tau = 0$ is z^{3k}.

At each seed zero z_0, use the local real coordinate $f = \tau u$, so $f^* = \tau\bar{u}$. Let $B_0 = B(z_0, \bar{z}_0)$ and $\eta_0 = \widetilde{A}(z_0, \bar{z}_0)/B_0$. Divide (19.27) in order j by τ^j. Because $G = \tau(\widetilde{A}u + B\bar{u})$, the local limit is precisely the recurrence for

$$\tau^{-k} P_{\epsilon,\tau} \longrightarrow S_k(u, \bar{u} + \eta_0 u)\big|_{b=B_0} \tag{19.29}$$

in C^1 on compact u-sets. In particular, the power τ^3 in the last recurrence coefficient is essential. By (19.25) and Lemma 19.9, ϵ can first be fixed so that every limit has k^2 simple zeros. For sufficiently small τ, these persist in disjoint neighborhoods of the N^2 seed zeros. This gives $(kN)^2$ nondegenerate zeros. Exactly N of the seed zeros are real. At each of these B_0, η_0 are real, and Lemma 19.9 supplies k simple real zeros of the local limit. Their continuations are real because the coefficients and the local inverse coordinate are real. There are therefore at least kN real zeros. The restriction $P_{\epsilon,\tau}(x, x)$ has degree kN for small τ (its leading coefficient tends to one), so this count is exact.

For $r \geq 2$, Lemma 2.1 proves exhaustion. For $r = 1$, write $p_{\tau,3k}$ for the highest homogeneous part. The pair $(p_{\tau,3k}, p^*_{\tau,3k})$ tends to (z^{3k}, w^{3k}), which has no common point in $\mathbb{CP}^1$. Nonvanishing of their homogeneous resultant is an open condition, so there is no intersection at infinity for small τ. A common projective component would meet the line at infinity; therefore the curves are coprime. Ordinary projective Bézout gives exactly $(3k)^2$ complex intersections counting multiplicity. The constructed simple real intersections exhaust them. □

The case $k = 1$ requires no recurrence: the seed f itself supplies N^2 simple zeros. This proves the $r \geq 2$ part of the pure-head assertion in Theorem 19.1. For $r = 1$ this particular recurrence has the term $\lambda z^5 w$ of total degree six in G. Those mixed highest terms are retained here; Theorem 19.16 supplies a separate pure-head formula.

19.6 The optimal quadratic shear radius and an analytic sextic seed

The general stability tolerance can be made exact in splitting order two. This removes the rational-certificate dependency for the pure-head endpoint $E_{2,6}$, although the higher splitting orders require a separate analysis.

Lemma 19.13. *For every $b \neq 0$ and $|\eta| < 1$,*

$$S_2(u, \bar{u} + \eta u) = u^2 + b\bar{u} + b\eta u$$

has exactly four nondegenerate zeros. The radius one is the largest open disk about zero in the η-plane for which this assertion holds uniformly in $b \neq 0$.

Proof. The origin is a negative simple zero because its Jacobian is $|b|^2(|\eta|^2 - 1) < 0$. The map is proper of degree two, and its zero set is finite. At every other zero its local index is at most one by Lemma 20.4, since its conjugate derivative is $b \neq 0$. The sum of the other indices must be three, so there are at least three other distinct zeros. The quadratic intersection bound gives at most four in total. Hence there are exactly four. Their complexified intersection multiplicities sum to at most four, so each is one and every zero is nondegenerate. At any $|\eta| = 1$ the origin is degenerate. Every larger centered disk contains such a parameter, proving optimality. □

Proposition 19.14 (Complete quadratic shear region)**.** *For fixed $b \neq 0$, put*

$$\Delta_b(\eta) = |\eta|^4 + 18|\eta|^2 + 8\,\mathrm{Re}\big((b/\bar{b})\eta^3\big) - 27.$$

The polynomial $u^2 + b\bar{u} + b\eta u$ has four distinct zeros, all nondegenerate, exactly when

$$\Delta_b(\eta) < 0, \qquad |\eta| \neq 1.$$

If $\Delta_b(\eta) > 0$, it has two nondegenerate zeros. On $\{|\eta| = 1\} \cup \{\Delta_b(\eta) = 0\}$ it has three distinct zeros, two nondegenerate, except at the six parameters

$$(b/\bar{b})\eta^3 \in \{1, -27\},$$

where it has two distinct zeros, one nondegenerate. For each fixed b, the unique largest open parameter disk with four distinct zeros at every parameter is $|\eta| < 1$. The parameter region giving four zeros uniformly for every $b \neq 0$ is exactly this open unit disk.

Proof. Choose $\alpha = |b|e^{i\arg(b)/3}$ and set $u = \alpha z$. After division by α^2 the polynomial is $z^2 + \bar{z} + \lambda z$, where $\lambda = (b/\alpha)\eta$, $|\lambda| = |\eta|$, and $\lambda^3 = (b/\bar{b})\eta^3$. The affine substitution $s = -2z - \lambda$ transforms its zero equation into

$$T_2(s, \bar{s}) = \lambda^2 + 2\bar{\lambda}.$$

The exact discriminant identity is

$$D(\lambda^2 + 2\bar{\lambda}) = (|\lambda|^2 - 1)^2 D(-\lambda), \qquad D(-\lambda) = \Delta_b(\eta).$$

Theorem 19.3, with $k = 2$, gives all stated counts and regularity assertions. The cusp targets have exactly two preimages: for target three these are $\lambda = 1, -3$; rotation gives precisely $\lambda^3 \in \{1, -27\}$. Thus the exceptional list is complete.

The four-zero region is contained in the rotated reflected deltoid $\{D(-\lambda) < 0\}$. Lemma 19.4 bounds every contained disk by radius one and forces center zero at equality. The open unit disk itself is contained in the four-zero region, proving the fixed-b assertion. Finally, if $r = |\eta| > 1$, choose the phase of b so that $(b/\bar{b})\eta^3 = r^3$. Then

$$\Delta_b(\eta) = r^4 + 8r^3 + 18r^2 - 27 = (r - 1)(r + 3)^3 > 0,$$

so there are only two zeros for that b. At $r = 1$ the origin is degenerate and the classification gives fewer than four. This proves the complete uniform region. □

Proposition 19.15 (An analytic 36-zero pure-head construction)**.** *Fix a rational number $0 < \kappa < 4/27$ and put*

$$f = z^3 + zw - \kappa, \qquad f^* = w^3 + zw - \kappa, \qquad G = w^2 f - z f^* = (z^3 - \kappa)w^2 - z^2 w + \kappa z.$$

For every sufficiently small positive τ,

$$P_{\kappa,\tau} = f^2 + \tau G \tag{19.30}$$

has exact conjugate degree two, pure highest term z^6, and exactly 36 nondegenerate zeros. It has exactly six real zeros. Rational τ gives real rational coefficients; clearing denominators gives an integer example. Thus $E_{2,6} = 36$ has a proof independent of the earlier rational root boxes.

Proof. The real equation $s^3 + s^2 - \kappa = 0$ has three simple roots in $(-1, -2/3)$, $(-2/3, 0)$, and $(0, 1/3)$. As in (19.22), its three rotations by cube roots of unity give all nine simple zeros of f. Every such zero z_0 satisfies $0 < |z_0| < 1$.

Use the local coordinate $f = \tau u$ at each seed zero. Then $f^* = \tau\bar{u}$, and

$$\tau^{-2}P_{\kappa,\tau} \longrightarrow u^2 + \bar{z}_0^2 u - z_0\bar{u} = u^2 + b(\bar{u} + \eta u), \qquad b = -z_0, \quad \eta = -\bar{z}_0^2/z_0.$$

The convergence is in C^1 on compact u-sets and $|\eta| = |z_0| < 1$. Lemma 19.13 gives four simple local zeros at each of the nine seed zeros. Their continuations are disjoint when τ is small. The explicit expression for G gives $\deg_w G \le 2$ and $\deg_{\mathrm{tot}} G = 5$. Thus (19.30) has pure highest term z^6 and exact conjugate degree two; the 36 constructed simple zeros exhaust the quadratic bound.

There are three real seed zeros. At each of them b, η are real and $|\eta| < 1$. The local real restriction is $u(u + b(1 + \eta))$, which has two distinct simple real zeros. Their real continuations give six real zeros of $P_{\kappa,\tau}$. Its real restriction has degree six, so there are exactly six. All coefficient claims follow from the displayed formula. □

This use of the unadjusted cofactor is specific to the exactly controlled quadratic local model. It does not give a corresponding unit shear disk for S_k when $k \ge 3$. The following translated construction handles every order while keeping the normalized error small.

19.7 A pure highest term at the cubic boundary, in every order

The mixed highest terms of (19.28) are a limitation of that recurrence. A translation of the local folding polynomial, followed by different reductions of its analytic and mixed coefficients, removes this restriction without requiring a unit shear tolerance.

Theorem 19.16 (The entire pure-head cubic boundary)**.** *For all integers $k \ge 1$ and $k \le m < 3k$,*

$$E_{m,3k} = 9k^2. \tag{19.31}$$

The maximum is attained with real rational coefficients, exact conjugate degree m, only nondegenerate zeros, and exactly $3k$ real zeros. In particular, $E_{4,12} = 144$. Clearing denominators gives integer-coefficient extremizers.

Here is a finite polynomial formula for every order. Fix $k \ge 2$. Using independent symbols U, T, C, define

$$\begin{aligned} R_0 &= 3, \qquad R_1 = U, \qquad R_2 = U^2 + \frac{2}{k}T, \\ R_j &= UR_{j-1} + \frac{1}{k}TR_{j-2} - \frac{C}{k^3}R_{j-3} \quad (j \ge 3). \end{aligned} \tag{19.32}$$

For two more independent symbols A, H, expand

$$Q_k(U, T; A, H, C) := R_k\left(U + \frac{A}{k},\ T - AU + \frac{H}{k}; C\right) = \sum_{i,j} c_{ij}(A, H, C)U^iT^j. \tag{19.33}$$

All these coefficients are rational polynomials. The relevant identities are

$$c_{k0} = 1, \qquad c_{k-1,0} = 0, \qquad c_{k-2,1} = 1, \qquad c_{ij} = \sum_{a+2b+3q=k-i-2j} \gamma_{abq}A^aH^bC^q. \tag{19.34}$$

The last sum is empty when its weight is negative. Indeed, (19.32) is homogeneous with weights $1, 2, 3$ on U, T, C, respectively, and the substitution preserves weights if A, H have weights $1, 2$. The

two contributions to U^{k-1} in (19.33) cancel: they are AU^{k-1} from the highest power and $-AU^{k-1}$ from TU^{k-2}.

Choose a small positive rational κ and put

$$\begin{aligned} &f = z^3 + zw - \kappa, \qquad f^* = w^3 + zw - \kappa, \\ &A = w^2, \quad B = -z, \quad D = -z^2 w, \quad H = BA^* = -z^3, \quad C = B^2 B^* = -z^2 w, \\ &G = Af + Bf^* = (z^3 - \kappa)w^2 - z^2 w + \kappa z. \end{aligned} \tag{19.35}$$

In the remaining formulas A, H, C, D denote these actual polynomials. Let ρ_κ be polynomial reduction modulo (f, f^*), with remainder in the span of $z^a w^b$, $0 \le a, b \le 2$. This reduction is well defined: in any graded monomial order the leading terms of f, f^* are the relatively prime monomials z^3, w^3. Their only overlap reduces to zero, so they form a Gröbner basis. Reduction uses $z^3 = \kappa - zw$ and $w^3 = \kappa - zw$ and preserves rational coefficients. Its remainder has total degree at most four and conjugate degree at most two, and agrees with the input at all seed zeros.

For $\tau > 0$ define the polynomial

$$\begin{aligned} P_{k,\kappa,\tau} = f^k + \sum_{j\ge 1,\ i\ge 0} \tau^{k-i-j} c_{ij}(D, H, C) f^i G^j \\ + \sum_{i=0}^{k-2} \tau^{k-i} \rho_\kappa\big(c_{i0}(A, H, C)\big) f^i. \end{aligned} \tag{19.36}$$

Only the finitely many nonzero coefficients in (19.33) are included. All powers of τ in the first sum are positive, since $i + 2j \le k$.

Proof of Theorem 19.16. We first verify every support restriction in (19.36). One has

$$\deg f = 3, \quad \deg_w f = 1, \qquad \deg G = 5, \quad \deg_w G = 2.$$

Consider a coefficient monomial in the first sum, of weight $\ell = k - i - 2j$. After substituting D, H, C, its total degree is $3(a + b + q) \le 3\ell$ and its conjugate degree is $a + q \le \ell$. Consequently

$$\deg\big(c_{ij}(D, H, C) f^i G^j\big) \le 3k - j < 3k, \qquad \deg_w\big(c_{ij}(D, H, C) f^i G^j\big) \le k. \tag{19.37}$$

Each term in the last sum has total degree at most $3i + 4 \le 3k - 2$ and conjugate degree at most $i + 2 \le k$. Thus the highest homogeneous part is exactly z^{3k}. At $\tau = 0$, the coefficient of $z^k w^k$ is one, so the conjugate degree is exactly k for sufficiently small τ.

For $0 < \kappa < 4/27$, the seed f has nine simple zeros. They are $z = s\omega^j$, where $\omega^3 = 1$, $j = 0, 1, 2$, and s ranges over the three real roots of $s^3 + s^2 = \kappa$; see Proposition 19.15. All have $0 < |z| < 1$. As $\kappa \downarrow 0$, the three real roots tend to $-1, 0, 0$, respectively. It follows that

$$\eta_\kappa := \max_{f(z,\bar z)=0} \left| \frac{A - D}{B} \right| = \max_{s^3+s^2=\kappa} |s + s^2| \longrightarrow 0. \tag{19.38}$$

The maximum in the last expression is over the three real roots.

Fix one seed zero. Its simplicity makes $f = \tau u$ a local real-analytic coordinate, with $f^* = \tau \bar u$ and $G = \tau(Au + B\bar u)$. Evaluate A, B, D, H, C at that zero in the following local limits, and set $t = Au + B\bar u$. The limit of $\tau^{-k} P_{k,\kappa,\tau}$ is

$$L(u) = u^k + \sum_{j\ge 1, i\ge 0} c_{ij}(D, H, C) u^i t^j + \sum_{i=0}^{k-2} c_{i0}(A, H, C) u^i. \tag{19.39}$$

The reduction operator disappears here because it preserves values at seed zeros. If D in the first sum were replaced by A, the result would be exactly

$$\begin{aligned} Q_k(u, Au + B\bar{u}; A, H, C) &= R_k\left(u + \frac{A}{k}, B\left(\bar{u} + \frac{\bar{A}}{k}\right); C\right) \\ &= S_k\left(u + \frac{A}{k}, \bar{u} + \frac{\bar{A}}{k}\right) \end{aligned} \tag{19.40}$$

with the parameter $b = B$ in (19.11).

We justify uniform smallness even at the six seed zeros approaching zero. Write $r = |B| = |z|$. On the seed set,

$$|A|/r = r, \quad |D|/r = r^2, \quad |H|/r^2 = r, \quad |C|/r^3 = 1, \quad |A - D|/r \le \eta_\kappa.$$

The weighted coefficient identity (19.34) therefore shows that replacing A by D changes a coefficient of weight ℓ by $O_k(\eta_\kappa r^\ell)$, uniformly over all nine zeros. Choose d as in (19.12), so $|d| = r/k$, and set $u = dv - A/k$. On every fixed compact v-set, u/r and t/r^2, together with their first real derivatives, are bounded independently of the seed zero and of small κ. Since $\ell + i + 2j = k$, equations (19.39)–(19.40) give

$$d^{-k} L(dv - A/k) = T_k(v, \bar{v}) + O_k(\eta_\kappa) \quad \text{in } C^1 \text{ on compact sets.} \tag{19.41}$$

The constant may depend on the compact set and k, which are fixed.

The k^2 zeros of $T_k(v, \bar{v})$ are simple by Theorem 19.3. Choose $\kappa > 0$ rational sufficiently small that (19.41) retains all of them at every seed zero. With this κ fixed, choose $\tau > 0$ sufficiently small. The local convergence to (19.39) is in C^1, so $P_{k,\kappa,\tau}$ has k^2 simple zeros near each of the nine seed zeros. The pure highest term and the quadratic bound in Lemma 2.1 show that these $9k^2$ zeros exhaust its zero set.

All coefficients are real when κ, τ are real. At each of the three real seed zeros, A, B, d are real and the normalized map in (19.41) preserves conjugation. The k simple real zeros of T_k therefore continue as real zeros, first to L and then to $P_{k,\kappa,\tau}$. This gives $3k$ real zeros. The restriction to $w = z = x$ has degree $3k$, so the real-axis count is exact. Rational τ retains all properties. The case $k = 1$ is the seed itself. Finally, Lemma 19.42 increases the conjugate degree from k to any $m < 3k$ while retaining the count and real rational coefficients. □

Together with the $r \ge 2$ construction, this proves the pure-head assertion of Theorem 19.1 for every $r, k \ge 1$. It settles the pure-head values at all multiples of three in $n \le 3m$. It does not settle the rectangular line $n = 3m - 1$ or the pure-head values at the other residues of n.

Lemma 19.17 (Increasing the analytic degree)**.** *Let P have pure highest term cz^d, $c \neq 0$, and only nondegenerate zeros. If $n > d$, then, for every sufficiently small positive ε, $P + \varepsilon z^n$ has exactly $N(P) + n - d$ zeros, all nondegenerate. The $n - d$ new zeros have positive Jacobian and approach infinity.*

Proof. The bounded zeros persist. For a large zero of the perturbed map, division by z^d gives $c + \varepsilon z^{n-d} + O(|z|^{-1}) = 0$. Choose a fixed sufficiently large radius. Outside it, this identity confines $\varepsilon|z|^{n-d}$ between $|c|/2$ and $3|c|/2$. After $z = \varepsilon^{-1/(n-d)} u$, all such zeros therefore lie in a fixed annulus. The rescaled equation converges in C^1 there to $cu^d + u^n = 0$. Its $n - d$ nonzero zeros are simple and analytic. The implicit function theorem, and a uniform lower bound away from their disjoint neighborhoods, give exactly these new zeros and exclude every intermediate scale. □

Corollary 19.18 (Uniform pure-head lower bounds)**.** *For $1 \le m < n$ with $n \ge 3$, put $q = \min\{m, \lfloor n/3 \rfloor\}$. Then*

$$E_{m,n} \ge 9q^2 + n - 3q. \tag{19.42}$$

The displayed count is attained exactly with real integer coefficients, exact conjugate degree m, and only nondegenerate zeros. In particular,

$$E_{m,n} \ge 9m^2 + n - 3m \quad (n \ge 3m), \qquad E_{4,13} \ge 145.$$

For $m < n \le 3m$, write $s = n \bmod 3$. One has the uniform bound

$$0 \le n^2 - E_{m,n} \le 2ns - s^2 - s \le 4n - 6. \tag{19.43}$$

Thus $E_{m,n}/n^2 \to 1$ uniformly as $n \to \infty$ within this cone, although the finite endpoints with $s = 1, 2$ are not all determined.

Proof. Start with the rational extremizer for $E_{q,3q}$ in Theorem 19.16. Apply Lemma 19.17 when $n > 3q$, choosing its small parameter rational. If necessary, add a sufficiently small real rational multiple of w^m after the analytic degree has reached n. The simple zeros persist, and the unchanged pure highest term excludes zeros escaping to infinity; compactness excludes any additional bounded zeros. The count is consequently unchanged and the conjugate degree becomes exactly m. Clearing denominators gives integer coefficients. In the cone, $q = (n - s)/3$, and (19.42) becomes $E_{m,n} \ge (n - s)^2 + s$, which is (19.43). □

19.8 Eighty-four zeros with conjugate degree three

The degree extension above gives $E_{3,10} \ge 82$. A shared seed and a compatible correction of the cubic local models improve this bound while retaining the same degrees.

Theorem 19.19. *There is a polynomial with real rational coefficients, pure highest term z^{10}, exact conjugate degree three, and exactly 84 nondegenerate zeros, of which 47 are positive and 37 negative, and exactly ten real-axis zeros. Consequently,*

$$84 \le E_{3,10} \le 100. \tag{19.44}$$

Real integer coefficients are possible after multiplication by a common denominator. This improves the lower bound; it does not determine the global maximum $E_{3,10}$.

We first record the local model used at the shared point.

Lemma 19.20. *If $-1 < \lambda < 1$ and $0 < t < 4/[27(1 + |\lambda|)^2]$, then*

$$Q_{\lambda,t}(u) = u^4 + \lambda u^3 \bar{u} + u\bar{u}^2 - t\bar{u}$$

has exactly sixteen nondegenerate zeros.

Proof. For $u = re^{i\theta} \ne 0$, divide by $\bar{u}$. The imaginary part gives $\sin 5\theta + \lambda \sin 3\theta = 0$, equivalently

$$\sin\theta\{16\cos^4\theta + (4\lambda - 12)\cos^2\theta + 1 - \lambda\} = 0.$$

The two roots in $\cos^2\theta$ are $[(3 - \lambda) \pm \sqrt{\lambda^2 - 2\lambda + 5}]/8$. They are distinct and lie in $(0, 1)$, so there are exactly ten simple angles. At each, $h = \cos 5\theta + \lambda \cos 3\theta$ is nonzero, since $|e^{5i\theta} + \lambda e^{3i\theta}| > 0$. The

angles $\theta, \theta+\pi$ pair opposite signs of h. For each of the five positive values, $hr^3+r^2-t=0$ has one positive root. For each negative value it has two: the maximum of $r^2-|h|r^3$ is $4/(27h^2)>t$. All radii are simple, as are the angular roots. The fifteen nonzero zeros are therefore nondegenerate. The origin supplies the sixteenth, with Jacobian $-t^2$. □

Proof of Theorem 19.19. Fix the rational numbers

$$a=\frac{100}{10101},\qquad b=-\frac{101}{10101},\qquad c=-\frac{10100}{10101},\qquad \kappa=a^2(1+a).$$

They are the three distinct roots of $r^3+r^2-\kappa=0$. Let $f=z^3+zw-\kappa$, $f^*=w^3+zw-\kappa$, and $g=z-a$. The nine seed points are

$$(z,w)=(r\omega^j, r\omega^{-j}),\qquad r\in\{a,b,c\},\quad j=0,1,2.$$

All are physical and simple. Indeed their Jacobians are $3r^3(3r+2)$, positive on the a,c orbits and negative on the b orbit. The leading terms z^3, w^3 show that these nine points exhaust the complexified seed intersection. Only (a,a) is shared with $g=0$.

Write

$$d=3a^2+a,\quad J=d^2-a^2,\quad A=d/J,\quad B=-a/J,\quad \beta=\frac{a}{1+3a},\quad \lambda=B/A=-\frac{3367}{3467}.$$

Near the shared point use the real local coordinate $u=f$, so $w=\bar z$ and $g=Au+B\bar u+O(|u|^2)$. Set

$$\begin{aligned} B_0(z,w)&=a^2(A-1)(z-a)+adA(w-a),\\ G&=w^2f-zf^*=(z^3-\kappa)w^2-z^2w+\kappa z,\\ H&=fGg+f^2B_0. \end{aligned}$$

This polynomial has total degree at most nine and conjugate degree at most three. Its shared jet is

$$H=A\beta u\bar u^2+O(|u|^4). \tag{19.45}$$

At any other seed i, with $g_i=z_i-a\neq 0$, define

$$\alpha_i=\bar z_i^2+\frac{B_0(z_i,\bar z_i)}{g_i},\qquad \nu_i=-z_i.$$

Then $H/g_i=\alpha_i u^2+\nu_i u\bar u+O(|u|^3)$.

The correction is concentrated on the negative seed orbit. Introduce the rational polynomials

$$\begin{aligned} \mathcal{E}(T)&=-\frac{3851837}{34343400}+\frac{389035537}{3434340000}T-\frac{41570361437}{343434000000}T^2,\\ \mathcal{C}(T)&=\frac{25+269T-59T^2}{2000}. \end{aligned} \tag{19.46}$$

For a nonshared seed put

$$(\delta_i,\chi_i)=\begin{cases}\big(b^2\mathcal{E}(z_i/b),\,(-b)^3\mathcal{C}(z_i/b)\big), & z_i^3=b^3,\\ \big(0,\,|\nu_i|^2\nu_i/9\big), & z_i^3\neq b^3.\end{cases}$$

There are real rational polynomials K,L in the space

$$\mathcal{V}=\operatorname{span}_{\mathbb{Q}}\{z^iw^j : 0\le j\le 3,\ 0\le i\le 9-j\}$$

with the following jets. At each nonshared seed,

$$K = g_i \left\{ \frac{\alpha_i^2 + \nu_i \bar{\alpha}_i}{3} u + \left(\frac{\nu_i \alpha_i}{3} + \delta_i \right) \bar{u} \right\} + O(|u|^2),$$
$$L(z_i, \bar{z}_i) = g_i \left\{ \frac{\alpha_i^3}{27} + \frac{\nu_i |\alpha_i|^2}{9} + \frac{\delta_i \bar{\alpha}_i}{3} - \chi_i \right\}. \tag{19.47}$$

At the shared point they satisfy

$$K = O(|u|^3), \qquad L = -\frac{A\beta^3}{100} \bar{u} + O(|u|^2). \tag{19.48}$$

For completeness, this is a compatible rational interpolation statement, not an inference from dimension alone. Order the 34 monomials in $\mathcal{V}$ by increasing j, then increasing i. Reduce values and first derivatives modulo (f, f^*) in the basis $z^i w^j$, $0 \le i, j \le 2$. The K matrix consists of these 27 rows and the three quadratic Taylor coefficients at (a, a); it has rank 27. The L matrix consists of nine value rows and two shared derivative rows; it has rank 11. The exact verifier solves both prescribed systems over $\mathbb{Q}$, checks every residual, and records rational coefficient lists for H, K, L. It separately verifies (19.45)–(19.48) and all support restrictions. Thus the displayed targets belong to the images of the two matrices despite the dependencies among the K rows.

Define

$$P_\tau = f^3 g + \tau H + \tau^2 K + \tau^3 L. \tag{19.49}$$

At a nonshared seed, rescale $f = \tau u$ and divide by $g_i \tau^3$. Translating $v = u + \alpha_i/3$ gives the local limit

$$v^3 + \nu_i |v|^2 + \delta_i \bar{v} - \chi_i. \tag{19.50}$$

For the five seeds outside the b orbit and the shared point, this is a scaled rotation of $v^3 + |v|^2 - 1/9$, which has nine simple zeros by the radial cubic argument. On the b orbit, rescale further by $v = (-b)\xi$. The three models become

$$\xi^3 + \zeta |\xi|^2 + \mathcal{E}(\zeta) \bar{\xi} - \mathcal{C}(\zeta), \qquad \zeta \in \{1, \omega, \omega^2\}. \tag{19.51}$$

Their exact simple zero counts are respectively $9, 7, 7$.

Here is a rational certificate for these last counts. Write $\xi = x + (1+2\omega)y = x + i\sqrt{3}y$ and reduce each polynomial in (19.51) modulo $\omega^2 + \omega + 1$. Its two coefficients are real rational polynomials $F(x, y)$, $G(x, y)$. Their quotient algebra has dimension nine. In a standard monomial basis the symmetric trace matrix is

$$\mathsf{H}_{ij} = operatornameTr(M_{q_i q_j}),$$

where M_q denotes multiplication by q in the quotient. Exact rational elimination gives inertia $(9, 0)$ for $\zeta = 1$ and $(8, 1)$ for $\zeta = \omega$; the latter case also covers ω^2 by conjugation. All nine rational elimination pivots are nonzero in both cases. A nondegenerate trace form makes the algebra reduced: every nilpotent belongs to its radical. Over $\mathbb{R}$, the resulting algebra is a product of real factors and complex conjugate-pair factors. Each real factor contributes one positive square; each complex factor contributes one positive and one negative square to the trace form. Its signature therefore counts the real points. Each local algebra has dimension one, so the two defining equations have independent differentials at every point. This proves the asserted counts and simplicity, without numerical root isolation. The verifier reconstructs the quotient, multiplication matrices and all rational pivots.

At the shared point, rescale $f = \tau\beta\xi$. Equations (19.45) and (19.48) give, after division by $A\beta^4\tau^4$, the limit $Q_{\lambda,1/100}$. Since

$$-1 < \lambda < 0, \qquad \frac{1}{100} < \frac{4}{27(1-\lambda)^2},$$

Lemma 19.20 supplies sixteen simple zeros.

All the local convergences are in C^1 on compact sets. There are no missing real zeros on an intermediate scale. At a nonshared seed, the leading term is bounded below by a positive multiple of $|f|^3$, while the three perturbations are $O(\tau|f|^2)$, $O(\tau^2|f|)$ and $O(\tau^3)$. At the shared seed the corresponding bounds are $|f|^4$, $O(\tau|f|^3)$, $O(\tau^2|f|^3)$ and $O(\tau^3|f|)$; the lower bound follows from $A > |B|$. In either case these inequalities confine all nearby zeros to $|f| \le C\tau$ with fixed C. The pure highest term z^{10} gives uniform properness as $\tau \to 0$, and compactness excludes zeros away from the seeds. The implicit function theorem therefore gives exactly

$$5 \cdot 9 + 9 + 2 \cdot 7 + 16 = 84$$

simple zeros for every sufficiently small positive τ. The exact conjugate degree is three, since it is already three at $\tau = 0$. Choosing τ rational proves the coefficient claim. The pure highest term gives signed count ten, hence 47 positive and 37 negative zeros. At the two nonshared real seeds the local cubics have three real zeros each; for the corrected cubic this is also checked by an exact Sturm count. On the real axis the shared quartic is $x\{(1+\lambda)x^3 + x^2 - 1/100\}$, with four simple real roots. Their real continuations give ten real zeros, the largest number permitted by the degree. □

Corollary 19.21 (Sharp degrees for the eighty-four-zero target)**.** *Among pure-head polynomials with exactly* 84 *nondegenerate zeros, the smallest total degree is ten. At that degree the smallest conjugate degree is three. In addition,*

$$E_{m,n} \ge n + 74 \qquad (n \ge 10,\ 3 \le m < n), \tag{19.52}$$

and the displayed count is attained exactly with real integer coefficients and nondegenerate zeros.

Proof. Degree at most nine permits at most 81 finite zeros. At total degree ten and conjugate degree at most two, (6.4) permits at most $9 \cdot 10 - 16 = 74$. Theorem 19.19 attains both degree thresholds. For the last assertion apply Lemma 19.17 and the exact-degree padding argument of Corollary 19.18. □

19.9 An explicit forty-one-zero construction in degree seven

The same shared-seed idea is simpler at splitting order two. Here the interpolation can be written explicitly, and every local zero count has an elementary proof.

Theorem 19.22. *There is a polynomial with real rational coefficients, pure highest term z^7, exact conjugate degree two, and exactly* 41 *nondegenerate zeros:* 24 *positive,* 17 *negative, and exactly seven on the real axis. Hence*

$$41 \le E_{2,7} \le 47. \tag{19.53}$$

Integer coefficients are possible after clearing denominators. For the forty-one-zero target, total degree seven is minimal; at that total degree, conjugate degree two is minimal.

Proof. Set $a = 1/4$, $\kappa = 5/64$, and

$$f = z^3 + zw - \kappa, \quad f^* = w^3 + zw - \kappa, \quad g = z - a, \quad G = w^2 f - z f^*.$$

The seed has nine simple physical zeros, as in the preceding construction; its three real zeros are a and $(-5 \pm \sqrt{5})/8$. The point a has positive Jacobian. Put

$$d = 7/16, \quad J = 33/256, \quad A = 112/33, \quad B = -64/33, \qquad T = 100.$$

In the coordinate $u = f$ at a, one has $g = Au + B\bar{u} + O(|u|^2)$. Define the linear polynomials

$$\ell = d(z-a) + a(w-a), \quad \ell^* = a(z-a) + d(w-a), \quad B_0 = -a^2 A\ell + \{A(T+a) - a^2 B\}\ell^*$$

and set $H = Gg + fB_0$. Direct differentiation gives

$$H = ATu\bar{u} - aB\bar{u}^2 + O(|u|^3). \tag{19.54}$$

At another seed z_i, with $g_i = z_i - a$, let

$$\alpha_i = \bar{z}_i^2 + B_0(z_i, \bar{z}_i)/g_i, \qquad \nu_i = -z_i.$$

There $H/g_i = \alpha_i u + \nu_i \bar{u} + O(|u|^2)$.

Here is an explicit polynomial for the required interpolation. Let ρ be reduction modulo (f, f^*) to the span of $z^i w^j$, $0 \le i, j \le 2$. Put

$$D(z) = (\kappa - z^3)^3 - z^6, \qquad e(z) = \frac{D(z)}{(z-a)D'(a)}, \qquad h(z) = \frac{1 - e(z)}{z - a}.$$

Both quotients are polynomials. On the eight nonshared seeds $h(z_i) = 1/g_i$, while $e(a) = 1$. Define

$$\alpha = \rho(w^2 + B_0 h), \quad \nu = -z, \qquad R = \rho\left(g\left\{\alpha^2/4 + \nu\alpha^*/2\right\}\right),$$

where $\alpha^*(z, w) = \alpha(w, z)$. Then $R(a, a) = 0$ and $R(z_i, \bar{z}_i) = g_i(\alpha_i^2/4 + \nu_i \bar{\alpha}_i/2)$. At (a, a) write

$$R_u = (dR_z - aR_w)/J, \qquad R_{\bar{u}} = (-aR_z + dR_w)/J.$$

These are rational constants. The polynomial

$$K = R - (R_u + aR_{\bar{u}})f + (R_{\bar{u}}/a)G \tag{19.55}$$

has the same values as R at all seed points and satisfies $K = O(|u|^2)$ at the shared point: there $G = a^2 u - a\bar{u} + O(|u|^2)$. This construction needs no full-rank interpolation assumption. All coefficients are rational, $\deg_{\mathrm{tot}} H \le 6$, $\deg_{\mathrm{tot}} K \le 5$, and both conjugate degrees are at most two.

Now let

$$P_\tau = f^2 g + \tau H + \tau^2 K - \frac{AT^3}{100}\tau^3. \tag{19.56}$$

At a nonshared seed, put $f = \tau u$ and divide by $g_i \tau^2$. The limiting quadratic, after $v = u + \alpha_i/2$, is $v^2 + \nu_i \bar{v}$. It has exactly four simple zeros: the origin and three points of radius $|\nu_i|$. Thus the eight nonshared seeds supply 32 zeros.

At a, put $f = \tau T \xi$ and divide by $AT^3\tau^3$. The limiting cubic is the fixed rational polynomial

$$F(\xi) = \xi^3 - \frac{4}{7}\xi^2\bar{\xi} + \xi\bar{\xi} + \frac{1}{700}\bar{\xi}^2 - \frac{1}{100}. \tag{19.57}$$

It has exactly nine simple zeros. To see this directly, write $\xi = x + iy$. Its imaginary part is

$$\frac{y}{7}\left(17x^2 - 11y^2 - \frac{x}{50}\right).$$

On $y = 0$, the real equation, multiplied by 700, is

$$p(x) = 300x^3 + 701x^2 - 7 = 0.$$

It has one root in each of

$$(-3, -2), \quad (-1/5, -1/20), \quad (1/20, 1/5),$$

by opposite endpoint signs. These are all its roots and all are simple. Off the axis,

$$y^2 = \frac{x(850x - 1)}{550}, \qquad q(x) = -1960000x^3 + 982200x^2 - 699x - 3850 = 0.$$

The latter cubic has one root in each of

$$(-1/5, -1/20), \quad (1/20, 1/5), \quad (1/5, 1),$$

again by opposite endpoint signs. All three yield positive y^2, so they give six further zeros. On the axis the imaginary derivative is nonzero at the three roots of p. Off the axis the imaginary derivative in y is $-22y^2/7$, and the reduced real derivative is nonzero because the roots of q are simple. Hence every one of the nine zeros is nondegenerate.

The local limits hold in C^1. At a nonshared seed the orders of the terms in (19.56) are $|f|^2$, $O(\tau|f|)$, $O(\tau^2)$ and $O(\tau^3)$. At the shared point they are $|f|^3$, $O(\tau|f|^2)$, $O(\tau^2|f|^2)$ and $O(\tau^3)$; the leading lower bound uses $A > |B|$. These estimates confine every nearby zero to $|f| \le C\tau$. The pure highest term z^7 bounds all zeros uniformly as $\tau \to 0$, and the unperturbed physical zero set consists only of the seeds. Compactness and the implicit function theorem therefore give exactly $8 \cdot 4 + 9 = 41$ simple zeros for all sufficiently small positive τ. The conjugate degree stays exactly two. Rational τ gives the stated coefficients. The signed count is seven, yielding 24 positive and 17 negative zeros.

At each of the two nonshared real seeds the limiting quadratic has two real zeros. The shared cubic has the three real zeros of p. Their real continuations give seven real zeros, exhausting the degree of the real restriction. Finally, degree at most six allows at most 36 finite zeros, whereas conjugate degree at most one and total degree seven allows at most $5 \cdot 7 - 6 = 29$ by (6.4). This proves both degree minimality statements. □

Corollary 19.23. *For $n \ge 7$ and $2 \le m < n$,*

$$E_{m,n} \ge n + 34.$$

The displayed count can be attained exactly with real integer coefficients, nondegenerate zeros and exactly n real zeros.

Proof. Apply Lemma 19.17 one degree at a time. Each step adds one real simple zero at infinity and preserves all existing real zeros. Then use sufficiently small real rational exact-degree padding. Clear denominators. □

19.10 A fixed integer polynomial with forty-one zeros

The preceding existence result also admits a fixed parameter certificate. In the formulas defining (19.56), replace $T = 100$ by $T = 2$ and set $\tau = 10^{-4}$. No sufficiently-small-parameter qualification is needed for the following specific polynomial.

Proposition 19.24. *Let* $D = 580800000000000000$ *and*

$$P_{41}(z, w) = D^{-1} \sum_{i=0}^{7} \sum_{j=0}^{2} c_{ij} z^i w^j,$$

where the integer coefficients are

i	c_{i0}	c_{i1}	c_{i2}
0	−880347286142071	−15129332671875	1084189687500
1	3535399696171875	22612116533437500	189498523500000
2	4541067750000	−90628046601000000	−145199962644000000
3	22612184288250000	135960000000000	580785244380000000
4	−90642750000000000	−290400000000000000	58080000000000
5	0	1161600000000000000	0
6	−145200000000000000	0	0
7	580800000000000000	0	0

Then $P_{41} \in \mathcal{E}_{2,7}$ *has exactly* 41 *regular zeros, with signs* $(24, 17)$ *and exactly seven real-axis zeros. Thus* DP_{41} *is an explicit real integer witness for the lower bound in* (19.53).

Proof. Exact quotient-ring reconstruction checks that the printed coefficients are the specialization $a = 1/4$, $T = 2$, $\tau = 1/10000$ of (19.56). The count has an independent rational certificate, so it does not rely on extending the small-parameter proof to this chosen value.

The file `pure_2_7_41_fixed_certificate.json` gives 41 physical boxes and eight complex boxes disjoint from $w = \bar{z}$. Every center has denominator 10^{40} and every coordinate radius is $r = 10^{-16}$. The standard-library verifier `verify_pure_2_7_41_fixed.py` reconstructs the normalized coefficients directly from the printed integer table. It checks pairwise disjointness, all support restrictions, the nonphysical status of the eight complex boxes, and the strict rational contraction inequality

$$q + \beta_0/r < 1/1024$$

in every box, using the inverse center Jacobian and polynomial Hessian bounds. Hence each box contains a regular zero. The $41 + 8 = 49$ disjoint intersections exhaust the complex Bézout count. Center Jacobian signs give $(24, 17)$. Exactly seven physical boxes are centered on the real axis and invariant under conjugation; uniqueness in those boxes forces their zeros to be real. The remaining boxes avoid the axis. This proves the full assertion.

The successful exact report is `pure_2_7_41_fixed_verification.json`. Independent tests reject a changed coefficient, duplicated boxes, a displaced center, and a complex box moved onto the physical slice. □

19.11 Recovering all eight cubic clusters: eighty-eight zeros

The zero quadratic jet imposed at the shared seed in Theorem 19.19 is unnecessarily restrictive. Allowing this jet to vary restores the full nine-zero count in both previously deficient cubic clusters, while retaining sixteen zeros in the shared quartic.

Theorem 19.25. *There is a polynomial in $\mathcal{E}_{3,10}$ with real integer coefficients and exactly 88 non-degenerate zeros, of which 49 are positive, 39 are negative, and exactly ten are real. Consequently,*

$$88 \le E_{3,10} \le 100. \tag{19.58}$$

Total degree ten is minimal for this target, and at that total degree the minimal conjugate degree is three. The global hundred-zero endpoint remains unresolved.

Proof. Retain $a, b, c, \kappa, f, f^*, g, G, d, J, A, B, \beta$ from the proof of Theorem 19.19; in particular

$$a = \frac{100}{10101}, \quad b = -\frac{101}{10101}, \quad c = -\frac{10100}{10101}, \quad \beta = \frac{100}{10401}, \quad \lambda = \frac{B}{A} = -\frac{3367}{3467}.$$

Keep its polynomial $H = fGg + f^2B_0$, where $B_0 = a^2(A-1)(z-a) + adA(w-a)$. At a seed other than a, keep $\alpha_i = \bar{z}_i^2 + B_0(z_i, \bar{z}_i)/(z_i - a)$ and $\nu_i = -z_i$. Replace the two correction polynomials in (19.46) by

$$\mathcal{E}_*(t) = -\frac{14493204979}{297671419500} + \frac{1463813702879}{29767141950000}t - \frac{167587774534279}{2976714195000000}t^2,$$
$$\mathcal{C}_*(t) = \frac{35 + 535t + 3t^2}{5000}.$$

For $z_i = b\zeta$, $\zeta^3 = 1$, set $\delta_i = b^2\mathcal{E}_*(\zeta)$ and $\chi_i = (-b)^3\mathcal{C}_*(\zeta)$. At the other five nonshared seeds set $\delta_i = 0$ and $\chi_i = |\nu_i|^2\nu_i/9$.

Choose K, L in the same 34-dimensional rational space $\deg_{\mathrm{tot}} \le 9$, $\deg_w \le 3$. At the eight nonshared seeds prescribe the jets in (19.47), with the new δ_i, χ_i. At the shared seed, in the local coordinates $u = f$, $v = f^*$, prescribe instead

$$K = A\beta^2\left(\frac{13}{125}u^2 + \frac{1}{20}uv - \frac{1}{5}v^2\right) + O(|u|^3), \tag{19.59}$$

$$L = A\beta^3\left(\frac{71}{2000}u - \frac{233}{2000}v\right) + O(|u|^2). \tag{19.60}$$

Here and below these local estimates are on the physical slice. Both systems are compatible over $\mathbb{Q}$, with ranks 27 and 11, respectively. This is an exact compatibility assertion, not an inference from the ranks. The independent verifier `verify_pure_3_10_88.py` forms the systems by reduction modulo (f, f^*), sets the free variables in rational row reduction to zero, and checks every equation. The complete coefficients and residual audit are supplied in `pure_3_10_88_verification.json`. For clarity, the extra shared quadratic condition has coefficients $(13/125, 1/20, -1/5)$ after division by $A\beta^2$; it is not the zero condition used in the previous construction.

Now set

$$P_\tau = f^3g + \tau H + \tau^2K + \tau^3L - \frac{A\beta^4}{500}\tau^4. \tag{19.61}$$

This has pure highest term z^{10} and exact conjugate degree three for sufficiently small nonzero τ. The five unchanged nonshared clusters have nine simple zeros each, by the same translated radial cubic calculation as before. The other three have normalized limits

$$\xi^3 + \zeta|\xi|^2 + \mathcal{E}_*(\zeta)\bar{\xi} - \mathcal{C}_*(\zeta), \qquad \zeta = 1, \omega, \omega^2. \tag{19.62}$$

For both $\zeta = 1$ and $\zeta = \omega$, the exact rational trace form described in the preceding proof has dimension nine and nine positive pivots in its LDL^{T} decomposition. Thus each real algebra is reduced and isomorphic to $\mathbb{R}^9$: all nine complex intersections lie on the real slice and are simple.

The verifier supplies every pivot. Conjugation gives the same count for ω^2. All eight cubic clusters therefore contribute 72 zeros.

At the shared point, put $f = \tau\beta\xi$ and divide (19.61) by $A\beta^4\tau^4$. Its limit is

$$Q(\xi) = \xi^4 - \frac{3367}{3467}\xi^3\bar{\xi} + \xi\bar{\xi}^2 + \frac{13}{125}\xi^2 + \frac{1}{20}\xi\bar{\xi} - \frac{1}{5}\bar{\xi}^2 + \frac{71}{2000}\xi - \frac{233}{2000}\bar{\xi} - \frac{1}{500}. \tag{19.63}$$

Here is a short rational certificate for its sixteen simple zeros. Write $\xi = x + iy$ and define

$$\begin{aligned} A_0(x) &= \frac{7134}{3467}x^3 - x^2 + \frac{76}{125}x + \frac{19}{125}, \\ B_0^*(x) &= -\frac{20602}{3467}x - 1, \\ R_0(x) &= \frac{100}{3467}x^4 + x^3 - \frac{23}{500}x^2 - \frac{81}{1000}x - \frac{1}{500}, \\ R_1(x) &= -6x^2 + x + \frac{73}{500}. \end{aligned}$$

Then

$$\Im Q = y(A_0 + B_0^* y^2), \qquad \Re Q = R_0 + R_1 y^2 + (1-\lambda)y^4.$$

The quartic R_0 changes sign on each of

$$(-35, -34), \quad (-.250, -.248), \quad (-.026, -.025), \quad (.317, .319).$$

The degree-six polynomial

$$S = (1-\lambda)A_0^2 - R_1 A_0 B_0^* + R_0 (B_0^*)^2$$

changes sign on each of

$$\begin{gathered} (-.214, -.213), \quad (-.192, -.191), \quad (-.181, -.180), \\ (.448, .449), \quad (.499, .500), \quad (.623, .624). \end{gathered}$$

All displayed decimals are exact rational endpoints. On each of the latter six intervals $A_0 B_0^* < 0$, as checked by rational interval evaluation; hence $y^2 = -A_0/B_0^* > 0$. The numbers of disjoint sign intervals equal the polynomial degrees, so the four axial roots and six off-axis abscissae are exhaustive and simple. Exact greatest-common-divisor checks give $\gcd(R_0, A_0) = \gcd(S, B_0^*) = \gcd(A_0, B_0^*) = 1$. On the axis, the two transverse derivatives are R_0' and A_0. Off the axis, solving the imaginary equation and differentiating the real equation gives a nonzero multiple of S', with nonzero denominator B_0^*. Thus all $4 + 2 \cdot 6 = 16$ zeros of Q are regular.

These local counts persist for sufficiently small $\tau > 0$. They exhaust the global zeros: in a nonshared seed disk the equation gives $|f|^3 \le C(\tau|f|^2 + \tau^2|f| + \tau^3)$, and at the shared point it gives

$$|f|^4 \le C(\tau|f|^3 + \tau^2|f|^2 + \tau^3|f| + \tau^4).$$

Both imply $|f| \le C'\tau$. Rescaling and regularity of the limiting zeros then exclude additional zeros in these disks. Outside them, compact convergence to $f^3 g$ excludes bounded zeros, while the uniform pure highest term excludes escape to infinity. Therefore $N(P_\tau) = 72 + 16 = 88$ exactly. The mapping degree ten gives the signs $(49, 39)$. The real b-cluster has three real roots, as verified by its real cubic; the unchanged real c-cluster has three; and the shared quartic has four. This attains all ten possible real-axis zeros. Taking a sufficiently small rational τ and clearing denominators gives real integer coefficients.

Finally, total degree nine permits at most 81 isolated zeros, and conjugate degree at most two at total degree ten permits at most $9\cdot 10-16=74$ by (6.4). These prove the two degree-minimality assertions. □

Corollary 19.26. *For $n\geq 10$ and $3\leq m<n$,*

$$E_{m,n}\geq n+78.$$

The count is attained exactly with real integer coefficients, all zeros nondegenerate, and exactly n real-axis zeros.

Proof. Apply the same one-degree extensions and real rational padding as in Corollary 19.23. □

19.12 Sharp maxima in the fixed-seed coefficient neighborhoods

The counts 41 and 88 exhaust what sufficiently small perturbations of their respective fixed seeds can achieve. This gives a sharp local result in the full coefficient space, including complex coefficients and singular competing polynomials.

Proposition 19.27. *Fix $0<a<1/3$, let $\kappa=a^2(1+a)$, and put*

$$P_0(z,w)=(z^3+zw-\kappa)^m(z-a),\qquad D=3m+1.$$

There is a coefficient neighborhood U of P_0 among polynomials with pure highest term z^D and conjugate degree at most m such that every $P\in U$ satisfies

$$N(P)\leq D^2-4m=9m^2+2m+1. \tag{19.64}$$

For $m=2,a=1/4$ and for $m=3,a=100/10101$, every sufficiently small such neighborhood has exact maximum 41 and 88, respectively. These maxima are attained by real integer polynomials after a common nonzero scaling of the coefficients.

Proof. For $P^*(z,w)=\overline{P(\bar w,\bar z)}$, the complexified system $P=P^*=0$ has exactly D^2 finite intersections counted with multiplicity. Indeed, the highest homogeneous parts z^D,w^D have no common projective zero or common factor; Bézout's theorem applies and no intersection lies at infinity.

Four intersections of $P_0=P_0^*=0$ lie off the physical slice. Two have $w=a$ and

$$z=\rho_\pm,\qquad \rho_\pm^2+a\rho_\pm+a+a^2=0,$$

and the other two are their images under $(z,w)\mapsto(\bar w,\bar z)$. The discriminant $-4a-3a^2$ is negative, so these four points are distinct and have positive distance from $w=\bar z$. At $(\rho_\pm,a)$, $f=z^3+zw-\kappa$ has a simple zero in z, while

$$f^*(\rho_\pm,a)=a(\rho_\pm-a)\neq 0.$$

Thus $P_0^*=(f^*)^m(w-a)$ has nonzero w-derivative there. For every sufficiently small coefficient perturbation, the holomorphic implicit function theorem solves $P^*=0$ locally as $w=w_P(z)$. Substitution into P gives a one-variable holomorphic perturbation of a zero of order exactly m. Rouché's theorem preserves precisely m zeros, counted with multiplicity, in a fixed small disk. Conjugation gives the same assertion in the other two neighborhoods. Choose all four neighborhoods disjoint from the physical slice. They then retain $4m$ nonphysical intersections for every P in a

common coefficient neighborhood U. Subtracting these from D^2 proves (19.64), even when physical zeros are singular.

The families in Theorems 19.22 and 19.25 converge coefficientwise to the specified P_0 and attain $D^2 - 4m$ regular zeros for every sufficiently small positive parameter. Hence each sufficiently small neighborhood attains the asserted maximum. Rational parameters permit clearing denominators; this scaling leaves all zeros unchanged. □

The neighborhood conclusion does not replace the global upper bounds $E_{2,7} \le 47$ and $E_{3,10} \le 100$. It proves that neither fixed-seed construction can be improved by an arbitrarily small coefficient perturbation. A varying seed or a different limiting polynomial is needed to exceed these local maxima.

19.13 Explicit real rational seeds for the restriction maps

The following specialization of the radial rational construction also supplies real coefficients for every transfer below. The maximal count $5d-6$ itself is the established endpoint in Proposition 6.8.

Lemma 19.28. *For every $d \ge 4$, there exists a real rational polynomial $f = p + wq \in \mathcal{E}_{1,d}$ with $\deg q = d-2$ and $\gcd(p,q) = 1$, having exactly $5d-6$ regular zeros. It can be chosen with four real-axis zeros when d is even, and five when d is odd, including a negative real zero. For $d = 4$, one explicit integer choice is*

$$\begin{aligned} f(z,w) = {} & 1197z^4 + 24000z^3 + 2400z^2 + 32000z + 160000 \\ & + w(-23880z^2 - 158400z + 8000). \end{aligned} \tag{19.65}$$

Here "negative" refers to the Jacobian sign.

Proof. Put $k = d-1 \ge 3$ and fix $r = 1/20$ in every degree. Define the real rational constants

$$A = -r^{k-2}[k - (k+1)r^2], \qquad e = \frac{r^2[(k-1) - kr^2]}{k - (k+1)r^2},$$

and the rational function

$$R(z) = \frac{p_0(z)}{q_0(z)}, \qquad p_0(z) = z^k - eA, \qquad q_0(z) = z(z^k - A).$$

Then $A < 0$ and $0 < e < r^2 < 1$, so p_0, q_0 are coprime. Direct substitution gives

$$R(r) = r, \qquad R'(r) = 0, \qquad q_0(r) = kr^{k-1}(1 - r^2) \ne 0.$$

Also

$$p_0''(r) - rq_0''(r) = kr^{k-2}[(k-1) - (k+1)r^2] \ne 0.$$

Thus r is a regular negative zero of $wq_0 - p_0$, and $R(z) - r$ has a zero of exactly order two at r.

The physical zero equation of $wq_0 - p_0$ is

$$z^k(|z|^2 - 1) - A(|z|^2 - e) = 0.$$

Write $C = -A > 0$. The two positive-radius equations are

$$h_+(\rho) = \rho^k(\rho^2 - 1) + C(\rho^2 - e) = 0, \qquad h_-(\rho) = \rho^k(\rho^2 - 1) - C(\rho^2 - e) = 0.$$

Descartes' rule permits at most three positive roots for h_+ and two for h_-. Our fixed parameter gives

$$2^k C \le \frac{4k}{10^{k-2}} \le \frac{6}{5} < 3 \qquad (k \ge 3);$$

the middle expression decreases with k. Consequently

$$h_+(r) = 0, \qquad h_+'(r) = kr^{k-1}(1-r^2) > 0, \qquad h_+(1/2) < 0, \qquad h_+(1) > 0.$$

Besides the simple root r, there is a root in $(r, 1/2)$ and one in $(1/2, 1)$. These exhaust the Descartes bound, so all three are simple. Similarly,

$$h_-(0) = Ce > 0, \qquad h_-(r) = 2r^k(r^2-1) < 0, \qquad h_-(1) = -C(1-e) < 0,$$

and $h_-(\rho) \to +\infty$. Thus its two simple positive roots lie in $(0, r)$ and $(1, \infty)$. The two radius sets are disjoint because a common radius would require $\rho^2 = 1 = e$. Each radius gives exactly k distinct arguments, and the radial and angular derivatives are nonzero. The original polynomial has exactly $5k$ regular physical zeros, for this one fixed rational parameter in every degree.

Move the critical fixed point r to infinity with $u = 1/(z-r)$, $v = 1/(w-r)$. Define the polynomials

$$Q(u) = u^d q_0(r + 1/u), \qquad P(u) = u^d p_0(r + 1/u),$$

and set

$$f(u, v) = Q(u) + v[rQ(u) - P(u)].$$

The two identities at r cancel the top two coefficients of $rQ - P$; the displayed second derivative shows that its degree is exactly $d-2$. The leading coefficient of Q is $q_0(r) \neq 0$, so f has pure highest term of degree d. The change of variables gives a bijection between the other $5k-1 = 5d-6$ physical zeros and those of f. Indeed, $f(0, 0) = 1$, so the original point at infinity contributes no further zero. Multiplication by the nonzero clearing factor at the remaining roots and the holomorphic change of variable preserve regularity and Jacobian signs. Coprimality is preserved as well.

If k is odd, every radial ring meets the real axis once; if k is even, the three h_+ rings meet it twice and the h_- rings do not. Removing r gives the asserted four or five real zeros. There is still a negative real zero: the outer h_+ root lies between $\sqrt{e}$ and 1, its radial derivative is positive, and its angular derivative $k\rho^k(\rho^2 - 1)$ is negative. It is different from the inner root r.

For the explicit quartic choose $r = 1/20$, $k = 3$. Then $A = -299/2000$ and $e = 797/478400$. The three roots of h_+ lie in

$$(1/25, 3/50), \quad (1/10, 1/5), \quad (4/5, 99/100),$$

and the two of h_- lie in $(1/100, 1/25)$ and $(1, 6/5)$. Direct rational endpoint signs and Descartes' rule prove these counts and simplicity. The two polynomials have no common positive root. Expanding the displayed inversion and multiplying by 160000 gives (19.65), proving all its assertions without a numerical root computation. □

19.14 A fourteen-point seed and fifty-seven pure-head zeros

A different seed and a sharp interpolation threshold improve the conjugate-degree-two bound at pure degree nine.

Theorem 19.29. *There is a polynomial in $\mathcal{E}_{2,9}$ with Gaussian integer coefficients and exactly 57 regular zeros, with signs $(33, 24)$. In particular,*

$$57 \le E_{2,9} \le 65. \tag{19.66}$$

Among pure-head polynomials of conjugate degree at most two, degree nine is the least total degree that can have at least 57 zeros. At total degree nine, conjugate degree two is also minimal for this target.

Proof. Use the explicit real rational seed in Lemma 19.28:

$$f(z, w) = p(z) + wq(z), \qquad \deg p = 4, \quad \deg q = 2, \quad \gcd(p, q) = 1,$$

with fourteen physical zeros $z_1, \dots, z_{14}$. Normalize the leading coefficient of p to one. The printed coefficient table and radial proof provide existence, regularity, coprimality, and exact degree two for q.

Consider the 24-dimensional space

$$V = \{H \in \mathbb{C}[z, w] : \deg_{\mathrm{tot}} H \le 8,\ \deg_w H \le 2\}$$

and its restriction map

$$\mathcal{R}(H)(z) = q(z)^2 H\big(z, -p(z)/q(z)\big) \in \mathbb{C}[z]_{\le 14}. \tag{19.67}$$

This map is onto. To see this, its kernel is exactly

$$f\{B : \deg_{\mathrm{tot}} B \le 4,\ \deg_w B \le 1\}.$$

The divisibility assertion follows from the coprimality of p, q and Gauss's lemma for the primitive polynomial $p + wq$. Total and w-degrees add under multiplication, so the kernel has dimension $5 + 4 = 9$. The image therefore has dimension $24 - 9 = 15$, equal to that of the target. The degree bound in (19.67) follows term by term: $q^2 z^i$, $-pqz^i$, and $p^2 z^i$ have maximum degrees $12, 13, 14$, respectively.

Choose $G \in V$ with

$$\mathcal{R}(G) = \prod_{i=1}^{14} (z - z_i). \tag{19.68}$$

Every $q(z_i)$ is nonzero. At each physical seed, use the invertible local coordinates $u = f$, $v = f^*$, where $f^*(z, w) = \overline{f(\bar{w}, \bar{z})}$. On the physical slice $v = \bar{u}$, and write

$$G = \alpha_i u + \beta_i v + O(|u|^2).$$

Crucially, $\beta_i \neq 0$ for all fourteen seeds. Indeed, restriction to the smooth curve $f = 0$ has a simple zero by (19.68), and v is a local coordinate on that curve because the zero of $f(z, \bar{z})$ is regular.

Surjectivity of (19.67) also makes arbitrary value interpolation at the fourteen seeds possible: interpolate the values $q(z_i)^2 K(z_i, \bar{z}_i)$ by a polynomial of degree at most thirteen. Choose $K \in V$ so that

$$K(z_i, \bar{z}_i) = \frac{\alpha_i^2}{4} + \frac{\beta_i \bar{\alpha}_i}{2}.$$

For $\varepsilon > 0$, define

$$F_\varepsilon = f^2 + \varepsilon G + \varepsilon^2 K. \tag{19.69}$$

Rescaling $f = \varepsilon u$ at a seed and dividing by ε^2 gives, after $v = u + \alpha_i/2$,

$$v^2 + \beta_i \bar{v}.$$

It has four simple zeros: the origin and three points of radius $|\beta_i|$. Thus all fourteen clusters split into 56 regular physical zeros. The annular estimate $|f|^2 \le C(\varepsilon|f| + \varepsilon^2)$ confines each cluster to $|f| = O(\varepsilon)$. The degree-eight homogeneous part of F_ε is $z^8 + O(\varepsilon)$ on the unit circle, uniformly with its first derivatives. It is bounded away from zero for small ε. Compactness outside the seed disks and this uniform bound at infinity prove that the count is exactly 56 for every sufficiently small positive ε.

The degree-eight homogeneous part generally has mixed terms. In fact the coefficient of $z^6 w^2$ in G is exactly one: it is the only monomial that can supply the leading coefficient of the degree-fourteen polynomial in (19.68). Consequently (19.69) is an intermediate total-degree-eight construction; its highest term is not pure.

Fix such a sufficiently small ε, and now set

$$P_\delta = F_\varepsilon + \delta z^9, \qquad \delta > 0. \tag{19.70}$$

For small δ the existing 56 zeros persist. There is exactly one additional zero at infinity. Here are the details of this count, since the intermediate highest part is mixed. Let $H_{8,\varepsilon}$ be that homogeneous part. With $\zeta = \delta z$,

$$\delta^8 P_\delta(\zeta/\delta) \longrightarrow \zeta^9 + H_{8,\varepsilon}(\zeta, \bar{\zeta})$$

in C^1 on compact sets. At $\varepsilon = 0$ the limit is $\zeta^8(\zeta + 1)$. For sufficiently small ε, every nonzero zero of the limit satisfies $|\zeta + 1| \le C\varepsilon$, obtained by division by ζ^8. On this disk the divided equation is a small C^1 perturbation of $\zeta + 1$, so it has exactly one simple zero, with positive Jacobian. Finally, at any zero of (19.70) with sufficiently large $|z|$, the leading-part estimate gives

$$\tfrac{1}{2} \le \delta|z| \le 2.$$

This excludes an intermediate scale or a faster escaping zero. The single nonzero limiting zero therefore accounts for all zeros entering from infinity. Hence $N(P_\delta) = 57$ exactly.

The polynomial in (19.70) has pure highest term δz^9, and its w-degree is exactly two because its $z^6 w^2$ coefficient is $\varepsilon + O(\varepsilon^2)$. The mapping degree nine gives signs $(33, 24)$. Approximate its coefficients sufficiently closely within this support by Gaussian rationals. Regularity, the exact count, and the pure highest term persist by properness. Clearing denominators gives Gaussian integer coefficients.

The general upper bound gives $E_{2,9} \le 65$. At conjugate degree at most two, total degree eight permits at most $9 \cdot 8 - 16 = 56$ zeros, and lower degrees permit even fewer. At degree nine, conjugate degree at most one permits at most $5 \cdot 9 - 6 = 39$. These prove the stated degree-minimality assertions. □

The interpolation threshold used here is itself sharp. If $\deg_{\mathrm{tot}} G \le 7$, $\deg_w G \le 2$, and G vanishes at all fourteen physical seeds, then $\mathcal{R}(G)$ has degree at most thirteen and fourteen distinct roots. It is identically zero, so f divides G and all the transverse coefficients β_i vanish. Thus this simultaneous four-way splitting needs degree eight in its first perturbation. This statement concerns the specified fourteen-point splitting mechanism.

Corollary 19.30. *For $n \geq 9$ and $2 \leq m < n$,*

$$E_{m,n} \geq n + 48.$$

The count is attained exactly with Gaussian integer coefficients and nondegenerate zeros.

Proof. Raise the pure degree one step at a time using Lemma 19.17, then apply sufficiently small exact-conjugate-degree padding and clear denominators. □

19.15 Two conjugate degrees suffice for eighty-four zeros

The same restriction-map argument extends to higher rational seeds, with two separate counts at infinity.

Proposition 19.31. *For every integer $d \geq 5$,*

$$E_{2,3d-3} \geq 20d - 19 + |d - 8|. \tag{19.71}$$

The displayed count is attained exactly with Gaussian integer coefficients and regular zeros. In particular,

$$84 \leq E_{2,12} \leq 92, \qquad 103 \leq E_{2,15} \leq 119. \tag{19.72}$$

Proof. Take a regular $f = p + wq \in \mathcal{E}_{1,d}$ with $N = 5d - 6$ physical zeros, p monic of degree d, $\deg q = d - 2$, and $\gcd(p, q) = 1$, using Lemma 19.28. Put $D = 3d - 4$ and

$$V_D = \{H : \deg_{\mathrm{tot}} H \leq D,\ \deg_w H \leq 2\}.$$

The restriction $\mathcal{R}(H) = q^2 H(z, -p/q)$ has target $\mathbb{C}[z]_{\leq D+2d-2} = \mathbb{C}[z]_{\leq N}$. Its kernel is f times the space of total degree at most $D - d$ and w-degree at most one. Thus its image dimension is

$$3D - \{2(D - d) + 1\} = D + 2d - 1 = N + 1,$$

so it is onto. Choose G, K by exactly the interpolation and translation conditions in the proof of Theorem 19.29. For G, use the monic product of the N physical abscissae; for K, a value-interpolating polynomial of degree at most $N - 1$ suffices. The transverse coefficient of G is nonzero at every seed, so $f^2 + \varepsilon G + \varepsilon^2 K$ has $4N$ regular zeros in the seed disks.

We can and do arrange

$$G = z^{D-2} w^2 + \{\text{terms of total degree at most } D - 1\}, \qquad \deg_{\mathrm{tot}} K \leq D - 1. \tag{19.73}$$

Indeed, the coefficient of $z^{D-2} w^2$ is the leading coefficient of $\mathcal{R}(G)$ and is therefore one; the corresponding coefficient for K is zero. The two other possible degree-D monomials, z^D and $z^{D-1} w$, can be removed by adding multiples of f with multipliers z^{D-d} and $z^{D-d-1} w$. Such additions preserve the restriction. Normalize G first, compute its new α_i and unchanged β_i, and then interpolate and normalize K. This order preserves all the required local conditions.

Let $F_\varepsilon = f^2 + \varepsilon G + \varepsilon^2 K$, where $\varepsilon > 0$. Since $D - 2d = d - 4 > 0$, the first infinite scale is $z = \varepsilon^{-1/(d-4)} \zeta$. After multiplication by $\varepsilon^{2d/(d-4)}$, the limiting equation is

$$\zeta^{2d} + \zeta^{D-2} \bar{\zeta}^2 = 0.$$

For nonzero $\zeta = re^{i\theta}$ this becomes

$$1 + r^{d-4}e^{i(d-8)\theta} = 0. \tag{19.74}$$

If $d \neq 8$, there are exactly $|d-8|$ roots, all simple, and their Jacobian sign is $\operatorname{sgn}(d-8)$. If $d = 8$, there are no nonzero roots because $1 + r^4 > 0$. These are all escaping zeros: for any sequence of zeros with $|z| \to \infty$ and $\varepsilon \to 0^+$, division by z^{2d} and (19.73) give

$$1 + O(|z|^{-1}) + \varepsilon|z|^{d-4}e^{i(d-8)\arg z}\{1 + O(|z|^{-1}) + O(\varepsilon/|z|)\} = 0.$$

Hence $\varepsilon|z|^{d-4} \to 1$, which confines every escaping zero to the displayed scale. Compact convergence, the simple limiting roots, and the seed-disk estimates prove

$$N(F_\varepsilon) = 4N + |d-8|$$

for sufficiently small positive ε, with all zeros regular. Its degree-D homogeneous part is precisely $\varepsilon z^{D-2}w^2$.

Fix such an ε. Adding δz^{D+1}, with $\delta > 0$ sufficiently small, supplies exactly five more positive zeros. To check this second count, set $z = (\varepsilon/\delta)\zeta$ and divide by $\varepsilon(\varepsilon/\delta)^D$. The limiting nonzero equation is

$$\zeta^{D+1} + \zeta^{D-2}\bar{\zeta}^2 = 0, \qquad \text{equivalently} \qquad re^{5i\theta} + 1 = 0.$$

Its five roots are simple and positive. The same leading-term balance excludes intermediate or faster escaping scales. The resulting pure-head polynomial has total degree $D + 1 = 3d - 3$, exact conjugate degree two, and exactly

$$4(5d-6) + |d-8| + 5 = 20d - 19 + |d-8|$$

regular zeros. Gaussian rational approximation and clearing denominators finish the arithmetic assertion. The upper bounds in (19.72) are (6.4). □

Corollary 19.32 (The complete degree tradeoff for eighty-four zeros)**.** *Let*

$$\nu_{84}(m) = \min\{n > m : E_{m,n} \geq 84\}.$$

Then

$$\nu_{84}(m) = \begin{cases} 18, & m = 1, \\ 12, & m = 2, \\ \max\{10, m+1\}, & m \geq 3. \end{cases} \tag{19.75}$$

Equivalently, the undominated degree pairs (m, n) for at least eighty-four zeros are exactly

$$(1, 18), \quad (2, 12), \quad (3, 10).$$

All three thresholds are attained by regular polynomials.

Proof. For $m = 1$, the exact formula $E_{1,n} = 5n - 6$ gives the threshold eighteen. For $m = 2$, $E_{2,11} \leq 83$, the upper bound excludes every smaller degree, and Proposition 19.31 attains 84 in degree twelve. For $m \geq 3$, degree nine permits at most 81 zeros, whereas Theorem 19.25 supplies at least 84 in degree ten. One-degree extensions and exact-degree padding cover every allowed $m \geq 3$ at $n = \max\{10, m+1\}$. The constraint $n > m$ accounts for the remaining minimum in (19.75). □

19.16 A globally sharp even-degree family in conjugate degree two

Selective splitting closes the linear bound in every sufficiently large even analytic degree, and leaves only two possible maxima in every odd analytic degree at least thirteen.

Lemma 19.33 (An odd cubic with nine regular zeros)**.** *For real λ with $|\lambda| > 1$, the polynomial*

$$Q_\lambda(\xi) = \xi^3 + \lambda\xi^2\bar{\xi} - \lambda\xi + \bar{\xi}$$

has exactly nine regular zeros. They are

$$0, \qquad \pm\sqrt{\frac{\lambda-1}{\lambda+1}}, \qquad \pm i\sqrt{\frac{\lambda+1}{\lambda-1}}, \qquad \frac{\pm 1 \pm i}{\sqrt{2}}.$$

Proof. For $\xi = x + iy$, its real and imaginary parts are

$$x[(1+\lambda)x^2 + (\lambda-3)y^2 - \lambda + 1], \qquad y[(3+\lambda)x^2 + (\lambda-1)y^2 - \lambda - 1].$$

The cases $xy = 0$ give the first five zeros. For $xy \neq 0$, the two linear equations in x^2, y^2 have determinant 8 and unique solution $x^2 = y^2 = 1/2$, giving four more. The Jacobian determinants are, respectively, $\lambda^2 - 1$ at zero, $-8(\lambda-1)/(\lambda+1)$ at the two nonzero real zeros, $-8(\lambda+1)/(\lambda-1)$ at the two imaginary zeros, and 8 at the other four. None vanishes. □

Theorem 19.34. *For every even integer $n \geq 10$,*

$$M_{2,n} = 9n - 10. \tag{19.76}$$

In particular, $M_{2,10} = 80$ and $M_{2,12} = 98$. For every odd integer $n \geq 13$,

$$M_{2,n} \in \{9n - 12,\ 9n - 10\}. \tag{19.77}$$

The smaller endpoints satisfy

$$M_{2,8} \in \{60, 62\}, \qquad 65 \leq M_{2,9} \leq 71, \qquad 85 \leq M_{2,11} \leq 89. \tag{19.78}$$

Every displayed lower count is attained exactly by a real integer polynomial with only nondegenerate zeros. The sixty- and eighty-zero witnesses have respectively ten and twelve real-axis zeros, the maximum allowed by their total degrees. The value $M_{2,7}$ remains unresolved.

Proof. For $d \geq 4$, take the real rational radial polynomial $f = p + wq$ in Proposition 5.2, with $m = 1$ and $k = d - 1$. It has bidegree $(1, d)$, $\deg q = d$, $\deg p \leq d$, $\gcd(p, q) = 1$, and

$$N := N(f) = 5d - 5$$

regular physical zeros. On $W_n = \{H : \deg_z H \leq n,\ \deg_w H \leq 2\}$, for $n \geq d$, the restriction map

$$\mathcal{R}(H) = q^2 H(z, -p/q)$$

is onto $\mathbb{C}[z]_{\leq n+2d}$. Its domain has dimension $3(n+1)$, and its kernel is f times the space of bidegree at most $(n - d, 1)$, of dimension $2(n - d + 1)$. Thus the image has exactly the target dimension $n + 2d + 1$. Divisibility follows from Gauss's lemma, and both degrees add under multiplication, so this calculation identifies the full kernel.

First let $n = 2d$. Select $s = \min\{N, 4d\}$ physical seeds, and choose $G \in W_{2d}$ with restriction the monic product of their abscissae. At a selected seed,

$$G = \alpha_i u + \beta_i \bar{u} + O(|u|^2), \qquad u = f, \quad \beta_i \neq 0.$$

The last inequality follows from the simple restricted zero. At every unselected seed G is nonzero. Interpolate $K \in W_{2d}$ with selected values $\alpha_i^2/4 + \beta_i \bar{\alpha}_i/2$, and put

$$F_\varepsilon = f^2 + \varepsilon G + \varepsilon^2 K.$$

Each selected seed splits into four regular zeros, by the translated model $v^2 + \beta_i \bar{v}$. Each unselected seed splits into two regular zeros: the rescaling $f = \sqrt{\varepsilon}\, u$ has limiting equation $u^2 + G(z_i, \bar{z}_i) = 0$, with nonzero constant term. The respective confinement estimates are $|f| = O(\varepsilon)$ and $|f| = O(\sqrt{\varepsilon})$. The corner coefficient stays close to the nonzero corner of f^2, giving uniform properness and excluding escape to infinity. Compactness outside the seed disks excludes additional bounded zeros. Consequently

$$N(F_\varepsilon) = 4s + 2(N - s) = 2N + 2s. \tag{19.79}$$

For $d \geq 5$, $s = 4d$, and this is $18d - 10 = 9n - 10$, exactly the upper bound in Theorem 1.2. For $d = 4$ all fifteen seeds are selected, giving 60 zeros at $n = 8$.

Here are the extra details for odd $n = 2d+1$. Choose a real zero a of f with negative Jacobian, and set $g = z - a$. Such a zero is available among the three positive roots of the radial restriction $B(r^2) + r^{d-1}A(r^2)$. Indeed, for the coefficients in (5.3), all three lie between the positive roots of $B(r^2)$ and $A(r^2)$, where $A > 0$. Their radial derivatives have signs $-, +, -$, while the angular derivative is positive. The first and third therefore have negative Jacobian. If necessary, a small real perturbation $t(z - a)$ makes $f_z(a, a) \neq 0$, retaining this seed, all regular counts, the nonzero corner, and coprimality.

In the local coordinate $u = f$ we have

$$g = Au + B\bar{u} + O(|u|^2), \qquad A, B \in \mathbb{R}, \quad |B| > |A| > 0, \quad \lambda = B/A, \quad |\lambda| > 1.$$

Select $s = \min\{N - 1, 4d\}$ other seeds. Prescribe $\mathcal{R}(H)$ to be a nonzero constant times the product of $z - a$ and the selected abscissae. This polynomial has degree $s + 1 \leq 4d + 1$, hence has a preimage in W_{2d+1}. Its simple restricted zero at a gives a nonzero transverse coefficient. Choose the constant so that this coefficient is A, and add a multiple of f to obtain

$$H = A(-\lambda u + \bar{u}) + O(|u|^2).$$

At each ordinary selected seed write $H/g = \alpha_i u + \beta_i \bar{u} + O(|u|^2)$, with $\beta_i \neq 0$, and interpolate $K \in W_{2d+1}$ with

$$\frac{K(z_i, \bar{z}_i)}{g(z_i)} = \frac{\alpha_i^2}{4} + \frac{\beta_i \bar{\alpha}_i}{2}.$$

There is no required value or jet of K at a. For

$$P_\varepsilon = f^2 g + \varepsilon H + \varepsilon^2 K,$$

the shared scale $f = \sqrt{\varepsilon}\, \xi$ has limit $AQ_\lambda(\xi)$ after division by $\varepsilon^{3/2}$. Lemma 19.33 gives nine regular zeros. The shared estimate $|f|^3 \leq C(\varepsilon|f| + \varepsilon^2)$ confines every local zero to $|f| = O(\sqrt{\varepsilon})$. The other selected seeds give four zeros each, and each unselected seed gives two. The stable corner excludes escape to infinity. The exact count is therefore

$$9 + 4s + 2(N - 1 - s) = 2N + 2s + 7.$$

For $d \geq 6$ this is $18d - 3 = 9n - 12$; for $d = 4, 5$ it is respectively 65 at $n = 9$ and 85 at $n = 11$.

All selections can be made invariant under conjugation. In the even-degree case the number of omitted seeds is $d - 5$ for $d \geq 5$; in the odd-degree case it is $d - 6$ for $d \geq 6$. If the number is even, omit conjugate pairs; if odd, omit one real seed (other than a in the odd case) and then conjugate pairs. The radial seeds have enough such pairs and real seeds, also after the small real perturbation. Consequently the restriction polynomials and interpolation data can be real, and all interpolants can be chosen real. Real rational approximation and clearing denominators preserve the regular counts and the nonzero corner. At $n = 8, 10$, the five and six real seed zeros each give exactly two real zeros, yielding the maximal real-axis counts asserted in the theorem.

The remaining upper bounds follow from Theorem 1.2. For bidegree $(2, 8)$ with nonzero corner, every regular count is even. Support-preserving regularization and a small generic corner perturbation show that the distinct maximum is attained by such a regular member. Thus $M_{2,8}$ is either 60 or 62. The same parity argument leaves exactly the two alternatives in (19.77). □

19.17 Pure-head bands with gaps two and four

Selective splitting is more efficient than splitting every seed when the perturbation must have degree strictly below the pure head. It gives almost matching bounds at every sufficiently large degree.

Theorem 19.35. *For every even $n \geq 8$ and every odd $n \geq 9$, respectively,*

$$E_{2,n} \in \{9n - 18,\ 9n - 16\}, \qquad n \text{ even}, \tag{19.80}$$

$$E_{2,n} \in \{9n - 20,\ 9n - 18,\ 9n - 16\}, \qquad n \text{ odd}. \tag{19.81}$$

The lower endpoint in each line is attained exactly by a real integer polynomial with only regular zeros. The witnesses in degrees eight, nine, and ten can have exactly eight, nine, and ten real-axis zeros, respectively. In particular,

$$\begin{aligned} &54 \leq E_{2,8} \leq 56, \qquad 61 \leq E_{2,9} \leq 65, \qquad 72 \leq E_{2,10} \leq 74, \\ &79 \leq E_{2,11} \leq 83, \qquad 90 \leq E_{2,12} \leq 92, \qquad 115 \leq E_{2,15} \leq 119. \end{aligned} \tag{19.82}$$

These statements leave the upper endpoints unresolved.

Proof. Use a real rational seed $f = p + wq \in \mathcal{E}_{1,d}$ from Lemma 19.28, with $N = 5d - 6$, $\deg p = d$, $\deg q = d - 2$, and $\gcd(p, q) = 1$. For $D \geq d + 1$, let

$$V_D = \{H : \deg_{\mathrm{tot}} H \leq D,\ \deg_w H \leq 2\}.$$

Its restriction $\mathcal{R}(H) = q^2 H(z, -p/q)$ is onto $\mathbb{C}[z]_{\leq D+2d-2}$. Indeed, the kernel is $f\{B : \deg_{\mathrm{tot}} B \leq D - d,\ \deg_w B \leq 1\}$, so its image has dimension $3D - [2(D-d)+1] = D + 2d - 1$. The degree bound follows because the three possible highest restricted degrees are $D + 2d - 4, D + 2d - 3, D + 2d - 2$.

For even $n = 2d$, $d \geq 4$, set $D = 2d - 1$ and select $s = 4d - 3$ of the N physical seeds. Choose $G \in V_D$ whose restriction is the product of their abscissae, and interpolate $K \in V_D$ with the quadratic translation values $\alpha_i^2/4 + \beta_i \bar{\alpha}_i/2$ there. Exactly as in (19.79), the polynomial

$$f^2 + \varepsilon G + \varepsilon^2 K$$

has four regular zeros per selected seed and two per unselected seed. Its pure degree-$2d$ head is unchanged, so no zero escapes. The exact count is

$$2(5d-6)+2(4d-3)=18d-18=9n-18.$$

For odd $n=2d+1$, $d\geq 4$, put $D=2d$. Choose a negative regular real seed a of f, supplied by the lemma, and set $g=z-a$. We may suppose $f_z(a,\bar{a})\neq 0$: adding a sufficiently small real multiple of $z-a$ preserves this seed, its negative sign, all regular counts, and the degree conditions, while making its z-derivative nonzero. Avoiding finitely many further parameters preserves $\gcd(p,q)=1$.

In the coordinate $u=f$ write $g=Au+B\bar{u}+O(|u|^2)$, where A,B are real and $|B|>|A|>0$. Thus $\lambda=B/A$ is real with $|\lambda|>1$, as required by Lemma 19.33. Select $s=4d-3$ other seeds. Since $s+1=4d-2$, the restriction map supplies $H\in V_D$ whose restriction vanishes simply at a and at all selected seeds, and nowhere at the other physical seeds. Multiply the restriction by a constant to make its transverse coefficient at a equal to A; adding a multiple of f then arranges

$$H=A(-\lambda u+\bar{u})+O(|u|^2).$$

At the selected ordinary seeds interpolate $K\in V_D$ so that K/g has the quadratic translation values for the linear jet of H/g. No condition on K at a is needed. For

$$P_\varepsilon=f^2g+\varepsilon H+\varepsilon^2K,$$

the shared scale $f=\sqrt{\varepsilon}\,\xi$ gives, after division by $A\varepsilon^{3/2}$, the nine-zero model of Lemma 19.33. The ordinary selected seeds give four zeros and the unselected seeds give two. The confinement estimates in the proof of Theorem 19.34 apply unchanged. The pure highest term remains that of f^2g, of degree $2d+1$, so there are no escaping zeros. Consequently

$$N(P_\varepsilon)=2N+2s+7=18d-11=9n-20.$$

All selections can be invariant under conjugation. The numbers of omitted seeds are $d-3$ in the even case and $d-4$ in the odd case. Use conjugate pairs, together with one real seed when the omitted number is odd, excluding a in the odd-degree construction. The four or five real seed zeros and the remaining conjugate pairs suffice. The restriction polynomials, interpolation data, and preimages can therefore all be real.

Both constructions retain exact conjugate degree two. For example, f^2 and f^2g have nonzero w^2 coefficients, which small perturbations preserve. At $n=8$, omit one of the four real seed zeros and choose the sign of the restriction of G to make $G<0$ there. The selected real seeds give six real zeros; the omitted seed gives two by $u^2+G=0$. At $n=10$, omit a conjugate pair, so all five real seeds split into two real zeros each. At $n=9$, all seeds except the shared negative real seed are selected: the three other real seeds give six real zeros, and Q_λ supplies three at the shared seed. These counts exhaust the respective real-axis degree bounds. Real rational approximation within the support and clearing denominators preserve every asserted exact regular count. The upper bound is $9n-16$ by (6.4). Finally, Corollary 3.4 keeps the maximizing polynomial in the pure-head support; its mapping degree is n. Thus $E_{2,n}\equiv n \pmod 2$, giving exactly the alternatives stated above. □

Corollary 19.36 (Sharp degree thresholds for whole intervals of targets)**.** *Fix $n\geq 8$ and an integer target L. If*

$$\begin{cases} 9n-24\leq L\leq 9n-18, & n \text{ even},\\ 9n-24\leq L\leq 9n-20, & n \text{ odd}, \end{cases}$$

then n is the least pure total degree, at conjugate degree at most two, that can have at least L distinct zeros. At that total degree, conjugate degree two is also minimal. In particular, every target from 84 *through* 90 *has minimum pure degree twelve at conjugate degree at most two.*

Proof. At every smaller degree the bound is at most $9(n-1)-16 = 9n-25 < L$, while Theorem 19.35 attains at least L in degree n. At degree n, conjugate degree one permits at most $5n-6 < 9n-24 \le L$. The constructed witnesses are regular. □

19.18 A critical shared seed and forty-five zeros in degree seven

Using a negative critical seed changes the shared cluster from nine zeros to thirteen. The new construction closes the local maximum at this limiting polynomial and leaves only two global possibilities.

Theorem 19.37. *There is a real integer polynomial in $\mathcal{E}_{2,7}$ with exactly* 45 *regular zeros, of signs* $(26, 19)$*, including exactly seven real-axis zeros. Consequently*

$$E_{2,7} \in \{45, 47\}. \tag{19.83}$$

Both total degree seven and conjugate degree two at that total degree are minimal for this target. The global endpoint 47 *is not decided here.*

Proof. Put $a = -1/3$, $\kappa = 2/27$, $g = z - a$, and

$$f = z^3 + zw - \kappa, \qquad D(z) = (\kappa - z^3)^3 - z^6.$$

The radial equations $\pm r^3 + r^2 - \kappa = 0$ have respectively one and two simple positive roots, so f has nine regular physical zeros. These exhaust the nine complex intersections of $f = f^* = 0$. The seed a is negative and satisfies $f_z(a, a) = 0$. Use the following fixed real rational polynomial of total degree six:

$$\begin{aligned} 2187H = {} & (6561z^4 - 243z - 162)w^2 \\ & + (-2187z^5 - 243z^2 + 621z - 18)w \\ & + 6318z^5 + 1782z^4 + 621z^3 + 18z^2 + 30z - 46. \end{aligned} \tag{19.84}$$

It satisfies the exact identity

$$z^2 H(z, (\kappa - z^3)/z) = -3(z - a)D(z).$$

In particular, its restricted zero is simple at each of the eight ordinary seeds and double at the shared seed a.

There is a real polynomial K of total degree at most six and conjugate degree at most two with the following properties. At each ordinary seed write $H/g = \alpha_i u + \beta_i \bar{u} + O(|u|^2)$, where $u = f$ and $\beta_i \neq 0$, and prescribe

$$K(z_i, \bar{z}_i) = g(z_i)(\alpha_i^2/4 + \beta_i \bar{\alpha}_i/2).$$

At a prescribe

$$K(a, a) = 0, \qquad K_z(a, a) = 2/27, \qquad K_w(a, a) = 5/9. \tag{19.85}$$

To justify compatibility, the restriction map $K \mapsto z^2 K(z, (\kappa - z^3)/z)$ is onto $\mathbb{C}[z]_{\le 10}$ by the kernel calculation in Theorem 19.35. The eight ordinary values, the shared value, and the shared derivative

impose ten Hermite conditions, hence can be interpolated even in degree nine. The derivative condition is $\mathcal{R}(K)'(a) = a^2(2/27)$, because the graph $w = (\kappa - z^3)/z$ has derivative zero at a. Adding a constant multiple of f then sets $K_w(a, a) = 5/9$ without altering any restriction, value, or z-derivative. Conjugate symmetry makes all these choices real. In fact, a convenient rational choice is

$$\begin{aligned} 26244K = {} & 7695w^2 + (-56862z^5 - 53568z + 1620)w \\ & + 32805z^6 - 55404z^5 + 20088z^4 \\ & - 46521z^3 - 13149z^2 - 3135z + 3887. \end{aligned} \tag{19.86}$$

All eight ordinary value conditions can be checked at once by reduction modulo $D(z)/(z - a)$, substituting $w = (\kappa - z^3)/z$; the three shared jets follow by direct differentiation. The accompanying exact verifier performs both checks on the printed coefficients.

Set

$$P_\varepsilon = f^2 g + \varepsilon H + \varepsilon^2 K.$$

The eight ordinary clusters give four regular zeros each. We count the shared cluster explicitly. Let $X = z - a$, $Y = w - a$. Then

$$f = aY - X^2 + XY + X^3,$$

and the terms of H of degree at most three are

$$-Y/9 - 2X^2/3 + 2XY/3 + 7X^3/3 - 29X^2Y/27 - 5XY^2/9.$$

Write $t = \sqrt{\varepsilon}$ and $X = t\xi$, $Y = t\bar{\xi}$. On bounded ξ-sets, division by t^3 gives

$$A_0(\xi) + tA_1(\xi) + t^2 A_2(\xi) + O(t^3), \tag{19.87}$$

with convergence of derivatives, where

$$A_0 = a^2\bar{\xi}(|\xi|^2 - 1), \qquad A_1 = 2a(|\xi|^2 - 1)(|\xi|^2 - \xi^2),$$

and the only needed value of A_2 is the exact identity

$$\frac{A_2(e^{i\theta})}{e^{-i\theta}} = e^{6i\theta} - \tfrac{1}{3}e^{4i\theta}. \tag{19.88}$$

This identity follows by substituting the displayed jets of H, K. The origin of the limiting equation is simple; in fact $X = 0$ is an exact regular negative zero for all sufficiently small $\varepsilon > 0$.

Near $|\xi| = 1$, divide (19.87) by

$$U_t(\xi) = a^2\bar{\xi} + 2at(|\xi|^2 - \xi^2),$$

which is bounded away from zero. Its real equation uniquely determines $|\xi| = 1 + O(t^2)$ as a smooth function of the angle. Divide its imaginary equation by t^2. Uniformly with its first derivative, the resulting angular equation tends to

$$\frac{1}{a^2}\left(\sin 6\theta - \tfrac{1}{3}\sin 4\theta\right) = \frac{1}{a^2}\sin 2\theta\left(4\cos^2 2\theta - \tfrac{2}{3}\cos 2\theta - 1\right).$$

It has exactly twelve simple zeros modulo 2π. Indeed, the quadratic has roots $(1 \pm \sqrt{37})/12$, both strictly between -1 and 1; neither coincides with the two endpoint values from $\sin 2\theta = 0$. The

radial derivative is nonzero, so all twelve continued zeros are regular. Compactness on the remaining arcs excludes additional zeros near the circle.

Finally, $|f^2 g| \geq c|X|^3$ near a, while $H = O(|X|)$ and $K = O(|X|)$. Every nonzero shared zero therefore has $|X| = O(\sqrt{\varepsilon})$. In the scaled disk the limiting zeros lie only at the origin or on the unit circle. Thus the shared count is exactly $1 + 12 = 13$, and the global count is $8 \cdot 4 + 13 = 45$. The pure highest term is z^7; uniform properness excludes escape to infinity or additional zeros away from the seed disks.

The shared cluster has three real zeros, namely its center and the continuations of angles $0, \pi$. The other two real seed zeros each contribute two, giving seven in total. The mapping degree seven gives signs $(26, 19)$. All coefficients are rational for rational ε. Every sufficiently small positive rational parameter gives the asserted regular counts; clearing denominators gives real integers. The upper bound is 47; regularization and degree parity leave the alternatives (19.83). Degree six permits at most 36 zeros, and conjugate degree one in degree seven permits at most 29, proving both minimality assertions. □

Proposition 19.38 (Sharpness in the critical seed neighborhood)**.** *For $f = z^3 + zw - 2/27$, $g = z + 1/3$, every sufficiently small coefficient perturbation of $f^2 g$ within $\mathcal{E}_{2,7}$ has at most 45 distinct physical zeros. Theorem 19.37 attains this bound in every such neighborhood.*

Proof. The two points $(-1/3, 2/3)$ and $(2/3, -1/3)$ are off the physical slice. At the first, $f \neq 0$, $f^* = 0$, and

$$f^*(-1/3, w) = (w + 1/3)^2 (w - 2/3).$$

The germ of $f^2 g$ is a unit times $z + 1/3$, and its star has a double zero in the transverse variable $w - 2/3$. The local complex intersection multiplicity is therefore two. It persists in a small bidisk off the physical slice, by the implicit function theorem followed by one-variable Rouché. The conjugate point contributes two more. The pure degree-seven head gives exactly 49 total complex intersections counted with multiplicity and none at infinity. At most $49 - 4 = 45$ distinct physical points remain. The constructed polynomials converge to $f^2 g$, proving local sharpness. □

19.19 When the conjugate degree itself is minimal

Maximizing the number of zeros and minimizing the degree needed to attain that number are different optimization questions. The upper bound already proved above also gives an obstruction to lowering the degree of the full-saturation constructions. For $n \geq 2$, define

$$\mu(n) = \min\{1 \leq m < n : E_{m,n} = n^2\}.$$

This minimum exists by Proposition 13.1 and padding.

Corollary 19.39 (A sharp degree obstruction through the eighth cubic order)**.** *For every $n \geq 2$,*

$$\mu(n) \geq L(n) := \left\lceil n - 1 - \sqrt{\frac{(n-1)(n-2)}{2}} \right\rceil = n - 1 - \left\lfloor \sqrt{\frac{(n-1)(n-2)}{2}} \right\rfloor. \tag{19.89}$$

Equivalently, a necessary condition for $E_{m,n} = n^2$ is

$$n \leq B(m) := \left\lfloor \frac{4m + 1 + \sqrt{8m^2 - 8m + 1}}{2} \right\rfloor. \tag{19.90}$$

In particular, the exact degree minima are

$$\mu(3k) = k \quad (1 \le k \le 8), \qquad \begin{array}{c|cccccccc} n & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ \hline \mu(n) & 1 & 1 & 2 & 2 & 2 & 3 & 3 & 3. \end{array} \tag{19.91}$$

The extremizers in Theorem 19.16 at total degrees $3, 6, \dots, 24$ therefore have the smallest possible conjugate degree, even when arbitrary complex coefficients and degenerate zeros are allowed in competing polynomials. More quantitatively, reducing that degree by one gives

$$E_{k-1,3k} \le 9k^2 - (9k - k^2 - 2), \qquad 1 \le k \le 8, \tag{19.92}$$

where $E_{0,3} = 3$ denotes the analytic case. The respective unavoidable deficits are

$$6, 12, 16, 18, 18, 16, 12, 6.$$

Proof. Full saturation and (6.4) imply

$$n^2 \le (4m+1)n - 2m(m+2) \quad \Longleftrightarrow \quad 2\big(m - (n-1)\big)^2 \le (n-1)(n-2).$$

Since $m \le n-1$, taking the lower square-root branch and rounding up proves (19.89). Solving the same quadratic inequality for n proves (19.90); only its upper root is needed for this necessary condition.

Theorem 19.16 supplies $\mu(3k) \le k$. For $k \ge 2$, substitution of $(m, n) = (k-1, 3k)$ into (6.4) gives

$$E_{k-1,3k} \le 10k^2 - 9k + 2 = 9k^2 - (9k - k^2 - 2).$$

The analytic case $k = 1$ gives the same displayed value, namely three. The deficit is positive exactly for the positive integers $1 \le k \le 8$. Monotonicity in the allowed conjugate degree excludes every smaller degree as well. This proves both (19.92) and the first assertion of (19.91).

For the remaining entries in its table, the lower bound is $L(n)$. The folding seeds give the matching upper bound for $n = 2, 4, 5, 7$, and Proposition 15.2 gives it for $n = 8$. The multiples of three were just proved. Thus every entry is an exact minimum, attained by regular examples. □

This obstruction also identifies where a further degree reduction is still logically possible. The proved results give

$$3 \le \mu(10) \le 4, \qquad 4 \le \mu(11) \le 5, \qquad 4 \le \mu(13) \le 6, \qquad 8 \le \mu(27) \le 9. \tag{19.93}$$

These are brackets, not assertions of attainability at their lower endpoints. In particular, total degree 27 is the first multiple of three at which the bound permits lowering the conjugate degree below the cubic construction. Asymptotically,

$$\liminf_{n\to\infty} \frac{\mu(n)}{n} \ge 1 - \frac{1}{\sqrt{2}}, \qquad \mu(3k) \le k.$$

The latter family alone does not prove a sharp universal degree ratio. The identities, integer roundings, and finite degree table are independently checked by `verify_minimum_conjugate_degree.py` and its report `minimum_conjugate_degree_verification.json`.

19.20 An explicit certified extremizer for the new endpoint

The arbitrary-order existence proof above can be supplemented by a fully specified degree-twelve example. This certificate fixes the parameters; no unspecified smallness threshold is used in the following assertion.

Proposition 19.40. *The polynomial in* (19.36) *with*

$$k = 4, \qquad \kappa = 10^{-12}, \qquad \tau = 10^{-30}$$

has exactly 144 *simple zeros. Of these,* 78 *have positive Jacobian and* 66 *have negative Jacobian; exactly* 12 *are real. It has* 43 *nonzero monomials, highest homogeneous part* z^{12}*, and exact conjugate degree four. Multiplication by* $256 \cdot 10^{144}$ *makes all its coefficients integral.*

Proof. The file `pure_4_12_certificate.json` gives all 43 rational coefficients and 144 centers in $10^{-220}\mathbb{Z}^2$. Every box is a square of radius $r = 10^{-160}$. The independent standard-library verifier `verify_pure_4_12.py` first reconstructs the coefficients from (19.36), using

$$Q_4 = U^4 + U^2T + \frac{H}{4}U^2 + \frac{A}{4}UT + \frac{AH - C}{16}U + \frac{1}{8}T^2 + \frac{A^2 + H}{16}T + \frac{A^4}{256} + \frac{A^2H}{64} - \frac{AC}{64} + \frac{H^2}{128},$$

and the monomial reductions $z^3 = \kappa - zw$, $w^3 = \kappa - zw$. It checks equality with the stored coefficients, the degree assertions, and the common integer multiplier.

Here is the exact contraction test used on every box. Write $F = (\operatorname{Re} P, \operatorname{Im} P)$, let c be its rational center, and set $Y = DF(c)^{-1}$, computed by rational arithmetic. On the box, bound $|z|$ by $R = |\operatorname{Re} c| + |\operatorname{Im} c| + 2r$ and put

$$M = \sum_{i+j\geq 2} |p_{ij}|\big(i(i-1) + ij + j(j-1)\big)R^{i+j-2}, \qquad P = \sum p_{ij} z^i w^j.$$

This bounds $|P_{zz}| + |P_{zw}| + |P_{ww}|$ throughout the box. The identities for F_{xx}, F_{xy}, F_{yy} in Wirtinger derivatives give

$$\|DF(z) - DF(c)\|_\infty \leq 4rM.$$

Consequently the map $z \mapsto z - YF(z)$ has contraction factor at most $q = 4r\|Y\|_\infty M$. The verifier proves, by exact rational comparisons in all 144 boxes,

$$q < 10^{-63}, \qquad \frac{\|YF(c)\|_\infty}{r} < 10^{-60}, \qquad \|YF(c)\|_\infty + qr < r.$$

Each box therefore contains exactly one zero, and its Jacobian is nonsingular. The boxes are pairwise disjoint. The pure highest parts z^{12}, w^{12} of P, P^* have no common point at infinity, so the two projective curves have no common component. The 144 certified real-slice intersections exhaust Bézout's bound.

The exact Jacobian signs give 78 positive and 66 negative zeros. Twelve box centers lie on the real axis, and the remaining boxes avoid that axis. A box centered on the axis is invariant under conjugation; the real coefficients and uniqueness of its zero force that zero to be real. This proves the real-axis count. The complete successful report is `pure_4_12_verification.json`. High-precision numerical iteration was used only to propose the centers; every acceptance test just described uses rational arithmetic. □

19.21 Inversion with a highest part close to a pure power

Proposition 19.41. *Let $1 \le m_0 \le m < n$. Suppose polynomials $P_\tau \in \mathbb{Q}(i)[z,w]$ exist for arbitrarily small positive rational τ, have total degree n, conjugate degree at most m_0, and n^2 nondegenerate real-slice zeros. Suppose also that their highest homogeneous parts converge coefficientwise to z^n. Then $M_{m,n} = m^2 + n^2$ is attained over $\mathbb{Q}(i)$ with exact bidegree (m,n), a nonzero coefficient of $z^n w^m$, and only simple zeros. If the starting family has real rational coefficients and n real zeros, the output can have real rational coefficients and $m+n$ real zeros.*

Proof. Choose the explicit integer-coefficient polynomial

$$H(u,v) = T_m(v-4, u-4) = v^m + \sum_{i+j<m} q_{ij} u^i v^j.$$

For $m \ge 2$, Theorem 19.3 at target zero gives m^2 simple zeros, and (19.16) gives m real zeros. All zeros lie in $|u-4| \le 3$, since the unshifted zeros are traces of three unit roots. Hence they are nonzero and $q_{00} \neq 0$. For $m = 1$ the same formula is $H = v - 4$. Write the highest part of P_τ as $\sum_{j=0}^{m_0} a_j(\tau) z^{n-j} w^j$. Then

$$H_\tau(u,v) = \sum_{j=0}^{m_0} a_j(\tau) u^j v^{m-j} + \sum_{i+j<m} q_{ij} u^i v^j$$

converges to H. For sufficiently small rational τ, it has at least m^2 nonzero simple zeros in fixed disjoint neighborhoods. Fix such a τ for which P_τ also has its n^2 simple zeros.

For a second small parameter $s > 0$ set

$$g_s = P_\tau + \sum_{i+j<m} q_{ij} s^{m-i-j} z^{n-i} w^{m-j}. \tag{19.94}$$

All exponents are nonnegative. The corner coefficient $q_{00} s^m$ is nonzero, so the bidegree is exactly (m,n). The n^2 bounded simple zeros of P_τ persist as $s \to 0$. With $G_s(u,v) = u^n v^m g_s(1/u, 1/v)$ one has

$$s^{-m} G_s(s\zeta, s\eta) \longrightarrow H_\tau(\zeta, \eta)$$

in C^1 on compact real-slice sets. Indeed, a monomial $z^i w^j$ of P_τ acquires the factor s^{n-i-j}: exactly its degree-n terms survive. The m^2 nonzero simple zeros of the limit give m^2 simple zeros escaping to infinity after inversion, distinct from the bounded zeros. Proposition 2.3 forces coprimality, since $m^2 + n^2 > (m-1)^2 + (n-1)^2$. The rectangular intersection bound then proves exhaustion. Taking s rational preserves all claims. If the input has real coefficients, so do H_τ and g_s. The m real zeros of H, lying in $(3,7)$, remain real for small τ and then small s, and give m real zeros escaping to infinity. Together with the n bounded real zeros this gives $m+n$, the degree of $g_s(x,x)$, so the real-axis count is exact. □

Apply Proposition 19.41 to Lemma 19.12, or to the constant seed family when $k = 1$. This proves (19.2) for every $r, k \ge 1$. Here the enlarged m is introduced during inversion. Theorem 19.16 independently supplies a pure-head starting family at $r = 1$.

Lemma 19.42 (Padding the conjugate degree)**.** *If $E_{m_0,n} = n^2$ is attained with simple zeros and $m_0 \le m < n$, then $E_{m,n} = n^2$ is attained with exact conjugate degree m and simple zeros. Gaussian rational coefficients can be retained. If the input has real rational coefficients and n real zeros, both properties can be retained as well.*

Proof. For $m > m_0$, add a sufficiently small nonzero Gaussian rational multiple of w^m. The simple zeros persist, the pure head remains, and the same bound excludes more zeros. In the real case choose the added coefficient real; the simple real zeros persist by conjugation symmetry. For $m = m_0$ nothing is needed. □

Lemma 19.43 (Circle addition over the real rationals)**.** *Suppose $P \in \mathbb{Q}[z, w]$ has bidegree (m, n), $n > m \geq 1$, a nonzero corner coefficient, and m^2+n^2 nondegenerate zeros. Then a polynomial with the same properties exists at bidegree $(m+1, n+1)$, still over $\mathbb{Q}$. If P has $m+n$ real zeros, the new polynomial has $m+n+2$ real zeros.*

Proof. The full count forces P, P^* to be coprime by Proposition 2.3. In the proof of Proposition 22.1, choose the center a real and the radius $R > 0$, both rational. This is possible even with the restricted center: each complex intersection excludes the zero set of the nonzero polynomial $(z-a)(w-a)-R^2$ in the real parameters (a, R). The top-coefficient exclusions are likewise proper, and rational pairs are dense. Thus, with $d = m + n$, the circle restrictions are

$$P = t^{-m}A(t), \quad A \in \mathbb{Q}[t], \quad \deg A = d, \quad A(0) \neq 0, \quad \gcd(A, A^{\#_d}) = 1.$$

Restrict now to real-coefficient perturbations $Q = t^{-(m+1)}B(t)$ with $B \in \mathbb{R}[t]$, $\deg B \leq d+2$. They span all such Laurent restrictions. The map $B \mapsto \operatorname{Im}(Q\overline{P})$ takes values in the odd real trigonometric polynomials of degree at most $d+1$, a space of dimension $d+1$. Its kernel condition is still $BA^{\#_d} = B^{\#_{d+2}}A$, hence $B = AS$ and

$$S = S^{\#_2}, \qquad S = a_0 + a_1 t + a_0 t^2, \quad a_0, a_1 \in \mathbb{R}.$$

The domain has dimension $d+3$ and the kernel has dimension two, so this restricted map is onto. In the sine basis its matrix has rational entries. Therefore a rational B, and then a rational polynomial Q, realize $\sin((d+1)\theta)$ exactly.

For sufficiently small nonzero rational ε, the polynomial $((z-a)(w-a)-R^2)P + \varepsilon Q$ retains the old simple zeros and has $2(d+1)$ new simple zeros by the normal–tangential argument of Proposition 22.1. Its corner coefficient remains nonzero. The resulting count is $(m+1)^2+(n+1)^2$, so coprimality and exhaustion follow from the rectangular bound; no further perturbation is necessary. The two points $\theta = 0, \pi$ give real zeros. All old real zeros and these two new ones persist as real by conjugation symmetry. If there were d old real zeros, the real restriction now has degree $d+2$, proving the last assertion. □

19.22 The linear region and its remaining boundary

Proof of Theorem 19.2. For a pair satisfying (19.3), put

$$s = n \bmod 3 \in \{0, 1, 2\}, \qquad k = (n-s)/3, \qquad a = m-k-s. \tag{19.95}$$

Then $k \geq 1$. To see $a \geq 0$, write $D = 3m - n \geq 0$. For $s = 0, 1, 2$, respectively, the integer D is congruent to $0, 2, 1$ modulo three. The exclusion $D \neq 1$ therefore implies $D \geq 2s$, and $3a = D - 2s \geq 0$. Also $k + a = m - s < n - s = 3k$. Theorem 19.1 with $r = 1$ supplies equality at $(m_0, n_0) = (k+a, 3k)$ with real rational coefficients. Apply Lemma 19.43 s times. The final bidegree is $(m_0 + s, n_0 + s) = (m, n)$, and its zero count is

$$m_0^2 + n_0^2 + 2s(m_0 + n_0 + s) = m^2 + n^2.$$

The real circle-addition lemma retains a nonzero corner coefficient, real rational coefficients, and all the constructed simple zeros. Corollary 2.4 proves irreducibility. The pairs $(1,2),(2,5),(3,8)$ follow, respectively, from the explicit five-zero example and Theorem 1.3's earlier low-degree construction (proved in Sections 12–16). □

The equation $D = 3a+2s$ explains the single missing boundary line: every nonnegative integer except 1 is a sum of a nonnegative multiple of three and a nonnegative multiple of two. It does not impose a restriction on other possible extremizers. At $m = 10000$ the proved range beyond $n = 2m$ now includes every $1 \le d \le 9998$ and $d = 10000$; the former explicit band ended at 46.

Corollary 19.44 (Simultaneous sharpness on the real axis)**.** *The pure-head extrema in Theorem 19.1, including padding to a larger conjugate degree, can be attained over $\mathbb{Q}$ with exactly n real zeros. Its rectangular extrema and all pairs in (19.3) can be attained over $\mathbb{Q}$ with exactly $m+n$ real zeros. The latter assertion also holds at $(1,2)$ and $(2,5)$. All zeros are nondegenerate, and multiplication by a nonzero integer makes the coefficients integral. These real-axis counts are maximal in the respective degree classes.*

Proof. Lemma 19.12 supplies n real zeros, and padding preserves them. The real form of Proposition 19.41 adds m real zeros. Lemma 19.43 adds two at every diagonal step. The explicit five-zero polynomial in Section 1.1 has three real zeros, and $(2,5)$ is the case $r = 2, k = 1$ of the inversion construction. The restrictions to $w = z = x$ have degrees n and $m+n$, respectively, so no higher real-axis counts are possible. □

Real coefficients here mean coefficients in $\mathbb{R}[z,w]$; $P(z,\bar z)$ is generally complex-valued. This conclusion does not upgrade the separate certified $(3,8)$ example from $\mathbb{Q}(i)$ to $\mathbb{Q}$, nor does it enlarge the degree range of the cone.

Corollary 19.45 (A sufficient arithmetic criterion)**.** *Let $d = n-2m \ge 1$ and $h = n-m$. If integers k, l, a satisfy*

$$k \ge 1, \qquad l \ge 2, \qquad a \ge 0, \qquad h + a = kl, \qquad d \le k - 2a, \tag{19.96}$$

then $M_{m,n} = m^2 + n^2$.

Proof. Use (19.2) with $r = l-1$ and conjugate degree $k(l-1)+a$. This degree is below $k(2l-1)$ because $a \le (k-d)/2 < k/2$. Add $s = k-d-2a$ circles. The resulting bidegree is (m,n), and quadratic saturation persists. □

The earlier condition $k \le l$ is no longer required. The direct parameters (19.95) are simpler for the continuous region; the arithmetic form is useful for matching individual seeds.

Corollary 19.46 (Sharp counts of the two orientations)**.** *At every degree pair in Theorem 19.2, among polynomials of bidegree (m,n) with nonzero corner coefficient and only nondegenerate zeros, the separate maxima are*

$$\max N_+ = \frac{m^2+n^2+n-m}{2}, \qquad \max N_- = \frac{m^2+n^2-n+m}{2}.$$

The same polynomial attains both maxima. At each pure-head degree pair proved in Theorem 19.1, the corresponding separate maxima among polynomials with only nondegenerate zeros are $(n^2+n)/2$ and $(n^2-n)/2$.

Proof. A nonzero rectangular corner is the unique term of total degree $m+n$. The map is proper and has degree $n-m$, giving $N_+ - N_- = n-m$ by (4.4). Combine this identity with $N_+ + N_- \le m^2+n^2$ and the constructed extremizers. For a pure head the degree is n and the total bound is n^2, which gives the second assertion in the same way. □

19.23 Exactness and verification scope

The full local fiber classification, the real-coefficient recurrence, the absence of intersections at infinity, and the separate small-parameter limits are proved above for all stated integers. The accompanying script `verify_exact_local_models.py` checks the discriminant–Jacobian identity for $2 \le k \le 15$, 80 smooth-boundary orders, 240 cusp orbit counts, real and nonreal scaling identities for $2 \le k \le 20$, 28 exact global degree instances, and 16 real circle-restriction ranks. Its report is `exact_local_models_verification.json`. These finite exact diagnostics do not replace the all-degree proofs or provide a certified numeric threshold for ϵ, τ, s. The radius in Corollary 19.5, in contrast, is an exact optimal threshold for constant-target translation of the local map.

The earlier `verify_linear_sharp_region.py` and `linear_sharp_region_verification.json` check the preceding shifted recurrence. Their seed, inversion, and integer-cone checks remain relevant; their global-polynomial checks are not claimed to test the present unshifted recurrence. The still older `verify_universal_splitting.py` concerns a correction-series version.

The rectangular cone and unrestricted splitting for all $r, k \ge 1$ have analytic proofs, independent of the older degree-eight certificate. The separate endpoint $(3, 8)$ retains its stated certificate dependency. The exact local target disk is not a disk for arbitrary coefficient perturbations. The translated construction keeps the normalized error small and proves pure-head saturation at every multiple of three in $n \le 3m$; the other residues in the full cone remain separate questions.

20 Sharp spectra of attainable zero counts

A sharp maximum leaves a second question: which smaller counts occur without changing the degree constraints? For unrestricted total degree, the range of distinct counts was studied in [SZ24, Theorem 2.16]. For fixed harmonic degrees, the absence of gaps and its caustic-path proof are already established by Sète–Zur [SZ21, Theorem 5.4 and Remark 5.5, author version, pp. 24–25]. We retain that attribution: the harmonic row below is not a new zero-count theorem. Here the argument is extended to the rectangular and pure-head coefficient spaces, with an explicit Gaussian-rational construction at the singular fibers. It does not determine an unknown maximal endpoint.

Write $\mathcal{R}^{\circ}_{m,n}$ for the exact rectangular class with a nonzero $z^n w^m$ coefficient. Its maps are proper of topological degree $n-m$. Pure-head and harmonic maps have degree n.

Theorem 20.1 (Complete nondegenerate spectra)**.** *For $1 \le m < n$, the attainable numbers of nondegenerate zeros, when every zero is required to be nondegenerate, are exactly*

Coefficient class	*Attainable counts*
$\mathcal{R}^{\circ}_{m,n}$	$n-m,\ n-m+2, \dots, M_{m,n}$
$\mathcal{E}_{m,n}$	$n,\ n+2, \dots, E_{m,n}$
harmonic, exact degrees (m,n)	$n,\ n+2, \dots, M^{\mathrm{harm}}_{m,n}$.

(20.1)

Every count can be attained with Gaussian integer coefficients. In the pure-head row, the examples can have exact conjugate degree m. For a row whose mapping degree is D and maximum is K, the

full set of orientation pairs is

$$(N_+, N_-) = (D + j, j), \qquad 0 \le j \le (K - D)/2. \tag{20.2}$$

Allowing degenerate zeros, the finite distinct-count *spectra are* $\{1, 2, \dots, M_{m,n}\}$ *for* $\mathcal{R}^{\circ}_{m,n}$ *and* $\{1, 2, \dots, E_{m,n}\}$ *for* $\mathcal{E}_{m,n}$*, again over Gaussian integers. The latter uses the stipulated bound* $\deg_w \le m$*. With exact positive conjugate degree, the pure-head distinct spectrum is* $\{2, \dots, E_{m,n}\}$*, as proved in Theorem 20.8. For the exact harmonic class, every integer from* n *to* $M^{\mathrm{harm}}_{m,n}$ *occurs. The remaining lower counts and the matching obstruction are determined in Theorem 20.5: the full distinct spectrum is* $\{n - m + 1, \dots, M^{\mathrm{harm}}_{m,n}\}$*.*

We supply the genericity details, since an arbitrary path of coefficients can pass through several singular zeros at once. In particular, continuity of the coefficients alone would not prove the absence of gaps.

Lemma 20.2 (Separating fold crossings)**.** *Suppose* $n \ge 3$ *and* P *belongs to any one of the three classes in* (20.1)*, with* K *simple zeros. An arbitrarily small perturbation within the same class has* K *simple zeros and admits a path of constant target translations from that fiber to a fiber of size* D*. Along the path, only one ordinary fold is crossed at a time. At a crossing the two neighboring regular counts differ by two, and the distinct count at the crossing is their average.*

Proof. The allowed variations include $1, z, w, z^2, z^3$ with arbitrary complex coefficients. For $n = 3$, varying z^3 is allowed because the nonzero leading coefficient need not be fixed to one. Small variations retain the nonzero leading coefficients. Properness and the nondegeneracy of the initial zeros make their total count locally constant.

At a source point, constant and complex-linear and conjugate-linear variations prescribe the value and the full real first derivative. Consequently a generic polynomial has no rank-zero critical points: that condition has real codimension four. At a rank-one critical point let v span the kernel and let ℓ annihilate the image. The fold condition is

$$\ell\big(D^2F(v, v)\big) \neq 0.$$

Adding $a(z - z_0)^2$ changes this real quantity freely while leaving the first jet at z_0 fixed. Thus the failure of the fold condition has codimension two, including the rank-one condition. Parametric transversality gives a zero-dimensional set of such exceptional points for a generic member; it is finite because it is semialgebraic. The same argument makes the remaining critical set smooth, with an immersed critical-value curve. These applications use the parametric transversality principle cited in Lemma 3.2.

There are also only finitely many targets with two distinct critical preimages. The cubic interpolation step parallels [SZ21, Lemma 5.2, author version, p. 23]. On $z_1 \neq z_2$ the equations

$$F(z_1) = F(z_2), \qquad J_F(z_1) = J_F(z_2) = 0$$

have codimension four. Complex Hermite interpolation by a cubic prescribes its values and analytic derivatives independently at z_1, z_2. At a rank-one point, varying the analytic derivative changes $|P_z|^2 - |P_w|^2$ nontrivially, since $P_z \neq 0$. Thus the displayed four real conditions have independent parameter directions. Transversality on the four-dimensional space of source pairs, followed by semialgebraic finiteness, proves the claim.

The leading term is either $az^n w^m$ or az^n. Its real Jacobian is strictly positive away from zero, and its lower-order terms are negligible at infinity. Hence the critical set is compact. The critical-value set is a compact semialgebraic curve with finitely many exceptional targets. Choose a piecewise smooth semialgebraic target path from zero to a large regular value, avoiding those targets and crossing the remaining arcs transversely. It has finitely many crossings. At each, the real fold normal form $(x, y) \mapsto (x, y^2)$, obtained from the implicit function theorem and the one-variable Morse lemma, gives exactly the asserted change.

For completeness, a sufficiently large target has precisely D preimages. If d is the total degree, rescale a target of modulus R^d by $z = Ru$. The rescaled map converges in C^1 to its leading monomial on a fixed annulus. That monomial has exactly D simple preimages of a nonzero target. The leading-term estimate puts every possible preimage in that annulus for large R. The implicit function theorem and compactness prove exhaustion. □

Proof of Theorem 20.1. First choose a polynomial attaining the relevant maximum. The derivative-preserving regularization in Corollary 3.4 stays within each of the three coefficient spaces and does not decrease the count. In the rectangular case, a further small generic perturbation can make the corner coefficient nonzero while retaining the zeros. Its count is then still the maximum. The pure-head conjugate degree can likewise be made exact by a small w^m perturbation. Thus all three maxima are achieved by proper maps with only simple zeros.

The signed-degree identity gives $N_+ - N_- = D$, and hence $N \geq D$ and $N \equiv D \pmod 2$. Apply Lemma 20.2 to a maximizer. A finite sequence of integer counts beginning at K, ending at D, and changing by two at each step visits every integer of that parity in the interval. The single-fold fibers give every intermediate opposite-parity integer. This also proves (20.2).

Regular fibers can be approximated by coefficients in $\mathbb{Q}(i)$ without changing their count. A fold fiber requires a separate argument. Let z_0 be its unique singular zero, choose a nearby $a \in \mathbb{Q}(i)$, and approximate its polynomial by $Q \in \mathbb{Q}(i)[z, w]$. Choose $\gamma \in \mathbb{Q}(i)$ with $|\gamma| = 1$ close to $P_z(z_0, \bar{z}_0)/P_w(z_0, \bar{z}_0)$; rational points on the unit circle are dense. Put

$$\begin{aligned} A &= \gamma Q_w(a, \bar{a}) - Q_z(a, \bar{a}), \\ B &= -Q(a, \bar{a}) - Aa, \qquad \widetilde{Q} = Q + Az + B. \end{aligned} \tag{20.3}$$

Then $\widetilde{Q}(a, \bar{a}) = 0$ and $J_{\widetilde{Q}}(a) = 0$ exactly. All coefficients are Gaussian rational, and the correction can be arbitrarily small. Its first derivative has rank one and its fold quadratic term stays nonzero. In fold coordinates it has exactly one zero near z_0; every other zero remains simple. Uniform properness excludes additional zeros far away. This gives the same distinct count over $\mathbb{Q}(i)$. Clearing denominators gives Gaussian integers in all these cases.

For $1 \leq h \leq D = n - m$, the rectangular polynomial

$$z^m w^m \, z^{D-h+1} \prod_{j=1}^{h-1} (z - j)$$

has exactly h distinct zeros and a nonzero corner. In the pure-head class use $z^{n-h+1} \prod_{j=1}^{h-1}(z-j)$ for $1 \leq h \leq n$. These fill the remaining lower counts. The nonzero mapping degree excludes a zero-free fiber.

It remains to treat $n = 2, m = 1$, where cubic interpolation is unavailable. The pure-head and harmonic maximum is four; $T_2(z, \bar{z}) - c = z^2 - 2\bar{z} - c$ has regular counts two and four and

a three-point smooth-boundary fiber by Theorem 19.3. The analytic polynomial z^2 supplies the one-point pure-head fiber. For the rectangle, the maximum is five. The polynomials

$$z^2w - tz - w \quad (t = -2, 0, 2)$$

have respectively one, three, and five simple zeros. The polynomial $z^2w - \frac{5}{2}z - \frac{1}{2}w - 2$ has exactly the two zeros $2, -1$. Indeed, its imaginary part is $y(x^2 + y^2 - 2)$, and the circle branch has no zero of the real part; the real-axis equation is $(x - 2)(x + 1)^2 = 0$. Finally, $z^2w - 4z - w - 2i$ has precisely $2i, -i, 2 + i, -2 + i$, as follows from its real part $x(x^2 + y^2 - 5)$ and imaginary part $y(x^2 + y^2 - 3) - 2$. In each of the two latter examples there is exactly one fold zero. All coefficients are Gaussian rational. □

The qualification "finite" in the rectangular spectrum matters: $z^{n-m}(zw - 1)^m$ has a nonzero corner and an entire zero circle. The pure-head and harmonic spaces with $n > m$ have finite fibers.

Corollary 20.3 (Numerical spectra in the equality region)**.** *At every rectangular pair for which this note proves $M_{m,n} = m^2 + n^2$, the complete regular spectrum with nonzero corner is*

$$\{n - m + 2j : 0 \le j \le (m^2 + n^2 - n + m)/2\},$$

and every finite distinct count from 1 to m^2+n^2 occurs. At each proved pure-head equality $E_{m,n} = n^2$, the regular spectrum is $\{n + 2j : 0 \le j \le n(n - 1)/2\}$, and its finite distinct spectrum is $\{1, \dots, n^2\}$.

For example, at $(m, n) = (4, 12)$ the rectangular regular counts are exactly $8, 10, \dots, 160$; all integers $1, \dots, 160$ occur when degeneracies are allowed. Theorem 19.16 also determines the pure-head regular spectrum there as $12, 14, \dots, 144$. At the pure-head pair $(8, 20)$, the corresponding spectra are $20, 22, \dots, 400$ and $1, \dots, 400$. For every harmonic pair, combining the theorem with the sharp minimum proved below realizes every integer from $n - m + 1$ to $(m + 2)n - m - \gcd(m, n)$. Among these counts, precisely those at least n and of parity n admit only simple zeros. The unknown harmonic maximum, and the unknown rectangular endpoints outside the equality region, remain unknown endpoints of these spectra.

Verification scope. The proofs of genericity, continuation, and rational realization above are analytic. The file `verify_spectra_and_cubic_radius.py` checks the cubic elimination and minimization identities, 14399 exact rational radius inequalities, the cubic Hermite determinant, the two low-degree fold examples, and 30 exact rational corrections enforcing both a zero and a vanishing Jacobian. Its report is `spectra_and_cubic_radius_verification.json`. These finite diagnostics do not prove transversality by sampling, nor do they establish any still-unknown maximal endpoint.

20.1 The sharp lower endpoint in the harmonic class

The regular lower endpoint n need not be the lower endpoint for distinct zeros. A multiple zero can carry positive index greater than one. The common critical points of the two analytic parts measure exactly how much a distinct count can fall below n. For comparison, the index bound $\{-1, 0, 1\}$ when the anti-analytic part has degree one is established in [LuSe21, Theorem 3.1, author version v2, pp. 6–7]. The elementary argument below also covers common critical points.

Lemma 20.4 (Local index and common vanishing order)**.** *Let a be an isolated zero of the harmonic germ $H(z) = p(z) + \overline{q(z)}$, with both analytic parts nonconstant. Then*

$$|\operatorname{ind}_a H| \le \min\{\operatorname{ord}_a(p - p(a)), \operatorname{ord}_a(q - q(a))\} = 1 + \min\{\operatorname{ord}_a p', \operatorname{ord}_a q'\}. \tag{20.4}$$

Proof. Translate a to zero and subtract the two constant terms. Put $s = \min\{\operatorname{ord}_0 p, \operatorname{ord}_0 q\}$. The first nonzero homogeneous part of H is $Az^s + B\bar{z}^s$, with $(A, B) \neq (0, 0)$. Choose a target rotation $e^{-i\phi}$ such that its real part is a nonzero real harmonic homogeneous polynomial. On $z = re^{i\theta}$, the real part of $r^{-s}e^{-i\phi}H(z)$ consequently converges in C^1 to $C\cos(s\theta + \psi)$, with $C \neq 0$. For sufficiently small r, it has exactly $2s$ simple zeros, and their crossing directions alternate. The image circle avoids zero by isolation. Its winding is the signed number of crossings of the positive imaginary ray. Among all $2s$ crossings of the imaginary axis there are s in each direction, so the winding has absolute value at most s. This proves the inequality even when the first homogeneous part itself vanishes in some directions. The last equality follows by differentiation of analytic germs. □

Theorem 20.5 (Sharp distinct minimum and a bound over all targets)**.** *Let $F = p + \overline{q}$, where $\deg p = n > m = \deg q \geq 1$, and put $G = \gcd(p', q')$, taken monic, and $g = \deg G$. For a target c define*

$$N_c = \#F^{-1}(c), \qquad b_c = \sum_{a \in F^{-1}(c)} \operatorname{ord}_a G.$$

Then every fiber is finite and

$$N_c \geq n - b_c \geq n - g \geq n - m + 1. \tag{20.5}$$

In addition,

$$\sum_{c \in \mathbb{C}} (n - N_c)_+ \leq g \leq m - 1. \tag{20.6}$$

Only finitely many summands can be nonzero. In particular, $N_c < n$ is possible only at the image of a common critical point of p and q. If p' and q' are coprime, every fiber has at least n distinct points, including singular fibers.

For each $0 \leq g \leq m - 1$, both the lower bound $n - g$ and the total-deficit bound g are attained by a polynomial with integer coefficients and exact degrees (m, n). Consequently, the minimum over the entire exact harmonic class is exactly $n - m + 1$.

Proof. Finiteness follows from the pure highest term of degree n. On a sufficiently large circle, $F - c$ has winding n. Additivity of the topological degree and Lemma 20.4 give

$$n = \sum_{a \in F^{-1}(c)} \operatorname{ind}_a(F - c) \leq \sum_{a \in F^{-1}(c)} (1 + \operatorname{ord}_a G) = N_c + b_c.$$

This proves (20.5). Every root of G belongs to exactly one fiber, so $\sum_c b_c = g$. Since $(n - N_c)_+ \leq b_c$, summation proves (20.6).

For attainment, take $k = g + 1$ and set

$$A = n + m + 1, \qquad F_{n,m,k}(z) = A(z^n - z^k) + \bar{z}^m, \qquad 1 \leq k \leq m. \tag{20.7}$$

The derivatives of its analytic parts have monic gcd z^{k-1}. We show directly that the origin has index k and that precisely $n - k$ other zeros occur, all with positive Jacobian. Divide by A, and put $\epsilon = 1/A$, $d = n - k$, $h = m - k$. At a nonzero zero with $r = |z|$, let $u = \epsilon\bar{z}^m/z^k$. Then

$$z^d = 1 - u, \qquad |u| = \epsilon r^h, \qquad J = r^{2k-2}\big(|d - nu|^2 - m^2|u|^2\big).$$

If $r \leq 1$, then $|u| \leq \epsilon$. If $r > 1$, the inequalities $r^h \leq r^d \leq 1 + \epsilon r^h$ imply $|u| \leq \epsilon/(1 - \epsilon) = 1/(n + m)$. When $h > 0$, one has $d \geq 2$, and hence $d > (n + m)|u|$. When $h = 0$, one instead has $|u| = \epsilon$ and

again $d > (n+m)|u|$. Thus $|d-nu| > m|u|$ at every nonzero zero, proving $J > 0$. At zero the initial part is $-z^k$ if $k < m$, or $-z^m + \epsilon\bar{z}^m$ if $k = m$; its index is k in either case. The sum of indices is n, so there are exactly $n-k$ nonzero zeros. Therefore $N_0 = n-k+1 = n-g$. All common critical points lie over zero, and every other target has $N_c \geq n$. The sum in (20.6) is exactly g. □

The proof also gives an exact equality criterion. A fiber has the absolute minimum $n-m+1$ if and only if

$$\deg G = m-1, \qquad Z(G) \subset F^{-1}(c), \qquad \operatorname{ind}_a(F-c) = 1 + \operatorname{ord}_a G \quad (a \in F^{-1}(c)). \tag{20.8}$$

Indeed, these are precisely the equality conditions in the three successive inequalities of (20.5).

Corollary 20.6 (The full harmonic distinct-count spectrum)**.** *For exact degrees $1 \leq m < n$, the attainable distinct zero counts are exactly*

$$\{n-m+1, n-m+2, \dots, M_{m,n}^{\mathrm{harm}}\}. \tag{20.9}$$

Every count can be realized over $\mathbb{Z}[i]$. The lower part $n-m+1, \dots, n$ is realized by the integer trinomials (20.7). In particular, the complete distinct-count spectra at the two known maximal endpoints are

$$m = 1: \quad \{n, n+1, \dots, 3n-2\}, \qquad m = n-1: \quad \{2, 3, \dots, n^2\}.$$

Proof. Theorem 20.5 excludes smaller counts, and (20.7), with $1 \leq k \leq m$, realizes all counts up to n. The interval from n to the maximum is the no-gap theorem of [SZ21, Theorem 5.4], with the Gaussian integer realization proved in Theorem 20.1. The classical maximal endpoints give the final two formulas. □

For example, for every $n \geq 2$ the polynomial

$$2nz^n - 2nz^{n-1} + \bar{z}^{n-1}$$

has exactly the two zeros 0 and $1-1/(2n)$. The first has index $n-1$, and the second is nondegenerate and positive. For $n > 2$ this explains why applying the regular lower bound n to an arbitrary distinct count would be incorrect. The extension below n and the proof of the common-critical-point bound are given here self-contained; the cited no-gap theorem is used only for counts at least n. No assertion of exhaustive priority for the lower-endpoint argument is needed.

20.2 Keeping the conjugate degree exact in the pure-head spectrum

The distinction between $\deg_w \leq m$ and $\deg_w = m$ changes the lowest distinct count. The following local observation gives the required obstruction without any genericity assumption.

Lemma 20.7 (Local index versus polynomial vanishing order)**.** *Let a be an isolated zero of a complex-valued polynomial in two real variables. If its first nonzero Taylor homogeneous term at a has degree s, then the absolute value of its local topological index is at most s.*

Proof. Translate a to zero, and write $F = H_s + H_{s+1} + \cdots + H_D$. Choose a target projection $\operatorname{Re}(e^{-i\alpha}F)$ such that $\operatorname{Re}(e^{-i\alpha}H_s)$ is not identically zero. On a small circle, after division by r^s, this projection converges to a nonzero real trigonometric polynomial $h(\theta)$ of degree at most s. Its zeros

on the circle have total multiplicity at most $2s$: multiplication of its Laurent form by ζ^s gives a nonzero polynomial of degree at most $2s$.

This zero bound remains valid for the projected small-circle map, counting multiplicities. To include multiple zeros of h, complexify θ in disjoint small neighborhoods of its real zeros and apply the argument principle, or equivalently Rouché's theorem, to the convergent finite trigonometric sums. Each neighborhood contains at most the corresponding multiplicity; away from them the limit is bounded away from zero. The same bounds hold for a sufficiently small change of α. For a fixed sufficiently small r, choose such a change for which the imaginary-axis crossings of the image are transverse. This is possible by the regular-value theorem for its argument; the image avoids zero by isolation. The crossings alternate directions, with at most s in each direction. Counting the oriented crossings of the positive imaginary ray bounds the absolute winding by s, as asserted. □

Theorem 20.8 (Exact conjugate degree and the one-point obstruction)**.** *For $1 \le m < n$, let $\mathcal{E}^{=}_{m,n} = \{f \in \mathcal{E}_{m,n} : \deg_w f = m\}$. Its complete distinct-count spectrum is*

$$\{2, 3, \dots, E_{m,n}\}. \tag{20.10}$$

Every count is attainable over $\mathbb{Z}[i]$, and every count $2, \dots, n$ is attainable over $\mathbb{Z}$. More generally, a pure-head polynomial of total degree n has just one distinct zero a if and only if it is $c(z-a)^n$, with $c \neq 0$.

Proof. A pure-head map has topological degree n, so a unique zero would have index n. Its vanishing order is at most n; Lemma 20.7 forces that order to equal n. After translation to the zero, all lower Taylor terms therefore vanish. Translation leaves the highest homogeneous part cz^n unchanged, and the polynomial is exactly $c(z-a)^n$. Conversely this polynomial has precisely the stated zero. In particular, exact positive conjugate degree excludes the one-point case.

For an explicit count $2 \le h \le n$, put

$$p = \max\{m+1, h\}, \qquad k = p - h + 1, \qquad 1 \le k \le m, \quad m < p \le n.$$

Using (20.7), define the integer polynomial

$$P_{m,n,h}(z) = z^{n-p}\left((p+m+1)(z^p - z^k) + \bar{z}^m\right). \tag{20.11}$$

Its highest term is $(p+m+1)z^n$. The other analytic exponent is $n-h+1 < n$, and the mixed term has total degree $n-p+m < n$ and conjugate degree exactly m. The inner trinomial has exactly $p-k+1 = h$ distinct zeros, including zero. Multiplication by z^{n-p} introduces no new zero. Thus (20.11) has exactly h distinct zeros, with local index $n-h+1$ at zero and $h-1$ other nondegenerate positive zeros.

For the interval from n to $E_{m,n}$, use the fold-path proof of Theorem 20.1, starting with exact conjugate degree m. The constant target path preserves that degree, and so do sufficiently small perturbations and its rational fold correction. Thus every count in that interval also has exact degree m. The same theorem and its maximum definition exclude larger counts. Clearing denominators gives the Gaussian integer assertion. □

For instance, when $E_{8,20} = 400$, allowing conjugate degree *at most* eight gives $1, \dots, 400$, whereas requiring degree *exactly* eight gives $2, \dots, 400$. The regular spectrum remains $20, 22, \dots, 400$ in both cases.

20.3 The exact parameter disk for the lower-endpoint trinomials

The integer choice above is convenient. Its stability region also has a sharp, explicit centered disk, valid for complex perturbations.

Theorem 20.9 (Exact first singular-parameter radius)**.** *For integers $1 \le k \le m < n$, define*

$$F_\epsilon(z) = z^n - z^k + \epsilon \bar{z}^m, \qquad R_{n,m,k} = \frac{n-k}{n+m}\left(\frac{n+m}{m+k}\right)^{(m-k)/(n-k)}.$$

For every complex $|\epsilon| < R_{n,m,k}$, the origin has index k and there are exactly $n-k$ other zeros, all nondegenerate with positive Jacobian. This radius is the largest centered open disk with that property. At $\epsilon = R_{n,m,k} > 0$ the nonzero point

$$r_* = \left(\frac{m+k}{n+m}\right)^{1/(n-k)} \tag{20.12}$$

is a singular zero. For $k = m$, the exact radius simplifies to $(n-m)/(n+m)$.

Proof. Put $d = n-k$ and $h = m-k$. At a nonzero zero use the variable $u = \epsilon\bar{z}^m/z^k$ as in the preceding proof. Singularity is equivalent to

$$d^2 - 2dn \operatorname{Re} u + (n^2 - m^2)|u|^2 = 0. \tag{20.13}$$

This is a circle with center $dn/(n^2-m^2)$ and radius $dm/(n^2-m^2)$. Consequently, with $x = |u|^2$,

$$\frac{d^2}{(n+m)^2} \le x \le \frac{d^2}{(n-m)^2}.$$

The zero equation gives $|z|^d = |1-u|$. Eliminating $\operatorname{Re} u$ from (20.13) yields

$$|1-u|^2 = C + Bx, \qquad C = \frac{k}{n}, \quad B = \frac{m^2-nk}{dn}, \qquad |\epsilon|^2 = \frac{x}{(C+Bx)^{h/d}}. \tag{20.14}$$

For $h = 0$ the last quantity is just x. Suppose $h > 0$. The circle does not contain $u = 1$, since substitution in (20.13) gives $k^2 - m^2 \neq 0$. The derivative of the last expression in (20.14) has the sign of

$$C + \frac{n-m}{d}Bx.$$

This is positive when $B \ge 0$. When $B < 0$, its minimum is at the upper endpoint, where it equals $m(m-k)/(n(n-m)) > 0$. Thus the expression is strictly increasing throughout the feasible interval. Its minimum is attained at $u = d/(n+m)$, and its square root is exactly $R_{n,m,k}$. The point (20.12) and the real parameter $\epsilon = R_{n,m,k}$ satisfy both the zero equation and (20.13), so this minimum is attained.

There are no nonzero singular zeros below that radius. At zero, the index remains k: if $k < m$ its initial term is $-z^k$; if $k = m$, use $R_{n,m,m} < 1$ and $-z^m + \epsilon\bar{z}^m$. On every compact parameter subdisk, zero is uniformly isolated and the leading term z^n confines all zeros to a common bounded set. Continuation from $\epsilon = 0$, where the nonzero zeros are the dth roots of unity, therefore preserves their number and their positive Jacobians. The exhibited singular zero prevents any enlargement of the disk. □

This is a radius for retaining the minimum number of distinct zeros with regular nonzero zeros; it is not a maximal-valence formula for all parameters. When $k > 1$, the origin is deliberately degenerate throughout the disk.

Corollary 20.10 (Exact boundary fibers and the first increase)**.** *Put $d = n-k$, $\ell = \gcd(d, m+k)$, and $R = R_{n,m,k}$. The only parameters on $|\epsilon| = R$ with a singular nonzero zero are the d/ℓ distinct values in*

$$\left\{ R\exp\left(\frac{2\pi i(m+k)j}{d}\right) : j = 0, \dots, d-1 \right\}. \tag{20.15}$$

Every parameter on that circle has exactly $d+1$ distinct zeros. At each exceptional parameter, precisely ℓ of the nonzero zeros are singular, each with index $+1$; the other $d-\ell$ nonzero zeros have positive Jacobian.

Fix an exceptional parameter ϵ_0. For every sufficiently small real $s > 0$, the parameter $(1+s)\epsilon_0$ has exactly $d+2\ell+1$ distinct zeros. Apart from the origin, there are $d+\ell$ positive and ℓ negative nondegenerate zeros. Thus R is also the largest centered open disk on which the distinct count is constantly $d+1$, even without a regularity requirement.

Proof. The strict minimization in the preceding proof shows that any singular zero on the first parameter circle must have $u = d/(n+m)$ and $z^d = (m+k)/(n+m)$. Thus its position is $r_* \exp(2\pi ij/d)$, and its parameter is exactly (20.15). The elementary residue map $j \mapsto (m+k)j \pmod d$ has kernel size ℓ, proving the asserted number of exceptional parameters and singular points.

It remains to check the local fibers, rather than infer their counts from continuity at a singularity. Rotating the source and the target reduces every listed point to the positive real one. With $z = r_* w$, division by r_*^k gives

$$f_s(w) = tw^n - w^k + a(1+s)\bar{w}^m, \qquad t = \frac{m+k}{n+m}, \quad a = \frac{n-k}{n+m}.$$

Write $w = 1 + x + iy$. Since $\partial_x \operatorname{Re} f_0(1) = 2ma > 0$, the real equation determines $x = X(y,s)$ analytically near zero. Conjugation symmetry makes X even in y. Substitution into the imaginary equation gives the exact Taylor form

$$\operatorname{Im} f_s(1 + X(y,s) + iy) = y\left(A_1 s + C_3 y^2 + O(s^2 + |s|y^2 + y^4)\right), \tag{20.16}$$

where

$$A_1 = -\frac{(m+k)(n-k)}{2m} < 0, \qquad C_3 = \frac{(m+k)(n-k)(km+3kn-m^2+mn)}{12m} > 0.$$

For an explicit check of these coefficients, put

$$L = 2ma, \quad S_2 = \frac{tn(n-1) - k(k-1) + am(m-1)}{2}, \quad T_2 = \frac{tn(n-1) - k(k-1) - am(m-1)}{2},$$

and $T_3 = (tn(n-1)(n-2) - k(k-1)(k-2) - am(m-1)(m-2))/6$. The real equation gives $X = -s/(2m) + (S_2/L)y^2 + O(s^2 + |s|y^2 + y^4)$, hence $A_1 = -T_2/m - ma$ and $C_3 = 2T_2S_2/L - T_3$, which simplify to the displayed formulas.

At $s = 0$, (20.16) has one zero of index $+1$. For small $s < 0$ it has one simple positive zero; for small $s > 0$ it has three simple zeros, the central one negative and the other two positive. Indeed, besides $y = 0$, the equation in y^2 has the unique small solution $y^2 = -A_1 s/C_3 + O(s^2)$, which is positive exactly on the latter side. The signs follow by differentiating the reduced equation, since the eliminated real derivative stays positive.

All other nonzero zeros are regular on the first circle and continue with positive Jacobian. The origin remains uniformly isolated, and the highest term bounds the remaining zeros uniformly. Thus no zeros occur on another scale. The ℓ local increases add exactly 2ℓ to the count $d+1$, with the stated orientation counts. On the circle itself every local singular cluster has exactly one point, proving its distinct count as well. □

Verification scope. The index bound, target summation, and continuation arguments are analytic proofs. The supplementary program `verify_harmonic_minimum.py` checks the critical-circle and radius identities and the first-increase expansion symbolically, exact rational inequalities, boundary orbit counts, derivative gcds, explicit two-zero examples, and the exact-degree padding formulas. It also checks the degree arithmetic in the completion corollary below. Its report is `harmonic_minimum_verification.json`; finite diagnostics do not replace the all-degree arguments.

20.4 A complete parameter-plane count for the trinomial family

The first critical circle is only the beginning of the parameter picture. For $k < m$ the same family has an exact global maximum and an explicit count on every regular parameter region. The related case with a linear analytic term has antecedents: Work studies $az^n + b\bar{z}^m + z$ for positive real a, b and obtains a range ending at $n + 2m + 2$ under his stated conditions [Work21]. The argument below treats complex parameters and all $1 \le k < m$; it is self-contained and is not a claim of exhaustive priority.

Throughout this subsection put

$$1 \le k < m < n, \quad d = n-k, \quad h = m-k, \quad s = n-m, \quad K = m+k, \quad L = n+m, \quad \ell = \gcd(d, K).$$

In particular $h, s > 0$ and $d = h + s$. The case $k = m$ is handled separately in Section 20.5: its origin changes index at $|\varepsilon| = 1$, so it cannot be obtained by substituting $h = 0$ in this argument.

Theorem 20.11 (Exact trinomial maximum and full distinct spectrum)**.** *For $F_\varepsilon(z) = z^n - z^k + \varepsilon\bar{z}^m$, $\varepsilon \in \mathbb{C}$, one has*

$$\max_{\varepsilon\in\mathbb{C}} N(F_\varepsilon) = n + 2m + k + 1. \tag{20.17}$$

The single integer choice $\varepsilon = 3$ attains this maximum for every $1 \le k < m < n$. All its nonzero zeros are nondegenerate; the origin has index k and is nondegenerate exactly when $k = 1$. The full distinct-count spectrum within this fixed family is

$$\{d + 1 + \ell j : 0 \le j \le 2K/\ell\}. \tag{20.18}$$

If every nonzero zero is required to be nondegenerate, the spectrum is $\{d + 1 + 2\ell j : 0 \le j \le K/\ell\}$. These are statements about the displayed trinomial family, not the unrestricted harmonic maximum $M_{m,n}^{\mathrm{harm}}$.

We first give the parameter curve and the counting rule that prove the theorem. They make its maximum computable without a zero search. For $t \in [0, 2\pi d]$, set

$$v(t) = \frac{k - me^{it}}{n - me^{it}}, \qquad z(t)^d = v(t), \qquad \Gamma(t) = \frac{z(t)^k(1 - z(t)^d)}{\overline{z(t)}^m}, \tag{20.19}$$

using one continuous choice of the dth root over that interval. It closes after d turns of v and traverses the critical curve in the z-plane exactly once. Write $\Gamma(t) = \rho(t)e^{i\Theta(t)}$ with a continuous argument. Then

$$\rho(t) = \frac{d}{|n - me^{it}|^{s/d}|k - me^{it}|^{h/d}}, \qquad \Theta(t) = \frac{K}{d}\arg(k - me^{it}) - \frac{L}{d}\arg(n - me^{it}). \tag{20.20}$$

Changing the initial root only rotates the parametrization through one of its existing symmetry angles.

Lemma 20.12 (Monotone argument of the critical values)**.** *The critical parameter curve never passes through zero. Its continuous argument is strictly increasing, with total increase $2\pi K$. More precisely, with $x = \cos t$,*

$$\Theta'(t) = \frac{m^2(1+x)\{(m-k)(n-m)+2kn(1-x)\}}{(k^2+m^2-2kmx)(n^2+m^2-2nmx)}. \tag{20.21}$$

It vanishes only at $t \equiv \pi \pmod{2\pi}$. Every ray from zero therefore has exactly K critical preimages, counting distinct points of the critical curve in the z-plane.

Proof. The numerator circle $k - me^{it}$ contains zero in its interior, whereas the denominator circle $n - me^{it}$ does not. Their arguments increase by 2π and zero, respectively, over each period. Formula (20.20) consequently gives the total increase $2\pi K$ over d periods. Differentiating the two arguments gives

$$\frac{1}{d}\left\{K\frac{m^2-km\cos t}{k^2+m^2-2km\cos t} - L\frac{m^2-nm\cos t}{n^2+m^2-2nm\cos t}\right\}.$$

Factoring its numerator gives (20.21). Both denominators are positive, and $(m-k)(n-m) > 0$. The derivative is positive away from its isolated stated zeros, which proves strict monotonicity and the ray count. □

For a fixed ray of argument α, let $\rho_1(\alpha), \dots, \rho_K(\alpha)$ be those K critical radii, with repetitions when different critical points have the same parameter value. The following formulas are exact.

Proposition 20.13 (All parameter regions and their boundaries)**.** *If $\varepsilon = re^{i\alpha} \neq 0$ is not on Γ, put $q = \#\{j : \rho_j(\alpha) < r\}$. Then*

$$N(F_\varepsilon) = d + 1 + 2q, \qquad (N_+^{\neq 0}, N_-^{\neq 0}) = (d+q, q). \tag{20.22}$$

Every singular nonzero zero away from the first-boundary cusp parameters of Corollary 20.10 is an ordinary fold. At a parameter with f such folds, put $q = \#\{j : \rho_j(\alpha) < r\}$ as before. Its distinct count is

$$N(F_\varepsilon) = d + 1 + 2q + f. \tag{20.23}$$

At each first-boundary cusp parameter the count is $d+1$. Also $N(F_0) = d+1$.

Proof. On $\mathbb{C} \setminus \{0\}$ consider the proper real-analytic map

$$H(z) = \frac{z^k - z^n}{\bar{z}^m}.$$

It tends to infinity at both ends of the punctured plane, because $h, s > 0$. Its real Jacobian has the sign of

$$|k - nz^d|^2 - m^2|1 - z^d|^2.$$

In the $v = z^d$ coordinate its critical circle has center $(nk-m^2)/(n^2-m^2)$ and radius $md/(n^2-m^2)$. The radius squared minus the center squared is $(m^2-k^2)/(n^2-m^2) > 0$. Thus the inverse image is a smooth Jordan curve around zero; it bounds a region Ω on which $J_H < 0$, while $J_H > 0$ outside. The parametrization (20.19) traverses it counterclockwise once. At $F_\varepsilon(z) = 0$, $z \neq 0$, one has $J_{F_\varepsilon} = |z|^{2m} J_H$.

Let $W(\varepsilon) = \operatorname{wind}(\Gamma, \varepsilon)$ for a parameter off the critical image. The argument principle for the real map $H - \varepsilon$ on Ω with a small disk about zero removed gives

$$-N_-^{\neq 0} = W(\varepsilon) - K.$$

Here the inner circle has counterclockwise winding K, since $H(re^{i\theta}) \sim r^{-h}e^{iK\theta}$ at zero. On a large circle the winding is L, so the degree on the punctured plane is $L - K = d$, giving $N_+^{\neq 0} - N_-^{\neq 0} = d$.

By Lemma 20.12, every intersection of Γ with the ray starting at ε and pointing away from zero has positive angular crossing. The winding is therefore

$$W(\varepsilon) = \#\{j : \rho_j(\alpha) > r\} = K - q.$$

For a ray through a cusp, use a nearby ray and continuity of the winding number; ε itself is off Γ. The origin contributes one distinct zero of index k to F_ε, proving (20.22).

The critical domain curve has a nonzero gradient everywhere. Away from $t \equiv \pi$, the derivative of its critical image does not vanish, by (20.21). These are ordinary fold points: the kernel is transverse to the critical curve, so the second derivative in the kernel direction is nonzero. The radial parameter direction is transverse to each such image arc, and (20.22) increases by two per critical preimage when that radius is crossed outward. At the fold itself the two zeros merge into one. Summing over its distinct critical preimages proves (20.23), also at intersections of several critical image arcs. The remaining points are exactly the first-boundary cusps already analyzed in Corollary 20.10. Finally $F_0 = z^k(z^d - 1)$. □

Corollary 20.14 (Optimal outer radius and integer maximizers)**.** *The smallest number S for which every $|\varepsilon| > S$ attains the maximum in (20.17) is*

$$S = \frac{d}{s^{s/d}h^{h/d}} \le 2. \tag{20.24}$$

Equality $S = 2$ holds exactly when $n + k = 2m$. On $|\varepsilon| = S$ the exceptional parameters are precisely

$$S \exp\left(\frac{\pi i K(2j+1)}{d}\right), \qquad 0 \le j < d. \tag{20.25}$$

There are d/ℓ different values, each with ℓ fold zeros and distinct count $n+2m+k+1-\ell$. All other parameters on that circle attain the maximum. The coefficient 3 is the smallest positive integer value of ε that maximizes every family covered by this subsection.

Proof. Formula (20.20) strictly increases with $\cos t$. Its minimum is the previous inner radius

$$R = \frac{d}{(n+m)^{s/d}(m+k)^{h/d}},$$

and its maximum is (20.24), attained precisely at $t \equiv 0$. The corresponding arguments give (20.25). The residue count gives the stated multiplicities; these points are folds because (20.21) is positive there. Equations (20.22)–(20.23) give the counts, so no smaller outer radius works.

Writing $p = s/d \in (0,1)$, one has $\log S = -p\log p - (1-p)\log(1-p) \le \log 2$; elementary differentiation gives equality exactly at $p = 1/2$. Thus $\varepsilon = 3$ always lies beyond the critical image and has $q = K$. To check the claimed integer optimality, take $(n,m,k) = (7,5,3)$, with $d = 4$, $K = 8$, and $\ell = 4$. On the positive real parameter ray the critical radii are $R = 1/\sqrt{6}$ and $S = 2$, each repeated four times. The first consists of cusps and the second of folds. Equations (20.22)–(20.23) give $N(F_1) = 13$, $N(F_2) = 17$, and $N(F_3) = 21$. Neither smaller positive integer is a uniform maximizer. □

Proof of Theorem 20.11. For regular parameters, $0 \le q \le K$ gives the two extremal counts. At folds, $q + f \le K$ makes $d + 1 + 2q + f \le d + 1 + 2K$. The cusp count is the minimum. Corollary 20.14 proves attainment of the maximum by the stated integer polynomial.

It remains to show that the full spectrum has no omissions. For $\zeta^\ell = 1$, $F_\varepsilon(\zeta z) = \zeta^k F_\varepsilon(z)$, so all nonzero zeros lie in orbits of size ℓ. Every count is therefore congruent to $d+1$ modulo ℓ. The critical image has intrinsic covering multiplicity ℓ: a shift by $2\pi j$ in t leaves its value unchanged precisely when Kj is divisible by d.

There is a ray on which no additional critical radii coincide. Indeed, equal moduli in (20.20) imply $t' \equiv t$ or $t' \equiv -t \pmod{2\pi}$. The first case gives the intrinsic repetitions just described. In the second, conjugation symmetry of the parametrization and its $2\pi K/d$ phase increment restrict a coincident ray to an angle in $(\pi\ell/d)\mathbb{Z}$ modulo 2π. The ray $\alpha = \pi/(2d)$ avoids all these angles and all cusps. It has exactly K/ℓ distinct critical radii, each repeated ℓ times. As its radius increases, the regular counts are $d+1, d+1+2\ell, \dots, d+1+2K$; at each intervening radius, (20.23) supplies the missing middle count. This proves (20.18). At cusp parameters there are singular nonzero zeros, so they add no counts to the stated nonzero-regular spectrum. □

For $k = 1$, the maximizing integer trinomial $z^n - z + 3\bar{z}^m$ has $n + 2m + 2$ nondegenerate zeros, of which $n + m + 1$ are positive and $m + 1$ negative. For example $z^3 - z + 3\bar{z}^2$ has exactly nine nondegenerate zeros. For $k > 1$, the origin is necessarily degenerate; neither the regularity claim nor the sharp family maximum should be silently transferred to a larger coefficient space.

20.5 The equal lower-degree endpoint and its arithmetic jump

The remaining endpoint $k = m$ has a different answer. Its critical arcs end at the origin, and zeros can pass through the origin when the parameter crosses the unit circle. The following result closes this endpoint without taking a singular limit in the previous theorem. Put

$$F_\varepsilon(z) = z^n - z^m + \varepsilon\bar{z}^m, \qquad d = n - m > 0, \quad \ell = \gcd(d, 2m), \quad J = 2m/\ell.$$

Here J is an integer, and ℓ is even whenever J is odd.

Theorem 20.15 (Exact endpoint maximum and spectrum)**.** *For every $n > m \geq 1$,*

$$\max_{\varepsilon\in\mathbb{C}} N(F_\varepsilon) = d + 1 + 2\ell\left\lceil \frac{m}{\ell} \right\rceil = \begin{cases} n+m+1, & J \text{ even}, \\ n+m+\ell+1, & J \text{ odd}. \end{cases} \tag{20.26}$$

Its full distinct-count spectrum is exactly

$$\left\{ d + 1 + \ell j : 0 \leq j \leq 2\left\lceil \frac{m}{\ell} \right\rceil \right\}. \tag{20.27}$$

Both extrema are attained with $\varepsilon \in \mathbb{Q}(i)$; multiplying by a common denominator gives Gaussian-integer polynomial coefficients. The maximum occurs on an open set of parameters with $|\varepsilon| < 1$. Its nonzero zeros are nondegenerate, and its origin has index m.

For every $|\varepsilon| > 1$, the distinct count is $n + m + 1$; all nonzero zeros are positive and the origin has index $-m$. When J is odd, no Gaussian-integer value of ε itself attains the family maximum.

Here is a count on the entire unit circle, including its exceptional parameters. Besides completing the maximum proof, it records precisely what survives when several zeros collapse into the origin.

Proposition 20.16 (The unit-circle fibers)**.** *Let $\varepsilon = e^{i\alpha}$ and, for $0 \leq j < 2n$, put*

$$\theta_j = \frac{\alpha + (2j+1)\pi}{2n}, \qquad c_j = 2(-1)^j \sin(m\theta_j - \alpha/2). \tag{20.28}$$

Every nonzero zero is positive and nondegenerate. There is exactly one on each ray θ_j for which $c_j > 0$, namely $c_j^{1/d} e^{i\theta_j}$, and there are no others. Thus $N(F_{e^{i\alpha}}) = 1 + \#\{j : c_j > 0\}$. As α varies, its possible values are exactly

$$U = \begin{cases} \{n-\ell+1, n+1\}, & J \text{ even}, \\ \{n+1-\ell/2, n+1+\ell/2\}, & J \text{ odd}. \end{cases} \tag{20.29}$$

For even J, the lower value occurs precisely when some $c_j = 0$; otherwise the upper value occurs. For odd J, a parameter with some $c_j = 0$ has the lower value, and the two values alternate on the complementary parameter arcs.

Proof. The nonzero equation, after multiplication by $e^{im\theta}$, is equivalent to

$$r^d e^{i(n\theta - \alpha/2)} = 2i \sin(m\theta - \alpha/2).$$

Its right side is purely imaginary and nonzero. Taking the two possible arguments and then the positive modulus gives exactly (20.28). At a zero, $v = z^d$ satisfies $|1 - v| = 1$, hence $2\Re v = |v|^2$. The Jacobian is

$$J_{F_\varepsilon}(z) = |z|^{2m-2}\big(|nv - m|^2 - m^2\big) = n(n-m)|z|^{2n-2} > 0.$$

Write $c_j/2 = \sin(\pi u_j)$, where

$$u_j = \frac{m - d\alpha/\pi}{2n} + \frac{n+m}{n} j.$$

Modulo 2, these $2n$ numbers form a regular grid of $T = 2n/\ell$ points, each repeated ℓ times, because $\gcd(n+m, 2n) = \ell$. Moreover $T = 2d/\ell + J$ has the parity of J. If T is even, an open semicircle contains $T/2$ grid points, or $T/2 - 1$ when its endpoints are grid points. If T is odd, it contains either $(T-1)/2$ or $(T+1)/2$; when an endpoint is a grid point, the count is $(T-1)/2$. Multiplication by ℓ proves the asserted possibilities. Varying α through 2π moves the grid through at least one full grid spacing, since $d/\ell \geq 1$, so both occur. □

For parameters inside the unit disk, the critical values can again be listed along each ray. There are d arcs, parametrized for $0 < t < 2\pi$ and $0 \leq j < d$ by

$$\begin{aligned} v(t) &= \frac{m(1-e^{it})}{n - me^{it}}, \qquad z_j(t)^d = v(t), \qquad \arg z_j(t) = \frac{\arg v(t) + 2\pi j}{d}, \\ \Gamma_j(t) &= \frac{z_j(t)^m (1 - z_j(t)^d)}{\overline{z_j(t)}^m} = \rho(t) e^{i\Theta_j(t)}, \qquad \rho(t) = \frac{d}{|n - me^{it}|}, \\ \Theta_j(t) &= \frac{m}{d}(t - \pi) - \frac{n+m}{d} \arg(n - me^{it}) + \frac{4\pi m j}{d}. \end{aligned} \tag{20.30}$$

The denominator has its continuous principal argument and $\arg v(t) = (t-\pi)/2 - \arg(n - me^{it})$. Direct differentiation gives

$$\Theta_j'(t) = \frac{mn(1+\cos t)}{n^2 + m^2 - 2mn\cos t} \geq 0. \tag{20.31}$$

It is positive except at $t = \pi$. The modulus is strictly below one, tends to one at both endpoints, and has its minimum $R = d/(n+m)$ at $t = \pi$. The argument range is

$$\left(\frac{4\pi m j}{d} - \frac{\pi m}{d}, \frac{4\pi m j}{d} + \frac{\pi m}{d} \right). \tag{20.32}$$

Proposition 20.17 (Counting inside the unit disk)**.** *For $\varepsilon = re^{i\alpha}$ with $0 < r < 1$ off the arcs, let q be the number of their critical preimages on the ray α whose parameter modulus is less than r. Preimages on different domain arcs are counted separately. Then*

$$N(F_\varepsilon) = d + 1 + 2q, \qquad (N_+^{\neq 0}, N_-^{\neq 0}) = (d + q, q). \tag{20.33}$$

At a parameter with f ordinary folds and q strictly smaller critical radii, the distinct count is $d+1+2q+f$. The only other nonzero singularities are the first-boundary cusps on $|\varepsilon| = R$, already counted by Corollary 20.10; their distinct count is $d+1$.

Proof. For $H(z) = e^{2im\arg z}(1 - z^d)$ on the punctured plane, the negative-Jacobian region is the union of d lobes given by

$$(n + m)|z^d|^2 < 2m\Re(z^d).$$

Their boundaries are (20.30) and their closures meet only at zero. These inequalities also imply $|H(z)| < 1$ in every negative lobe. After deleting a small disk at zero, apply the argument principle to $H - \varepsilon$ on their union. The image of each outer boundary is Γ_j, with increasing argument. The inner boundary tends to the unit-circle arc with the same endpoint arguments but opposite orientation. For a generic ray α, the latter contributes one negative crossing for each critical arc on that ray. The outer boundary contributes one positive crossing exactly when that critical radius exceeds r. Their difference is $-q$; hence $N_-^{\neq 0} = q$. Continuity extends this calculation to any ray on which the parameter itself is a regular value. Since the origin has index m, total degree n gives $N_+^{\neq 0} - N_-^{\neq 0} = d$.

The domain critical curve is smooth away from the origin. By (20.31) its image derivative does not vanish except at $t = \pi$. Every other critical point is therefore a fold, and the outward radial crossing creates two zeros. At the fold they merge into one. At $t = \pi$ use the cusp calculation of Corollary 20.10. □

Proof of Theorem 20.15. Put $D = d/\ell$, so $\gcd(D, J) = 1$. The centers in (20.32), reduced modulo 2π, form a regular D-point grid, each repeated ℓ times. Each interval has length $\pi J/D$. A ray which avoids the endpoints is consequently covered by either $\ell\lfloor J/2\rfloor$ or $\ell\lceil J/2\rceil$ arcs. The maximum

$$I_{\max} = \ell\lceil J/2\rceil = \ell\,\lceil m/\ell\rceil$$

occurs on an open set of ray directions. Along any such ray, take $r < 1$ larger than all its finitely many critical radii. Proposition 20.17 gives $d + I_{\max}$ positive and $I_{\max}$ negative nonzero zeros, together with the origin. The distinct count is therefore $d + 1 + 2I_{\max}$. At a fold, $q + f \le I_{\max}$ bounds its count by the same number; cusp fibers have the minimum count.

For $|\varepsilon| > 1$ the negative lobes have no preimages and there are no critical preimages. The origin has index $-m$; degree n therefore gives $n+m$ positive nonzero zeros. Proposition 20.16 covers $|\varepsilon| = 1$ and bounds its count by $n + m + 1$. Finally $F_0 = z^m(z^d - 1)$ has $d + 1$ distinct zeros. These facts prove the global maximum and minimum. The maximizing open set inside the disk contains points in $\mathbb{Q}(i)$, while a sufficiently small positive rational parameter attains the minimum. If J is odd, parameters of modulus at least one cannot attain the larger maximum; a nonzero Gaussian integer has modulus at least one, and zero attains only the minimum.

For the spectrum, $F_\varepsilon(\zeta z) = \zeta^m F_\varepsilon(z)$ for $\zeta^\ell = 1$. All nonzero zeros therefore occur in free orbits of size ℓ. Choose a ray having $I_{\max}$ critical preimages and avoiding endpoints, cusps, and self-intersection directions. Only finitely many directions are excluded: equality of two critical moduli

forces $t' = t$ or $t' = 2\pi - t$; the first produces only the intrinsic ℓ repetitions, and the second forces the ray into a finite set by the strict argument monotonicity. The chosen ray thus has $I_{\max}/\ell$ distinct fold radii. At its regular intervals the counts increase by 2ℓ, and each fold supplies the intervening increment ℓ. This realizes every number in (20.27). The proved bounds and the orbit congruence exclude all other numbers. □

Corollary 20.18 (Nonzero-regular spectrum and real parameters)**.** *If every nonzero zero is required to be nondegenerate, the exact spectrum is*

$$\{d+1+2\ell j : 0 \le j \le \lceil m/\ell \rceil\} \ \cup \ \{n+m+1\} \ \cup \ U, \tag{20.34}$$

where U is (20.29). The origin is also nondegenerate exactly when $m = 1$ and $|\varepsilon| \neq 1$.

The maximum restricted to real ε equals the complex-parameter maximum, except when $J \equiv 3 \pmod 4$ and $D = (n-m)/\ell$ is even. In that exceptional case,

$$\max_{\varepsilon \in \mathbb{R}} N(F_\varepsilon) = n+m+1, \qquad \max_{\varepsilon \in \mathbb{C}} N(F_\varepsilon) = n+m+\ell+1. \tag{20.35}$$

Thus allowing a complex parameter can strictly increase the maximum.

Proof. Inside the unit disk, regular counts are precisely the first set in (20.34), by the generic ray just used. Outside it the count is $n+m+1$, and on its boundary all nonzero zeros are regular with exactly the values in U. The assertion about the origin follows from its homogeneous part $-z^m + \varepsilon \bar{z}^m$.

For odd J, the number of covering arcs on the positive real ray is the number of grid centers within $J/4$ grid spacings of zero, multiplied by ℓ. It is $\ell\lceil J/2 \rceil$ if $J \equiv 1 \pmod 4$, and $\ell\lfloor J/2 \rfloor$ if $J \equiv 3 \pmod 4$. The negative real ray shifts the position by $D/2$ grid spacings: it changes between these two counts exactly when D is odd. Thus a real ray reaches the larger maximum except in the stated case. In that case all real parameters inside the disk have at most $n+m-\ell+1$ zeros; the unit circle has at most $n+m+1$, and every real $|\varepsilon| > 1$ attains $n+m+1$. For even J, the outside count is already the global maximum. □

For example, $z^5 - z^3 + \varepsilon\bar{z}^3$ has maximum 11, attained for negative real ε sufficiently close to -1 from inside the disk. In contrast, $z^7 - z^3 + \varepsilon\bar{z}^3$ has real-parameter maximum 11 and complex-parameter maximum 13. At $m = 1$, the endpoint maximum is $n+4$ for odd n and $n+2$ for even n. These sharp family values do not assert sharpness of the unrestricted harmonic maximum.

20.6 A sharp three-term bound when the conjugate degree is one

The endpoint theorem settles the lower analytic term z. For a higher lower term there is a different gcd formula. The negative-Jacobian components can be proved injective in logarithmic coordinates, which also controls singular parameters.

Theorem 20.19 (A gcd formula for the linear-conjugate family)**.** *Let $n > k \ge 2$, $d = n-k$, and $\ell = \gcd(d, k+1) = \gcd(n+1, k+1)$. For $F_\varepsilon = z^n - z^k + \varepsilon\bar{z}$,*

$$\max_{\varepsilon \in \mathbb{C}} N(F_\varepsilon) = n+2+2\ell. \tag{20.36}$$

Every real parameter in the nonempty interval

$$\frac{d(k+1)^{(k-1)/d}}{(n+1)^{(n-1)/d}} < \varepsilon < \frac{d(k-1)^{(k-1)/d}}{(n-1)^{(n-1)/d}} \tag{20.37}$$

attains this maximum, with all zeros nondegenerate. For $\varepsilon \neq 0$ the complete distinct-count spectrum is

$$\{n+2,\ n+2+\ell,\ n+2+2\ell\}. \tag{20.38}$$

Only the first and last values occur when all zeros are nondegenerate. At $\varepsilon = 0$ the distinct count is $d+1$. In particular, the maximum has a real-integer coefficient example after choosing a rational parameter in (20.37) and clearing denominators.

Proof. For nonzero z let $H(z) = (z^k - z^n)/\bar{z}$. Its Jacobian has the sign of $|k - nz^d|^2 - |1 - z^d|^2$. In $v = z^d$, the negative region is the disk Δ with diameter

$$a = \frac{k-1}{n-1}, \qquad b = \frac{k+1}{n+1}, \qquad 0 < a < b < 1.$$

It lifts to d disjoint components Ω_j in the z-plane. We prove that H is injective even on each closed component.

Put $c = (a+b)/2$, $r = (b-a)/2$, and use the principal logarithm $\xi = \log v = x + iy$ on $\overline{\Delta}$. Since $c^2 - r^2 = ab > 0$, its logarithmic image is

$$\left\{(x,y) : |y| < \frac{\pi}{2}, \quad \log\cosh\left(x - \tfrac{1}{2}\log(ab)\right) - \log\cos y < \log\frac{c}{\sqrt{ab}}\right\}. \tag{20.39}$$

The defining function has positive definite Hessian, so this domain is strictly convex. On the branch Ω_0,

$$\log H = P(\xi) - \frac{1}{d}\bar{\xi}, \qquad P(\xi) = \frac{k}{d}\xi + \log(1 - e^{\xi}), \qquad |P'(\xi)| = \frac{|k - nv|}{d|1 - v|} < \frac{1}{d}. \tag{20.40}$$

For any two different points of the closed logarithmic domain, the open segment between them lies in its interior. Integrating P' along that segment gives $|P(\xi_1) - P(\xi_2)| < |\xi_1 - \xi_2|/d$. Consequently (20.40) is injective on the closure.

To pass from $\log H$ to H, and to separate its different component images, we need an argument bound. Parametrize the critical circle by $v(t) = (k - e^{it})/(n - e^{it})$, $0 \le t \le 2\pi$. The argument of its critical image on the first component is

$$\Theta(t) = \frac{k+1}{d}\arg(k - e^{it}) - \frac{n+1}{d}\arg(n - e^{it}).$$

It is zero at $t = 0, \pi, 2\pi$ and negative for $0 < t < \pi$. Indeed its derivative has the sign of $(1 + \cos t)\{-(k-1)(n-1) + 2kn(1 - \cos t)\}$, so there is exactly one interior minimum on $(0, \pi)$. Writing $A_j = -\arg(j - e^{it}) > 0$ on that interval, we obtain

$$0 < -d\Theta(t) = (k+1)A_k - (n+1)A_n < (k+1)\arcsin(1/k) \le \frac{\pi}{2}.$$

The last inequality follows from the increase of $\arcsin u/u$ on $(0,1)$ and $1/k \le 1/2$. Reflection treats the other half-circle. The harmonic maximum principle applied to $\Im \log H$ gives the same strict bound on the closed component. Hence its image lies in $|\arg H| < \pi/(2d)$, and exponentiation preserves injectivity.

The other images are rotations by $2\pi(k+1)j/d$. There are d/ℓ different images, each repeated ℓ times. Their closed angular sectors are disjoint because their centers are spaced by $2\pi\ell/d$ and their half-widths are less than $\pi/(2d)$. For a regular nonzero parameter, therefore, the number of nonzero negative zeros is either zero or ℓ. The origin is always a negative nondegenerate zero when $\varepsilon \neq 0$. Total degree n now gives exactly $n+2$ or $n+2+2\ell$ distinct zeros.

For attainment, along the real diameter $a < v < b$,

$$H(v^{1/d}) = v^{(k-1)/d}(1-v)$$

strictly decreases: its logarithmic derivative is $\{(k-1)-(n-1)v\}/\{dv(1-v)\} < 0$. Its endpoint values are respectively the right and left bounds in (20.37). Every intervening parameter has exactly ℓ negative nonzero preimages and no critical preimages; the other component groups have disjoint sectors. This proves the nonempty maximizing interval.

It remains to include singular parameters in the upper bound and spectrum. Injectivity on the closed components shows that each nonzero critical value has exactly ℓ critical preimages, all related by the same rotation symmetry. They have the same local index ν. For a nearby parameter outside that closed image, all zeros in each local neighborhood are positive, so $\nu \geq 0$. Lemma 20.4, applied to the harmonic map with conjugate degree one, gives $\nu \leq 1$. At the critical parameter the only negative regular zero is the origin. The degree identity then yields

$$N(F_\varepsilon) = n + 2 + \ell(1-\nu) \in \{n+2, n+2+\ell\}.$$

There are ordinary folds: at $t = 0$ the critical domain curve is smooth and the derivative of its image is nonzero, since $\Theta'(0) < 0$. Their local index is zero, so the middle value is attained. Small nonzero parameters lie outside all closed component images and give $n+2$. Together with the maximizing interval this proves (20.38) and the global upper bound. The exceptional zero parameter is immediate. □

This result permits an exact optimization over all three-term harmonic polynomials with conjugate degree one. Count coefficients after combining the two constant terms in $p + \overline{q}$; the available monomials are $1, z, \dots, z^n, \bar{z}$. Let $H^{(3)}_{1,n}$ denote the maximum distinct count with $\deg p = n$, $\deg q = 1$, and at most three nonzero coefficients. For an integer $N \geq 2$, write $D(N)$ for its largest proper positive divisor, with $D(N) = 1$ when N is prime.

Theorem 20.20 (The sharp three-term harmonic maximum)**.** *One has*

$$H^{(3)}_{1,2} = 4, \qquad H^{(3)}_{1,3} = 7, \qquad H^{(3)}_{1,n} = n + 2 + 2D(n+1) \quad (n \geq 4). \tag{20.41}$$

All these maxima admit real-integer coefficients and nondegenerate zeros. For odd $n \geq 5$ the last formula is $2n+3$; when $n+1$ is prime it is $n+4$. Compared with the unrestricted sharp value $M^{\mathrm{harm}}_{1,n} = 3n-2$, three nonzero coefficients suffice exactly for $n \in \{2,3,5\}$. For every other $n \geq 2$, any unrestricted extremizer requires at least four nonzero coefficients. This is a sparsity statement for the harmonic class with conjugate degree one, separate from the total-degree polyanalytic quantity s_n.

Proof. Apart from the two mandatory terms az^n and $b\bar{z}$, the only possible third term is cz^k, $0 \leq k < n$. For $1 \leq k < n$ and $c \neq 0$, a complex source dilation and a nonzero target factor normalize this polynomial to $z^n - z^k + \varepsilon\bar{z}$ with $\varepsilon \neq 0$. For $k \geq 2$, Theorem 20.19 applies. For $k = 1$, Theorem 20.15 gives $n+4$ for odd n and $n+2$ for even n.

For $k = 0$, the maximum is $n+2$. Here is a direct justification including singular targets. The negative-Jacobian region of $az^n + b\bar{z}$ is a disk, and on its interior $|naz^{n-1}| < |b|$. Integrating this analytic derivative along segments proves that $az^n + b\bar{z}$ is injective on the closed disk. A regular target has at most one negative preimage, hence at most $n+2$ preimages in total. At a critical target there is at most one singular preimage. Moving the constant term slightly into the image of

the negative disk preserves every regular zero and supplies a zero near that singular preimage. Thus the same upper bound holds there. The binomial $z^n+\bar{z}$ already has $n+2$ nondegenerate zeros, and a small nonzero rational constant preserves them if exactly three terms are desired.

For $n \geq 4$, the maximum of $\gcd(n+1, k+1)$ over $2 \leq k < n$ is $D(n+1)$. If $n+1$ is prime this is immediate. Otherwise $D(n+1) \geq 3$ and the choice $k = D(n+1) - 1$ realizes it. The endpoint and constant-term families never exceed the result. The cases $n = 2, 3$ follow separately from the same formulas. All maximizing constructions above permit rational parameters and real-integer coefficients after multiplication by a denominator.

For $n \geq 4$, equality with $3n-2$ would require $D(n+1) = n-2$. Since $n-2$ must divide $n+1$, it divides three; the only such $n \geq 4$ is $n = 5$, which does attain equality. The two smaller degrees also attain it. This proves the final claims. □

For an independently transparent example, consider

$$12z^5 - 12z^2 + 5\bar{z}.$$

After division by 12, set $z = re^{i\theta}$ in its nonzero equation and divide by z^2. Its imaginary part is $\sin(3\theta)(r^3 - 5/(12r)) = 0$. The branch $\sin(3\theta) = 0$ gives two positive solutions of $r^4 - r + 5/12 = 0$, each with three angles. The other branch has $r^4 = 5/12$ and $\cos(3\theta) = 1/(2r^3)$, giving six more zeros. The strict inequalities

$$2^{-4/3} < 5/12 < 3/4^{4/3}$$

ensure these branches are distinct and nondegenerate. Together with the origin this is a thirteen-zero harmonic trinomial attaining $3n-2$ in degree five. Likewise $8z^7 - 8z^3 + 3\bar{z}$ attains the three-term maximum of seventeen in degree seven, while the unrestricted maximum there is nineteen.

20.7 Higher conjugate degree under angular separation

The logarithmic injectivity argument extends beyond conjugate degree one. The extra issue is that the rotated images of different negative components can overlap. The following explicit criterion prevents that overlap, and it holds for every triple with $m = 2 < k < n$. Define, for real $k > m > 0$,

$$B(k,m) = (k+m)\arctan\frac{m\sqrt{3k+m}}{(2k+m)\sqrt{k-m}} - \frac{m\sqrt{(k-m)(3k+m)}}{2k}. \tag{20.42}$$

Theorem 20.21 (A sharp separated-component formula)**.** *Let $n > k > m \geq 1$, $d = n-k$, and $\ell = \gcd(d, k+m) = \gcd(n+m, k+m)$. Suppose*

$$B(k,m) \leq \pi\ell. \tag{20.43}$$

For $F_\varepsilon = z^n - z^k + \varepsilon\bar{z}^m$ one then has

$$\max_{\varepsilon\in\mathbb{C}} N(F_\varepsilon) = n + m + 1 + 2\ell. \tag{20.44}$$

Every real parameter in the nonempty interval

$$\frac{d(k+m)^{(k-m)/d}}{(n+m)^{(n-m)/d}} < \varepsilon < \frac{d(k-m)^{(k-m)/d}}{(n-m)^{(n-m)/d}} \tag{20.45}$$

attains the maximum. For $\varepsilon \neq 0$, the complete spectrum of distinct counts is

$$\{n+m+1,\ n+m+1+\ell,\ n+m+1+2\ell\}. \tag{20.46}$$

The first and last values are exactly those attained with every nonzero zero regular. The origin always has index $-m$; it is nondegenerate exactly when $m=1$. At $\varepsilon=0$ the count is $d+1$. Rational parameters in (20.45) give real-integer examples after clearing denominators.

Condition (20.43) holds whenever $\ell \geq m$, or $k-m \geq m^2/(2\ell)$. In particular, for every $n>k>2$,

$$\boxed{\max_{\varepsilon\in\mathbb{C}} N(z^n - z^k + \varepsilon\bar{z}^2) = n+3+2\gcd(n+2,k+2).} \tag{20.47}$$

Proof. For nonzero z, put $H(z)=(z^k-z^n)/\bar{z}^m$ and $v=z^d$. The sign of J_H is the sign of $|k-nv|^2-m^2|1-v|^2$. Its negative region is the inverse image of the disk with diameter

$$a=\frac{k-m}{n-m}, \qquad b=\frac{k+m}{n+m}, \qquad 0<a<b<1.$$

It has d disjoint components. On the first one, use $\xi=\log v$ to write

$$\log H = P(\xi) - \frac{m}{d}\bar{\xi}, \qquad P(\xi)=\frac{k}{d}\xi+\log(1-e^{\xi}), \qquad |P'(\xi)|<\frac{m}{d}.$$

The logarithmic disk is strictly convex by (20.39). The strict segment inequality in the proof of Theorem 20.19, with $1/d$ replaced by m/d, proves injectivity of $\log H$ on its closure.

Parametrize the critical circle by $v(t)=(k-me^{it})/(n-me^{it})$. The continuous argument on the first critical component is

$$\Theta(t)=\frac{k+m}{d}\arg(k-me^{it})-\frac{n+m}{d}\arg(n-me^{it}).$$

It vanishes at $0,\pi,2\pi$ and is negative on $(0,\pi)$. For completeness, its derivative has the sign of

$$(1+\cos t)\{-(k-m)(n-m)+2kn(1-\cos t)\}.$$

There is one interior minimum on this half-circle, and reflection treats the other half. For $0<t<\pi$, set $s=\sin t$, $c=\cos t$, and $A_j=\arctan(ms/(j-mc))$. Then

$$0<-d\Theta(t)=(k+m)A_k-(n+m)A_n<(k+m)A_k-m\sin t. \tag{20.48}$$

Indeed, $\arctan x \geq x/(1+x^2)$ for $x\geq 0$, and

$$(n+m)(n-mc)-(n^2+m^2-2nmc)=m(n-m)(1+c)\geq 0.$$

The first inequality is strict for $x>0$, which proves the strict last inequality in (20.48).

Write $f_{k,m}(t)=(k+m)A_k-m\sin t$. Direct differentiation gives

$$f'_{k,m}(t)=\frac{m^2(1+\cos t)\{2k\cos t-(k+m)\}}{k^2+m^2-2km\cos t}.$$

Its maximum is exactly $B(k,m)$, attained when $\cos t=(k+m)/(2k)$. Therefore $d\max|\Theta|<B(k,m)$. The harmonic maximum principle for $\Im\log H$ gives the same bound inside each component. Under (20.43) its closed image lies in a sector of half-width strictly less than $\pi\ell/d\leq\pi$. Exponentiation preserves the injectivity just proved. The other images are rotations by $2\pi(k+m)j/d$: there are d/ℓ distinct closed images, each repeated ℓ times, and their sectors are pairwise disjoint.

A regular nonzero target thus has either zero or ℓ negative nonzero preimages. The origin of F_ε has local leading term $\varepsilon\bar{z}^m$ and index $-m$. Since the degree at infinity is n, the two regular-nonzero counts are $n+m+1$ and $n+m+1+2\ell$. On the positive diameter $a<v<b$,

$$H(v^{1/d}) = v^{(k-m)/d}(1-v)$$

strictly decreases from the right to the left endpoint of (20.45). Every intervening target is in the interior of exactly ℓ component images and is not a critical value. This proves attainment and the asserted interval.

Each nonzero critical value has exactly ℓ critical preimages, related by the rotation symmetry. They have a common local index $\nu \in \{0,1\}$: a nearby target outside the local negative image shows $\nu \geq 0$, while Lemma 20.4 gives $\nu \leq 1$ because the conjugate derivative of F_ε is nonzero there. No other negative nonzero zeros occur at that target. Consequently its distinct count is $n+m+1+\ell(1-\nu)$. There are ordinary folds at $t=0$, where the critical-image derivative is nonzero. Thus the middle value is attained. Small nonzero parameters give the first value; this proves the full spectrum, including singular fibers.

We verify the simpler sufficient conditions. For fixed $0<t<\pi$, the function $f_{k,m}(t)$ strictly decreases as the real variable $k>m$ increases. This follows from $\arctan x < x$ and

$$(k+m)(k-mc) - (k^2+m^2-2kmc) = m(k-m)(1+c) \geq 0.$$

Its limit at $k=m$ is $m(\pi - t - \sin t)$, so $B(k,m) < \pi m$. Also

$$f_{k,m}(t) < \frac{m^2 \sin t(1+\cos t)}{k-m} \leq \frac{3\sqrt{3}\,m^2}{4(k-m)}.$$

Since $3\sqrt{3}/2 < \pi$, both sufficient conditions follow. Finally monotonicity in k gives, for $m=2$ and $k \geq 3$,

$$B(k,2) \leq B(3,2) = 5\arctan\frac{\sqrt{11}}{4} - \frac{\sqrt{11}}{3} < \frac{5\pi}{4} - \frac{\sqrt{11}}{3} < \pi.$$

Here $\sqrt{11}/4 < 1$ and $\pi/4 < \sqrt{11}/3$ justify the last two inequalities. Since $\ell \geq 1$, this proves (20.47) without any restriction on $n > k > 2$. □

For example, $7z^8 - 7z^3 + 4\bar{z}^2$ has exactly 21 distinct zeros. In (20.45), its parameter $4/7$ satisfies the exact rational tests

$$\frac{5^6}{10^6} < \left(\frac{4}{7}\right)^5 < \frac{5^5}{6^6}.$$

Its twenty nonzero zeros are regular, while the origin has index -2. The criterion is a sufficient condition for the displayed gcd formula in higher conjugate degree; no conclusion about overlapping component images is inferred when it fails. Nor does (20.47) alone optimize over every harmonic trinomial of conjugate degree two: the alternative support $az^n + b\bar{z}^2 + c\bar{z}$ must also be considered.

20.8 The complete three-term maximum in conjugate degree two

To optimize over all supports, the third monomial can also be $\bar{z}$ or a constant. We first give a small regularization fact that allows critical parameter values in these remaining families.

Lemma 20.22 (Two opposite target perturbations)**.** *Let H be real analytic near a finite fiber $H^{-1}(b)$ in the plane. Suppose its differential has rank at least one at every point of that fiber. There is a*

vector $v \neq 0$ such that, for every sufficiently small $t > 0$, the two fibers at $b + tv$ and $b - tv$ have only regular points near the original fiber, and the sum of their local distinct counts is twice the original count. In particular, one perturbation preserves at least the original number of points. If all nearby fibers remain in a common compact set, the perturbed fibers are globally regular and the same assertion applies to their total counts.

Proof. At a regular point both counts are one. At a rank-one point, linear target coordinates and analytic source coordinates put the map, after subtracting b, in the form $(u, y) \mapsto (u, g(u, y))$, where $Dg(0, 0) = 0$. Isolation of the fiber implies $g(0, y) = cy^r + O(y^{r+1})$ for some $r \geq 2$, $c \neq 0$. Choose v whose second target component at every rank-one point is nonzero; only finitely many lines are excluded. For a target $\pm t(a, b_0)$, substitute $u = \pm ta$ and $y = t^{1/r}s$. The remaining equation converges in C^1 on bounded s-sets to $cs^r = \pm b_0$. For odd r there is one simple real root for each sign; for even r there are two for one sign and none for the other. Every sufficiently small local solution has $|y|$ comparable to $t^{1/r}$: this follows by comparing cy^r with $\pm tb_0$ and the error $O(y^{r+1})+O(t|y|)+O(t^2)$. Compactness and local uniqueness therefore account for all local solutions, and they are regular. Sum over disjoint neighborhoods. The final compactness assumption excludes additional zeros away from them. □

Proposition 20.23. *For $n \geq 5$,*

$$\max_{\varepsilon \in \mathbb{C}} N(z^n - \bar{z}^2 + \varepsilon \bar{z}) = n + 4.$$

Every sufficiently small nonzero parameter attains this value with regular zeros. In degree four every member of the same family has at most ten distinct zeros.

Proof. For $z \neq 0$ write the equation as $H(z) = \varepsilon$, where

$$H(z) = \bar{z} - \frac{z^n}{\bar{z}}, \qquad u = \frac{z^n}{\bar{z}^2}.$$

The negative region of H is $|nu| < |1 + u|$. The corresponding disk in the u-plane contains the origin, and its inverse image is one radial domain Ω about zero. Its boundary is a smooth simple closed curve. Extend H continuously to $H(0) = 0$. It is differentiable there with negative Jacobian, since $n > 2$.

Parametrize the critical boundary by $u(t) = (ne^{it} - 1)^{-1}$, lifting its argument continuously to $\arg z = \arg u/(n + 2)$. As t runs from 0 to $2\pi(n + 2)$ the domain boundary is traversed clockwise once. The continuous argument of its image is

$$\Theta(t) = \arg(ne^{it} - 2) - \frac{n + 1}{n + 2} \arg(ne^{it} - 1), \qquad \Theta(0) = 0.$$

Its derivative is

$$\Theta'(t) = \frac{n^2(1 + \cos t)\{4 \cos t + n^2 - 3n - 2\}}{(n + 2)(n^2 + 4 - 4n \cos t)(n^2 + 1 - 2n \cos t)}. \tag{20.49}$$

For $n \geq 5$ this is positive except at isolated points. The image is therefore a star-shaped Jordan curve, traversed once counterclockwise. Its winding number about any regular target is zero or one. The degree formula on Ω, with the clockwise boundary orientation, identifies that number with the number of negative preimages of H. It is thus at most one.

For $\varepsilon \neq 0$ the original polynomial has a negative simple zero at the origin, and its degree at infinity is n. If its nonzero zeros are regular, its total count is $n + 2 + 2N_-(H - \varepsilon) \leq n + 4$. At a

nonzero critical zero the differential of H has rank one, because $H_z = -nz^{n-1}/\bar{z} \neq 0$. Lemma 20.22 extends this upper bound to critical targets. Nearby nonzero fibers stay away from zero and infinity. At $\varepsilon = 0$ the polynomial has $n+3$ distinct zeros. Since H is a local diffeomorphism at zero, small nonzero ε gives one small negative nonzero solution of $H = \varepsilon$, in addition to the negative simple origin of the polynomial. The $n+2$ nonzero old zeros persist, and compactness excludes further zeros. This proves attainment and the assertion for $n \geq 5$.

For $n = 4$, put $L = 2\pi/(n+2) = \pi/3$. On one interval $0 \leq t \leq 2\pi$, (20.49) is positive up to $t_* = 2\pi/3$, negative from t_* to $2\pi - t_*$ apart from its zero at π, and positive afterward. Moreover $\Theta(2\pi) = L$ and $\Theta(2\pi - t) = L - \Theta(t)$. At its maximum,

$$\Theta(t_*) = \frac{\pi}{6} - \arctan\frac{\sqrt{3}}{2} + \frac{5}{6}\arctan\frac{2}{\sqrt{3}} < \frac{5\pi}{18} < L.$$

We used $\arctan(\sqrt{3}/2) > \pi/6$ and $\arctan(2/\sqrt{3}) < \pi/3$. Thus the open image of this argument interval stays in $(0, L)$. For any generic direction in that sector the critical image meets the ray either once or three times; the argument derivatives have signs $+$, or $+, -, +$, respectively. The successive periods occupy disjoint angular sectors. The winding number about any regular target, computed by intersections with its outward radial ray, is consequently at most two: at most two positive crossings are available. Hence $N_-(H - \varepsilon) \leq 2$ and the polynomial count is at most $4 + 2 + 4 = 10$. The same target-perturbation lemma handles critical targets, while the zero parameter has seven zeros. □

Let $H^{(3)}_{2,n}$ denote the maximum for harmonic polynomials of exact degrees $\deg p = n$, $\deg q = 2$, with at most three nonzero monomials after combining constants. Let $H^{(3),\mathrm{reg}}_{2,n}$ impose, in addition, that every zero be nondegenerate. As before, $D(N)$ denotes the largest proper positive divisor of N.

Theorem 20.24 (Sharp quadratic-conjugate sparsity bounds)**.** *The complete distinct maxima are*

$$H^{(3)}_{2,3} = 9, \qquad H^{(3)}_{2,4} = 10, \qquad H^{(3)}_{2,5} = 11, \qquad H^{(3)}_{2,6} = 13, \tag{20.50}$$

and, for every $n \geq 7$,

$$\boxed{H^{(3)}_{2,n} = \max\{n+6,\ n+3+2D(n+2)\}.} \tag{20.51}$$

All these maxima admit real-integer coefficients. If every zero is required to be nondegenerate, the sharp answer is instead

$$\boxed{H^{(3),\mathrm{reg}}_{2,n} = n+6 \quad (n \geq 3),} \tag{20.52}$$

attained by $z^n - z + 3\bar{z}^2$. In particular, $H^{(3)}_{2,n} = 2n+5$ for even $n \geq 8$; if $n+2$ is prime, its value is $n+6$.

Proof. The mandatory monomials are az^n and $b\bar{z}^2$, with $ab \neq 0$. The third is either cz^k for $0 \leq k < n$, or $c\bar{z}$. Source dilations and nonzero target factors reduce each nonzero third coefficient to the corresponding normalized family below, without changing the number or regularity of zeros.

For $k = 1$, Theorem 20.11 gives the sharp value $n+6$, with the displayed integer maximizer. For $k = 2$, Theorem 20.15 gives $n+7$ when $n \equiv 2 \pmod 4$, and $n+3$ otherwise. For $k \geq 3$, Theorem 20.21 gives $n + 3 + 2\gcd(n+2, k+2)$.

The constant-term family has maximum $n+4$; this is the $m = 2$ instance of the known monomial-plus-conjugate-monomial bound [BBN22, Corollary 1.4, author version v2, p. 3]. One may also see

the upper bound directly. Normalize the map before translating the target to $T(z) = z^n + \bar{z}^2$. Its negative region is the disk of radius $R = (2/n)^{1/(n-2)}$, and the continuous argument of

$$T(Re^{i\theta}) = R^2 e^{-2i\theta}\left(1 + \frac{2}{n}e^{i(n+2)\theta}\right)$$

decreases monotonically through 4π. Differentiation gives a positive factor times $\cos((n+2)\theta) - 1$. Every regular target has at most two negative preimages; degree then bounds the total by $n+4$. Lemma 20.22 treats nonzero critical targets, and the zero target has $n+3$ distinct preimages. Small generic nonzero targets attain $n+4$. The alternative third term $c\bar{z}$ is bounded by $n+4$ for $n \geq 5$ and by ten for $n = 4$, by Proposition 20.23. In degree three the universal harmonic bound $n^2 = 9$ suffices. Omitting the third coefficient gives the binomial count $n+3$.

For $n \geq 7$, the maximum of $\gcd(n+2, k+2)$ over $3 \leq k < n$ is $D(n+2)$. If $n+2$ is prime all these gcds are one. If $D(n+2) \geq 5$, choose $k = D(n+2) - 2$. The only remaining composite case is $n+2 = 9$, where $k = 4$ gives gcd three. The endpoint value $n+7$ when $n \equiv 2 \pmod 4$ is already dominated by the gcd value: in this range $n \geq 10$ and $D(n+2) = (n+2)/2$. This proves (20.51). The four smaller degrees follow by checking the same finite list of supports. All maximizing families above allow real rational parameters.

If $k \geq 2$, the origin has zero differential for every member of the analytic-third-term family. Such a polynomial cannot have only nondegenerate zeros. All remaining supports have just been bounded by $n+6$, and the $k = 1$ integer example attains that value with regular zeros. This proves (20.52). □

For concrete low-degree and even-degree examples,

$$32z^6 - 32z^4 + 7\bar{z}^2, \qquad 16z^7 - 16z^4 + 5\bar{z}^2, \qquad 7z^8 - 7z^3 + 4\bar{z}^2$$

have respectively $13, 16, 21$ distinct zeros, attaining their three-term maxima. Each has a degenerate origin of index -2. In degree eight the largest count with all zeros regular and at most three terms is only 14. These are exact restrictions on coefficient support, not a determination of the full harmonic or polyanalytic maximum.

20.9 A further lower bound for the pure-head maximum

The rectangular equality region can improve the pure-head lower bound even at degrees where it does not prove pure-head saturation.

Proposition 20.25. *If $1 \leq a \leq m < n$, $b > a$, and $a + b < n$, then*

$$E_{m,n} \geq M_{a,b} + n - b + a. \tag{20.53}$$

The count on the right is attained exactly by a pure-head polynomial of exact conjugate degree m, over $\mathbb{Q}(i)$, with only nondegenerate zeros. In particular, for every $m \geq 1$ and $n \geq 4m + 1$,

$$E_{m,n} \geq n + 10m^2 - 2m. \tag{20.54}$$

In (20.54) the coefficients can be real rational, and hence integer after a common rescaling.

Proof. Regularization and a small generic perturbation supply an extremizer for $M_{a,b}$ with a nonzero corner coefficient and only simple zeros. There can be no gain beyond $M_{a,b}$, and approximation by

Gaussian rationals preserves all these properties. Its unique highest term is $cz^b\bar{z}^a$, of total degree $a+b$. Lemma 5.6, with the new analytic degree n, adds exactly $n-b+a$ nondegenerate zeros. Choose its small coefficient rational. If $a<m$, add a sufficiently small nonzero Gaussian rational multiple of $\bar{z}^m$. The highest term z^n is unchanged; uniform control near infinity and local uniqueness at the simple zeros show that the exact count is unchanged. The conjugate degree is now m.

For (20.54), take $(a,b)=(m,3m)$. Theorem 19.1 gives $M_{m,3m}=10m^2$, over the real rationals with a nonzero corner and simple zeros, and $a+b=4m<n$. Substitution proves the stated count and arithmetic claim. □

For $m\geq 2$ and $n=4m+1$ the new count is $10m^2+2m+1$. The optimized radial expression (5.11) there is $9m^2+4m+1$ for even m and $9m^2+4m$ for odd m. Thus the respective gains over that expression are $m(m-2)$ and $(m-1)^2$. Concrete improvements are

$$\begin{aligned} E_{3,13} &\geq 97 && \text{(radial: 93)},\\ E_{10,41} &\geq 1021 && \text{(radial: 941)},\\ E_{100,401} &\geq 100201 && \text{(radial: 90401)}. \end{aligned}$$

These are lower bounds for the unrestricted pure-head maximum. They do not determine the still-open rectangular endpoints. The separate construction in Theorem 19.16 proves $E_{4,12}=144$.

20.10 Optimizing completion over the whole proved cone

Fix $n\geq 5$ and $1\leq m<n$. Let $\mathcal{B}_{m,n}$ be the finite set of integer pairs

$$1\leq a\leq m, \qquad a<b\leq 3a, \qquad a+b<n,$$

excluding $b=3a-1$ when $a\geq 4$. Thus the three separately proved cases on the missing line remain admissible. Define

$$C_{m,n}=\max_{(a,b)\in\mathcal{B}_{m,n}}\{n+a^2+b^2+a-b\}. \tag{20.55}$$

The set is nonempty because it contains $(1,2)$.

Theorem 20.26. *One has $E_{m,n}\geq C_{m,n}$. The count $C_{m,n}$ is attained exactly over $\mathbb{Q}(i)$, with exact conjugate degree m and only nondegenerate zeros. It is the largest count supplied by Proposition 20.25 using a seed in $\mathcal{B}_{m,n}$.*

The maximum in (20.55) needs at most three candidate values of a. Put

$$a_0=\min\left\{m,\left\lfloor\frac{n-1}{4}\right\rfloor\right\}.$$

Test $a\in\{a_0,a_0+1,a_0+2\}\cap\{1,\dots,m\}$, taking

$$b_a=\min(3a,n-a-1), \qquad b_a\leftarrow b_a-1 \text{ if } a\geq 4 \text{ and } b_a=3a-1. \tag{20.56}$$

Discard a candidate with $b_a\leq a$, and take the largest $n+a^2+b_a^2+a-b_a$ among the remaining candidates.

Proof. Every admitted seed has $M_{a,b}=a^2+b^2$, so Proposition 20.25 gives the asserted exact constructed count and coefficient properties. This is an optimization of a construction, not an upper bound for the whole pure-head class.

For fixed a, the quantity $b^2 - b$ strictly increases for $b > a$; therefore (20.56) gives the best admissible b. Put $q = \lfloor (n-1)/4 \rfloor$. When $a \le q$, the value is $n + 10a^2 - 2a$, which strictly increases with a. When $a > q$, apart from the excluded line, it is

$$f(a) = a^2 + (n - a - 1)^2 + 2a + 1.$$

Feasibility requires $a \le \lfloor (n-2)/2 \rfloor$. The successive difference is $f(a+1) - f(a) = 4a - 2n + 6 < 0$ whenever both arguments are feasible. The exceptional reduction of b can occur only when $n = 4a$, hence only at the first integer $a = q + 1$. It suffices to test that first integer and the next one, as well as the best value below them. This proves the three-candidate rule. □

Corollary 20.27. *For every $m \ge 4$, the following counts are attained by integer-coefficient pure-head polynomials of exact conjugate degree m, with all zeros nondegenerate and exactly n real-axis zeros:*

$$\begin{aligned} E_{m,4m-2} &\ge 10m^2 - 16m + 10, \\ E_{m,4m-1} &\ge 10m^2 - 10m + 5, \\ E_{m,4m} &\ge 10m^2 - 10m + 6. \end{aligned} \tag{20.57}$$

Each displayed right side equals $C_{m,n}$ at that degree.

Proof. The three-candidate rule selects respectively $(a, b) = (m, 3m-3), (m, 3m-2), (m, 3m-2)$. Direct substitution gives (20.57). All three seeds lie in the cone away from its missing line. By Corollary 19.44 they have real rational coefficients and precisely $a + b$ nondegenerate real zeros. In the first two cases $n - (a + b) = 1$, so completion adds one real zero at infinity. In the last case this difference is two; multiply the seed by -1 if necessary to make its corner coefficient $c < 0$. The scaled limiting equation on the real axis then has $x^2 = -c$, and completion adds two real simple zeros. Conjugation symmetry preserves the real zeros under a sufficiently small real perturbation. Thus there are exactly n real zeros, the largest possible number for a pure-head degree-n restriction. Clear denominators to obtain integer coefficients. □

For $m \ge 11$, all three inequalities improve the corresponding optimized radial bounds. More precisely, the gains at $n = 4m - 2$ and $4m$ are $(m-1)(m-10)$ and $m^2 - 11m + 6$; at $n = 4m - 1$ they are $m^2 - 8m + 6$ for even m and $m^2 - 8m + 5$ for odd m. For example,

(m, n)	optimized radial count	cone-completion count
$(100, 398)$	89500	98410
$(100, 399)$	89799	99005
$(100, 400)$	90100	99006

For $n \ge 4m + 1$ the same envelope reduces to $C_{m,n} = n + 10m^2 - 2m$, recovering (20.54). The optimization is sharp for the specified set of seed transfers; the generally larger endpoint $E_{m,n}$ is not determined by it.

21 A paired seed improves the first missing rectangle

The first unresolved pair on $n = 3m - 1$ admits a further strict improvement. The construction below uses a cubic and a quadratic with shared zeros, including two tangencies. Its interpolation steps have exact finite-field certificates; its zero count follows from the local limits proved below.

Theorem 21.1. *There is a polynomial with real integer coefficients, exact bidegree* $(4,11)$*, and exactly* 125 *zeros, all nondegenerate. There are* 66 *positive and* 59 *negative zeros. Consequently,*

$$125 \le M_{4,11} \le 137. \tag{21.1}$$

The lower bound improves the radial count 121 *in* (5.8)*. It does not prove the upper endpoint.*

21.1 The shared cubic and quadratic

Put $\ell = (\sqrt{5}-1)/2$ and choose

$$\frac{1-\ell}{4} < c < \frac{1-\ell}{3+\ell}. \tag{21.2}$$

Define

$$\begin{aligned} &g = z^2 + \ell zw + z + c, && L = z - w, \qquad B = \ell z, \\ &f = Lg + Bg^* = z^3 + z^2 + (\ell-1)zw + c(1+\ell)z - cw. \end{aligned} \tag{21.3}$$

There are four shared physical zeros, two real and two off the real axis, and five further zeros of f. All nine zeros of f and all four zeros of g are nondegenerate.

Here is an explicit verification. The real shared zeros solve $x^2 + \ell x + \ell c = 0$; the off-axis shared zeros have $x = -1/2$ and $y^2 = c/(1-\ell) - 1/4 > 0$. The real restriction of f is $x(x^2 + \ell x + \ell c)$. For $y \neq 0$, its imaginary equation gives $y^2 = 3x^2 + 2x + c(2+\ell)$, after which its real equation is

$$-(1+2x)\{4x^2 + (4-2\ell)x + c(3+\ell)\} = 0.$$

The quadratic has two distinct negative roots, neither equal to $-1/2$. Writing either as $x = -s$, its equation implies $s > c(3+\ell)/(4-2\ell)$, and hence

$$y^2 = \frac{2-3\ell}{2}s - \frac{1-\ell}{4}c > 0.$$

Thus the displayed roots are distinct. The highest forms of f, f^*, and separately of g, g^*, are coprime; Bézout's theorem proves simplicity and exhausts their intersections.

Since $(f, g^*) = (Lg, g^*)$, the cross intersection number is two at each real shared zero and one at each off-axis shared zero. These numbers sum to six, exhausting the cross intersections. For

$$P_0 = f^3 g$$

the affine intersection multiplicities with P_0^* are therefore

$$\begin{aligned} 9 &\quad \text{at each of five } f\text{-only zeros,} \\ 16 &\quad \text{at each off-axis shared zero,} \\ 22 &\quad \text{at each real shared zero.} \end{aligned} \tag{21.4}$$

Their sum is 121. These are multiplicities, not distinct zero counts for the unperturbed product.

At a shared zero use the complexified coordinates $u = g$, $v = g^*$, which satisfy $v = \bar{u}$ on the physical slice, and write $\varphi_a(u,v) = f(z(u,v), w(u,v))$. Define the convergent local series

$$Q_a(u,v) = \frac{\varphi_a(u,v)^3 - \varphi_a(u,0)^3}{v}. \tag{21.5}$$

At a real shared zero $a = (x, x)$,

$$\varphi_a(u, 0) = k_a u^2 + O(u^3), \qquad k_a = (2x + 1)^{-1}, \qquad (Q_a)_2 = B(a)^3 v^2. \tag{21.6}$$

Indeed, $g - g^* = (z - w)(z + w + 1)$. At an off-axis shared zero,

$$\varphi_a(u, v) = B(a)v + L(a)u + O(2), \qquad (Q_a)_2 = B(a)^3\{v^2 + 3\alpha_a uv + 3\alpha_a^2 u^2\}, \quad \alpha_a = L(a)/B(a). \tag{21.7}$$

As c decreases to the left endpoint in (21.2), both $|\alpha_a|$ tend to zero.

21.2 An exact certificate for the required interpolation

Let $V = \operatorname{span}\{z^i w^j : 0 \le i \le 11,\ 0 \le j \le 4\}$. At an f-only zero use coordinates $u = f$, $v = f^*$ instead. For any local expansion write $[ij]_a H = [u^i v^j]H$. The notation $O(d)$ denotes terms of total local degree at least d.

Lemma 21.2. *Outside a finite subset of (21.2), the following three assertions hold. There is a real-coefficient $H \in V$ satisfying:*

(i) its constant and linear jets vanish at all nine zeros;

(ii) its quadratic jet equals $(Q_a)_2$ at the four shared zeros;

(iii) at each real shared zero, $[d0]_a H = [d0]_a Q_a$ and $[0d]_a H = [0d]_a Q_a$ for $d = 3, 4, 5$.

It can be chosen with

$$[z^{11}w^4]H \neq 0, \qquad [21]_a(Q_a - H) \neq 0 \quad \text{at both real shared zeros.} \tag{21.8}$$

Moreover, arbitrary compatible real data can be prescribed by $K \in V$ for the following 37 complex jet entries: its constant and linear jets at all nine zeros, and additionally its $20, 02, 30, 03, 11$ jets at the two real shared zeros. Arbitrary compatible real data can likewise be prescribed by $C \in V$ for its values at all nine zeros and its two first derivatives at the two real shared zeros. The latter list has 13 entries.

Proof. The H list has $27 + 12 + 12 = 51$ entries. Order the 60 columns by $5i + j$, starting with column zero. At $c = 1/10$, an exact nonvanishing certificate uses the prime 461, with $\ell \mapsto 21$ and $c \mapsto 415$. In row order the nine points reduce to

$$(251, 251), (189, 189), (416, 44), (44, 416), (0, 0), (5, 21), (21, 5), (83, 371), (371, 83).$$

The first two are the real shared points, the next two are the off-axis shared points, and the last five are f-only points. All required coordinate Jacobians are nonzero modulo 461. The following square minors have the indicated determinants:

Matrix	Selected columns	Determinant modulo 461
H	all except $45, 46, 50, 51, 52, 55, 56, 57, 58$	79
K	$0, \dots, 30, 32, 33, 34, 39, 44, 49$	260
C	$0, \dots, 10, 13, 14$	350

For each point, the constant and linear entries are ordered $00, 10, 01$, followed for H by $20, 11, 02$ at shared points and by $30, 40, 50, 03, 04, 05$ at real shared points. The additional K entries have order $20, 02, 30, 03, 11$. Setting the nine omitted H coefficients to zero, the unique solution has corner coefficient 47 modulo 461; the two entries $[21]_a(Q_a - H)$ reduce to $228, 163$.

The standard-library program `verify_golden_4_11_rank.py` reconstructs every entry by formal inversion through degree six and verifies these determinants and nonzero values using integer arithmetic modulo 461. Its full report is `golden_4_11_rank_verification.json`. This is a certificate of algebraic nonvanishing, not a numerical rank test. All denominators in the formal inversion are units at the displayed reduction. Thus each witnessed minor is nonzero in characteristic zero. The same holds for the three nonzero solution entries in (21.8). As algebraic functions of c, they have only finitely many zeros or poles in the interval. The node sets and their roles are invariant under permutations of their algebraic roots, so the selected-column solution is well defined independently of their ordering. Finally, complex conjugation permutes the rows and conjugates their data. Averaging a solution with its coefficientwise conjugate gives real coefficients; for the selected-column H solution, uniqueness already gives this property. □

One free direction will be essential. Put

$$\begin{aligned} q(z) &= (\ell - 1)z - c, & t(z) &= z^2 + \ell z + \ell c, \\ G &= w^2 f - q(z) f^*, & A(z) &= \operatorname{rem}_{t(z)}\big(\ell^2 z q(z)((2+\ell)z + 1)\big), \\ H_0 &= gf\{t(z)G + A(z)f\}. \end{aligned} \tag{21.9}$$

Then $H_0 \in V$, its total degree is at most 12, and it satisfies the homogeneous version of every H condition. At either real shared point it also has $[21]_a H_0 = 0$. To check the only higher-order condition, restrict to $v = 0$: $f = k_a u^2 + O(u^3)$, $G = -q(a)B(a)u + O(u^2)$ and $t(z(u, 0)) = t_u u + O(u^2)$, where $t_u/k_a = \ell((2 + \ell)x + 1)$. The chosen remainder A cancels the coefficient of u^5 in H_0. The $u = 0$ restriction vanishes identically. Its other stated jets follow directly from its factors. At an f-only point,

$$[02]_a H_0 = 0, \qquad [11]_a H_0 = -g(a)t(z_a)q(z_a) \neq 0. \tag{21.10}$$

Here $g \neq 0$, the two roots of t are the real shared points, and the real root of q cannot be an off-axis f-only point or the remaining real point $z = 0$. Thus replacing H by $H + \lambda H_0$ preserves (21.8) and makes the ratio $|[02]_a H/[11]_a H|$ arbitrarily small at all five f-only points as $\lambda \to +\infty$.

21.3 All finite clusters and the four escaping zeros

Choose c near the left endpoint of (21.2), outside the finite exceptional set, so that $0 < |\alpha_a| < 1/10$ at the off-axis shared points. Choose H and then λ as above. We construct

$$P_\varepsilon = f^3 g - \varepsilon H - \varepsilon^2 K - \varepsilon^3 C, \qquad \varepsilon > 0. \tag{21.11}$$

The following prescriptions are all permitted by Lemma 21.2.

At a real shared point, the pure jets imply a formal identity

$$Q_a - H = uvR_a + O(6), \qquad R_a(0) = 0.$$

Prescribe the pure u and v coefficients of K through degree three to equal those of R_a, and set its value to zero. Write $a_1 = [10]_a K \neq 0$, $b_1 = [01]_a K$ and $q_0 = B(a)^3$. Put

$$c_1 = \frac{b_1^2}{4q_0} + \frac{a_1 \bar{b}_1}{2\bar{q}_0}, \qquad [11]_a K = [11]_a R_a - c_1.$$

Now $R_a - K = uvT_a + O(4)$. Prescribe the constant and linear jets of C to equal those of T_a; in particular $C(a) = c_1$. An exact jet rearrangement gives

$$P_\varepsilon = (uv - \varepsilon)\{Q_a + \varepsilon K + \varepsilon(uv + \varepsilon)C\} + u\varphi_a(u, 0)^3 + \varepsilon O(6). \tag{21.12}$$

At scale $u = \varepsilon\zeta$, division by $-\varepsilon^3$ converges in C^1 to

$$q_0\bar{\zeta}^2 + a_1\zeta + b_1\bar{\zeta} + c_1.$$

Translation reduces this exactly to $q_0\bar{\xi}^2 + a_1\xi$, which has four simple zeros.

At scale $|u| \asymp \sqrt{\varepsilon}$, the braced factor in (21.12) is $q_0v^2(1 + o(1))$ and is nonzero. After division by this factor, the leading nonreal term is

$$\frac{k_a^3}{q_0}\rho^5 e^{9i\theta}, \qquad u = \rho e^{i\theta}.$$

All the displayed remainder terms are of smaller order in C^1 after the annular scaling. The imaginary equation therefore has 18 simple angular branches. On each branch the radial equation has derivative $2\rho(1 + o(1))$ and a root $\rho/\sqrt{\varepsilon} \to 1$. These give 18 further regular zeros, disjoint from the four inner zeros. Each real shared point consequently contributes its full multiplicity 22.

At an off-axis shared point, set $K(a) = 0$, $[10]_aK = -B(a)^3$ and $[01]_aK = 0$. The inner limit, divided by $B(a)^3$, is

$$\bar{\zeta}^2 + 3\alpha_a|\zeta|^2 + 3\alpha_a^2\zeta^2 - \zeta + \delta_a, \qquad \delta_a = C(a)/B(a)^3.$$

For $\delta_a = 0$ the origin is a positive simple zero. The highest form has winding -2, since $3|\alpha_a| + 3|\alpha_a|^2 < 1$; it is proper and its two complexified highest forms are coprime. A sufficiently small generic δ_a preserves that positive zero and makes all zeros regular. The degree formula then requires at least three negative zeros, and the quadratic Bézout bound gives exactly four. Choose the two δ_a compatibly with conjugation.

For the off-axis outer limit, division by $(Q_a)_2$ gives

$$\rho^2 - 1 + \frac{\alpha_a^3\rho^2 e^{6i\theta}}{1 + 3\alpha_a e^{2i\theta} + 3\alpha_a^2 e^{4i\theta}} = 0.$$

The denominator D satisfies $|D| \geq 67/100$ and $|\partial_\theta D| \leq 72/100$. Thus $6\theta - \arg D$ is strictly increasing and winds six times. There are exactly 12 simple angular solutions; the radial coefficient is positive because $|\alpha_a|^3/|D| < 1$. They are nonzero, regular, and persist for small ε. Together with the four inner zeros they exhaust multiplicity 16.

At an f-only point put $g_0 = g(a)$ and

$$a_2 = -[20]_aH/g_0, \qquad b_2 = -[11]_aH/g_0, \qquad c_2 = -[02]_aH/g_0.$$

The ratio $|c_2/b_2|$ can be made uniformly small by the choice of λ. Prescribe $K(a) = 0$ and its two linear coefficients, and then $C(a)$, so that the scale $f = \varepsilon\zeta$ of $P_\varepsilon/(g_0\varepsilon^3)$ is

$$(\zeta + a_2/3)^3 + b_2(\zeta + a_2/3)(\bar{\zeta} + \bar{a}_2/3) + c_2(\bar{\zeta} + \bar{a}_2/3)^2 - \tfrac{1}{8}|b_2|^2 b_2.$$

There is no conflict with any previous jet prescription. After translation and the usual cubic dilation, this is $\xi^3 + |\xi|^2 - 1/8 + e\bar{\xi}^2$, with $|e| = |c_2/b_2|$. The nine simple zeros at $e = 0$ persist for sufficiently small $|e|$. Each of the five f-only points therefore contributes nine.

We have constructed $2 \cdot 22 + 2 \cdot 16 + 5 \cdot 9 = 121$ finite regular zeros. Local conservation of complex intersection multiplicity and (21.4) exclude any additional zeros in these neighborhoods.

Finally invert the rectangle: $\widehat{P}_\varepsilon(z,w) = z^{11}w^4 P_\varepsilon(1/z, 1/w)$. The leading part of $\widehat{P}_0$ at zero is $w^3(w+\ell z)$. If $h = [z^{11}w^4]H \neq 0$, then

$$\varepsilon^{-1}\widehat{P}_\varepsilon(\varepsilon^{1/4}z, \varepsilon^{1/4}w) \longrightarrow w^3(w+\ell z) - h. \tag{21.13}$$

On the physical circle the homogeneous term is $\rho^4 e^{-4i\theta}(1+\ell e^{2i\theta})$. Its argument is strictly decreasing: its derivative is

$$-4 + 2\,\mathrm{Re}\,\frac{\ell e^{2i\theta}}{1+\ell e^{2i\theta}} < 0,$$

and its total change is -8π. Every nonzero target therefore has exactly four regular preimages. The bound $|w^3(w+\ell z)| \geq (1-\ell)|z|^4$ on $w = \bar{z}$ confines every nearby zero to this scale, so exactly four zeros escape to infinity. Compactness excludes zeros away from the nine finite clusters and infinity. This proves exactly 125 regular zeros for all sufficiently small positive ε, with nonzero corner coefficient and exact bidegree $(4, 11)$.

All choices can be made with real coefficients. A sufficiently small real rational approximation preserves the simple zeros, the nonzero corner, and the exact count: the common highest corner term supplies a uniform properness estimate. Clearing denominators proves the integer-coefficient assertion of Theorem 21.1. The remaining gap to 137 is at infinity in this construction; the finite clusters already realize all 121 affine intersection multiplicities. The nonzero corner gives winding $11 - 4 = 7$ on a large circle, so the two Jacobian counts are $(125+7)/2 = 66$ and $(125-7)/2 = 59$.

21.4 A vanishing corner and eight escaping zeros

The finite splitting above is compatible with a smaller first perturbation at infinity. An intermediate perturbation scale then increases the number of escaping zeros from four to eight.

Theorem 21.3. *There is a polynomial with real integer coefficients, exact bidegree* $(4, 11)$*, and exactly* 129 *zeros, all nondegenerate. Its positive and negative zero counts are* 68 *and* 61*. In particular,*

$$129 \leq M_{4,11} \leq 137. \tag{21.14}$$

The upper endpoint remains undetermined.

We retain f, g, L, B, t, q, Q_a, V from the preceding construction. Write the lower off-axis shared point as $z = -1/2 - iy$, $w = -1/2 + iy$, where $y > 0$ tends to zero as c decreases to the left endpoint of (21.2). At either off-axis point put

$$s = zw = \tfrac{1}{4} + y^2, \qquad D_0 = \ell^2 s + L^2, \qquad \delta = -\frac{L^3(3+L^2)}{4\ell^2 s^2}, \qquad \alpha = L/B. \tag{21.15}$$

These prescriptions at the two points are conjugate. Modify only the off-axis quadratic target for H, replacing it by

$$[H]_2 = B^3\{v^2 + 3\alpha uv + (3\alpha^2 + \delta)u^2\}. \tag{21.16}$$

All other 51 interpolation conditions are unchanged.

Lemma 21.4. *Every solution of these modified jet conditions satisfies*

$$[z^{11}w^4]H = 0, \qquad [z^{10}w^4]H = b, \qquad b = -\frac{8(1+L^2)}{\ell^5 D_0^2 (1-L^2)^2} < 0$$

when y is sufficiently small. It can be chosen with real coefficients so that $[z^{11}w^3]H = b$ and $[21]_a(Q_a - H) \neq 0$ at both real shared points.

Proof. Write $f = p(z) + q(z)w$, put $W(z) = -p(z)/q(z)$, and define $E(z) = q(z)^3 f^*(z, W(z))$. The polynomial E has degree nine, leading coefficient -1, and nine simple roots, corresponding to the nine zeros of f. None is a root of q. Since the value and first derivatives of H vanish at each,

$$q^4 H(z, W(z)) = E(z)^2 S(z), \qquad \deg S \leq 5. \tag{21.17}$$

At a real shared point, the curve $f = 0$ has $v = -(k_a/B(a))u^2 + O(u^3)$. The unchanged pure jets give $H = Q_a + uvR + O(6)$ with $R(0) = 0$; hence its restriction to this curve vanishes to order at least four. Consequently

$$S(z) = t(z)^2(hz + b_0).$$

Comparison of the degree-23 coefficients in (21.17) gives $h = [z^{11}w^4]H$.

At an off-axis point, temporarily write the quadratic target as $B^3(Au^2 + Duv + Cv^2)$ and set $\Delta = A - \alpha D + \alpha^2 C$. Along $f = 0$ one has $v = -\alpha u + O(u^2)$ and $f^* = (D_0/B)u + O(u^2)$. Differentiating $E = q^3 f^*(z, W)$ at the point therefore gives $u' = BE'/(D_0 q^3)$. Comparing the quadratic coefficients in (21.17), and using the exact identities $q = B^2$ and $t = \ell BL$ at these points, yields

$$hz + b_0 = \frac{\Delta}{\ell^2 B D_0^2 L^2}. \tag{21.18}$$

For (21.16), $\Delta = \alpha^2 + \delta$. The right-hand side becomes

$$\frac{1}{\ell^5 D_0^2}\left\{\frac{1}{z^3} - \frac{L(3+L^2)}{4zs^2}\right\} = -\frac{8(1+L^2)}{\ell^5 D_0^2 (1-L^2)^2} = b.$$

The equality follows from $z = (L-1)/2$ and $s = (1-L^2)/4$. It has the same value at the two distinct off-axis points, so $h = 0$ and $b_0 = b$. Comparing degree 22 in (21.17) now gives $[z^{10}w^4]H = b$. For small y, $L^2 = -4y^2$ and $D_0 \neq 0$, proving $b < 0$.

The interpolation matrix is unchanged, so the same 51-column minor is invertible away from finitely many c. Among its nine omitted columns is 58, corresponding to $z^{11}w^3$. Prescribe this coefficient to be b and all other omitted coefficients to be zero, then solve using that minor. At the same exact specialization modulo 461, the solution has

$$[z^{11}w^4]H = 0, \quad [z^{10}w^4]H = [z^{11}w^3]H = 349, \quad \big([21]_a(Q_a - H)\big)_{a\ mreal} = (247, 251). \tag{21.19}$$

The determinants of the H, K, C minors remain $79, 260, 350$. The standalone integer-arithmetic verifier `verify_golden_4_11_129_rank.py` reconstructs the modified targets and these entries; its report is `golden_4_11_129_rank_verification.json`. The argument of Lemma 21.2 lifts the nonzero entries to characteristic zero and excludes only finitely many parameters. Uniqueness and conjugation give real coefficients. □

We next check that the modified targets preserve all finite zeros. At an off-axis point the inner quadratic limit becomes

$$\bar{\zeta}^2 + 3\alpha|\zeta|^2 + (3\alpha^2 + \delta)\zeta^2 - \zeta + \delta_0.$$

For small y, the quadratic highest part is close to $\bar{\zeta}^2$, and the origin is a positive simple zero when $\delta_0 = 0$. The same degree and Bézout argument as above gives exactly four regular zeros for a small generic target.

For the outer scale $u = \sqrt{\varepsilon}\rho e^{i\theta}$, put $X = e^{2i\theta}$. The limiting equation is

$$\rho^2 = R_y(X) := \frac{1 + 3\alpha X + (3\alpha^2 + \delta)X^2}{(1 + \alpha X)^3} = 1 + \frac{\delta X^2 - \alpha^3 X^3}{(1 + \alpha X)^3}. \tag{21.20}$$

At the lower point, as $y \downarrow 0$,

$$\frac{\alpha^3}{|\alpha|^3} \longrightarrow -i, \qquad \frac{\delta}{\alpha^3} \longrightarrow \kappa := \frac{3\ell}{2} \in (0, 1).$$

Thus, uniformly in C^1 for $X = e^{i\phi}$,

$$\frac{\operatorname{Im} R_y(e^{i\phi})}{|\alpha|^3} \longrightarrow \cos 3\phi - \kappa \cos 2\phi.$$

The cubic $4x^3 - 3x - \kappa(2x^2 - 1)$ has respective signs $-, +, -, +$ at $-1, -1/2, 1/2, 1$. It has exactly three simple roots in $(-1, 1)$, so the limiting angular function has six simple zeros. For sufficiently small fixed y, these give six simple angles ϕ, with $R_y > 0$ since $R_y \to 1$ uniformly. Each has two lifts to θ, giving twelve regular outer zeros. Conjugation handles the other shared point. Together with the inner four, each still supplies sixteen.

Add λH_0 with a sufficiently large fixed real λ. Its total degree is at most twelve, so it changes neither displayed coefficient at infinity. Its homogeneous jets preserve the real shared-point conditions and their nonzero inner coefficients, and (21.10) again gives nine zeros at each f-only point. The real shared-point argument is unchanged and gives twenty-two at each. Thus the modified construction still splits all 121 finite intersection multiplicities.

To arrange the intermediate scale, take any real solution J of the original H conditions. Formula (21.18) with $\delta = 0$ shows that its corner coefficient is

$$j = [z^{11}w^4]J = -\frac{1 - s}{\ell^5 D_0^2 s^3} \neq 0.$$

Here the divided difference of z^{-3} at z, w is $-(1 - s)/s^3$, since $z + w = -1$. Set $k = -b > 0$ and, for small positive ε, define

$$\beta = \frac{k^{4/3}}{10j}, \qquad \tau = \varepsilon + \beta\varepsilon^{4/3} > 0, \qquad H_\varepsilon = \frac{\varepsilon H + \beta\varepsilon^{4/3} J}{\tau}. \tag{21.21}$$

At the real shared points H_ε satisfies exactly the original pure-jet conditions. At the off-axis points its quadratic target has δ replaced by $(\varepsilon/\tau)\delta$, which tends to δ. Moreover $H_\varepsilon \to H$ coefficientwise. Choose $K_\varepsilon, C_\varepsilon$ by the preceding jet prescriptions, using H_ε and parameter τ. The fixed invertible minors make these choices continuous and bounded as $\varepsilon \downarrow 0$. The nonzero inner coefficients, the four-root inner targets, and the twelve-root outer targets persist. All finite-scale estimates above are uniform in this bounded family. Consequently

$$P_\varepsilon = f^3 g - \tau H_\varepsilon - \tau^2 K_\varepsilon - \tau^3 C_\varepsilon \tag{21.22}$$

has exactly 121 regular zeros near the nine finite clusters.

Its inverted polynomial, at scale $(z, w) = \varepsilon^{1/3}k^{1/3}(Z, W)$, satisfies

$$\frac{\widehat{P}_\varepsilon(\varepsilon^{1/3}k^{1/3}Z, \varepsilon^{1/3}k^{1/3}W)}{\varepsilon^{4/3}k^{4/3}} \longrightarrow \Psi(Z, W) := W^3(W + \ell Z) + Z + W - \frac{1}{10} \tag{21.23}$$

in C^1 on compact sets. Indeed, the inverted H begins with $b(z + w)$, and the intermediate J term contributes the constant $-1/10$. The K, C terms are smaller.

Here the eight-zero count is elementary and exact. On writing $Z = x + iy$, $W = x - iy$, the imaginary equation is

$$-2xy\{(2+\ell)x^2 - (2-\ell)y^2\} = 0.$$

On $x = 0$ the real equation is $(1-\ell)y^4 = 1/10$, giving two roots. On $y = 0$ it is $(1+\ell)x^4+2x-1/10 = 0$, also giving exactly two simple roots. On either of the other two lines it is

$$-\frac{16}{(2-\ell)^2}x^4 + 2x - \frac{1}{10} = 0.$$

This has exactly two positive simple roots and no negative ones: it is negative at zero, positive at $x = 1/4$, tends to $-\infty$, and has one critical point on the positive half-line. The two lines therefore give four further roots. All eight are distinct. Away from the origin, the imaginary equation has a nonzero transverse derivative on each of its four lines; the real restrictions just checked have nonzero derivatives. Hence all eight zeros are nondegenerate.

The estimate $|w^3(w+\ell z)| \geq (1-\ell)|z|^4$ confines every zero near inverted infinity to $|z| = O(\varepsilon^{1/3})$. The compact limit (21.23) and its simple zeros then give exactly eight such zeros. None is at the origin: the corner of P_ε is $-\beta j\varepsilon^{4/3} + O(\varepsilon^2) \neq 0$. Compactness excludes other physical zeros. Thus the exact total is $121+8 = 129$. As before, stability and properness allow real rational approximation and then clearing of denominators. The large-circle winding is $11 - 4 = 7$, giving 68 positive and 61 negative zeros and proving Theorem 21.3.

22 Two perturbations that increase the bidegree

The preceding constructions prescribe many zeros at once. There is also value in modifying an existing example while retaining its zeros. We give two such modifications. The first increases both degrees and preserves the deficit from the quadratic bound. The second increases only the analytic degree, under a local contact condition.

22.1 Adding a circle

The basic observation is that a real circle becomes a rational curve in the complexified surface. Restrictions to this curve allow enough freedom to prescribe the tangential equation of a perturbation.

Proposition 22.1. *Suppose a polynomial of bidegree (m, n), with $n > m \geq 1$, has at least N nondegenerate zeros. There is a polynomial of bidegree $(m+1, n+1)$ with finitely many zeros, all nondegenerate, and with at least $N + 2(m+n+1)$ zeros.*

Proof. An arbitrarily small generic perturbation preserves the chosen N zeros and makes P and

$$P^*(z, w) = \overline{P(\bar{w}, \bar{z})}$$

coprime. Choose a circle with complexified equation

$$L_0(z, w) = (z - a)(w - \bar{a}) - R^2 = 0, \qquad R > 0,$$

whose complexification avoids all intersections of $P = 0$ and $P^* = 0$. This is possible: each of their finitely many intersection points excludes a proper real algebraic subset of the parameters (a, R). Require also that the circle avoid the selected zeros. With

$$z = a + Rt, \qquad w = \bar{a} + R/t,$$

put $d = m+n$. Avoiding the zeros of the highest w-coefficient at $z = a$ and of the highest z-coefficient at $w = \bar{a}$ gives

$$P|_{L_0} = t^{-m}A(t), \qquad \deg A = d, \qquad A(0) \neq 0.$$

For a polynomial V of degree at most e define its reversal, with respect to the specified degree e, by $V^{\#_e}(t) = t^e\overline{V(1/\bar{t})}$. The avoidance condition gives $\gcd(A, A^{\#_d}) = 1$. In particular, P is nonzero on the real circle $|t| = 1$.

The restrictions of polynomials Q of bidegree at most $(m+1, n+1)$ are precisely

$$Q|_{L_0} = t^{-(m+1)}B(t), \qquad \deg B \leq d+2.$$

Indeed, the monomials in $z-a$ and $w-\bar{a}$ span every Laurent power from $-(m+1)$ to $n+1$. Consider the real linear map

$$B \longmapsto \operatorname{Im}(Q\overline{P})\big|_{|t|=1}.$$

Its target is the real vector space of trigonometric polynomials of degree at most $d+1$, which has dimension $2d+3$. Its kernel is specified by

$$BA^{\#_d} = B^{\#_{d+2}}A.$$

Coprimality implies $B = AS$ with $\deg S \leq 2$, and the remaining condition is $S = S^{\#_2}$. Thus

$$S(t) = \gamma + \lambda t + \bar{\gamma}t^2, \qquad \gamma \in \mathbb{C}, \quad \lambda \in \mathbb{R}.$$

The kernel has real dimension three, while the domain has dimension $2d+6$. The map is therefore onto.

Choose Q so that the target function is $\sin((d+1)\theta)$, where $t = e^{i\theta}$. Then $\operatorname{Im}(Q/P)$ has exactly $2(d+1)$ simple zeros on the circle. For real $\varepsilon \neq 0$ put

$$P_\varepsilon = L_0P + \varepsilon Q.$$

Every selected old zero persists. Near the circle, division by P reduces the equation to

$$u + \varepsilon \operatorname{Re}(Q/P) = 0, \qquad \operatorname{Im}(Q/P) = 0, \qquad u = L_0.$$

At $\varepsilon = 0$ the Jacobian in normal and tangential coordinates is nonzero at each of the $2(d+1)$ simple circle zeros. They therefore give that many new nondegenerate zeros for sufficiently small nonzero ε. Their neighborhoods can be chosen disjoint from those of the old zeros. A final generic perturbation gives exact bidegree $(m+1, n+1)$ and a finite, entirely nondegenerate zero set, without losing any of these zeros. □

The increment in Proposition 22.1 agrees exactly with the increment of the bidegree intersection number. This gives a useful monotonicity statement for the unknown exact maxima.

Corollary 22.2. *For $n > m \geq 1$ and every integer $r \geq 0$,*

$$M_{m+r,n+r} \geq M_{m,n} + 2r(m+n+r).$$

In particular, if $M_{m,n} = m^2 + n^2$, then $M_{m+r,n+r} = (m+r)^2 + (n+r)^2$ for every $r \geq 0$.

Proof. Lemma 3.2 gives a nondegenerate extremal polynomial. Iterate Proposition 22.1; the sum of its r increments is $2r(m+n+r)$. This is also the difference between the two quadratic bounds. Proposition 2.2 then proves the equality assertion. □

22.2 Increasing only the analytic degree

Perturbations by a pole can create several nearby zeros of a rational harmonic function. A detailed analysis of that phenomenon is given by Sète, Luce, and Liesen [SLL15, Theorem 3.1, author version, p. 7]. The following polynomial modification has the same local origin, but keeps the degree in w fixed and gives an exact global count.

Proposition 22.3. *Let $n > m \geq 1$, $q \geq 2$, and*

$$P(z,w) = wA(z,w) - z^q B(z), \qquad A(0,0)B(0) \neq 0.$$

Suppose that P has bidegree (m,n), its $w^m z^n$ coefficient is nonzero, and all zeros of F_P are nondegenerate. For every sufficiently small real $\varepsilon > 0$, the polynomial

$$P_\varepsilon(z,w) = (wz - \varepsilon)A(z,w) - z^{q+1}B(z) = zP(z,w) - \varepsilon A(z,w) \tag{22.1}$$

has bidegree $(m, n+1)$, all its zeros are nondegenerate, and

$$N(P_\varepsilon) = N(P) + 2q + 1.$$

Proof. The nonzero highest corner coefficient makes F_P proper, so its isolated zeros form a finite set. The origin is a negative nondegenerate zero of F_P, since its linear part is $A(0,0)\bar{z}$. Every nonzero old zero remains nondegenerate for $zF_P(z)$ and therefore persists under (22.1).

In a fixed small disk about zero, divide by $A(z,\bar{z})$ and set $b(z) = B(z)/A(z,\bar{z})$. The local equation becomes

$$|z|^2 - \varepsilon - z^{q+1}b(z) = 0, \qquad b(0) \neq 0. \tag{22.2}$$

Write $t = \sqrt{\varepsilon}$ and $z = tre^{i\theta}$. For $t, r > 0$, this is equivalent to

$$\begin{aligned} r^2 - 1 - t^{q-1}r^{q+1}\operatorname{Re}\bigl(e^{i(q+1)\theta}b(tre^{i\theta})\bigr) &= 0, \\ \operatorname{Im}\bigl(e^{i(q+1)\theta}b(tre^{i\theta})\bigr) &= 0. \end{aligned}$$

At $t = 0$ there are exactly $2q+2$ solutions: $r = 1$ and $(q+1)\theta + \arg b(0) \in \pi\mathbb{Z}$. The Jacobian in (r,θ) at each is

$$2(q+1)\operatorname{Re}\bigl(e^{i(q+1)\theta}b(0)\bigr) \neq 0.$$

The implicit function theorem gives $2q+2$ distinct nondegenerate zeros for small positive t.

To exclude other local zeros, shrink the disk so that $|b(z)| \leq C$ and $C|z|^{q-1} \leq 1/2$. Equation (22.2) implies

$$\bigl||z|^2 - \varepsilon\bigr| \leq \tfrac{1}{2}|z|^2, \qquad \sqrt{2/3} \leq |z|/\sqrt{\varepsilon} \leq \sqrt{2}.$$

Every possible scaled limit lies in this compact annulus and solves the limiting system. Local uniqueness at its $2q+2$ solutions therefore accounts for all zeros in the disk.

Finally, if $c \neq 0$ is the $w^m z^n$ coefficient of P, then

$$P_\varepsilon(z,\bar{z}) = c\bar{z}^m z^{n+1} + O(|z|^{m+n}) \quad (|z| \to \infty),$$

uniformly for ε in a bounded interval. Thus zeros cannot escape to, or enter from, infinity. On a fixed compact set outside disjoint neighborhoods of the old zeros, zF_P is bounded away from zero. There are no other new zeros. Replacing the old origin by $2q+2$ zeros proves the exact increment $2q+1$. The highest corner coefficient also proves the asserted bidegree. □

The contact condition in Proposition 22.3 is restrictive; it cannot be imposed on an arbitrary extremal polynomial by a change of coordinates. The linear bound measures that restriction.

Corollary 22.4. *Under the hypotheses of Proposition 22.3,*

$$N(P) \le \min\{m^2+n^2,\ (4m+1)n-m-2q\}.$$

Proof. Apply Theorem 1.2 to P_ε, and subtract $2q+1$ from its linear bound. Apply Proposition 2.2 to P for the other term. □

For $q > 2m$, Corollary 22.4 improves the general linear bound by $2(q-2m)$. The local Jacobian signs in the proof of Proposition 22.3 alternate around the small circle. Of the $2q+2$ new local zeros, $q+1$ are positive and $q+1$ are negative. Since the old origin was negative, the total increments in the positive and negative counts are $q+1$ and q, respectively. Their difference agrees with the increase of the mapping degree by one.

22.3 Pure-head contact bounds and a sharp endpoint

The contact splitting argument also works with a pure analytic highest term. Its properness hypothesis can be checked directly, without inserting a rectangular corner that would destroy that term.

Proposition 22.5. *Let $P \in \mathcal{E}_{m,n}$ have exact conjugate degree m, only nondegenerate zeros, and*

$$P(z,w) = wA(z,w) - z^q B(z), \qquad q \ge 2, \quad A(0,0)B(0) \ne 0.$$

For every sufficiently small positive ε,

$$P_\varepsilon = (zw-\varepsilon)A - z^{q+1}B \in \mathcal{E}_{m,n+1}$$

has exact conjugate degree m, only nondegenerate zeros, and $N(P_\varepsilon) = N(P)+2q+1$. Consequently

$$N(P) \le \min\{n^2,\ (4m+1)n - 2m^2 - 2q\}. \tag{22.3}$$

The linear term improves the unrestricted pure-head bound by $2(q-2m)$ whenever $q > 2m$.

Proof. The support condition gives $\deg_{\mathrm{tot}} A \le n-2$ and $\deg B = n-q$. Thus $q \le n$, and if cz^n is the highest term of P, then cz^{n+1} is the unique highest term of $P_\varepsilon = zP - \varepsilon A$. Its exact conjugate degree is still m. The local proof of Proposition 22.3 applies without change. It replaces the old negative origin by $2q+2$ alternating regular zeros and preserves every other old zero. Uniformly for bounded ε,

$$P_\varepsilon(z,\bar z) = cz^{n+1} + O(|z|^n) \qquad (|z| \to \infty).$$

This excludes zeros entering from infinity, so the increment is exact. Apply (6.4) in degree $n+1$ and subtract $2q+1$ to obtain the linear term in (22.3); the initial polynomial gives the quadratic term. □

For example, a pure-head $(m,n) = (2,6)$ polynomial with this sixth-order contact has at most 34 zeros, so it cannot be a 36-zero pure-head extremizer. At $(m,n,q) = (3,10,10)$ the bound is 92, excluding a 100-zero contact seed. These obstructions concern the prescribed contact class.

22.4 A sharp obstruction for cyclic contact seeds

The contact bound need not be attainable after imposing rotational symmetry. A reciprocal-map argument gives an exact smaller maximum for the full threefold-symmetric contact-five space. The dynamical mechanism is the period-two obstruction discussed in [KLP24, Section 3, author version, pp. 8–9]; the proof below also treats singular zeros in the prescribed coefficient space.

Lemma 22.6 (A shared zero of the logharmonic factors)**.** *Let $p \in \mathbb{C}[z]$ have degree $d \geq 2$, let $p(0) = 0$, and let $c \neq 0$. The sharp maximum of the number of distinct zeros of*

$$G(z) = \bar{z}\, p(z) + c$$

is $3d-3$, including degenerate maps. For every d the integer example $p(z) = z^d - 3z$, $c = 1$, attains this maximum with only nondegenerate zeros.

Proof. First suppose all zeros are nondegenerate. Write $r(z) = -c/p(z)$ and $h(z) = \overline{r(z)}$. At a zero of G,

$$J_G = |p|^2\big(|r'|^2 - 1\big).$$

Thus a negative zero is an attracting fixed point of h. Each such fixed point attracts a critical point of h, by Fatou's theorem for anti-rational maps [KLP24, Lemma 2.1, author version, p. 5]. The critical point at infinity belongs to the two-cycle $\infty \mapsto 0 \mapsto \infty$, so it cannot be assigned to a fixed-point basin. Every finite critical point lies among the zeros of p', including multiple poles. Hence there are at most $d-1$ negative zeros. The leading term of G has winding number $d-1$, and G is proper. Consequently

$$N_+ - N_- = d - 1, \qquad N(G) \leq 3d - 3.$$

The zeros are always finite: each is a fixed point of the holomorphic rational map h^2 of degree $d^2 > 1$. At any zero, $p(z) \neq 0$, so the real derivative of G has rank at least one. Choose a constant $b \in \mathbb{C}$ outside the images of the rank-one derivatives at all singular zeros. By Lemma 3.1, the two maps $G + tb$ and $G - tb$ have at least $2N(G)$ nondegenerate zeros in total near the old zeros for every sufficiently small $t > 0$. This conclusion persists in a cone of directions. Sard's theorem permits a parameter in that cone for which both maps are regular; their constants remain nonzero and $p(0) = 0$ is unchanged. One sign therefore has at least $N(G)$ zeros. Applying the regular bound proves the distinct bound.

For the stated example, write $z = re^{i\theta}$. The nonzero solutions satisfy $\sin((d-1)\theta) = 0$. On the positive angular branch their radii solve $r^{d+1} - 3r^2 + 1 = 0$; this polynomial has exactly two positive roots, since its derivative has one positive zero, its value at 0 is positive, and its value at 1 is negative. On the negative branch, $-r^{d+1} - 3r^2 + 1 = 0$ has exactly one positive root by strict monotonicity. Each branch supplies $d-1$ angles at every radius. The radial roots and the angular equations are simple, giving exactly $3(d-1)$ nondegenerate zeros. □

Proposition 22.7 (The exact cyclic contact-five maximum)**.** *For all $a, b, c \in \mathbb{C}$ with $c \neq 0$, the polynomial*

$$P(z, w) = z^5 + az^3w + bzw^2 + cw \tag{22.4}$$

has at most 19 distinct real-slice zeros. The bound is sharp: $z^5 - 3z\bar{z}^2 + \bar{z}$ has exactly 19 zeros, all nondegenerate. More generally, equality in the bound is equivalent to the reduced equation

$$\bar{u}\,(u^3 + au^2 + bu) + c = 0$$

having six distinct solutions. In the regular case this is equivalent to the associated anti-rational map $u \mapsto \overline{-c/[u(u^2+au+b)]}$ *having two attracting fixed points.*

Proof. Away from the origin, set $u = z^2/\bar{z}$. This is the regular three-sheeted covering $re^{i\theta} \mapsto re^{3i\theta}$ of $\mathbb{C}^*$, and

$$\frac{P(z,\bar{z})}{\bar{z}} = |u|^2(u^2+au+b)+c = \bar{u}\,(u^3+au^2+bu)+c.$$

The reduced map has at most six zeros by Lemma 22.6, and none is zero. Every such zero lifts to exactly three distinct zeros. The origin contributes one additional zero and has Jacobian $-|c|^2 < 0$. This proves the bound, the equality criterion, and, with $(a,b,c) = (0,-3,1)$, its regular integer-coefficient attainment. For a regular reduced map, its degree is two, so it has six zeros exactly when two are negative, equivalently attracting. □

Indeed, (22.4) is the whole pure-head contact-five subspace of conjugate degree at most two satisfying $F(\omega z) = \omega^2 F(z)$, $\omega^3 = 1$: enumerating its permitted monomials leaves exactly the four terms displayed. Thus even singular parameters in this symmetry class cannot reach the pure-head bound 25. This is a sharp obstruction inside that class, not a smaller bound for $E_{2,5}$.

22.5 A certified contact seed in degree six

The contact stratum also gives a strict improvement at the next unresolved pure-head degree. Define

$$P(z,w) = z^6 + \frac{1}{D}\sum_{i,j}(U_{ij}+\mathrm{i}V_{ij})z^i w^j, \qquad D = 2^{56}, \tag{22.5}$$

where the sum runs over the following nine pairs.

i	j	U_{ij}	V_{ij}
0	1	70091836005420736	-16001690119397516
0	2	14122997105099940	-2986034004005576
1	1	42161821745008496	-46945701215175096
1	2	-219916042949887776	80025661292566688
2	1	-98771991024976304	-15075947084773352
2	2	-89735945237992768	137562448325045232
3	1	-19449207227161472	-50229378538153040
3	2	298863005077113024	58885059195421552
4	1	-105070499099672416	-8448110744614189

Proposition 22.8 (A regular contact-six seed)**.** *The Gaussian rational polynomial* (22.5) *has exactly* 26 *real-slice zeros, all nondegenerate:* 16 *have positive Jacobian and* 10 *have negative Jacobian. Its pure highest term is* z^6*, its exact conjugate degree is two, and*

$$P(z,0) = z^6, \qquad P_w(0,0) \neq 0.$$

Thus the origin is a negative regular zero of analytic contact order six.

Proof. The support and the nonzero $(0,1)$ coefficient in the table prove the degree and contact assertions directly. The count is certified by exact rational contractions. The file `pure_contact_2_6_26_certificate.json` contains these exact coefficients, 26 real two-dimensional centers and 10 nonreal four-dimensional centers, all with denominator 10^{180} and common box radius $r = 10^{-80}$.

For a real box use the real system $(\Re F_P, \Im F_P)$; for a nonreal box use the four real components of (P, P^*) in independent complex variables (z, w). Let Y be the exact inverse of the center Jacobian and let β be the infinity norm of Y times the center residual. If R bounds the complex moduli throughout the box, set

$$H = \sum_{i+j\geq 2} (i+j)(i+j-1)(|\Re p_{ij}| + |\Im p_{ij}|)R^{i+j-2}.$$

One may take R to be the maximum of the real-coordinate ℓ^1 norms of the center variables, plus $2r$. The contraction bounds are $q = 4r\|Y\|_\infty H$ in real boxes and $q = 16r\|Y\|_\infty H$ in complex boxes. In every box the independently executed verifier proves, using fractions only,

$$q + \beta/r < 2^{-10}.$$

Hence every box contains one simple zero. It also checks all boxes are disjoint after embedding the real centers as $(z, \bar{z})$, and that the ten other boxes avoid the real slice. The center Jacobians give the stated 16 positive and 10 negative signs.

The highest homogeneous forms of P and P^* are z^6 and w^6. They have no common factor or projective zero at infinity. Their 36 simple intersections supplied by the boxes therefore exhaust the degree-six Bézout count. This proves both completeness and nondegeneracy of the real-slice zero set. The verifier `verify_pure_contact_2_6_26.py` independently checks the printed coefficient table, support, contact, exact inequalities and exhaustion; its successful report is `pure_contact_2_6_26_verification.json`. No root finder is used in verification. □

Corollary 22.9 (Improved pure-head endpoint bracket)**.** *One has*

$$39 \leq E_{2,7} \leq 47. \tag{22.6}$$

The lower count 39 *is attained exactly by a polynomial with Gaussian integer coefficients, pure highest term up to a nonzero constant, exact conjugate degree two, and only nondegenerate zeros. More explicitly, with P from (22.5), put $A = (P - z^6)/w$. Then*

$$Q = z^7 + (zw - 10^{-24})A \tag{22.7}$$

has exactly 39 *nondegenerate zeros, with signs* $(23, 16)$*. The scalar multiple $2^{56}10^{24}Q$ has Gaussian integer coefficients.*

Proof. Write $P = z^6 + wA$ in Proposition 22.8. Then $\deg_{\mathrm{tot}} A \leq 4$ and $A(0,0) \neq 0$. For sufficiently small positive rational ε, Proposition 22.5 applies to

$$Q_\varepsilon = z^7 + (zw - \varepsilon)A.$$

It has exact conjugate degree two, pure head z^7, and exactly $26 + 2 \cdot 6 + 1 = 39$ regular zeros. More precisely, the negative origin is replaced by fourteen alternating simple zeros, giving 23 positive and 16 negative zeros in total. Clear the Gaussian rational denominators. The upper bound is (6.4) with $(m, n) = (2, 7)$. The former degree-extension lower bound was 37; the upper endpoint 47 and the rectangular value $M_{2,7}$ remain unresolved.

For the fixed parameter in (22.7), the certificate `pure_2_7_39_certificate.json` supplies 39 real two-dimensional boxes and 10 nonreal four-dimensional boxes, with center denominator 10^{300}

and radius 10^{-150}. The independently executed `verify_pure_2_7_39.py` reconstructs every coefficient directly from the printed contact-six table and the exact value 10^{-24}. Using the contraction estimates in the preceding proposition, it proves $q + \beta/r < 2^{-10}$ in every box using fractions only, checks all pairwise separations and nonreal-slice exclusions, and obtains exactly 23 positive and 16 negative real boxes. The heads z^7, w^7 give a total intersection number of 49, which these boxes exhaust. Thus this specific parameter, as well as the stated integer multiple, is certified without an unspecified smallness threshold. The successful report is `pure_2_7_39_verification.json`. □

Corollary 22.10 (The maximal-contact endpoint in conjugate degree one)**.** *For every $n \geq 2$, among pure-head polynomials*

$$P(z, w) = wA(z) - z^n, \qquad \deg A \leq n - 2, \quad A(0) \neq 0,$$

the sharp distinct maximum, including singular polynomials, is $3n - 2$. It is attained with nondegenerate zeros and Gaussian rational, and after a nonzero scalar multiplication Gaussian integer, coefficients. The contact modification then supplies a regular pure-head polynomial of degree $n + 1$ with exactly $5n - 1$ zeros. This recovers the known endpoint $E_{1,n+1} = 5(n + 1) - 6$.

Proof. For a regular polynomial the upper bound is (22.3) with $m = 1, q = n$. For an arbitrary member, write $A(z) = a_0 + \widetilde{A}(z)$, where $a_0 \neq 0$ and $\widetilde{A}(0) = 0$. Its nonzero zeros are the fiber at a_0 of

$$H(z) = \frac{z^n}{\bar{z}} - \widetilde{A}(z), \qquad z \neq 0.$$

Here $H_{\bar{z}} = -z^n/\bar{z}^2 \neq 0$, so every differential has rank at least one. Nearby fibers stay away from zero and infinity. Lemma 20.22 perturbs only the constant coefficient of A to produce a regular member with at least as many nonzero zeros. The negative simple origin persists. Hence the same upper bound covers singular members. For attainment, Geyer's critically fixed construction [Gey08, Theorem 1 and proof of Corollary 2, author version, pp. 1, 3–4], after translating one critical fixed point to zero, gives

$$g(z) = \sum_{j=2}^{n} c_j z^j, \qquad c_2 c_n \neq 0,$$

such that $g(z) - \bar{z}$ has $3n - 2$ nondegenerate zeros. Set $A(u) = u^n g(1/u)$. This is a polynomial of degree $n - 2$ with $A(0) = c_n \neq 0$. For $u \neq 0$,

$$\bar{u}A(u) - u^n = u^n \bar{u} \left(g(1/u) - \overline{1/u} \right).$$

Inversion therefore bijects the nonzero zeros and preserves their regularity. The old negative origin is omitted and a new negative simple origin is added, leaving the same total count. The resulting polynomial has exactly the required pure head and contact order. A sufficiently close Gaussian rational approximation of the coefficients of A preserves all zeros, the support, and the contact. Properness gives the exact count; clearing denominators gives the stated integral version. Finally choose a sufficiently small positive rational contact parameter in Proposition 22.5. Its count is $(3n - 2) + (2n + 1) = 5n - 1$, which equals the known upper bound. □

23 Earlier near-extremal constructions and their limits

The following radial and angular construction was an intermediate step. Its count is correct, but for $m \geq 2$ it is not the maximum: the exact range proved above supplies two additional zeros. We retain the proof because its three distinct scales make the limitation of that ansatz transparent.

Proposition 23.1. *For every $m \geq 1$ there is a polynomial of bidegree $(m, 2m+2)$ with exactly $5m^2+8m+2$ zeros, all nondegenerate.*

Proof. Set $k = m+2$, and define

$$\theta_j = \frac{j\pi}{k} \quad (0 \leq j < k), \qquad \phi_\nu = \frac{(\nu+1/2)\pi}{k} \quad (1 \leq \nu \leq m), \qquad \ell_\nu(x,y) = \sin\phi_\nu\, x - \cos\phi_\nu\, y.$$

For later use put

$$a_{\nu j} = \sin\frac{(\nu+1/2-j)\pi}{k}, \qquad L_j = \prod_{\nu=1}^{m} a_{\nu j}. \tag{23.1}$$

None is zero, $L_0 > 0$, and $\operatorname{sgn} L_j = (-1)^{j-1}$ for $1 \leq j < k$. Indeed, in the latter product exactly $j-1$ factors are negative. Let

$$\sigma = \sin\frac{\pi}{2k}, \qquad R_\nu = \frac{\nu\sigma}{4m}, \qquad c_\nu = (k+1)^\nu, \qquad A(\rho) = \prod_{\nu=1}^{m}(R_\nu^2 - \rho).$$

The real plane polynomial

$$F_t(z) = z^k A(|z|^2) + t^2 \prod_{\nu=1}^{m}\bigl(|z|^2 + \ell_\nu(\operatorname{Re} z, \operatorname{Im} z) - c_\nu t^2\bigr) \tag{23.2}$$

is the restriction of a polynomial of bidegree $(m, m+k)$. Its imaginary part is $A(|z|^2)\operatorname{Im} z^k$. Thus zeros lie on the m circles $|z| = R_\mu$ or the k lines $z = e^{i\theta_j}s$, $s \in \mathbb{R}$.

On $|z| = R_\mu$, each factor of the product in (23.2) gives the line $\ell_\nu = c_\nu t^2 - R_\mu^2$, which cuts the circle transversely in two points for small t. Different factors give different points. To see this at $t = 0$, the nonzero common intersection of $|z|^2 + \ell_\nu = 0$ and $|z|^2 + \ell_\eta = 0$ has modulus $|\cos((\phi_\nu - \phi_\eta)/2)| \geq \sin(3\pi/(2k)) > \sigma$, whereas $R_\mu \leq \sigma/4$. The other common intersection is 0. There are no overlaps when $m = 1$ either. These separations persist for small t. The circle zeros also avoid the k lines because $|a_{\nu j}| \geq \sigma > R_\mu$. The circles contribute exactly $2m^2$ distinct zeros.

On the jth line the real equation is

$$f_{j,t}(s) = (-1)^j s^k A(s^2) + t^2 \prod_{\nu=1}^{m}(s^2 + a_{\nu j}s - c_\nu t^2) = 0. \tag{23.3}$$

Its degree is $3m+2$. There are $2m$ simple real roots near $\pm R_\nu$. At scale $s = t^2 u$,

$$t^{-(2m+2)} f_{j,t}(t^2 u) \longrightarrow \prod_{\nu=1}^{m}(a_{\nu j}u - c_\nu).$$

This gives m distinct nonzero real roots. Distinctness follows from $\sigma \leq |a_{\nu j}| \leq 1$, $\sigma \geq 1/k$, and $c_{\nu+1}/c_\nu = k+1$. At scale $s = tu$,

$$t^{-k} f_{j,t}(tu) \longrightarrow u^m\bigl((-1)^j A(0)u^2 + L_j\bigr).$$

Since $A(0) > 0$, (23.1) gives two nonzero real roots for $j \geq 1$, and two nonreal roots for $j = 0$. All are simple. The complex implicit function theorem preserves the last two roots, and real coefficients preserve reality of the real simple roots. The three scales are disjoint and account for all $3m+2$ roots of (23.3). Its number of real roots is therefore $3m$ for $j = 0$ and $3m+2$ for every other j.

The lines meet only at 0, where $F_t(0) \neq 0$, and their zeros are disjoint from the circle zeros. Their total contribution is $k(3m+2)-2$. Thus (23.2) has exactly $2m^2+(m+2)(3m+2)-2 = 5m^2+8m+2$

zeros. On the lines, simplicity of the restricted real equation and the nonzero normal derivative of the imaginary equation give a nonzero Jacobian. On the circles, the same conclusion follows from their transverse intersections with the product factors and from $\operatorname{Im} z^k \neq 0$. These checks also show directly that all zeros are isolated and nondegenerate. □

For $m = 2$, Proposition 23.1 gives 38 zeros. Section 24 will exhibit a different polynomial with 40. The loss of two in this construction is therefore not a universal obstruction for the bidegree problem. It is, however, an exact obstruction in the total-degree class considered next.

For $m \geq 2$, Theorem 1.3 supersedes this lower bound by $M_{m,2m+2} = m^2 + (2m+2)^2$. In particular the two missing points in this specific construction were not a universal obstruction. The first total-degree-gap bound n^2-2 and its trace proof likewise remain valid, but Theorem 7.1 retains more information when the gap is larger.

24 Exact certificates and the remaining extremal problem

Constructions with many zeros may involve coefficients for which floating-point root finding is unreliable. A list of approximations alone does not prove either that the roots exist or that the list is complete. We use rational coefficients and exact contraction estimates. This serves the same purpose as the certified counts in [HLLM15, Section 1.3, author version, p. 3], although the verification below uses only rational arithmetic and the contraction mapping theorem.

24.1 Two explicitly certified polynomials

The coefficient arrays are printed in Appendix A. The supplementary files give the rational centers and independently executable verification programs.

Theorem 24.1. *The polynomial P_{40} in Appendix A has bidegree $(2,6)$ and exactly 40 zeros on $w = \bar{z}$, all nondegenerate, with 22 positive and 18 negative zeros. The polynomial P_{49} has bidegree $(2,7)$ and exactly 49 such zeros, all nondegenerate, with 27 positive and 22 negative zeros. Both polynomials are irreducible over $\mathbb{C}$.*

We give the verification argument in detail before drawing extremal conclusions. Every inequality in this argument is checked using exact fractions by the accompanying programs; numerical approximations are not used in the verification.

For either polynomial put $P^*(z,w) = \overline{P(\bar{w},\bar{z})}$. The programs first prove coprimality over $\mathbb{C}$. Multiply the coefficients by their common denominator and reduce modulo 101, sending i to 10, whose square is -1 in $\mathbb{F}_{101}$. Writing $P = \sum_j p_j(z) w^j$, the tests are

$$\gcd(p_0, p_1) = 1 \quad \text{in } \mathbb{F}_{101}[z], \qquad \gcd(P(0,w), P^*(0,w)) = 1 \quad \text{in } \mathbb{F}_{101}[w].$$

The relevant leading coefficients are checked to remain nonzero. The first nonzero resultant excludes a common factor depending only on z. The second shows that the resultant in w of P, P^* is not identically zero, by specialization at $z = 0$; it excludes any common factor of positive w-degree. Nonvanishing modulo 101 implies nonvanishing of the Gaussian integer resultant and hence of the characteristic-zero resultant. The two curves are therefore coprime and have at most $m^2 + n^2$ complex intersections, counted with multiplicity.

For the real zeros, define

$$F(x,y) = \big(\operatorname{Re} P(x+iy, x-iy),\ \operatorname{Im} P(x+iy, x-iy)\big).$$

A certificate box has rational center c and is $B_c = c + [-r,r]^2$, where $r = 10^{-20}$. At its center the verifier computes, exactly,

$$A = DF(c)^{-1}, \qquad T(v) = v - AF(v), \qquad \eta = \|AF(c)\|_\infty.$$

Here vector norms are maximum norms and matrix norms are the induced maximum-row-sum norms. Write $P(z,w) = \sum_{i,j} a_{ij} z^i w^j$, put $R = |c_1| + |c_2| + 2r$, and define

$$H = \sum_{i+j\ge 2} (i+j)(i+j-1)(|\operatorname{Re} a_{ij}| + |\operatorname{Im} a_{ij}|)R^{i+j-2}.$$

Every second partial derivative of either real component of F is bounded in absolute value by H on B_c. Consequently

$$\|DF(v) - DF(c)\|_\infty \le 4rH, \qquad \|DT(v)\|_\infty \le q := 4r\|A\|_\infty H.$$

For every box the exact inequalities verified are

$$q + \eta/r < 10^{-7}. \tag{24.1}$$

In particular, T is a contraction and $\|T(v) - c\|_\infty \le \eta + qr < r$ for $v \in B_c$. It maps the closed box into its interior, so that box contains a unique zero. The estimate $\|I - A\,DF(v)\|_\infty < 1$ also shows that its Jacobian is invertible. All boxes are checked to be pairwise disjoint.

For P_{40} there are 40 such boxes. They exhaust the complex intersection bound $2^2 + 6^2 = 40$, and hence certify the exact count. For P_{49} there are 49 real boxes and four additional boxes for the complex system $P = P^* = 0$. In the latter boxes we regard (z,w) as four real coordinates and apply the same contraction argument to $(\operatorname{Re} P, \operatorname{Im} P, \operatorname{Re} P^*, \operatorname{Im} P^*)$. The Hessian bound uses $R = \max\{|\operatorname{Re} z_c| + |\operatorname{Im} z_c|, |\operatorname{Re} w_c| + |\operatorname{Im} w_c|\} + 2r$ and the contraction constant $q = 16r\|A\|_\infty H$. These boxes also satisfy (24.1). The four boxes are checked to avoid $w = \bar z$, and all 53 boxes are disjoint after the real boxes are embedded in four coordinates. They exhaust $2^2 + 7^2 = 53$ complex intersections. Exactly 49 intersections are fixed, proving the asserted real count.

The signs of the real Jacobians at the rational centers are computed exactly. They cannot change within a box because the Jacobian is invertible there, and yield the stated orientation counts. Irreducibility of P_{40} follows from Corollary 2.4. For P_{49} it follows from the next proposition, whose proof does not assume that irreducibility. This completes the count and nondegeneracy part of Theorem 24.1.

24.2 A twelve-term real integer extremizer

Proposition 24.2. *The real integer polynomial*

$$\begin{aligned} R(z,w) = {} & 10000(10z^3 + 10zw - 1)^2 \\ & + 5000\big((10z^3 - 1)w^2 - 10z^2 w + z\big) - z^5 w^2 (z + 2000) \end{aligned} \tag{24.2}$$

has exactly forty zeros on $w = \bar z$, all nondegenerate: twenty-two positive and eighteen negative. Exactly eight lie on the real axis. Its expansion has twelve nonzero terms, bidegree $(2,6)$, and coefficients of absolute value at most 2000000. In particular it is an explicit real integer extremizer for $M_{2,6} = 40$ and is irreducible over $\mathbb{C}$.

Proof. Apply the preceding exact contraction argument to $P = 10^{-6}R$. The file `real_2_6_40_certificate.json` supplies forty centers in $10^{-80}\mathbb{Z}^2$ and radius $r = 10^{-40}$. The independent standard-library verifier `verify_real_2_6_40.py` reconstructs all coefficients directly from (24.2), and checks $q + \eta/r < 10^{-7}$ in every box using rational arithmetic only. The boxes are pairwise disjoint; their center Jacobians have the stated signs. Thus they contain forty distinct nondegenerate zeros.

For completeness the coprimality check uses the integer polynomial R modulo 101. The degrees of $R(z,0)$, $[w]R(z,w)$, $R(0,w)$, and $R^*(0,w)$ remain respectively $6, 4, 2, 6$. The first two have gcd one in $\mathbb{F}_{101}[z]$, and the last two have gcd one in $\mathbb{F}_{101}[w]$. The resultant argument above proves that R, R^* are coprime. The forty real zeros therefore exhaust all $6^2 + 2^2$ complex intersections. Eight centers have second coordinate zero. Their boxes are invariant under conjugation, so uniqueness and the reality of the coefficients place these eight zeros on the real axis. Every other box misses the real axis. Irreducibility follows from Corollary 2.4. □

25 Sharp factorization bounds and reducible extrema

Irreducibility at the quadratic maximum can be strengthened to an exact deficit and a classification of every reducible equality case. Write $M^{\mathrm{red}}_{m,n}$ for the rectangular maximum restricted to reducible polynomials. In this section only, extend the notation by $M_{0,b} = M_{b,0} = b$, $M_{a,b} = M_{b,a}$, and $M_{a,a} = 2a^2$. The last equality is elementary: take

$$R(z,w) = \prod_{j=1}^{a}(zw - r_j^2), \qquad I(z,w) = \prod_{j=1}^{a}(z + w - 2c_j), \qquad f = R + iI,$$

where $0 < r_1 < \cdots < r_a$ and the distinct real numbers c_j satisfy $|c_j| < r_1$. The a circles and a vertical lines have exactly $2a^2$ transverse intersections. Proposition 2.2 supplies the matching upper bound. Rational choices give Gaussian integer coefficients after multiplication by a common denominator.

25.1 The optimal quadratic deficit and its equality cases

Theorem 25.1. *For $n > m \geq 2$, every reducible polynomial of bidegree (m,n) with finite zero set satisfies*

$$N(f) \leq (m-1)^2 + n^2 + 1 = m^2 + n^2 - 2(m-1). \tag{25.1}$$

Equality holds if and only if, up to a nonzero scalar,

$$f(z,w) = (w - \bar{a})R(z,w), \qquad \operatorname{bideg} R = (m-1, n), \qquad N(R) = (m-1)^2 + n^2,$$

and $R(a,\bar{a}) \neq 0$. In particular, R is irreducible, and

$$M^{\mathrm{red}}_{m,n} = (m-1)^2 + n^2 + 1 \quad \Longleftrightarrow \quad M_{m-1,n} = (m-1)^2 + n^2. \tag{25.2}$$

Proof. If f has a nonconstant factor depending only on z, extract one linear factor. The remaining polynomial has bidegree $(m, n-1)$, so its count plus one is at most $m^2 + (n-1)^2 + 1$, strictly less than the right side of (25.1).

Otherwise choose any factorization $f = uv$ and write its bidegrees as (a, b) and $(m - a, n - b)$, where $1 \le a \le m - 1$. If $1 \le b \le n - 1$, the intersection bounds give

$$N(f) \le a^2 + (m - a)^2 + b^2 + (n - b)^2 \le (m - 1)^2 + 1 + (n - 1)^2 + 1,$$

again strictly below (25.1). If $b = 0$, the univariate factor has at most a distinct zeros, and hence

$$N(f) \le a + (m - a)^2 + n^2 \le 1 + (m - 1)^2 + n^2.$$

Indeed, the difference in the last inequality is $(a - 1)(2m - a - 2)$; since $1 \le a \le m - 1$, it vanishes only at $a = 1$. The case $b = n$ follows by interchanging the factors. Equality therefore requires a conjugate-linear factor, a remaining factor attaining its rectangular bound, and disjoint zero sets. These conditions also suffice. Corollary 2.4 proves the remaining factor is irreducible. □

Consequently the reducible maximum is explicitly $n^2 + (m - 1)^2 + 1$ whenever the neighboring pair $(m - 1, n)$ belongs to a proved quadratic range. In particular this holds for

$$m \ge 3, \qquad m < n \le 3m - 3, \qquad n \ne 3m - 4.$$

The omitted pairs $(m, n) = (3, 5), (4, 8)$ also satisfy the formula, as does $(2, 3)$, by the separately established neighboring endpoints. For example $M^{\mathrm{red}}_{4,9} = 91$, $M^{\mathrm{red}}_{5,12} = 161$, and $M^{\mathrm{red}}_{10,27} = 811$. All these maxima have nondegenerate Gaussian integer extremizers: choose a rational complex a outside the finite zero set of the corresponding rational extremizer and clear denominators. The excluded infinite line is not being declared unattainable.

25.2 A linear bound for split conjugate degree

Define a symmetric, explicit bound for all nonnegative degrees by

$$\widehat{U}_{a,b} = \begin{cases} a + b, & \min(a, b) = 0, \\ 2a^2, & a = b > 0, \\ \min\{a^2 + b^2, (4u + 1)v - 5u\}, & 0 < u := \min(a, b) < v := \max(a, b). \end{cases}$$

The diagonal convention is necessary; the off-diagonal surface formula must not be substituted at $(1, 1)$.

Lemma 25.2. *If $n > m \ge 2$, $1 \le a \le m - 1$, and $0 \le b \le n$, then*

$$\widehat{U}_{a,b} + \widehat{U}_{m-a,n-b} \le \widehat{U}_{m-1,n} + 1. \tag{25.3}$$

Proof. The quadratic part $(m-1)^2+n^2+1$ was proved in the preceding argument, which used only the same individual upper bounds. For the linear part use, for every $a, b \ge 0$,

$$\widehat{U}_{a,b} \le H(a, b) := 4ab + a + b - 6\min(a, b) + 21_{(a,b)=(1,1)}.$$

On the axes this is exact; off the diagonal it is the linear surface bound; on the diagonal it follows from $2a^2 \le 4a^2 - 4a$ for $a \ge 2$, with the displayed correction at $a = 1$.

Put $s(b) = \min(a, b) + \min(m - a, n - b)$. First omit the two possible correction terms in H. Subtracting the sum of the remaining terms from $(4m - 3)n - 5m + 6$ gives

$$D(b) = 4\{(a - 1)n + (m - 2a)b\} + 6\{s(b) - m + 1\}.$$

This is a concave function of $b \in [0, n]$, with endpoint values $(a-1)(4n-6)$ and $(m-a-1)(4n-6)$, both nonnegative. Thus $D(b) \geq 0$. At most one correction can occur because $n > m$. If one factor has degrees $(1, 1)$, direct subtraction including that correction gives $4(m-1) > 0$. This proves the linear bound $(4m-3)n - 5m + 6$ as well as the quadratic one. Their minimum is exactly the right side of (25.3). □

Theorem 25.3. *For $n > m \geq 2$,*

$$\begin{aligned} 1 + \max\{M_{m,n-1}, M_{m-1,n}\} &\leq M^{\mathrm{red}}_{m,n} \\ &\leq 1 + \max\{M_{m,n-1}, \widehat{U}_{m-1,n}\} \\ &\leq 1 + \max\{\widehat{U}_{m,n-1}, \widehat{U}_{m-1,n}\}. \end{aligned} \tag{25.4}$$

For every $m \geq 2$ and $n \geq m^2 + 2$ one has the exact reduction

$$M^{\mathrm{red}}_{m,n} = M_{m,n-1} + 1. \tag{25.5}$$

Every extremizer in this range has an analytic linear factor $z - a$ whose zero is disjoint from the zeros of a maximal bidegree-$(m, n-1)$ factor. Furthermore,

$$(4m+1)n - 4m^2 - 5m \leq M^{\mathrm{red}}_{m,n} \leq (4m+1)n - 9m, \qquad n \geq m^2 + 2. \tag{25.6}$$

Thus the reducible class also has the sharp fixed-m leading coefficient $4m + 1$.

Proof. Multiplication of an extremizer at either neighboring degree by $z - a$ or $w - \bar{a}$, with a outside its zero set, proves the lower bound. If a reducible polynomial has an analytic factor, extract a linear one to bound its count by $M_{m,n-1} + 1$. Otherwise its conjugate degree splits into two positive parts, and Lemma 25.2 applies. This proves (25.4).

If $n \geq m^2 + 2$, then $n - 1 \geq 2m + 1$ and the radial construction in Theorem 5.5 gives

$$M_{m,n-1} + 1 \geq (4m+1)(n-m-1) + 1.$$

Its difference from the split-conjugate upper bound $(4m-3)n - 5m + 6$ is $4n - 4m^2 - 6 \geq 2$. The analytic-factor branch is therefore strictly larger. Equality in its count requires both a maximal remaining factor and disjoint zero sets, proving (25.5) and the stated structure. Combining this equality with the rectangular linear upper bound proves (25.6). This reduction does not assign a numerical value to an unknown $M_{m,n-1}$. □

For conjugate degree one, every reducible polynomial already has an analytic linear factor. Hence

$$M^{\mathrm{red}}_{1,2} = 3, \qquad M^{\mathrm{red}}_{1,n} = 5n - 9 \quad (n \geq 3).$$

For conjugate degree two, (25.4) and the known small endpoints give the sharper starting point

$$M^{\mathrm{red}}_{2,3} = 11, \qquad M^{\mathrm{red}}_{2,4} = 16, \qquad M^{\mathrm{red}}_{2,n} = M_{2,n-1} + 1 \quad (n \geq 5).$$

In particular the previously isolated certificate consequence becomes

Proposition 25.4. *$M^{\mathrm{red}}_{2,7} = 41$. It is attained by $(z - a)P_{40}$ for any a outside the zero set of the certified 40-zero polynomial.*

Proof. Use $M_{2,6} = 40$ in the preceding formula. □

Since $49 > 41$, this supplies the remaining irreducibility assertion for P_{49} in Theorem 24.1. It also gives $M^{\mathrm{red}}_{2,8} = M_{2,7} + 1 \in \{50, 52, 54\}$; the three alternatives remain unresolved together with $M_{2,7}$.

25.3 Sharp factor-count bounds in the pure-head class

In this subsection the conjugate degree is *exactly* $m > 0$. Factors are counted with multiplicity, so repeated factors cannot evade the degree budget.

Theorem 25.5. *Let $f \in \mathcal{E}_{m,n}$ have exact conjugate degree $m > 0$ and a factorization into ℓ nonconstant factors. Then*

$$\ell \le n - m, \qquad N(f) \le (n - \ell + 1)^2 + \ell - 1. \tag{25.7}$$

In particular every such polynomial with $m = n - 1$ is irreducible. For $2 \le \ell \le n - m$, put $d = n - \ell + 1 > m$. The upper bound in (25.7), among polynomials having at least ℓ factors, is attained if and only if $E_{m,d} = d^2$. Every equality case has the form

$$f(z, w) = c \prod_{j=1}^{\ell-1} (z - a_j) R(z, w), \tag{25.8}$$

where R is an irreducible pure-head extremizer of total degree d and exact conjugate degree m, the a_j are distinct, and none is a zero of $R(z, \bar{z})$. Conversely every polynomial of this form attains the bound.

Proof. The product of the highest homogeneous forms of the factors is cz^n. Unique factorization in $\mathbb{C}[z, w]$ therefore makes the highest form of every factor a nonzero pure power of z. If its total degree is $d_j \ge 1$ and its conjugate degree is m_j, then $m_j \le d_j - 1$. Degrees add under multiplication, so $m \le n - \ell$.

Each factor has at most d_j^2 distinct zeros, and therefore

$$N(f) \le \sum_{j=1}^{\ell} d_j^2 \le (n - \ell + 1)^2 + \ell - 1, \qquad \sum_j d_j = n.$$

For the last inequality, transferring one degree from a smaller $d_j > 1$ to a largest degree strictly increases the sum of squares. Equality forces the degrees $d, 1, \dots, 1$. Every degree-one pure-head factor is analytic. The remaining factor consequently has exact conjugate degree m, and it must attain d^2 with zero set disjoint from the distinct linear-factor zeros. An additional factorization of that degree-d factor would give strictly fewer than d^2 zeros. Likewise, more than ℓ factors strictly decreases the bound because $m > 0$ ensures $d \ge 2$. This proves necessity and the stated structure. For sufficiency, an extremizer attaining $E_{m,d} = d^2$ can be taken with exact conjugate degree m, by a small generic w^m perturbation if needed, and with all zeros nondegenerate by the regularization result. Choose distinct a_j outside that finite set. □

For example, whenever $m \ge 2$, $m \le n - 2$, and $n \le 2m + 3$, the reducible pure-head maximum with exact conjugate degree is

$$E_{m,n}^{\mathrm{red,ex}} = (n - 1)^2 + 1. \tag{25.9}$$

The further pure-head saturation families extend this equality whenever they contain $(m, n-1)$. For $m = 1$, all reducible members have an analytic linear factor, giving instead the complete formula $E_{1,n}^{\mathrm{red,ex}} = 5n - 10$ for $n \ge 3$. At the largest possible factor count $\ell = n - m$, the exact answer for *every* $n > m > 0$ is

$$\max N(f) = (m + 1)^2 + n - m - 1. \tag{25.10}$$

Indeed, the degree-$(m+1)$ harmonic endpoint attains $(m+1)^2$, and multiplication by $n-m-1$ disjoint analytic linear factors attains (25.10). These sharp factorization strata do not determine a missing maximum in the irreducible class.

26 A fixed rational fourteen-term extremizer of bidegree $(3,9)$

The family in Section 17 already proves $M_{3,9} = 90$ and also proves the stronger pure-head assertion $E_{3,9} = 81$. The following separate, fourteen-term family is retained because it has a certified fixed rational parameter and an independently checkable local-to-global proof. Its total degree is twelve, so it is not an element of $\mathcal{E}_{3,9}$. Section 2 recalls why the same bound holds whenever the real zero set is finite, without a coprimality assumption.

Here the intersection bound is attained in bidegree $(3,9)$.

Theorem 26.1. *The exact maximum is*

$$M_{3,9} = 90.$$

It is attained by a polynomial with rational real coefficients and ninety nondegenerate zeros.

The construction is explicit. Put $c = 1/8$ and define

$$f = z^3 + zw - c, \qquad f^* = w^3 + zw - c, \qquad A = z^2w^2, \qquad B = -z^3. \tag{26.1}$$

Set

$$T_1 = f(Af + Bf^*), \tag{26.2}$$

and

$$\begin{aligned} T_2 = &-\frac{w^3}{1536} - \frac{7}{24}z^8w^2 + \frac{103}{192}z^7w + \frac{11}{384}z^4w - \frac{11}{12288}zw \\ &+ \frac{13}{48}z^9 - \frac{83}{1536}z^6 - \frac{23}{6144}z^3 + \frac{13}{98304}. \end{aligned} \tag{26.3}$$

The final correction is

$$T_3 = -\frac{1}{8}z^9w^3 + \frac{169}{6912}z^2w^2 - \frac{29}{3072}zw + \frac{67}{98304}. \tag{26.4}$$

For a positive real parameter t, let

$$P_t = f^3 + tT_1 + t^2T_2 + t^3T_3. \tag{26.5}$$

The exact cancellations in (26.2) are important: although its unexpanded expression appears to allow w^4, its conjugate degree is only three. Table 2 records the full coefficient array of (26.5) and provides another definition of the family.

The following stronger statement gives both an analytic construction and a fixed rational example.

Theorem 26.2. *There exists $t_0 > 0$ such that, for every $0 < t < t_0$, P_t has exact bidegree $(3,9)$ and exactly ninety nondegenerate zeros. The same conclusion holds for $t = 10^{-8}$. In both statements the numbers of positive and negative zeros are 48 and 42, respectively.*

The coefficient identities in Section 26.2 lead to the analytic small-parameter proof in Section 26.3. The particular value 10^{-8} is verified independently in Section 26.4; the analytic proof does not assume that an unspecified smallness threshold exceeds this number. Theorem 26.1 follows from Theorem 26.2 and Proposition 2.2. This settles a single bidegree beyond the range $n \le 2m+2$; it is not a formula for arbitrary $M_{m,n}$.

Table 2: The nonzero coefficients $[z^i w^j]P_t$. At $t = 10^{-8}$ all fourteen displayed coefficients are nonzero.

i	j	$[z^i w^j]P_t$
9	3	$-t^3/8$
9	0	$(13t^2+48)/48$
8	2	$-t(7t-24)/24$
7	1	$(103t^2-192t+576)/192$
6	3	t
6	0	$-(83t^2-192t+576)/1536$
5	2	$-(5t-12)/4$
4	1	$(11t^2+96t-288)/384$
3	3	$-(t-8)/8$
3	0	$-(23t^2+96t-288)/6144$
2	2	$(169t^3+108t-2592)/6912$
1	1	$-(116t^3+11t^2-576)/12288$
0	3	$-t^2/1536$
0	0	$(67t^3+13t^2-192)/98304$

26.1 A cubic with nine zeros

The elementary model used at both scales is the seed in (26.1).

Lemma 26.3. *The function $h(u) = u^3 + |u|^2 - 1/8$ has exactly nine nonzero, nondegenerate zeros. Six have positive Jacobian and three have negative Jacobian. If*

$$\omega = e^{2\pi i/3}, \qquad r_1 = -\frac{1}{2}, \qquad r_2 = \frac{-1+\sqrt{5}}{4}, \qquad r_3 = \frac{-1-\sqrt{5}}{4},$$

its zero set is $\{r_a\omega^j : 1 \le a \le 3,\ 0 \le j < 3\}$.

Proof. The imaginary part is $\operatorname{Im} u^3$, so every zero lies on $\bigcup_{j=0}^{2} \omega^j\mathbb{R}$. On each of these three lines the equation is $r^3 + r^2 - 1/8 = 0$, whose factorization is

$$r^3 + r^2 - \frac{1}{8} = \left(r + \frac{1}{2}\right)\left(r^2 + \frac{1}{2}r - \frac{1}{4}\right).$$

The three roots displayed in the statement are distinct and nonzero. The three lines meet only at zero, which is not a root. At a real root r, the Jacobian is $3r^3(3r+2)$; it is negative for $r = -1/2$ and positive at the other two roots. Rotating the source by ω preserves the function and the Jacobian sign. □

We shall also need translated and rescaled copies of this model. No inequality between the quadratic coefficients is required.

Lemma 26.4. *Let $a, b \in \mathbb{C}$ with $b \neq 0$. The real-analytic polynomial*

$$\mathcal{M}_{a,b}(u) = u^3 + au^2 + bu\bar{u} + \frac{a^2 + b\bar{a}}{3}u + \frac{ba}{3}\bar{u} + \frac{a^3}{27} + \frac{b|a|^2}{9} - \frac{1}{8}|b|^2 b \tag{26.6}$$

has exactly nine nondegenerate zeros, six positive and three negative.

Proof. Choose $\lambda \neq 0$ such that $|\lambda| = |b|$ and $\lambda^3 = |b|^2 b$. With $v = (u + a/3)/\lambda$, direct expansion gives

$$\mathcal{M}_{a,b}(u) = |b|^2 b \left(v^3 + |v|^2 - \frac{1}{8} \right).$$

Both the affine change of source and multiplication of the target by a nonzero complex number preserve nondegeneracy and Jacobian signs. Apply Lemma 26.3. □

26.2 Two polynomial identities

The corrections (26.3)–(26.4) were chosen to prescribe the linear and constant terms of (26.6) at every seed zero without increasing the bidegree. Lemma 26.5 states all the algebra needed for the construction.

Let $I = (f, f^*) \subset \mathbb{Q}[z, w]$. Define

$$D = \frac{A^2 + BA}{3}, \qquad E = \frac{BA}{3}, \qquad G = \frac{A^3}{27} + \frac{BA^2}{9} - \frac{1}{8} B^2 B^*. \tag{26.7}$$

Here $A^* = A$. At a real slice point this is precisely the occurrence of $\bar{a}$ and $|b|^2 b$ in (26.6).

The first identity is a statement about first derivatives, not just values.

Lemma 26.5. *The polynomials above satisfy*

$$T_2 - Df - Ef^* \in I^2, \tag{26.8}$$

and

$$T_3 - G \in I. \tag{26.9}$$

Moreover, with $\widehat{P}_t(u, v) = u^9 v^3 P_t(1/u, 1/v)$,

$$t^{-3} \widehat{P}_t(tu, tv) \longrightarrow v^3 + uv - \frac{1}{8} \tag{26.10}$$

in C^1 *on every compact subset of* $\mathbb{C}^2$ *as* $t \to 0^+$.

Proof. For (26.8), an explicit identity is

$$T_2 - Df - Ef^* = U(f^*)^2 + Vff^* + Wf^2, \tag{26.11}$$

where

$$\begin{aligned} U &= -\frac{z^3}{3}, \\ V &= \frac{z^2 w^2}{3} - \frac{z^4 w}{3} + \frac{zw}{24} - \frac{z^3}{12} + \frac{1}{192}, \\ W &= \frac{3}{8} z^2 w^2 + \frac{7}{192} zw + \frac{13}{48} z^3 + \frac{5}{1536}. \end{aligned}$$

Expanding the right side of (26.11), with (26.1) and (26.7), gives exactly (26.3).

For (26.9), reduce by the two relations $z^3 = 1/8 - zw$ and $w^3 = 1/8 - zw$. Their leading monomials are coprime, so these are a Gröbner basis in a graded order. The remainder of $T_3 - G$ is

zero. This assertion may also be checked as a one-variable calculation: at $f = f^* = 0$, put $s = zw$. Then

$$z^3 = w^3 = \frac{1}{8} - s, \qquad s^3 = \left(\frac{1}{8} - s\right)^2,$$

and reduce $G + z^9 w^3/8$ to degree at most two in s. The remainder is $169s^2/6912 - 29s/3072 + 67/98304$. The two-variable remainder calculation proves the ideal membership, and in particular its required value consequence.

For (26.10), consider a term $t^k z^i w^j$ of (26.5). After inversion, rescaling, and division by t^3, it becomes $t^{k+9-i-j} u^{9-i} v^{3-j}$. All these powers of t are nonnegative. The terms with exponent zero are z^9 in f^3, tz^8w^2 in tT_1, and $-t^3 z^9 w^3/8$ in $t^3 T_3$. They give $v^3 + uv - 1/8$. All remaining terms have a positive integral power of t, which proves convergence together with all derivatives on compact sets. □

The accompanying verifier checks (26.11) by polynomial arithmetic over $\mathbb{Q}$, and checks (26.9) by the explicit degree-reducing substitutions in the proof. The construction below uses these identities, not a numerical approximation to them.

26.3 Eighty-one finite zeros and nine zeros at infinity

We prove the small-parameter part of Theorem 26.2. By Table 2, every term of P_t has $i \le 9$ and $j \le 3$, and its corner coefficient is $-t^3/8$. Thus its bidegree is exactly $(3, 9)$ for every $t > 0$.

26.3.1 The finite clusters

Let z_0 be any of the nine zeros of F_f. Lemma 26.3 and the real inverse function theorem give a real-analytic local inverse $z = \psi(\xi)$ of $\xi = f(z, \bar{z})$, with $\psi(0) = z_0$. On the real slice, $f^*(z, \bar{z}) = \bar{\xi}$. Set

$$a = A(z_0, \bar{z}_0), \qquad b = B(z_0, \bar{z}_0).$$

The seed zeros are nonzero, so $b = -z_0^3 \neq 0$.

Substitute $z = \psi(tu)$ and divide $P_t(z, \bar{z})$ by t^3. The first two terms of (26.5) become

$$u^3 + A(\psi(tu), \overline{\psi(tu)})u^2 + B(\psi(tu), \overline{\psi(tu)})u\bar{u}.$$

By (26.8), the third term converges to $D(z_0, \bar{z}_0)u + E(z_0, \bar{z}_0)\bar{u}$: its error is $O(t)$ in C^1 on bounded sets. Indeed, the remainder belongs to $(\xi, \bar{\xi})^2$. By (26.9), the last term converges to $G(z_0, \bar{z}_0)$. Hence

$$t^{-3} P_t\big(\psi(tu), \overline{\psi(tu)}\big) \longrightarrow \mathcal{M}_{a,b}(u) \tag{26.12}$$

in C^1 on compact subsets of the local u-plane.

By Lemma 26.4, the right side of (26.12) has nine simple zeros, each of which persists for small positive t. Choose their neighborhoods disjoint, and choose the nine seed neighborhoods disjoint as well. There are finitely many of these choices, so a single positive threshold for t works for all of them. We obtain $9 \cdot 9 = 81$ distinct nondegenerate zeros, all in a fixed bounded set. At this point only a lower count is needed; no assertion about other scales is being made.

26.3.2 The unbounded cluster

Restrict (26.10) to $v = \bar{u}$. Its limit is

$$\bar{u}^3 + |u|^2 - \frac{1}{8} = \overline{h(u)}.$$

It has the same nine nonzero simple zeros as h. Apply the real implicit function theorem in disjoint neighborhoods of these nine points. For small positive t this gives nine simple zeros of $\widehat{P}_t(tu, t\bar{u})$ with u bounded away from zero. They correspond to zeros of P_t under

$$z = \frac{1}{tu}.$$

The corresponding $|z|$ tend to infinity. They are thus distinct from all eighty-one bounded zeros. Inversion is holomorphic off zero; multiplication by a nonvanishing complex factor at a zero also preserves the sign and nonvanishing of its real Jacobian. All nine new zeros are nondegenerate.

26.3.3 Completeness and signs

We have constructed ninety distinct nondegenerate zeros. By Lemma 2.3, P_t, P_t^* are coprime, and their bidegree intersection number is $9^2 + 3^2 = 90$. This proves that the zero set is finite and contains no additional zeros.

For the signs, the seed has six positive and three negative zeros. At a positive seed zero, (26.12) supplies six positive and three negative zeros; at a negative seed zero, the signs are reversed by the local coordinate f. The bounded clusters therefore contain $6{\cdot}6{+}3{\cdot}3 = 45$ positive and $6{\cdot}3{+}3{\cdot}6 = 36$ negative zeros. The model at infinity is the conjugate of h, so it contributes three positive and six negative zeros. The total is 48 positive and 42 negative zeros, as stated in Theorem 26.2.

26.4 An exact certificate for a fixed rational parameter

This section proves the assertion in Theorem 26.2 for $t = 10^{-8}$ independently of any unspecified smallness threshold. The certificate contains ninety centers in $10^{-100}\mathbb{Z}^2$ and uses the common box radius $r = 10^{-40}$. The polynomial has the fourteen rational coefficients in Table 2, evaluated at this t. The file `bidegree_3_9_certificate.json` records these data exactly.

Here is the inclusion argument checked for every box. Let

$$V(x, y) = \big(\operatorname{Re} P_t(x + iy, x - iy), \operatorname{Im} P_t(x + iy, x - iy)\big)$$

and let c_0 be the rational center of a box $\mathcal{B} = c_0 + [-r, r]^2$. All entries of $V(c_0)$ and $DV(c_0)$ are rational. Put

$$L = DV(c_0)^{-1}, \qquad \eta = \|LV(c_0)\|_\infty.$$

Writing $P_t(z, w) = \sum a_{ij} z^i w^j$, define the rational number

$$H = \sum_{i+j\geq 2} |a_{ij}|(i + j)(i + j - 1)\big(|(c_0)_x| + |(c_0)_y| + 2r\big)^{i+j-2}. \tag{26.13}$$

Every second real derivative of either component of V is bounded by (26.13) on $\mathcal{B}$. Thus

$$\|DV(v) - DV(c_0)\|_\infty \leq 4rH \qquad (v \in \mathcal{B}).$$

For $T(v) = v - LV(v)$, let $q = 4r\|L\|_\infty H$. The exact verifier checks

$$q + \frac{\eta}{r} < 10^{-12} \tag{26.14}$$

for each of the ninety boxes. It follows that T maps each box strictly into itself and is a contraction there. Its unique fixed point is a zero of V. Moreover, $\|I - LDV(v)\|_\infty < 1$, so DV is invertible throughout the box.

The verifier also checks that the boxes are pairwise disjoint, that the center Jacobian signs are 48 positive and 42 negative, and that the coefficient data agree exactly with (26.5). All comparisons in (26.14) use integer fractions. The resulting ninety simple zeros force coprimality by Lemma 2.3, and therefore exhaust the zero set.

The certificate is independently executable with the Python standard library:

```
python verify_bidegree_3_9.py
```

No numerical root finder is called by this program. The role of approximation was only to locate the rational boxes; existence, nondegeneracy, and completeness follow from exact inequalities and the intersection argument.

27 Audit of the successive claims and the remaining questions

The research record contains a sequence of improving bounds, several changes of coefficient space, and both analytic and computer-assisted constructions. The following ledger states what is retained. A claim superseded by a stronger theorem is not thereby false; conversely, sharpness inside a restricted family is not evidence of sharpness in a larger one.

27.1 The original problem and assertions that must not be conflated

Problem 1.1 asks only for a coefficient-uniform bound linear in n for fixed m. Theorem 1.2 proves that statement. Its asymptotic coefficient is sharp by Proposition 5.2. Determining $M_{m,n}$ for every pair is a strictly stronger problem. The polynomial part of Eremenko's note is addressed; the elliptic systems on its later pages are not solved by any argument in this note.

The finite equality range in Theorem 1.3, including the unrestricted splitting family and the linear sharp region, does not cover $(m, n) = (2, 7)$. The certified example of Theorem 24.1, the intersection bound, and the parity result in Section 4 imply precisely

$$49 \le M_{2,7} \le 53, \qquad M_{2,7} \in \{49, 51, 53\}.$$

None of the analytic constructions or exact certificates supplied here chooses among these three possibilities. In particular, the candidate formula $M_{m,n} = \min\{m^2 + n^2, (4m + 1)n - 5m\}$ remains a question outside the proved equality range, not a theorem inferred from small examples.

Inside $m < n \le 3m$, the remaining unproved line is $n = 3m - 1$, $m \ge 4$. The integer decomposition used in the new proof misses this line; that is not an upper-bound obstruction. Theorem 21.3 now improves the first missing case to $129 \le M_{4,11} \le 137$, with an exact regular 129-zero construction. The region $n > 3m$ and the pure-head values at the other residues of n also remain separate questions. Theorem 19.16 now proves $E_{m,3k} = 9k^2$ for $k \le m < 3k$, including $E_{4,12} = 144$; the corresponding rectangular value is $M_{4,12} = 160$.

The maximum in the harmonic class is a different question again. Its best proved general bounds here are (1.4). The minimum and the lower part of the distinct-count spectrum, in contrast, are now exact by Theorem 20.5; this does not determine an unknown maximum. For each fixed $m \geq 2$ the maximal-valence bounds leave a gap between the constructed coefficient $m+2$ and the upper coefficient $3m+2$. Neither existence of a limit of $M^{\mathrm{harm}}_{m,n}/n$ nor its value for arbitrary $m \geq 2$ is proved. The suggested lower bound $2m(n-1)$ cannot hold throughout all $m < n$: for $n = m+1$ and $m \geq 3$, it exceeds the classical upper bound n^2. For fixed m and sufficiently large n, that observation supplies no obstruction, so it does not settle the corresponding asymptotic question.

27.2 A claim-by-claim ledger

Assertion in the successive drafts	Audited disposition and proof in this note
The initial quadratic estimate m^2+n^2	Correct for finite zero sets, including common conjugate factors. Proposition 2.2 gives a direct factorwise proof independent of regularization.
The intermediate bounds $6mn-5m$ and $5m(n-1)$	Valid but superseded by (1.1); they are not advertised as the strongest result. A constant perturbation alone is not sufficient for the reduction.
Unrestricted splitting and a linear sharp region	Theorems 19.1 and 19.2 prove pure-head saturation for every $r,k \geq 1$, and rectangular saturation for $m < n \leq 3m$, $n \neq 3m-1$. The old restriction $r \geq k-1$ and the square-root-width limitation are removed. The missing line is not proved unattainable, and the rectangular cone is not asserted for the pure-head class.
Exact local fibers and real-coefficient extrema	Theorem 19.3 counts every target, including smooth boundary and cusp targets. The phase criterion is necessary and sufficient, and the centered target radius is optimal. The unshifted model realizes the principal infinite families and rectangular cone over $\mathbb{Q}$ and $\mathbb{Z}$, simultaneously attaining the real-axis bounds. This does not close the missing global degrees.
Complete attainable-count spectra	Theorem 20.1 fills every admissible regular count below each maximum and every positive finite distinct count in the rectangular corner and pure-head classes. It does not calculate unknown maximal endpoints. The fixed-degree harmonic no-gap theorem is attributed to [SZ21].
Sharp harmonic minimum and target deficits	Theorem 20.5 proves the minimum $n-m+1$ and the sharp bound $\sum_c (n-N_c)_+ \leq \deg \gcd(p',q')$. Integer trinomials attain each lower count. Combining this with the attributed no-gap theorem completes the distinct spectrum, without identifying an unknown maximum.
Pure-head spectrum with exact conjugate degree	Theorem 20.8 proves the distinct spectrum $2,\dots,E_{m,n}$ for exact $m > 0$. A pure-head map has one-point zero fiber only when it is an analytic pure power. Integer trinomials realize every lower count with the requested exact degree.
Exact trinomial parameter disk	Theorem 20.9 gives the largest centered disk retaining $n-k$ regular nonzero zeros. Corollary 20.10 counts all first-boundary fibers and the exact first increase $2\gcd(n-k,m+k)$. The origin has index k and is degenerate when $k > 1$.

Assertion in the successive drafts	Audited disposition and proof in this note
Pure-head completion from the rectangular cone	Proposition 20.25 gives $E_{m,n} \geq n + 10m^2 - 2m$ for $n \geq 4m + 1$. This improves a range of lower bounds and is not a claim of an exact maximum.
Optimal cubic stability parameter	Theorem 17.2 gives the exact radius for every real constant term and its unique maximum $(5\sqrt{5} - 11)/2$. The rational constant $9/100$ and all existing numerical inputs remain valid.
Quartic splitting on $n = 2m + 4$	The earlier worked proof covers $m = 4r$, $r \geq 3$. Theorem 19.1 now also gives $E_{8,20} = 400$, $M_{8,20} = 464$, and $M_{4,12} = 160$. The translated boundary construction now also proves $E_{4,12} = 144$.
Complete trinomial parameter-plane count	For $1 \leq k < m < n$, Theorem 20.11 proves the family maximum $n + 2m + k + 1$, attained by $\varepsilon = 3$, and its full arithmetic-step spectrum. The critical-value winding formula covers regular regions, folds, and first-boundary cusps. The endpoint $k = m$ has its separate sharp maximum and full spectrum in Theorem 20.15; Proposition 20.16 resolves its origin transition. These results do not identify the unrestricted harmonic maximum.
Globally optimal target disks	Lemma 19.4 makes the radius-one disk the unique optimum over every complex center, simultaneously for all folding orders. Corollary 19.6 gives the unique largest quartic stability disk $\lvert\tau - 1/9\rvert < 1/27$ and the exact radius for each real center.
Sharp unit-triple power constants	Lemma 13.10 proves the optimal uniform coefficient $4/3$, with equality exactly at order three and the stated triples. Theorem 13.7 determines the individual best constants in every order not divisible by three, with all equality traces and the sharp asymptotic ratio $\Gamma \approx 1.11562380$. Orders $3, 6, 9, 12, 15$ are also solved, with every equality case; Proposition 13.9 gives the last two. Multiples of three starting at eighteen remain open. These constants do not determine an unknown global zero-count endpoint.
Optimal cubic shear and equality cases	Theorem 19.10 gives the exact analytic-quadratic shear radius $\min\{\sqrt{c/27}, 1 - (c/4)^{1/3}\}$ for $0 < c < 4$, uniquely maximized by $c = 27/16$ with radius $1/4$. It identifies all first-boundary parameters and their eight-point fibers.
Optimal quadratic shear and analytic sextic seed	Lemma 19.13 gives the largest uniform centered shear disk $\lvert\eta\rvert < 1$. Proposition 19.15 proves $E_{2,6} = 36$ with a rational-coefficient analytic family, replacing the certificate dependency for this existence statement. Theorem 19.16 subsequently treats every order at the cubic boundary.
Translated pure-head cubic boundary	Theorem 19.16 proves $E_{m,3k} = 9k^2$ for every $k \leq m < 3k$, including $E_{4,12} = 144$. Its weighted coefficient substitution and seed-ideal reduction preserve the pure highest part in arbitrary order; all examples can have real integer coefficients and maximal real-axis counts.
Paired cubic–quadratic seed	Theorem 21.3 improves $M_{4,11}$ from the radial lower bound 121 to 129. All 121 affine intersection multiplicities split into regular physical zeros; eight additional zeros occur at infinity after canceling the first corner and introducing an intermediate scale. Exact finite-field minors certify the bounded-degree jet interpolation. The upper endpoint 137 remains open.

Assertion in the successive drafts	Audited disposition and proof in this note
Globally sharp even rectangular degrees	Theorem 19.34 proves $M_{2,n} = 9n - 10$ for every even $n \geq 10$. For odd $n \geq 13$ only $9n - 12$ and $9n - 10$ remain possible. It also gives $M_{2,8} \in \{60, 62\}$, $65 \leq M_{2,9} \leq 71$, and $85 \leq M_{2,11} \leq 89$. All constructed counts have regular real integer witnesses.
Sharp degree tradeoff for eighty-four zeros	Corollary 19.32 proves that the complete undominated pairs are $(1, 18), (2, 12), (3, 10)$. In particular, conjugate degree two suffices in pure degree twelve, and degree eleven is impossible for that target at conjugate degree two.
Nearly matching pure-head bands	Theorem 19.35 proves $9n - 18 \leq E_{2,n} \leq 9n - 16$ for even $n \geq 8$, and $9n - 20 \leq E_{2,n} \leq 9n - 16$ for odd $n \geq 9$. Thus $E_{2,9} \geq 61$, $E_{2,12} \geq 90$, and $E_{2,15} \geq 115$. These regular real integer constructions leave the upper endpoints open. Corollary 19.36 gives sharp degree thresholds, including degree twelve for every target from 84 through 90.
Fixed forty-one-zero integer witness	Proposition 19.24 prints all coefficients at $a = 1/4$, $T = 2$, $\tau = 10^{-4}$. Exact rational contractions in 41 physical and eight nonphysical boxes exhaust all 49 complex intersections and verify signs $(24, 17)$ and seven real-axis zeros.
Forty-five zeros from a critical shared seed	Theorem 19.37 leaves $E_{2,7} \in \{45, 47\}$. Its shared cluster splits into a negative central zero and twelve regular zeros on a shrinking circle; the eight ordinary clusters contribute four each. The real integer witness has signs $(26, 19)$ and seven real-axis zeros. Proposition 19.38 proves the exact local maximum 45.
Forty-one zeros in total degree seven	Theorem 19.22 gives the earlier bracket $41 \leq E_{2,7} \leq 47$ by an explicit rational construction; Theorem 19.37 now raises the lower endpoint to 45. Its eight quadratic clusters have four zeros each and its shared cubic has nine, with all counts proved analytically. Total degree seven and conjugate degree two are minimal for this target; all seven possible real-axis zeros occur.
Eighty-eight zeros in total degree ten	Theorem 19.25 improves the bracket to $88 \leq E_{3,10} \leq 100$. Allowing a nonzero shared quadratic jet restores all eight nine-zero cubic clusters; the shared quartic has sixteen simple zeros. The real integer construction has signs $(49, 39)$ and ten real-axis zeros. Total degree ten and conjugate degree three are minimal for this target; the hundred-zero endpoint remains open.
Minimal conjugate degree for full saturation	Corollary 19.39 inverts the existing upper bound and proves $\mu(3k) = k$ for $1 \leq k \leq 8$, with exact losses after decreasing the degree. It does not claim new full-saturation examples at $n = 10, 11, 13, 27$.
Sharp three-term harmonic maximum in conjugate degree one	Theorem 20.20 gives $H^{(3)}_{1,n} = n + 2 + 2D(n + 1)$ for $n \geq 4$, with the two smaller values separately. Three terms attain the unrestricted maximum $3n - 2$ exactly in degrees $2, 3, 5$. The underlying gcd-family spectrum includes singular parameters; all maximizing examples can have real-integer coefficients and regular zeros.
Sharp three-term harmonic maximum in conjugate degree two	Theorem 20.24 treats every support and every degree. The distinct maximum is $\max\{n + 6, n + 3 + 2D(n + 2)\}$ for $n \geq 7$, with four explicit smaller values; the fully nondegenerate maximum is always $n + 6$. All extrema have real-integer examples.

Assertion in the successive drafts	Audited disposition and proof in this note
Higher lower-term trinomial families	Theorem 20.21 gives an exact gcd maximum, complete nonzero-parameter spectrum, and rational maximizing interval under an explicit angular criterion. It covers all $n > k > 2 = m$; unrestricted overlap cases for $m \geq 3$ are not declared solved.
Certified improvement at pure degree seven	Proposition 22.8 certifies exactly 26 regular zeros for a sixth-order contact seed, including all 36 complex intersections. Corollary 22.9 gives the exact attainable count 39; Theorem 19.22 subsequently improves the global lower bound to 41. The fixed parameter 10^{-24} has a separate certificate exhausting all 49 intersections. The maximum remains unresolved.
Cyclic contact obstruction	Proposition 22.7 gives the sharp distinct maximum 19 for the entire threefold-symmetric contact-five subspace, including singular parameters. The reduction uses the shared-zero logharmonic bound $3d - 3$ and retains the known period-two dynamical obstruction; this does not reduce $E_{2,5} = 25$.
Pure-head contact bounds	Proposition 22.5 gives the exact contact-splitting increment and upper bound $\min\{n^2, (4m+1)n - 2m^2 - 2q\}$. The $m = 1, q = n$ stratum has sharp distinct maximum $3n - 2$ and recovers the known $E_{1,n+1}$ endpoint by Geyer's construction and inversion.
Sharp reducible bounds and equality structure	Theorem 25.1 gives the exact quadratic deficit $2(m-1)$ whenever the neighboring rectangle saturates. Theorem 25.3 proves $M^{\mathrm{red}}_{m,n} = M_{m,n-1} + 1$ for $n \geq m^2 + 2$. This is an exact reduction, not a numerical evaluation of an unknown endpoint.
Pure-head factor-count strata	Theorem 25.5 bounds the number of factors by $n - m$ and the zero count by $(n - \ell + 1)^2 + \ell - 1$. It characterizes equality and determines the maximum at $\ell = n - m$ for every $n > m > 0$.
Optimized cone completion	Theorem 20.26 finds the largest count among all transfers from its explicitly stated seed set using at most three candidates. Corollary 20.27 improves three pure-head lower bounds and simultaneously attains the real-axis bound. It does not determine the full pure-head maxima.
The two-frequency and optimized radial constructions	Theorems 5.3–5.7 give a full sharp subfamily formula, stronger lower bounds, and uniform errors in all degrees. They do not identify the unrestricted maxima in every degree.
The rectangular coefficient $4m + 1$	Sharp asymptotically. The radial family is generally not harmonic; its mixed terms are essential to that particular construction.
The first equality band $m < n \leq 2m + 1$, and individual 40-, 64-, and 73-zero examples	Correct; subsumed by the larger band $m \geq 2$, $m < n \leq 2m + 2$. The general proof of the extra boundary uses the certified degree-eight input in Appendix B.
The earlier $(m, 2m + 2)$ construction with two fewer zeros	Its exact count is correct, as proved in Section 23, but the deficit is not universal. It is suboptimal for every $m \geq 2$.
The explicit 90-zero family at $(3, 9)$	Correct, including the fixed parameter 10^{-8}; Section 26. It has total degree twelve and does not itself prove $E_{3,9} = 81$.
The later cubic construction	Gives the stronger pure-head equality and an infinite rectangular family: (17.1). The exceptional first parameter $r = 1$ is treated separately rather than substituted into an invalid degree inequality.

Assertion in the successive drafts	Audited disposition and proof in this note
The 49-zero example at $(2, 7)$	Exactly 49, not merely at least 49, by certification of all 53 complex intersections. It does not prove extremality in the full rectangle. The reducible maximum is 41, now a case of the general factorization bounds in Section 25.
The monomial-head harmonic formula	Exact for $S_{m,n}$ only. The zero count is not invariant under arbitrary imaginary translations of the target within that restricted family.
The Chebyshev construction	Correct exact count for the constructed family; supplies a lower bound, not a general harmonic maximum. The resonant directions at infinity are included in its proof.
The real-valued perturbation formula	Exact for $A_{m,n}$, including its leading-form refinement. Allowing an independent imaginary lower part or an arbitrary analytic head changes the problem.
The harmonic linear improvement and the Newton-support extension	The former is a special case of (6.7). The proof requires the negative-double-point version of the genus inequality and a separate check of the link at infinity; the embedded statement alone is not enough.
The $n^2 - 2$ total-degree statement	Correct and sharp for a gap of two. Theorem 7.1 strengthens its upper bound for larger gaps; general equality for those gaps is not claimed.
The quintic sparsity counterexample	Correct, but the trinomial quartic gives an earlier and stronger counterexample. Degree four and three terms are both optimal for the first failure.
The even- and odd-degree sparse families	Attain the quadratic zero bound. Their number of terms is an upper bound for s_n, not a proved minimum in every degree.
The cyclic symmetry formula	Exact for the entire cyclic coefficient space, including complex radial coefficients. It is not a formula for all polyanalytic polynomials with the same degrees.
Circle addition, contact splitting, and zeros at infinity	Correct deformation tools, with the degree conditions explicitly stated. They do not guarantee the existence of an extremal starting polynomial with an arbitrarily prescribed contact order.
The rational degree-one endpoints	Already covered by [KN06, LSL15, LZ18, LM14]; they are included for completeness, not claimed as new.
The use of Descartes signs and Hermitian traces	Standard antecedents are cited. The simultaneous extremal counts, the particular low-degree trace identity, and the isotropic-space estimate are proved rather than attributed to the general framework alone.

27.3 The exact certificate inputs

Scope of the strengthening revision. The full trinomial parameter-plane theorem in Section 20.4 is proved analytically using a monotone critical-value argument. Its exact identities, symmetry checks, and one independent cubic elimination are in `verify_trinomial_global.py` and `trinomial_global_verification.json`. The same report labels the separate Newton samples as numerical diagnostics, not certificates. The endpoint checks are in `verify_trinomial_endpoint.py` and `trinomial_endpoint_verification.json`. Their exact finite checks test the unit-circle sign formula and the lifted critical-arc covering count; floating-point root samples remain explicitly separate from the analytic proof. The conjugate-degree-one sparse theorem is checked by `verify_`

`linear_harmonic_trinomials.py` and `linear_harmonic_trinomial_verification.json`; its interval witnesses, integer gcd optimization, and quintic radial certificate use exact arithmetic.

The higher-conjugate formulas and full degree-two support optimization are checked by `verify_separated_harmonic_trinomials.py`, `verify_quadratic_conjugate_sparsity.py`, and their `separated_harmonic_trinomial_verification.json` and `quadratic_conjugate_sparsity_verification.json` reports. The optimal quadratic shear and the analytic sextic seed are checked by `verify_quadratic_shear_sextic.py` and `quadratic_shear_sextic_verification.json`. The sharp unit-triple power constants for orders two through five are checked by `verify_low_order_power_constants.py` and `low_order_power_constants_verification.json`; all tests use exact symbolic identities and algebraic inequalities. The sixth-order optimum is checked independently by `verify_sixth_power_constant.py` and `sixth_power_constant_verification.json`. This audit verifies Newton's identity, the full stationary resultant, the reduced ideal, all boundary Bernstein coefficients, and rational isolating intervals; it gives $6.053587379096 < C_6 < 6.053587379097$. The seventh- and eighth-order constants are checked by `verify_seventh_eighth_power_constants.py` and `seventh_eighth_power_constants_verification.json`. All 52 angular Bernstein coefficients, both radial concavity certificates, Newton identities, and rational root isolations are verified exactly. The resulting certified intervals are $7.593974047523 < C_7 < 7.593974047524$ and $9.223593505382 < C_8 < 9.223593505383$. The ninth-order optimum is checked by `verify_ninth_power_constant.py` and `ninth_power_constant_verification.json`. The exact resultant, stationary ideal, Sturm variations, all boundary Bernstein coefficients, and rational critical-value bounds pass; the certified interval is $8.770533141272 < C_9 < 8.770533141273$. The infinite family in Theorem 13.7 is checked by `verify_nonmultiple_power_constants.py` and `nonmultiple_power_constants_verification.json`. The verifier checks eighteen finite orders, all 39292 coefficients in the uniform tensor certificate, and the rational geometric bounds for the entire remaining tail. It also compares the normalized coefficient formula with integer Newton recurrence in 4004 independent coefficient checks. The infinite-order conclusion uses the analytic tail proof, not an extrapolation from the checked orders. Orders twelve and fifteen are checked by `verify_twelfth_fifteenth_power_constants.py` and `twelfth_fifteenth_power_constants_verification.json`. The integer recurrence identities, complete resultant factorizations, all Sturm root counts, stationary ideal equalities, boundary Bernstein coefficients and rational critical-value intervals pass. The explicit degree-seven construction in Theorem 19.22 is checked by `verify_pure_2_7_41.py` and `pure_2_7_41_verification.json`. Exact reductions verify the polynomial interpolation formula, all shared jets and support bounds. Six rational sign intervals give the nine simple shared zeros, and the other eight quadratic counts are explicit. The forty-five-zero critical-seed construction is checked by `verify_pure_2_7_45.py`, with report `pure_2_7_45_verification.json`. It verifies the restriction identity, the shared jets, the scaled radial factorization, and the twelve simple angular roots by exact symbolic arithmetic. The proof of the global count is analytic; no fixed small parameter is claimed to have been certified numerically. The fixed forty-one-zero witness is verified independently by `verify_pure_2_7_41_fixed.py`, using `pure_2_7_41_fixed_certificate.json`; its report is `pure_2_7_41_fixed_verification.json`. The verifier uses only standard-library rational arithmetic, compares every coefficient with the printed integer table, and exhausts all 49 intersections. Four deliberate certificate corruptions are rejected. The family specialization is also reconstructed independently by exact quotient-ring

arithmetic. The restriction-map results in Section 19.14 and the following interpolation sections, and Theorem 19.35 are justified by analytic proofs: the exact kernel determines the full image, simple restricted zeros give the nonzero transverse jets, and separate scale balances count every escaping zero. Their seed inputs are the explicit radial construction and Lemma 19.28, including the printed fourteen-zero quartic. The cubic model has nine explicitly listed zeros and nonzero Jacobian determinants. These are not assertions based on sampled roots or numerical ranks. The improved degree-ten construction in Theorem 19.25 is checked independently by `verify_pure_3_10_88.py` and `pure_3_10_88_verification.json`. Every rational interpolation residual vanishes; both representative cubic trace forms have nine positive pivots. Ten exact sign intervals and rational interval evaluations prove the shared quartic's sixteen simple zeros. The final small-parameter confinement argument is analytic; no numerical value for that final parameter is claimed. The earlier construction in Theorem 19.19 is checked by `verify_pure_3_10_84.py` and `pure_3_10_84_verification.json`. Both interpolation systems are solved over the rationals and every residual is zero; the ranks are 27 and 11. Exact trace forms have inertia $(9, 0)$ and $(8, 1)$, proving local counts nine and seven without sampled root searches. The shared jets and all degree restrictions are also checked. The small-parameter confinement proof is analytic; no particular numerical value of its final parameter is asserted. The short real integer extremizer in Proposition 24.2 has its own forty-box certificate, `real_2_6_40_certificate.json`; the exact verifier `verify_real_2_6_40.py` records the successful checks in `real_2_6_40_verification.json`. It gives a compact fixed formula attaining the already sharp value $M_{2,6} = 40$. The complete quadratic shear classification in Proposition 19.14 is checked by `verify_quadratic_shear_region.py` and `quadratic_shear_region_verification.json`. The exact affine substitution and discriminant factorization reduce every parameter, including all six cusp cases, to the complete local fiber theorem. The global deltoid inradius and quartic stability disks are checked by `verify_deltoid_inradius.py` and `deltoid_inradius_verification.json`. Exact identities verify the three-contact circle polynomial, coefficient comparison, boundary parametrization, distance derivative, affine discriminant relation, and all nearest quartic parameters. The maximal-circle argument itself is the analytic proof of Lemma 19.4. The optimal all-order coefficient in Lemma 13.10 is proved analytically from phase control, sine concavity, and the strict monotonicity of $\sin t/t$; no finite-order test is substituted for that uniform argument. The sharp cubic shear disk, its optimal constant, and all nearest fold parameters are checked by `verify_cubic_shear_radius.py` and `cubic_shear_radius_verification.json`. The all-order translated cubic boundary is checked by `verify_translated_cubic_boundary.py` and `translated_cubic_boundary_verification.json`; orders two through twelve pass exact coefficient, weighted-degree, reduction, and support checks. The arbitrary-order stability argument is analytic. The printed rectangular 40- and 49-zero examples are independently reverified by `verify_rectangular_certificates.py`; both full complex intersections are exhausted, and the results and hashes are in `rectangular_certificate_verification.json`. The fixed-parameter degree-twelve extremizer has the separate exact certificate `pure_4_12_certificate.json`, the standard-library verifier `verify_pure_4_12.py`, and the successful report `pure_4_12_verification.json`; see Proposition 19.40. The separate contact-six example is checked by `verify_pure_contact_2_6_26.py`, with coefficient and box data in `pure_contact_2_6_26_certificate.json` and successful results in `pure_contact_2_6_26_verification.json`. Its 26 real and ten nonreal boxes exhaust all 36 complex intersections using exact fractions. This supplies a fixed thirty-nine-zero example; the stronger lower bound 41 is proved by Theorem 19.22. The pure-head contact checks are in

`verify_pure_contact_bounds.py` and `pure_contact_verification.json`. All numerical winding and Newton samples are marked as diagnostics; none is used to certify the remaining global endpoints.

The factorization bounds, equality classifications, and optimized completion envelope have analytic proofs in Sections 20.10 and 25. The independent finite degree enumerations and symbolic identities are in `verify_factorization_and_completion.py` and `factorization_and_completion_verification.json`. These diagnostics are not certificates for an unresolved global maximum.

The harmonic minimum, common-critical-point bound, exact-degree pure-head spectrum, exact trinomial radius, and completion corollary have analytic proofs in Sections 20.1–20.9. Their finite exact checks are in `verify_harmonic_minimum.py` and `harmonic_minimum_verification.json`.

The attainable-count spectra and optimal cubic-radius theorem have analytic proofs. Their exact diagnostics are in `verify_spectra_and_cubic_radius.py` and `spectra_and_cubic_radius_verification.json`. The harmonic no-gap theorem is an existing result of [SZ21]; the coefficient-space extensions and rational fold construction are proved explicitly in Section 20.

The exact local fiber classification, real-coefficient splitting, strengthened inversion, and linear rectangular cone in Section 19 have complete analytic proofs. The script `verify_exact_local_models.py` and its report `exact_local_models_verification.json` give finite exact-algebra and orbit diagnostics for the current construction. The earlier `verify_linear_sharp_region.py` checks the shifted recurrence; `verify_universal_splitting.py` checks the still earlier correction-series construction. Their historical checks are not presented as reruns of the present unshifted recurrence.

The further quartic theorem in Section 18 has a complete analytic proof. The file `verify_quartic_splitting.py` checks its local expansion, degree cancellations, phase calculation, and normalized error identities; `quartic_splitting_verification.json` records the completed checks. It is not a numerical root certificate for a fixed choice of the small parameters.

The newly added claims in Section 5.1 have analytic proofs. Their separate exact-arithmetic diagnostics are recorded in `new_results_verification.json` and reproduced by `verify_new_results.py`. The original supplementary certificate archive was not among the four files supplied for this revision. The pre-existing verification statements below are retained from the source note; they are not statements that those older programs were rerun during the strengthening revision.

The logical role of computation is limited and explicit. The symbolic proofs of the rectangular upper bound, its asymptotic coefficient, the support and gap bounds, the harmonic and real-valued counts, and the cubic, quartic, and arbitrary-order families do not depend on floating-point searches. The completion of the entire boundary $n = 2m + 2$ uses one finite rational input: the 64-zero polynomial and two auxiliary inequalities in Appendix B. All subsequent degrees in that argument follow by analytic iteration and odd-degree composition.

Other exact certificates provide explicit finite-parameter examples, not new logical assumptions for those analytic theorems. The reproduction archive contains the original rational inputs, independently executable programs, and the logs of the reruns performed during this consolidation. Its machine-readable manifest records the SHA-256 digest of each file. Hashes identify the actual input; they do not replace a proof of the verification algorithm, which is given in Section 24 and the appendices.

The following table retains the original verification record and adds the new diagnostics. The older entries have the historical scope stated above; finite checks are not treated as universal proofs.

Check	Content and scope
Global trinomial family	Four symbolic identities, 20300 exact positivity checks, 161700 symmetry-orbit checks, and one exact cubic elimination. Nineteen Newton samples independently agree with the parameter-plane counts; those samples are diagnostics only.
Equal lower-degree trinomial endpoint	Three symbolic identities, 22200 exact unit-circle counts, 2775 orbit/parity checks, 13875 lifted-arc counts, and 921 real-parameter comparisons. Separately labelled diagnostics check 612 explicit root substitutions and eleven Newton counts.
Linear-conjugate harmonic trinomials	Four exact identities, 2701 rational maximizing-interval witnesses, and 1997000 gcd-optimization candidates. An independent radial/Sturm certificate gives thirteen zeros for $12z^5 - 12z^2 + 5\bar{z}$. Thirty-six Newton samples are separately labelled diagnostics.
Separated higher-conjugate trinomials	Ten symbolic identities and 9696 exact rational interval/orbit checks, including all 1653 tested conjugate-degree-two pairs. The displayed degree-eight parameter satisfies exact rational inequalities. Twenty-seven Newton samples are diagnostics.
All three-term supports in conjugate degree two	Three symbolic identities, 1994997 exact support-optimization candidates, 997 even-degree formulas, and three rational maximizing examples. Thirteen Newton examples and caustic winding grids in six degrees are separately labelled diagnostics.
Translated pure-head cubic boundary	Exact checks in eleven orders, from two through twelve: translated coefficient identities, divisibility of coefficient changes, seed-ideal reductions, every term's support, and the pure highest part. No numerical root search enters the proof.
Certified pure-head degree twelve	An independent reconstruction of all 43 coefficients and exact rational contraction tests in 144 disjoint boxes. All boxes pass $q < 10^{-63}$ and $\|YF(c)\|_\infty / r < 10^{-60}$; the verified orientation counts are $(78, 66)$ and the real-axis count is 12.
Cubic shear disk	Twenty-nine exact symbolic identities verify the critical-angle and modulus elimination, the feasible intervals, the positive-quartic certificate, the unique optimal radius, and both fold expansions. The proof covers every real constant term and every complex shear parameter in the sharp disk.
Quadratic shear and sextic seed	An exact cofactor expansion and support check verify the pure term z^6 and conjugate degree two; eighteen rational Sturm intervals verify representative cubic seeds. The count 36 follows analytically from the optimal local disk and the implicit function theorem.

Check	Content and scope
Fixed pure-head degree-seven example	An independent reconstruction from the printed contact-six coefficients and $\varepsilon = 10^{-24}$; exact contractions in 39 real and 10 nonreal boxes; all 49 intersections exhausted; signs $(23, 16)$. The independent standard-library verifier uses no numerical root finder.
Cyclic contact-five obstruction	The exact covering identity and complete monomial enumeration, with rational Sturm checks of the integer logharmonic witness in degrees 2 through 20. The all-degree bound and its singular extension are analytic; `verify_cyclic_contact.py` records the algebraic audit in `cyclic_contact_verification.json`.
Pure-head contact	22350 exact bound comparisons, 23 polynomial inversion identities, and a Sturm/radial check of the nine-zero cubic contact modification. The arbitrary-degree sharp endpoint uses the cited critically fixed construction.
Factorization and optimized completion	Four symbolic identities; 2262070 split-degree inequalities; 1143801 seed transfers in 4266 independently optimized envelopes; 2991 subdiagonal formulas, 2970 strict gain checks, and 5049 factor-degree partition optimizations. The universal statements and realization arguments are proved analytically.
Harmonic and exact-degree minima	Nine symbolic identities; 20825 rational radius comparisons; 680 derivative gcds; 4495 boundary orbit counts; 79 two-zero-family checks and one exact elimination; 167480 exact-degree padding checks; 2400 completion degree checks and 299 gain comparisons. The local-index and continuation proofs are analytic.
Spectra and optimal cubic radius	Exact critical-parameter identities; 14399 rational radius inequalities; the Hermite interpolation determinant; two explicit fold fibers; 30 rational zero-and-Jacobian corrections. These diagnose algebraic steps, not the analytic genericity argument.
Exact local fibers and real-coefficient splitting	Fourteen discriminant–Jacobian identities; 80 smooth-boundary orders; 240 cusp cases; 19 orders of real and nonreal scaling; 28 exact global degree cases; 16 real circle-restriction ranks. Finite diagnostics, not numerical root certificates or substitutes for all-degree proofs.
Earlier shifted splitting and linear cone	Thirty exact global degree instances; 19 local scaling identities; eight seed-ideal identities; 81345 inversion monomials; 1000000 degree pairs with $m \leq 1000$. Checks of the preceding shifted construction and of the unchanged degree arithmetic.

Check	Content and scope
Earlier arbitrary-order diagnostics	Twelve degree instances; eleven cyclotomic model orders; 780 intermediate cancellation identities; 56 weighted-error checks; 250000 integer constructions; 17920 fourth-power band pairs. Historical checks of the preceding correction-series proof.
Paired-seed rectangle $(4, 11)$	Exact formal inversion and minors of sizes 51, 37, 13 modulo 461, with determinants 79, 260, 350; the modified solution has zero corner, two linear infinity coefficients 349, and inner coefficients $247, 251$. The original corner is 47. These certify algebraic interpolation. The exact 129-zero count, including eight escaping zeros, and the intermediate scale are proved analytically.
Rectangular 40 and 49	All real boxes; the four additional nonfixed complex boxes for 49; modular coprimality tests; orientation signs.
Pure-head 36; pure-head 64 and rectangular 73	Exact contraction inequalities on every rational box.
Full-band auxiliary input	All 64 seed boxes and both auxiliary inequalities throughout each box, not just at its center.
Fourteen-term 90	Both polynomial-ideal identities and all 90 boxes at $t = 10^{-8}$.
Harmonic sign counts	2016 degree pairs and 11466 sparse coefficient-sign patterns; diagnostic checks in addition to the all-degree proof.
Harmonic 20 and Chebyshev 17	Exact Sturm calculations over the indicated real quadratic fields.
Degree-gap trace calculation	20 coefficient sets, 470 trace identities, and 149 isotropic matrix entries, using exact arithmetic.
Newton formulas	7006 integer triples, checking lattice, genus, and constant expansions; not a substitute for the surface proof.
Real-valued construction	Exact quartic check; the rerun used the reduced exhaustive ranges of 1225 degree/form triples and 8184 binary cycles. The default larger exhaustive run was not completed and is not recorded as passed.
Sparse constructions	Exact quintic factorization; odd-degree Sturm tests; recurrence identities and cyclotomic quotient checks; the quartic and cyclic checks from the final sparse supplement.
Cubic family	The cubic discriminant identities, 1860 rational endpoint comparisons, cancellation checks for $2 \le r \le 8$, and every special $r = 1$ correction.

27.4 What the source audit does and does not establish

A source locator is meaningful only together with a version. For example, page 16 of the published Sète–Zur article contains Remark 4.4; the corresponding arXiv version has different pagination. Likewise, Lazebnik–Lundberg is cited as its 2026 published article, whereas the Lee–Makarov pole-prescription theorem is cited as the specified 2014 preprint. Lu–Wang is a 2025 publication despite its 2026 arXiv posting. No DOI was identified in the checked sources for the Peretz–Schmid

conference article or the Passare–Rojas–Shapiro journal article; no DOI has been manufactured from a journal's usual URL pattern.

The most consequential source distinctions are these: the polynomial conjecture is on the first page of Eremenko's note, not only in a MathOverflow discussion; the stronger orientation-reversing conjecture in [LLL15] remains separate; and average Gaussian zero counts do not settle maximal valence. The current public status notice in [Ere26b] records a solution of the qualitative estimate, but does not by itself establish attribution or validate every sharper formula in this note. This review found no cited theorem that makes all of the new constructions and bounds redundant. That statement is not an exhaustive priority search through unpublished work.

27.5 Conclusion

The distinction between the resolved growth problem and the unresolved finite-degree maximum should remain visible in any later manuscript. The exact results for smaller coefficient spaces are substantial in their own right; they need not be overstated as answers to larger questions.

Conclusion. Fixed conjugate degree gives a sharp asymptotically linear bound for the full rectangular class. Unrestricted splitting and inversion now prove $M_{m,n} = m^2+n^2$ throughout $m < n \le 3m$, except for $n = 3m-1$ (whose first three cases are already known). Pure-head saturation holds at $(m,n) = (kr, k(2r+1))$ for every $r, k \ge 1$. In particular, the entire pure-head boundary $E_{k,3k} = 9k^2$ is sharp, including $E_{4,12} = 144$, by the translated coefficient construction. The pure-head deficit is at most $4n-6$ throughout $m < n \le 3m$, so the quadratic bound is also uniformly asymptotically sharp there. The principal infinite families and rectangular cone admit integer-coefficient extremizers with maximal real-axis counts. Their local folding model has a complete sharp fiber classification, and both cubic perturbation families have optimized uniform stability radii. For the analytic quadratic shear, the best radius is exactly $1/4$, and every exceptional first-boundary fiber has exactly eight zeros. Every admissible regular count below the proved maxima is attained; the rectangular and pure-head finite distinct spectra are complete. For exact positive conjugate degree the latter starts at two; the one-point case is characterized by analytic pure powers. The harmonic distinct spectrum now has the exact lower endpoint $n-m+1$, with a sharp bound on the total deficit over all targets. The cases $m = n-1$ have the complete spectrum $2, 3, \dots, n^2$. The trinomial family with $1 \le k < m < n$ now has a complete parameter-plane count and exact maximum $n+2m+k+1$, attained by the integer parameter 3. The remaining endpoint $k = m$ also has an exact maximum and full spectrum; its gcd-dependent jump and real-versus-complex distinction are resolved, including the singular-origin fibers on the unit circle.
For harmonic polynomials with conjugate degree one and at most three terms, a divisor formula gives the exact maximum in every degree, including singular polynomials. The unrestricted maximum can be attained with three terms only in degrees $2, 3, 5$. The analogous three-term problem in conjugate degree two is also completely sharp: the distinct maxima depend on the largest proper divisor of $n+2$, while the fully regular maximum is exactly $n+6$ in every degree. Pure-head contact splitting gives additional upper bounds and a sharp maximal-contact stratum. Sharp factorization bounds classify reducible equality cases and determine the pure-head maximum at each proved factor-count stratum. For large analytic degree the reducible rectangular maximum is $M_{m,n-1}+1$. Optimized cone completion also

improves pure-head lower bounds at $n = 4m-2, 4m-1, 4m$. The individual unit-triple constants are determined in every order not divisible by three, including their sharp asymptotic ratio $\Gamma \approx 1.11562380$. Individual multiples of three starting at eighteen remain open; orders twelve and fifteen are also sharp. The paired seed also gives the new bound $129 \le M_{4,11} \le 137$. The critical degree-seven construction gives $E_{2,7} \in \{45, 47\}$, with minimal degrees for forty-five zeros and all seven possible real-axis zeros. Its neighborhood maximum 45 is sharp; the global forty-seven-zero endpoint remains unresolved. The shared cubic–linear seed, with its quadratic jet retained, improves the bracket to $88 \le E_{3,10} \le 100$. Its eighty-eight-zero target has minimal total degree ten and, at that degree, minimal conjugate degree three. The construction also attains all ten possible real-axis zeros. The hundred-zero endpoint remains open. The entire even-degree rectangular family is now globally sharp: $M_{2,n} = 9n - 10$ for every even $n \ge 10$. For odd $n \ge 13$ the remaining gap is at most two. Also $M_{2,8} \in \{60, 62\}$, $65 \le M_{2,9} \le 71$, and $85 \le M_{2,11} \le 89$. The selective pure-head constructions attain $9n - 18$ zeros in every even degree $n \ge 8$ and $9n - 20$ in every odd degree $n \ge 9$, leaving gaps two and four. In particular, they give 61 at $(2, 9)$, 90 at $(2, 12)$, and 115 at $(2, 15)$. For at least eighty-four zeros, the complete degree tradeoff is sharp: the undominated pairs are $(1, 18), (2, 12), (3, 10)$. The remaining global boundary, the full bidegree formula, and the optimal fixed-degree harmonic coefficient remain undetermined here.

A Rational coefficients and verification files

The following arrays completely specify the two polynomials in Theorem 24.1. With row index ℓ and column index j, write

$$P(z, w) = D^{-1} \sum_{\ell,j} (u_{\ell j} + i v_{\ell j}) z^{\ell} w^{j}.$$

All entries are integers, not rounded decimal approximations.

The polynomial P_{40}

Here $D = 10^{10}$ and $0 \le \ell \le 6$, $0 \le j \le 2$.

Real numerators

ℓ	$u_{\ell 0}$	$u_{\ell 1}$	$u_{\ell 2}$
0	9989148	−1437879431	42110613567
1	−1488721221	85880077969	44531370007
2	41495821383	181048684005	−8945073549
3	127109446221	24463634439	−54584415742
4	70927784116	−90270783729	−13319637790
5	−30642879880	−58978230217	12977971008
6	171904696	8861758778	10000000000

Imaginary numerators

ℓ	$v_{\ell 0}$	$v_{\ell 1}$	$v_{\ell 2}$
0	2386743	−536302879	19849756729
1	−170102520	20587381168	26848909014
2	1758707049	27892403146	9999884021
3	−18670288098	19443937659	−6680374702
4	−10494108981	26168926864	−13516927630
5	7007756920	1936949230	−10640650833
6	4236054207	1869520737	0

The polynomial P_{49}

Here $D = 10^3$ and $0 \le \ell \le 7$, $0 \le j \le 2$.

Real numerators

ℓ	$u_{\ell 0}$	$u_{\ell 1}$	$u_{\ell 2}$
0	−42	−176	−139
1	−227	933	−432
2	−179	−1457	737
3	794	3712	353
4	−2958	3048	−1571
5	7455	1348	−1396
6	−448	−5235	−166
7	−68	158	1000

Imaginary numerators

ℓ	$v_{\ell 0}$	$v_{\ell 1}$	$v_{\ell 2}$
0	−33	−479	−2287
1	−262	−1987	653
2	−1501	−8252	965
3	−4456	4656	3427
4	−7424	1665	−1101
5	1057	8157	−195
6	−1669	−264	−2045
7	363	697	0

Reproducibility

The two data files

```
sharp_2_6_40_certificate.json
sharp_2_7_49_certificate.json
```

contain these same integer coefficients, all rational box centers, their common denominator 10^{40}, and the radius 10^{-20}. The second file also contains the four nonreal centers. These boxes have been regenerated directly from the printed coefficients and independently reverified. Place both data files beside the common verifier and run

```
python verify_rectangular_certificates.py
```

Only the Python standard library is used. The arithmetic in the verification is performed with exact fractions and finite fields. The script rejects execution with Python's optimization option, since disabling assertions would invalidate the verification. It checks all hypotheses used in Section 24, including exact agreement with both printed coefficient arrays, bidegree, coprimality, invertibility of the center Jacobians, box separation, contraction, and orientation. For P_{49} it also checks separation of the four nonreal boxes from the fixed plane and exhausts all 53 complex intersections. The successful report is `rectangular_certificate_verification.json`; it includes SHA-256 hashes of both data files and the verifier.

The harmonic example of Section 8 has a separate verifier, `verify_harmonic_20.py`, which requires SymPy and uses exact arithmetic over $\mathbb{Q}(\sqrt{3})$. The supplementary program `check_harmonic_signs.py` checks the finite sign counts for all $1 \le m < n \le 64$ using integer and rational arithmetic. This last check is a regression test, not a substitute for the proof for arbitrary degrees. Neither harmonic construction depends on numerical root finding.

B The finite rational certificate

This appendix makes the computer-assisted input in Proposition 14.2 reproducible. The only starting polynomial requiring certification is

$$f_0(z,w) = z^8 + 10^{-18}\sum_{j=0}^{3}\sum_{i=0}^{7-j}(a_{ij}+ib_{ij})z^i w^j, \tag{B.1}$$

with integers from Table 5. The two new auxiliary polynomials are specified by

$$A_0(z,w) = 10^{-8}\sum(\alpha_{ij}+i\beta_{ij})z^i w^j, \qquad S_0(z,w) = 10^{-8}\sum(\sigma_{ij}+i\tau_{ij})z^i w^j, \tag{B.2}$$

using Tables 6 and 7. Every omitted coefficient is zero. In particular, the coefficient of w^5 in A_0 and the coefficient of w in S_0 are exactly one, not approximations.

B.1 Existence of the 64 seed zeros

The file `E3_8_64_certificate.json` contains the integer coefficients in (B.1) and 64 centers $c_\nu = (X_\nu/10^{60}, Y_\nu/10^{60})$. Put

$$B_\nu = c_\nu + [-\rho,\rho]^2, \qquad \rho = 10^{-30}.$$

The verifier checks that the boxes are pairwise disjoint and avoid the origin. For

$$V(x,y) = (\operatorname{Re} f_0(x+iy, x-iy), \operatorname{Im} f_0(x+iy, x-iy)),$$

write $J_\nu = DV(c_\nu)$, $K_\nu = J_\nu^{-1}$, and $\eta_\nu = \|K_\nu V(c_\nu)\|_\infty$. All these quantities are rational. If $f_0 = \sum a_{ij} z^i w^j$, set

$$R_\nu = |c_{\nu,x}| + |c_{\nu,y}| + 2\rho, \quad H_\nu = \sum_{i+j\ge 2}(i+j)(i+j-1)(|\operatorname{Re} a_{ij}| + |\operatorname{Im} a_{ij}|)R_\nu^{i+j-2}.$$

Every second real partial derivative of either component of V is bounded by H_ν on the box. Thus

$$\|DV(v) - DV(c_\nu)\|_\infty \le 4\rho H_\nu, \qquad q_\nu = 4\rho\|K_\nu\|_\infty H_\nu.$$

For every box, the exact program checks

$$q_\nu + \eta_\nu/\rho < 10^{-12}. \tag{B.3}$$

It follows from (B.3) that $T_\nu(v) = v - K_\nu V(v)$ maps B_ν strictly into itself and is a contraction. Its unique fixed point is a zero of V. Moreover, $\|I - K_\nu DV(v)\|_\infty < 1$ on the whole box, so the Jacobian is invertible there. This proves nondegeneracy as well as existence. The orientation counts are 36 positive and 28 negative, consistent with degree eight. They are not used as a substitute for the contraction certificate.

B.2 The auxiliary inequalities on entire boxes

For a polynomial $P = \sum p_{ij} z^i w^j$, a box center $c = (x, y)$, and $R = |x| + |y| + 2\rho$, define the rational number

$$L_P = \sum_{i+j\geq 1} (i+j)(|\operatorname{Re} p_{ij}| + |\operatorname{Im} p_{ij}|)R^{i+j-1}, \qquad \delta_P = 2\rho L_P.$$

Each real component of $P(z,\bar z) - P(c,\bar c)$ has absolute value at most δ_P in the box. If $P(c,\bar c) = u + iv$, the rational numbers

$$\begin{aligned} \ell_P &= \max(0, |u| - \delta_P)^2 + \max(0, |v| - \delta_P)^2, \\ u_P &= (|u| + \delta_P)^2 + (|v| + \delta_P)^2 \end{aligned}$$

therefore satisfy $\ell_P \leq |P(z,\bar z)|^2 \leq u_P$ throughout the box. The symbol u_P here denotes a bound, not a polynomial variable.

Using the exact polynomials B_0, C_0, D_0, e_0, d_0 in Proposition 14.2, the verifier checks, in every box,

$$\ell_{B_0} > 0, \quad \ell_{d_0} > 0, \quad u_{A_0} < \left(\frac{3}{50}\right)^2 \ell_{B_0}, \quad \left(\frac{53}{50}\right)^2 u_{S_0} u_{B_0} < \left(\frac{7}{10}\right)^2 \ell_{d_0}. \tag{B.4}$$

The first modulus inequality in (14.7) follows immediately. It implies $|A_0| + |B_0| < (53/50)|B_0|$, so the last comparison in (B.4) gives the second modulus inequality. This proves the exact finite hypotheses needed for the iteration, including nonvanishing of B_0 and d_0.

B.3 Verification files and scope

The programs `verify_seed.py` and `verify_auxiliary.py` verify the seed boxes and the auxiliary inequalities, respectively. Both JSON input files are supplied. Running the wrapper

```
python verify_all.py
```

checks all degree conditions, the 64 root boxes, and every comparison in (B.3) and (B.4). It uses Python's standard-library `Fraction` type for every mathematical comparison. The auxiliary verifier refuses to run with Python's optimization flags that would disable its assertions. Informational decimal ratios printed at the end do not enter the proof. The complete exact run has been performed and its output is included.

The verifier supplies only the finite assertion in Proposition 14.2. Proposition 14.1, Corollary 14.3, and Proposition 15.1 are the proofs in all higher degrees. No extrapolation from finite numerical experiments is involved.

Table 5: Exact integers in (B.1). The coefficient of z^8 is one; every coefficient not specified here or by that term is zero.

i	j	a_{ij}	b_{ij}
0	0	325069001102644	67963801878226
1	0	450588204151335	-808429770939430
2	0	14923899097622392	-5215338168864671
3	0	-4193995275619317	-5607909027127370
4	0	-2285437933666029	1137114477688244
5	0	-256070329168631650	47437728771068840
6	0	39523562708151494	64053725075318900
7	0	-5060183014488084	-65745045532666410
0	1	4177129780392874	-1023578682503584
1	1	-3393291315201153	1031467504051507
2	1	1430170989562624	19247381823938400
3	1	-344982929090280050	106363039160048020
4	1	34205364104215395	44197369842432850
5	1	-33034647699973135	-122080649425548670
6	1	2542780791248466000	-411360496357260800
0	2	1072493974482530	2260721271437080
1	2	-87741687798084990	29388032582878587
2	2	-8388736302391480	-16179666096049084
3	2	-44302401572865020	-131522667442191320
4	2	1973476089687808700	-614554296250387700
5	2	88015603513326930	68500967995269020
0	3	-1048236220001323	-1254186823454144
1	3	-12625398733567206	-19057910533194857
2	3	449481778934072500	-191881909553323800
3	3	84177992300826730	54870976261917030
4	3	7730029550774110	45273336743991960

Table 6: Exact integers defining A_0 in (B.2); the denominator is 10^8.

i	j	α_{ij}	β_{ij}
0	0	-365212	-192995
1	0	-3524147	-1005031
2	0	-1938007	-2579631
3	0	-1503160	4757283
4	0	-1735632	-1234227
5	0	959672	20316700
0	1	-2286231	-3022935
1	1	3964573	2244562
2	1	38217049	14420232
3	1	17236977	41263280
4	1	1994439	-18949910
0	2	-11343430	-4404664
1	2	21665754	32721321
2	2	-12502311	3057533
3	2	-11767481	12691914
0	3	5185651	-3728884
1	3	133662149	42271688
2	3	8234474	7491773
0	4	18517574	20072885
1	4	-12040293	25284423
0	5	100000000	0

Table 7: Exact integers defining S_0 in (B.2); the denominator is 10^8.

i	j	σ_{ij}	τ_{ij}
0	0	-10030551	-894598
1	0	6150734	-2984218
2	0	-134397669	-948638
3	0	15813796	87467028
0	1	100000000	0

C Two additional fixed-parameter certificates

The following examples were used at intermediate stages of the research. Their extremal values are subsumed by Theorem 1.3, but the shorter fixed-parameter representatives remain useful for independent reproduction. These are polyanalytic, not harmonic, examples.

First put

$$\alpha = 3 - 6i, \qquad \beta = -4 + 32i, \qquad H = z^3 + \alpha z w + \beta,$$

and define

$$D = z^3 w^2 + \beta w^2 - 45 z^2 w + (204 + 72i) z, \qquad F_{36} = H^2 + \frac{1}{1000} D.$$

The explicit identity $D = w^2 H - \alpha z H^*$ cancels the forbidden highest conjugate term. The resulting polynomial has total degree six, unique highest homogeneous term z^6, and conjugate degree two.

The file `sharp_even_diagonal_36_certificate.json` and its verifier certify 36 disjoint simple real-slice zeros by the contraction argument of Section 24. There are 21 positive and 15 negative zeros. The pure-head bound proves that these exhaust the zero set, giving a fixed rational representative of $E_{2,6} = 36$.

Second, use precisely the polynomial f_0 of (B.1), and put

$$G_{73} = f_0 + (3 + 6i)10^{-20} z^7 w^2 - (4 + 32i)10^{-60} z^8 w^3.$$

The companion certificate `M3_8_73_certificate.json` gives 73 disjoint simple real-slice zeros, 39 positive and 34 negative. Since $73 > (8-1)^2 + (3-1)^2 = 53$, the transverse-zero criterion in Proposition 2.3 forces G_{73} and G_{73}^* to be coprime. Their bidegree intersection number is 73, so the certified zeros are the complete zero set. Thus this is a fixed rational representative of $M_{3,8} = 73$.

The programs for these two assertions are `verify_sharp_even_diagonal_36.py` and `verify_extremal_3_8.py`. Each is supplied with its original JSON input in the reproduction archive. The first also checks the coefficient formula above; the second verifies both the 64- and the 73-zero inputs. No approximation is used in the contraction comparisons.

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